



CLC

Educational Institute

DSAT FOUNDATION COURSE

Math Theory & Practice Companion

Building the Foundation for the PSAT — and Beyond

All 29 skills across Algebra • Problem-Solving & Data Analysis • Advanced Math • Geometry & Trigonometry

Written for students beginning their SAT/PSAT journey — explanatory, bullet-point theory (no dense paragraphs), Greek terminology for every skill, and 8 practice problems per skill (4 Easy + 4 Medium) with full worked solutions.

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The CLC Digital SAT Team

CLC Educational Institute



Welcome to the Foundation Course

Why This Course Exists

- You're building the math skills that show up on the PSAT — and later, the full Digital SAT.
- Every skill starts with clear, bullet-point theory — no dense paragraphs to wade through.
- Each skill gives you 8 practice problems: 4 Easy to build confidence, then 4 Medium to start training the kind of thinking the real test rewards.
- Every problem is broken down the same way: Formula, step-by-step Solution, Tip, Trap, Process of Elimination, Shortcut, and a Desmos calculator tip.
- Key vocabulary is given in Greek for every skill, since precise vocabulary is the first step to understanding a question.

The 29 Skills

Domain	Skills Covered
Algebra	Lines · Systems of Linear Equations · Linear Inequalities · Absolute Value · Exponent Rules · Radicals, Rational Exponents & Scientific Notation
Problem-Solving and Data Analysis	Ratios & Proportions · Probability · Unit Conversions & Rates · Statistics · Data Displays · Scatterplots · Sampling, Bias & Margin of Error
Advanced Math	Functions · Composite & Inverse Functions · Transformations · Polynomial Operations & Factoring · Polynomial Division & Rational Expressions · Quadratics · Radicals & Rational Exponents · Exponential Functions · Radical & Rational Equations
Geometry and Trigonometry	Angles, Parallel Lines & Triangle Fundamentals · Right Triangles & Trigonometry · Radians & the Unit Circle · Circles · Volume, Surface Area & Similarity of Solids

How to Use This Book

- Read the terminology and theory bullets for a skill before attempting its problems.
- Try each problem yourself first — cover the solution and only check it after you've committed to an answer.
- Read the Tip and Trap even for problems you got right — they train the pattern recognition that speeds you up later.
- Use the Process of Elimination table to check your scratch work against the most common slips.
- Practice the Desmos tip on the actual Digital SAT calculator so it becomes automatic by test day.



How to Use the Desmos Calculator

The Digital SAT gives you a built-in Desmos graphing calculator for the entire Math section, inside the Bluebook testing app. Learn these six moves now, so that every 'Desmos Tip' in this book makes sense the moment you read it.

Typing & Basic Calculation

- Click into the input line and type any expression — Desmos calculates it instantly, just like a scientific calculator.
- Use ^ for exponents, sqrt() for square roots, and / for fractions — Desmos formats them automatically as you type.
- Press Enter to confirm a line and start a new one; each line keeps its own value or graph.

Graphing a Line or Curve

$$y = mx + b$$

- Type an equation with x and y (e.g. $y = 2x + 3$) and Desmos plots it immediately on the grid.
- Drag the graph to pan, and pinch or scroll to zoom — use this to read intercepts and check an answer visually.
- Click a point on the graph, or type the x-value on its own line, to see its exact coordinates.

Making a Table of Values

- Click the + menu and choose 'table,' or type x1, y1 on a new line to start one.
- Enter x-values in the left column; link the table to a typed function to see matching y-values fill in automatically.
- A table is the fastest way to test a few values without doing the arithmetic by hand.

Finding an Intersection (Solving Systems)

- Graph both equations on the same screen, then click directly on the point where the two graphs cross.
- Desmos labels the exact coordinates of that intersection — that point is the solution to the system.
- This works for two linear equations, and for finding where a line meets a parabola or a circle.

Using a Slider

- Type an equation with an undefined constant (e.g. $y = ax + 1$) and Desmos offers to turn 'a' into a slider.
- Drag the slider to watch how changing that value moves or reshapes the graph in real time.
- Use this to build intuition for how a, m, b, or other constants affect a graph's shape.

Regression — Line of Best Fit

$$y1 \sim mx1 + b$$



- Enter your data points into a table, one point per row.
- Type $y_1 \sim mx_1 + b$ beneath the table and Desmos fits the best line through the data and reports the values of m and b .
- Use this on scatterplot questions to find the equation of the line of best fit directly, instead of estimating by eye.

Golden Rules for Test Day

- The Desmos calculator is built into the Digital SAT's Bluebook app — practice on the real tool, not a lookalike.
- Reach for Desmos when a problem is faster to graph or table than to solve by hand, not for every single question.
- Always re-read the actual question after Desmos gives you a number — it is easy to correctly answer the wrong thing.
- Every worked problem in this book that benefits from Desmos includes a Desmos Tip telling you exactly what to type.

PART 1 — ALGEBRA

1. Lines

Lines — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Slope	Κλίση
Y-intercept	Σημείο τομής με τον άξονα y
Slope-intercept form	Μορφή κλίσης-τομής
Parallel lines	Παράλληλες ευθείες
Perpendicular lines	Κάθετες ευθείες
Point-slope form	Μορφή σημείου-κλίσης

Lines — What It Is

What It Is

- A line is any relationship where y changes by the same fixed amount every time x increases by 1.
- That fixed rate of change is called the slope, and it's the single most important number for any line.
- The SAT tests lines because they model steady, predictable change, like a constant hourly rate or a fixed fee.
- Almost every Lines question is really just asking you to find or use slope, an intercept, or both.
- You'll see lines as equations, as tables of points, as graphs, and as word problems — same math, different disguise.

Lines — Core Rules

Slope Formula

$$m = (y_2 - y_1) / (x_2 - x_1)$$

- Slope measures steepness: how much y changes for every 1 unit x increases.
- Pick two points, subtract the y-values on top and the x-values on the bottom.
- Keep the point order consistent — if y₂ comes from point 2, x₂ must too.
- Example: points (1, 4) and (3, 10) give $m = (10 - 4) / (3 - 1) = 6/2 = 3$.

Reading Slope's Sign and Special Cases



- Positive slope means the line rises as you move left to right.
- Negative slope means the line falls as you move left to right.
- A horizontal line has slope 0 — y never changes no matter what x does.
- A vertical line has undefined slope — you'd be dividing by zero.

Slope-Intercept Form

$$y = mx + b$$

- This is the most common way a line's equation is written on the SAT.
- m is the slope, telling you the rate of change (rise over run).
- b is the y -intercept, the value of y when $x = 0$ — where the line crosses the y -axis.
- Once an equation is in this form, you can just read off m and b directly.

Point-Slope Form

$$y - y_1 = m(x - x_1)$$

- Use this when you know the slope and just one single point on the line.
- Plug the point's coordinates in for x_1 and y_1 , and the slope in for m .
- Distribute and rearrange to convert it into $y = mx + b$ if the question needs that form.
- Example: slope 2 through $(3, 5)$ gives $y - 5 = 2(x - 3)$, which simplifies to $y = 2x - 1$.

Standard Form

$$Ax + By = C$$

- This form doesn't show slope or intercept directly, so it can be tricky to read at a glance.
- To find the slope, solve the equation for y to rewrite it as $y = mx + b$.
- Once solved for y , the slope is $-A/B$ and the y -intercept is C/B .
- Don't guess the slope from A and B without rearranging — it's easy to mix up the sign.

Parallel and Perpendicular Lines

$$\text{Parallel: } m_1 = m_2. \quad \text{Perpendicular: } m_1 \times m_2 = -1$$

- Parallel lines never meet, so they must have the exact same slope.
- Perpendicular lines cross at a perfect right angle (90 degrees).
- Perpendicular slopes are negative reciprocals — flip the fraction and switch the sign.
- Example: a slope of $2/3$ has a perpendicular slope of $-3/2$.

Finding Intercepts

$$\text{x-intercept: set } y = 0. \quad \text{y-intercept: set } x = 0.$$

- The x -intercept is where the line crosses the x -axis, so its y -value is always 0.



- The y-intercept is where the line crosses the y-axis, so its x-value is always 0.
- Substitute the 0 into the equation and solve for the one remaining variable.
- In $y = mx + b$, the y-intercept is always just b , with no extra work needed.

How to Recognize This on the Test

- The words 'slope,' 'rate of change,' or 'steepness' almost always mean this is a Lines question.
- Two coordinate points given together, like (2, 5) and (6, 13), signal you'll likely need the slope formula.
- Phrases like 'per hour,' 'per item,' or 'starting fee plus a charge per unit' describe a linear word problem.
- 'Parallel to' or 'perpendicular to' a given line means you need the slope relationship rules.
- An equation shown in $Ax + By = C$ form is a hint you'll need to rearrange it first.

Common Mistakes — and the Fix

- Subtracting coordinates in mismatched order ($y_1 - y_2$ over $x_2 - x_1$) flips the sign of the slope — always match the order.
- Reading slope by counting grid squares on a graph instead of using real coordinate values — axes can be scaled unevenly.
- Mixing up which number is the slope and which is the y-intercept in $y = mx + b$.
- Forgetting to rearrange standard form into $y = mx + b$ before trying to read off the slope.
- Using the same slope for a perpendicular line instead of flipping it and changing its sign.

Lines — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Lines — Problem 1 (Easy)

A bakery in Thessaloniki charges a setup fee plus a per-cake price for catering orders. The total cost in euros is $C = 15n + 40$, where n is the number of cakes ordered. The bakery raises its setup fee by 10 euros but keeps the per-cake price the same. What is the new total cost for an order of 6 cakes?

Formula / Rule Used: Slope-intercept form: $y = mx + b$

Solution

Step 1: Compare $C = 15n + 40$ to slope-intercept form $y = mx + b$.

Step 2: Here the slope is $m = 15$ (price per cake) and the intercept is $b = 40$ (setup fee).

Step 3: Raising the setup fee by 10 changes only b , not the slope m .

Step 4: New equation: $C = 15n + (40 + 10) = 15n + 50$.

Step 5: Substitute $n = 6$ into the new equation: $C = 15(6) + 50$.

Step 6: Multiply first: $C = 90 + 50$.

Step 7: Add: $C = 140$.

Answer: 140 euros

Tip

When a rate stays the same but a fixed fee changes, only update b in $y = mx + b$, then plug in.

Trap

It's tempting to add the 10 euro increase to the per-cake rate instead of the flat fee, which changes the wrong number.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
130 euros	Forgot to add the fee increase at all and used the original equation: $15(6) + 40 = 130$.
190 euros	Added the 10 euro increase to the slope instead of the intercept: $(15 + 10)(6) + 40 = 190$.
100 euros	Added only the 10 euro increase and left out the original 40 euro fee: $15(6) + 10 = 100$.

Shortcut

You can skip rebuilding the equation: just compute the old cost, $15(6) + 40 = 130$, then add the 10 euro fee increase to get 140.

Desmos Tip

Type $y = 15x + 50$ in Desmos and trace the graph to $x = 6$ to confirm y lands on 140.



A line passes through the points $(-3, 4)$ and $(5, -2)$. What is the slope of this line?

Formula / Rule Used: Slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$

Solution

Step 1: Label the points: $(x_1, y_1) = (-3, 4)$ and $(x_2, y_2) = (5, -2)$.

Step 2: Subtract the y-values on top: $y_2 - y_1 = -2 - 4 = -6$.

Step 3: Subtract the x-values on the bottom: $x_2 - x_1 = 5 - (-3) = 5 + 3 = 8$.

Step 4: Write the slope as a fraction: $m = -6 / 8$.

Step 5: Simplify by dividing top and bottom by 2: $m = -3 / 4$.

Answer: $-3/4$

Tip

Always write the two points directly under the slope formula before subtracting, so you don't lose track of signs.

Trap

The negative coordinate (-3) is easy to mishandle — subtracting $5 - (-3)$ is not the same as $5 - 3$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-3	Computed the denominator as $5 - 3 = 2$ instead of $5 - (-3) = 8$, then simplified $-6/2$ incorrectly.
$3/4$	Dropped the negative sign somewhere in the subtraction and reported a positive slope.
$-4/3$	Flipped the fraction, computing run over rise (x-change over y-change) instead of rise over run.

Shortcut

No real shortcut here — the slope formula is already the fastest path, just track the negative signs carefully.

Desmos Tip

Enter the two points as a table in Desmos, draw a line through them, and confirm the slope shown matches $-3/4$.

Lines — Problem 3 (Easy)



A line is given by the equation $3x + 4y = 24$. What is the x-intercept of this line?

Formula / Rule Used: x-intercept: set $y = 0$ in the equation.

Solution

Step 1: The x-intercept is the point where the line crosses the x-axis, so $y = 0$ there.

Step 2: Substitute $y = 0$ into the equation: $3x + 4(0) = 24$.

Step 3: Simplify: $3x + 0 = 24$, so $3x = 24$.

Step 4: Divide both sides by 3 to solve for x : $x = 24 / 3$.

Step 5: $x = 8$.

Answer: $x = 8$ (the point $(8, 0)$)

Tip

To find the x-intercept fast, just set $y = 0$ and solve — you don't need the y-intercept at all.

Trap

It's easy to set $x = 0$ instead of $y = 0$ by mixing up which intercept the question is asking for.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
6	Set $x = 0$ by mistake and solved for the y-intercept instead: $4y = 24$, $y = 6$.
24	Correctly set $y = 0$ but forgot to divide by 3, stopping at $3x = 24$.
-8	Misread the equation's sign and solved $3x = -24$ instead of $3x = 24$.

Shortcut

Cover-up trick: ignore the y-term entirely and divide the constant by x's coefficient, $24 / 3 = 8$, straight away.

Desmos Tip

Type $3x + 4y = 24$ into Desmos and read where the graph crosses the x-axis to confirm $x = 8$.

Lines — Problem 4 (Easy)



Line k has equation $y = 4x - 7$. Line p is parallel to line k and passes through the point (2, 9). What is the y-intercept of line p?

Formula / Rule Used: Parallel lines: $m_1 = m_2$, then $y = mx + b$.

Solution

Step 1: Parallel lines always have the same slope, so line p also has slope $m = 4$.

Step 2: Use slope-intercept form for line p: $y = 4x + b$.

Step 3: Substitute the point (2, 9) in for x and y: $9 = 4(2) + b$.

Step 4: Multiply: $9 = 8 + b$.

Step 5: Subtract 8 from both sides to solve for b: $b = 9 - 8$.

Step 6: $b = 1$.

Answer: 1 (line p crosses the y-axis at (0, 1))

Tip

Remember parallel lines share slope but almost always have a different y-intercept — always solve for the new b.

Trap

It's tempting to just copy line k's y-intercept (-7) instead of recalculating a new b for line p.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-1/4	Used the perpendicular slope rule by mistake, flipping and negating 4 to get -1/4 instead of keeping it 4.
17	Made a sign error solving $9 = 8 + b$, adding instead of subtracting to get $b = 9 + 8$.
-7	Copied the y-intercept from line k directly instead of computing the new intercept for line p.

Shortcut

No shortcut beyond plugging into $y = mx + b$ with the parallel slope — it's already quick.

Desmos Tip

Type $y = 4x - 7$ and $y = 4x + 1$ in Desmos and confirm the second line passes through (2, 9) while staying parallel.

A vintage record shop in Berlin tracks its weekly profit. In week 3, its profit was -120 euros (a loss). In week 9, its profit was 240 euros. Assuming profit changes at a constant rate, in which week does the shop first break even (profit = 0)?

Formula / Rule Used: Slope formula + point-slope form, then set $y = 0$.

Solution

Step 1: Treat the two situations as points: (3, -120) and (9, 240), where x is the week and y is profit.

Step 2: Find the slope: $m = (240 - (-120)) / (9 - 3)$.

Step 3: Simplify the top: $240 + 120 = 360$, so $m = 360 / 6$.

Step 4: Divide: $m = 60$ (profit grows by 60 euros each week).

Step 5: Use point-slope form with (3, -120): $y - (-120) = 60(x - 3)$.

Step 6: Simplify: $y + 120 = 60x - 180$.

Step 7: Solve for y : $y = 60x - 300$.

Step 8: Set $y = 0$ to find break-even: $60x = 300$, so $x = 5$.

Answer: Week 5

Tip

Whenever two data points are given and the question asks 'when does it hit zero,' build the full linear equation first.

Trap

It's tempting to just average the two given weeks (3 and 9) instead of actually solving for where profit equals zero.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Week 6	Averaged the two given weeks, $(3 + 9)/2 = 6$, instead of solving the equation for profit = 0.
Week 1	Made a sign error simplifying $y + 120 = 60x - 180$, writing $y = 60x - 60$ instead of $y = 60x - 300$.
Week 9	Mistook the second data point, where profit is already positive, for the break-even week.

Shortcut

The direct method is fastest here — building the linear equation and setting it to zero avoids error-prone repeated addition.

Desmos Tip

Type $y = 60x - 300$ in Desmos and trace the graph to find where it crosses the x -axis, confirming week 5.

Line r has equation $2x + 5y = 10$. Line s is perpendicular to line r and passes through the point $(4, -3)$. What is the x -intercept of line s ?

Formula / Rule Used: Perpendicular slopes (negative reciprocal) + point-slope form.

Solution

Step 1: First rewrite line r in slope-intercept form: $5y = -2x + 10$.

Step 2: Divide by 5: $y = (-2/5)x + 2$, so the slope of line r is $-2/5$.

Step 3: Perpendicular slope is the negative reciprocal: flip $-2/5$ and switch its sign to get $5/2$.

Step 4: Use point-slope form for line s with $(4, -3)$: $y - (-3) = (5/2)(x - 4)$.

Step 5: Simplify the left side: $y + 3 = (5/2)x - 10$.

Step 6: Solve for y : $y = (5/2)x - 13$.

Step 7: Find the x -intercept by setting $y = 0$: $0 = (5/2)x - 13$.

Step 8: Solve: $(5/2)x = 13$, so $x = 13 \times (2/5) = 26/5$.

Answer: $x = 26/5$ (or 5.2)

Tip

For a perpendicular-line question, find and flip the given line's slope before doing anything else.

Trap

It's tempting to use line r 's own slope ($-2/5$) for line s instead of switching to the perpendicular slope.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$-7/2$ (-3.5)	Used line r 's own slope, $-2/5$, as if the lines were parallel instead of perpendicular.
$23/2$ (11.5)	Took the reciprocal of $-2/5$ correctly as $5/2$ in size, but forgot to apply the negative sign flip and used $2/5$ instead.
-13	Correctly found the equation of line s but reported its y -intercept instead of the x -intercept the question asked for.

Shortcut

No shortcut beyond flipping and negating the slope first — the direct method is already fastest for this type of question.

Desmos Tip

Type $y = (5/2)x - 13$ in Desmos and trace the graph to where it crosses the x -axis to confirm $x = 26/5$.

A furniture company in Milan models its delivery cost with the equation $6x + 3y = 18$, where x is distance in kilometers and y is cost in euros. A competing company charges according to a line parallel to this one, and that line happens to pass through the point $(5, 1)$. While reviewing this, the company's designer, Elena, separately updates the delivery van's logo. What is the slope of the competing company's line?

Formula / Rule Used: Standard form to slope-intercept, plus parallel slope rule.

Solution

Step 1: The detail about Elena updating the logo has nothing to do with slope — ignore it.

Step 2: Rewrite $6x + 3y = 18$ in slope-intercept form: $3y = -6x + 18$.

Step 3: Divide every term by 3: $y = -2x + 6$.

Step 4: The slope of this line is -2 .

Step 5: Parallel lines always share the same slope, no matter which point they pass through.

Step 6: So the competing company's line also has slope -2 — the point $(5, 1)$ doesn't change this.

Answer: -2

Tip

Before using a given point, check whether you actually need it — parallel lines share slope no matter what point they pass through.

Trap

It's tempting to spend time using the point $(5, 1)$ to calculate slope, when parallel lines share slope by definition alone.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2	Made a sign error dividing $-6x$ by 3, dropping the negative sign and getting slope 2 instead of -2 .
$1/2$	Used the perpendicular slope rule (negative reciprocal) instead of the parallel slope rule.
$6/5$	Tried to compute slope using the point $(5, 1)$ and the y-intercept $(0, 6)$ instead of recognizing parallel lines just share slope.

Shortcut

Once you divide to get $y = -2x + 6$, you can stop right there — the y-intercept and the extra point aren't needed for this question.

Desmos Tip

Type $y = -2x + 6$ in Desmos and check the slope shown matches -2 , confirming the parallel line's slope regardless of any point.

**Lines — Problem 8 (Medium)**

A line has an x-intercept of 8 and a y-intercept of -4. What is the value of y when $x = 20$ on this line?

Formula / Rule Used: Intercepts to slope, then slope-intercept form.

Solution

Step 1: An x-intercept of 8 means the line passes through the point (8, 0).

Step 2: A y-intercept of -4 means the line passes through the point (0, -4).

Step 3: Find the slope using these two points: $m = (0 - (-4)) / (8 - 0)$.

Step 4: Simplify the top: $0 - (-4) = 4$, so $m = 4 / 8$.

Step 5: Simplify the fraction: $m = 1/2$.

Step 6: Write the equation using slope-intercept form with the known y-intercept: $y = (1/2)x - 4$.

Step 7: Substitute $x = 20$: $y = (1/2)(20) - 4$.

Step 8: Multiply: $y = 10 - 4$, so $y = 6$.

Answer: $y = 6$

Tip

When both intercepts are given, you already have two points — use them to find slope, then plug straight into $y = mx + b$.

Trap

It's tempting to mix up which value is the x-intercept and which is the y-intercept, which flips the whole equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
48	Swapped the intercepts, treating -4 as the x-intercept and 8 as the y-intercept, giving a very different line.
10	Correctly found the slope but forgot to add the y-intercept term after substituting x, stopping at $(1/2)(20) = 10$.
-14	Made a sign error simplifying $0 - (-4)$ as $0 - 4 = -4$ instead of 4, getting slope $-1/2$ instead of $1/2$.

Shortcut

Since you already know $b = -4$, skip point-slope form entirely — just find m from the two intercepts and plug directly into $y = mx + b$.

Desmos Tip

Type $y = (1/2)x - 4$ in Desmos, trace to $x = 20$, and confirm y reads 6.

2. Systems of Linear Equations

Systems of Linear Equations — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
System of equations	Σύστημα εξισώσεων
Solution (of a system)	Λύση συστήματος
Substitution method	Μέθοδος αντικατάστασης
Elimination method	Μέθοδος απαλοιφής
No solution / infinite solutions	Καμία λύση / άπειρες λύσεις

Systems of Linear Equations — What It Is

What It Is

- A system of linear equations is just two straight-line equations considered together.
- Solving the system means finding the (x, y) point that makes BOTH equations true at once.
- On a graph, that point is exactly where the two lines cross.
- The SAT tests this two ways: actually solving for x and y, and reasoning about how many crossing points exist.
- You'll see this skill in both bare algebra questions and real-world 'two unknowns' word problems.

Systems of Linear Equations — Core Rules

Substitution Method

Solve one equation for a variable, plug it into the other

- Pick the equation where a variable is alone or easy to isolate, like $y = \dots$ or $x = \dots$
- Substitute that expression into the OTHER equation, replacing that variable completely.
- You now have one equation with one variable — solve it normally.
- Plug that number back into either original equation to find the second variable.

Elimination Method

Add or subtract the equations to cancel a variable

- Stack the two equations and look at the coefficients of x and y.
- Multiply one or both equations by a number so one variable has opposite coefficients.
- Add the equations together — that variable cancels out to zero.
- e.g. $2x + y = 10$ and $x - y = 2$: add them, y cancels, $3x = 12$, so $x = 4$.

Setting Equal (when both solved for y)**If $y = \dots$ and $y = \dots$, set the right sides equal**

- Works only when both equations are already written as $y = (\text{something with } x)$.
- Since both expressions equal the same y , they must equal each other.
- Solve that one equation for x , then substitute back for y .
- This is really just substitution in disguise, but faster to spot.

Solving Directly for a Combined Expression**Add, subtract, or scale the equations to build the target**

- Sometimes the question asks for a combo like $2x + 3y$, not x and y separately.
- Check whether adding, subtracting, or scaling the two equations gives that combo directly.
- If it does, you never have to solve for x and y individually — much faster.
- e.g. if the equations add straight to $2x + 3y = 14$, that IS the answer.

Number of Solutions**One: different slopes. None: same slope, different y-intercept.
Infinite: same line.**

- One solution means the two lines cross at exactly one point — different slopes.
- No solution means the lines are parallel — same slope, but different y-intercepts.
- Infinite solutions means the two equations describe the exact same line.
- In standard form $Ax + By = C$, compare the ratios A_1/A_2 , B_1/B_2 , and C_1/C_2 to tell which case it is.

Coefficient-Ratio Shortcut**No solution: $A_1/A_2 = B_1/B_2 \neq C_1/C_2$. Infinite: $A_1/A_2 = B_1/B_2 = C_1/C_2$.**

- Use this when a question gives an unknown constant and asks for no solution or infinite solutions.
- Line up both equations in standard form $Ax + By = C$ first.
- Set the coefficient ratios equal (or not equal, matching the case asked) and solve for the unknown.
- This turns a messy system into one simple proportion, no full solving needed.

How to Recognize This on the Test

- Two separate equations, each with the same two variables (usually x and y), given together.
- Phrases like 'system of equations,' 'which ordered pair,' or 'for what value(s) of x and y .'
- A question asking how many solutions a system has, without asking you to actually find them.
- A word problem describing two different relationships between the same two unknown quantities.
- An equation with an unknown constant (like k) paired with instructions about 'no solution' or 'infinitely many.'

Common Mistakes — and the Fix

- Solving for only one variable and forgetting to finish by finding the second one.
- Substituting an expression into the SAME equation it came from instead of the other equation.
- Forgetting to distribute a negative sign when subtracting one whole equation from another.
- Mixing up which variable the question actually wants (x , y , or a combined expression like $x + y$).
- Assuming a system has one solution without checking whether the lines might be parallel or identical.

Systems of Linear Equations — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Systems of Linear Equations — Problem 1 (Easy)

At a bakery in Thessaloniki, a croissant costs c euros and a muffin costs m euros. Two croissants and one muffin cost 7 euros total. One croissant and three muffins cost 11 euros total. What is the cost of one muffin, in euros?

Formula / Rule Used: Substitution: isolate one variable, plug it into the other equation.

Solution

Step 1: Write the two equations: $2c + m = 7$ and $c + 3m = 11$.

Step 2: Solve the first equation for m , since it's easiest to isolate: $m = 7 - 2c$.

Step 3: Substitute this expression for m into the second equation: $c + 3(7 - 2c) = 11$.

Step 4: Distribute the 3 across both terms: $c + 21 - 6c = 11$.

Step 5: Combine like terms on the left: $-5c + 21 = 11$.

Step 6: Subtract 21 from both sides: $-5c = -10$.

Step 7: Divide both sides by -5 : $c = 2$.

Step 8: Substitute $c = 2$ back into $m = 7 - 2c$: $m = 7 - 4 = 3$.

Answer: 3 euros (one muffin costs 3 euros)

Tip

When one variable already has a coefficient of 1 in either equation, isolate it first — substitution will go much faster.

Trap

It's tempting to stop once you find $c = 2$ and report it, but the question actually asks for the muffin's price, m .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2	Correctly solved for $c = 2$ but reported the croissant's price instead of the muffin's price, m , which the question asked for.



11	Made a sign error dividing -10 by -5, writing $c = -2$ instead of $c = 2$, which then gave $m = 7 - 2(-2) = 11$.
-13	Forgot to distribute the 3 across both terms in $3(7 - 2c)$, only multiplying the 7 and leaving $-2c$ unchanged, which threw off both c and m .

Shortcut

None faster here — with these coefficients, isolating m first and substituting is already the quickest path.

Desmos Tip

Graph $y = 7 - 2x$ and $y = (11 - x)/3$ (using x for c and y for m), then use the intersection point tool to confirm $x = 2$, $y = 3$.

Systems of Linear Equations — Problem 2 (Easy)

If $2x + 3y = 12$ and $2x - y = 4$, what is the value of y ?

Formula / Rule Used: Elimination: subtract equations that share a matching term.

Solution

Step 1: Write the two equations: $2x + 3y = 12$ and $2x - y = 4$.

Step 2: Notice both equations have the exact same x -term, $2x$ — that's a signal to eliminate x by subtracting.

Step 3: Subtract the second equation from the first: $(2x + 3y) - (2x - y) = 12 - 4$.

Step 4: On the left side, the $2x$ terms cancel, leaving $3y - (-y)$.

Step 5: Simplify $3y - (-y)$ to $3y + y = 4y$.

Step 6: So the equation becomes $4y = 8$.

Step 7: Divide both sides by 4: $y = 2$.

Answer: 2

Tip

When two equations share the exact same x -term (or y -term), subtracting them instantly eliminates that variable.

Trap

It's easy to forget to distribute the subtraction across the entire second equation, especially flipping $-y$ into $+y$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
8	Correctly subtracted to get $4y = 8$ but stopped there without dividing both sides by 4.
4	Forgot that subtracting a negative y adds it, computing $3y - y = 2y$ instead of $3y + y = 4y$, then solved $2y = 8$ as $y = 4$.



3	Finished the problem correctly but reported the value of x (3) instead of y, the variable the question asked for.
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Shortcut

Since both equations already share the term $2x$, subtracting directly is faster than solving for x first.

Desmos Tip

Graph $y = (12 - 2x)/3$ and $y = 2x - 4$, then use the intersection point tool to read the y -value, 2, off the point (3, 2).

Systems of Linear Equations — Problem 3 (Easy)

Two lines are given by $y = 3x - 1$ and $y = -2x + 9$. At what point (x, y) do these two lines intersect?

Formula / Rule Used: Setting equal: if both equations are solved for y , set the right sides equal.

Solution

Step 1: Since both equations are already solved for y , set the right-hand sides equal to each other: $3x - 1 = -2x + 9$.

Step 2: Add $2x$ to both sides to gather the x -terms on one side: $5x - 1 = 9$.

Step 3: Add 1 to both sides: $5x = 10$.

Step 4: Divide both sides by 5: $x = 2$.

Step 5: Substitute $x = 2$ into either original equation, say $y = 3x - 1$: $y = 3(2) - 1$.

Step 6: Multiply first: $y = 6 - 1$.

Step 7: Subtract: $y = 5$.

Step 8: The intersection point is (2, 5).

Answer: (2, 5)

Tip

Whenever both equations are already written as $y = \dots$, just set them equal to each other — no substitution step needed.

Trap

After finding x , it's tempting to stop there, but the question wants the full point, including y .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 2$	Found x correctly but reported it alone instead of the full ordered pair the question asked for.
(10, 29)	Subtracted $2x$ from both sides instead of adding, turning $3x - 1 = -2x + 9$ into $x - 1 = 9$ instead of $5x - 1 = 9$, giving $x = 10$.
(2, 13)	Substituted $x = 2$ back in but dropped the negative sign on $-2x$, computing



$$y = 2(2) + 9 = 13 \text{ instead of } y = -2(2) + 9 = 5.$$

Shortcut

None needed — setting the two y -expressions equal is already the fastest way to find where these lines meet.

Desmos Tip

Type $y_1 = 3x - 1$ and $y_2 = -2x + 9$, then click the intersection point Desmos highlights to confirm $(2, 5)$.

Systems of Linear Equations — Problem 4 (Easy)

The equations $4x + 6y = 18$ and $2x + 3y = 9$ are graphed in the xy -plane. How many solutions does this system have?

Formula / Rule Used: Number of solutions: compare the equations after matching their coefficients.

Solution

Step 1: To compare the equations, try to write both with matching coefficients.

Step 2: Divide every term of the first equation by 2: $4x/2 + 6y/2 = 18/2$.

Step 3: This simplifies to $2x + 3y = 9$.

Step 4: Compare this to the second equation, which is already $2x + 3y = 9$.

Step 5: Both equations are now identical: $2x + 3y = 9$.

Step 6: Since the two equations describe the exact same line, every point on that line is a shared solution.

Step 7: A system where both equations represent the same line has infinitely many solutions.

Answer: Infinitely many solutions

Tip

Before deciding how many solutions a system has, simplify or scale one equation and compare it directly to the other.

Trap

It's tempting to see two 'different-looking' equations and assume they cross at exactly one point, without checking if one is just a multiple of the other.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
One solution	Assumed two different-looking equations must cross at a single point without checking whether one is a multiple of the other.
No solution	Noticed the equations have the same slope but incorrectly assumed the y -intercepts differ, without simplifying to check they're actually identical.
Two solutions	Confused 'a system of two equations' with 'two solutions,' not realizing the number of equations doesn't determine the number of solutions.



Shortcut

Divide the equation with larger coefficients by the common factor (here, 2) — if it matches the other equation exactly, that instantly confirms infinite solutions.

Desmos Tip

Graph $y = (18 - 4x)/6$ and $y = (9 - 2x)/3$ — Desmos will show only one line on screen since they overlap completely, confirming infinite solutions.

Systems of Linear Equations — Problem 5 (Medium)

The system of equations $6x + 9y = 15$ and $4x + ky = 8$ has no solution. What is the value of k ?

Formula / Rule Used: Coefficient-ratio shortcut: for no solution, $A_1/A_2 = B_1/B_2$, but not equal to C_1/C_2 .

Solution

Step 1: A system has no solution when the two equations represent parallel lines — same slope, different y-intercept.

Step 2: In standard form $Ax + By = C$, this means the ratio of the A's equals the ratio of the B's, but NOT the ratio of the C's.

Step 3: Set up the ratio of the x-coefficients: $6/4$, which simplifies to $3/2$.

Step 4: For no solution, the y-coefficients must share this same ratio: $9/k = 3/2$.

Step 5: Cross-multiply: $9 \times 2 = 3 \times k$.

Step 6: Simplify the left side: $18 = 3k$.

Step 7: Divide both sides by 3: $k = 6$.

Step 8: Quick check: with $k = 6$, both equations simplify to $2x + 3y = 5$ and $2x + 3y = 4$ — same left side, different right side, confirming no solution.

Answer: $k = 6$

Tip

For a 'no solution' or 'infinite solutions' question with an unknown constant, set up the coefficient ratio instead of solving the whole system.

Trap

It's tempting to set the y-coefficient ratio equal to the constant ratio ($9/k = 15/8$) instead of the x-coefficient ratio — that's the infinite-solutions setup, not the no-solution one.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 4.8$	Used the constant ratio $15/8$ instead of the x-coefficient ratio $6/4$ for the proportion, mixing up the no-solution and infinite-solutions conditions.
$k = 13.5$	Set up the proportion upside down, writing $k/9 = 6/4$ instead of $9/k = 6/4$.
$k = 1.5$	Mistook the simplified ratio $3/2$ itself for the value of k , without actually



	solving the proportion $9/k = 3/2$.
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Shortcut

Skip solving the full system entirely — setting the coefficient ratios equal (or unequal) as a proportion is always faster for these constant-finding questions.

Desmos Tip

Type the first equation as $6x+9y=15$ and the second as $4x+6y=8$ (using $k=6$) — Desmos draws two parallel, non-overlapping lines, confirming no intersection.

Systems of Linear Equations — Problem 6 (Medium)

At a bookstore in Lisbon, a paperback costs p euros and a hardcover costs h euros. Two paperbacks and three hardcovers cost 28 euros. Three paperbacks and two hardcovers cost 27 euros. What is the combined cost of one paperback and one hardcover, $p + h$, in euros?

Formula / Rule Used: Solving directly for a combined expression by adding the equations.

Solution

Step 1: Write the two equations: $2p + 3h = 28$ and $3p + 2h = 27$.

Step 2: The question doesn't ask for p or h separately — it asks for $p + h$ combined, so look for a shortcut before solving fully.

Step 3: Add the two equations together: $(2p + 3h) + (3p + 2h) = 28 + 27$.

Step 4: Combine like terms on the left: $2p + 3p + 3h + 2h = 5p + 5h$.

Step 5: Combine the right side: $28 + 27 = 55$.

Step 6: So the equation becomes $5p + 5h = 55$.

Step 7: Every term on the left shares a factor of 5 — divide the whole equation by 5: $p + h = 11$.

Step 8: That is exactly the combined value the question asked for.

Answer: 11 euros

Tip

Before grinding through substitution or elimination, check whether adding or subtracting the equations directly gives the combo the question actually wants.

Trap

It's tempting to solve fully for p and h individually first — that works too, but takes longer and gives more chances for an arithmetic slip.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
55	Added the equations correctly to get $5p + 5h = 55$, but forgot the final step of dividing both sides by 5.
1	Subtracted the equations instead of adding them, getting $-p + h = 1$, and



	mistakenly treated that result as $p + h$.
30	Solved correctly for $p = 5$ and $h = 6$ individually, but multiplied them instead of adding, misreading 'combined' as a product.

Shortcut

Adding the two equations directly gives $5(p + h) = 55$ in one step — much faster than solving for p and h separately and adding.

Desmos Tip

Graph $y = (28 - 2x)/3$ and $y = (27 - 3x)/2$ (using x for p , y for h), find the intersection point, then add its coordinates to confirm $p + h = 11$.

Systems of Linear Equations — Problem 7 (Medium)

A school in Vienna is ordering notebooks and pens for a stationery drive, and the shop also offers free delivery on orders over 50 euros. Notebooks cost 2 euros each and pens cost 1 euro each. The school buys n notebooks and p pens, spending exactly $2n + p = 40$ euros. The school buys 4 more pens than notebooks, so $p = n + 4$. How many notebooks did the school buy?

Formula / Rule Used: Substitution, with one extra irrelevant fact to filter out.

Solution

Step 1: Write the two relevant equations, ignoring the delivery detail, which doesn't affect the math: $2n + p = 40$ and $p = n + 4$.

Step 2: Since the second equation already isolates p , substitute it into the first equation: $2n + (n + 4) = 40$.

Step 3: Combine like terms: $2n + n + 4 = 3n + 4$.

Step 4: So the equation becomes $3n + 4 = 40$.

Step 5: Subtract 4 from both sides: $3n = 36$.

Step 6: Divide both sides by 3: $n = 12$.

Answer: 12 notebooks

Tip

Read word problems for the actual math relationships first — extra details like delivery thresholds or store policies are often just context, not numbers to use.

Trap

It's easy to try to work the '50 euros free delivery' detail into an equation — it's irrelevant information meant to test if you can tell what matters.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
16	Solved correctly but reported p (16), the number of pens, instead of n , the



	number of notebooks the question asked for.
13.3	Divided 40 by 3 directly without first subtracting the 4, skipping the step that isolates $3n = 36$.
15.3	Mistakenly used the irrelevant 50-euro free-delivery threshold as the total spent instead of the actual 40, solving $3n + 4 = 50$.

Shortcut

Since p is already isolated, substitution is instant here — there's no faster path than plugging $p = n + 4$ directly into the total equation.

Desmos Tip

Graph $y = 40 - 2x$ and $y = x + 4$ (using x for notebooks, y for pens), then find the intersection point to confirm $n = 12$, $p = 16$.

Systems of Linear Equations — Problem 8 (Medium)

The system of equations $ax + 2y = 22$ and $3x - y = 5$ has the solution $x = 4$. What is the value of a ?

Formula / Rule Used: Reverse setup: use the known solution to find a missing coefficient.

Solution

Step 1: Since the first equation contains the unknown a , start with the second equation instead, which doesn't: $3x - y = 5$.

Step 2: Substitute the given $x = 4$ and multiply: $3(4) - y = 5$, so $12 - y = 5$.

Step 3: Subtract 12 from both sides: $-y = -7$.

Step 4: Multiply both sides by -1 to solve for y : $y = 7$.

Step 5: Now substitute both $x = 4$ and $y = 7$ into the first equation: $a(4) + 2(7) = 22$.

Step 6: Multiply: $4a + 14 = 22$.

Step 7: Subtract 14 from both sides: $4a = 8$.

Step 8: Divide both sides by 4: $a = 2$.

Answer: $a = 2$

Tip

When one equation has an unknown coefficient and the other doesn't, solve the 'clean' equation first to find the missing y -value.

Trap

It's tempting to substitute $x = 4$ straight into the first equation before knowing y , which leaves you with two unknowns in one equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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8	Correctly reached $4a = 8$ but forgot the final step of dividing both sides by 4.
9	Made a sign error solving $-y = -7$, writing $y = -7$ instead of $y = 7$, which threw off the final value of a .
0	Misread the first equation's right-hand side as 14 instead of 22, getting $4a + 14 = 14$ and solving $a = 0$.

Shortcut

None faster here — since a is trapped in the one equation with two unknowns, you must find y from the other equation first.

Desmos Tip

Type $3x - y = 5$ and trace to $x = 4$ to confirm $y = 7$, then type $4a + 2(7) = 22$ and use a slider for a to find where the equation balances at $a = 2$.

3. Linear Inequalities

Linear Inequalities — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Inequality	Ανίσωση
Solution set	Σύνολο λύσεων
Compound inequality	Σύνθετη ανίσωση
Flip the inequality sign	Αναστροφή συμβόλου ανίσωσης
Boundary line / shaded region	Οριακή ευθεία / σκιασμένη περιοχή

Linear Inequalities — What It Is

What It Is

- An inequality compares two expressions using $<$, $>$, $\leq$, or $\geq$ instead of an equals sign.
- It describes a whole range of possible values, not just one single exact answer.
- The SAT tests inequalities because many real situations have limits, minimums, or maximums, not exact targets.
- You solve an inequality almost exactly like an equation, with one key exception to watch for.
- The answer is usually a range like $x > 3$, or a shaded region on a graph.

Linear Inequalities — Core Rules

Solving Like an Equation

Add or subtract the same value on both sides; the sign stays the same.

- Adding or subtracting any number on both sides never changes the direction of the sign.
- Move variable terms to one side and plain numbers to the other side, just like an equation.
- e.g. $x - 3 > 5$ becomes $x > 8$ by adding 3 to both sides.

The Flip Rule

Multiply or divide both sides by a negative number → flip the sign.

- This is the one rule that makes solving inequalities different from solving equations.
- Multiplying or dividing both sides by a positive number never flips the sign.
- e.g. $-2x < 6$ becomes $x > -3$ after dividing by -2 and flipping $<$ to $>$.
- Forgetting this flip is the single most common inequality mistake on the SAT.

Compound (Three-Part) Inequalities

$a < x < b$

- This notation means x is squeezed between two boundary values at the same time.
- Whatever operation you do, apply it to all three parts of the inequality at once.
- e.g. $-1 < 2x + 3 < 9$ starts by subtracting 3 from all three parts.
- If you multiply or divide all three parts by a negative, flip both signs, not just one.

Graphing on a Number Line

- An open circle means the endpoint is NOT included, used for strict $<$ or $>$ symbols.
- A closed, filled-in circle means the endpoint IS included, used for $\leq$ or $\geq$ symbols.
- Draw an arrow or shading in the direction of every value that makes the inequality true.

Graphing Inequalities in the xy -Plane

$y > mx + b$, $y < mx + b$, $y \geq mx + b$, $y \leq mx + b$

- First graph the boundary line exactly as if it were the equation $y = mx + b$.
- Use a solid line for $\leq$ or $\geq$, since the line itself counts as part of the solution.
- Use a dashed line for $<$ or $>$, since the line itself is not part of the solution.
- Shade above the line for ' $y >$ ' or ' $y \geq$ ', and shade below the line for ' $y <$ ' or ' $y \leq$ '.

Systems of Inequalities

- A system means two or more inequalities that must all be true at the same time.
- Graph each inequality separately, then look for where their shaded regions overlap.
- Any point sitting inside that overlapping region is a valid solution to the whole system.

Translating Word Problems

'at least' $\rightarrow \geq$, 'at most' $\rightarrow \leq$, 'more than' $\rightarrow >$, 'fewer/less than' $\rightarrow <$

- Underline the key phrase in the sentence before deciding which symbol to use.
- 'At least' and 'no less than' both mean $\geq$, so the minimum value itself is allowed.
- 'At most' and 'no more than' both mean $\leq$, so the maximum value itself is allowed.
- e.g. 'Maria needs at least 20 points' translates directly to points ≥ 20 .

How to Recognize This on the Test

- The question uses phrases like 'at least,' 'at most,' 'more than,' or 'maximum/minimum.'
- Inequality symbols ($<$, $>$, $\leq$, $\geq$) appear directly in the problem, a graph, or an answer.
- The question asks for a whole range of possible values instead of one exact number.



- A graph shows a shaded region, or a dashed or solid boundary line instead of just a plotted line.
- The context is a real-world limit, like a budget, capacity, weight limit, or speed limit.

Common Mistakes — and the Fix

- Forgetting to flip the sign when multiplying or dividing both sides by a negative number.
- Flipping the sign when only adding or subtracting, which is never actually required.
- Mixing up 'at least' ($\geq$) with 'more than' ($>$), so the boundary value gets included or excluded wrongly.
- Using a solid line or closed circle for a strict inequality ($<$ or $>$) instead of a dashed line or open circle.
- Forgetting to round to a sensible whole number in real-world contexts, like people, boxes, or tickets.

Linear Inequalities — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Linear Inequalities — Problem 1 (Easy)

Solve the inequality for x : $-4x + 9 \leq 21$.

Formula / Rule Used: Isolate x like an equation, then flip the sign when dividing by a negative.

Solution

Step 1: Subtract 9 from both sides to start isolating the x -term: $-4x \leq 21 - 9$.

Step 2: Simplify the right side: $-4x \leq 12$.

Step 3: Divide both sides by -4 to solve for x .

Step 4: Because you divided by a negative number, flip the inequality sign from $\leq$ to $\geq$.

Step 5: Simplify $12 / -4 = -3$, so the solution is $x \geq -3$.

Answer: $x \geq -3$

Tip

Handle the inequality exactly like an equation until the very last step, then check whether you divided by a negative.

Trap

It's easy to flip the sign the moment you see any negative number in the problem, but the flip only happens when you multiply or divide by a negative, not when you subtract one.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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$x \leq -3$	Divided both sides by -4 correctly but forgot to flip the inequality sign.
$x \geq 3$	Divided 12 by -4 but dropped the negative sign, getting positive 3 instead of -3.
$x \geq -7.5$	Added 9 instead of subtracting it, turning $-4x \leq 21$ into $-4x \leq 30$ before dividing.

Shortcut

None needed — with only one x-term, isolating it in two moves is already the fastest path.

Desmos Tip

Type $y_1 = -4x + 9$ and $y_2 = 21$ in Desmos, then check where the first graph stays at or below the second to confirm it's true for $x \geq -3$.

Linear Inequalities — Problem 2 (Easy)

A bakery in Vienna sells croissants in boxes of 6 for €7 per box. Elena has €50 to spend, but she wants to keep at least €10 aside for her tram ticket home. What is the greatest number of boxes she can buy?

Formula / Rule Used: Translate 'at least' into an inequality, then solve and round down to a whole box.

Solution

Step 1: Let b = the number of boxes Elena buys, which cost $7b$ euros total.

Step 2: She needs at least 10 euros left over, so she can spend at most $50 - 10 = 40$ euros on boxes.

Step 3: Set up the inequality: $7b \leq 40$.

Step 4: Divide both sides by 7, which is positive, so the sign stays the same: $b \leq 40/7$.

Step 5: Simplify: $b \leq 5.71$ (rounded to two decimals).

Step 6: Since Elena can't buy part of a box, round down to the nearest whole number: $b = 5$.

Answer: 5 boxes

Tip

Turn 'keep at least X aside' into 'at most (total - X) to spend,' and always round DOWN for a maximum number of items.

Trap

The algebra gives $b \leq 5.71$, but rounding up to 6 boxes would leave her without enough money for the tram.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
6 boxes	Solved correctly to $b \leq 5.71$ but rounded up instead of down, leaving less than 10 euros for the tram.
7 boxes	Forgot to subtract the 10 euros kept aside and solved $7b \leq 50$ instead,



	getting $b \leq 7.14$, rounded to 7.
5.71 boxes	Solved the inequality correctly but never rounded to a whole number of boxes.

Shortcut

Estimate that $40/7$ is a little less than 6, since $7 \times 6 = 42$ is already over 40, so 5 must be the answer.

Desmos Tip

Type $y1 = 7x + 10$ and $y2 = 50$ in Desmos and find the largest whole x where the first stays at or below the second — it lands at $x = 5$.

Linear Inequalities — Problem 3 (Easy)

Solve the compound inequality for x : $-5 < 3x - 2 < 10$.

Formula / Rule Used: Apply the same operation to all three parts of the compound inequality.

Solution

Step 1: Add 2 to all three parts to start isolating x : $-5 + 2 < 3x < 10 + 2$.

Step 2: Simplify each side: $-3 < 3x < 12$.

Step 3: Divide all three parts by 3, which is positive, so the signs stay the same.

Step 4: Simplify: $-3/3 < x < 12/3$.

Step 5: Final answer: $-1 < x < 4$.

Answer: $-1 < x < 4$

Tip

Treat all three parts as one unit and do the exact same operation to each part, every single time.

Trap

It's easy to apply an operation to only two of the three parts and accidentally break the inequality.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$-1 < x < 12$	Divided the middle and right parts by 3 but forgot to divide the left part.
$-7 < x < 8$	Subtracted 2 from all three parts instead of adding, undoing the -2 in the wrong direction.
$-1.67 < x < 4$	Added 2 to the right side but forgot to add 2 to the left side before dividing by 3.

Shortcut

None genuinely faster — handling all three parts together in two clean moves is already the quickest method.

**Desmos Tip**

Type $y1 = 3x - 2$ in Desmos and trace the graph to see it's between -5 and 10 exactly when x is between -1 and 4.

Linear Inequalities — Problem 4 (Easy)

Twice a number, decreased by 7, is more than 15. What is the smallest integer that could be the number?

Formula / Rule Used: Translate the words into an inequality, then solve for the variable.

Solution

Step 1: Let n = the number. 'Twice a number decreased by 7' translates to $2n - 7$.

Step 2: 'Is more than 15' means this expression is greater than 15: $2n - 7 > 15$.

Step 3: Add 7 to both sides: $2n > 15 + 7$.

Step 4: Simplify: $2n > 22$.

Step 5: Divide both sides by 2, which is positive, so the sign stays the same: $n > 11$.

Step 6: Since n must be greater than 11 and not equal to it, the smallest integer that works is 12.

Answer: 12

Tip

Translate the sentence into an inequality piece by piece before doing any solving — get the symbol right first.

Trap

Because the boundary value 11 is a whole number, it's tempting to answer 11 itself, but the strict ' $>$ ' symbol means 11 does NOT work.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
11	Solved correctly to $n > 11$ but included the boundary value even though the inequality is strict.
22	Divided by 2 too early, before adding 7, mixing up the correct order of operations.
4	Misread 'decreased by 7' as 'divided by 7' and solved $2n/7 > 15$ instead.

Shortcut

None needed — translating directly into $2n - 7 > 15$ and solving is already the fastest route.

Desmos Tip

Type $y1 = 2x - 7$ and $y2 = 15$ in Desmos and check where the first graph rises above the second — it crosses right after $x = 11$.

Linear Inequalities — Problem 5 (Medium)

The inequality $2x + k \leq 10$ has the solution set $x \leq 3$. What is the value of k ?

Formula / Rule Used: The boundary of the solution set makes the inequality true as an equation.

Solution

Step 1: Since the solution set is $x \leq 3$, the boundary value $x = 3$ must make the inequality an equation:
 $2(3) + k = 10$.

Step 2: Multiply: $6 + k = 10$.

Step 3: Subtract 6 from both sides: $k = 10 - 6$.

Step 4: Simplify: $k = 4$.

Step 5: Check the answer: $2x + 4 \leq 10$ leads to $2x \leq 6$, which gives $x \leq 3$, matching the given solution set.

Answer: 4

Tip

Whenever you're given the solution set instead of the constant, plug the boundary value in as if it were an equation.

Trap

Don't try to guess-and-check random values for k — setting the boundary value equal is faster and guaranteed to work.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-4	Set up $6 + k = 10$ but made a sign error, writing $k = 6 - 10$ instead of $k = 10 - 6$.
10	Plugged $x = 3$ in but forgot to multiply 2 times 3, treating the equation as $x + k = 10$.
16	Added 6 and 10 together instead of subtracting, treating the equation as $k = 6 + 10$.

Shortcut

Since you're given the exact boundary value, plugging it in as an equation is already the fastest possible method.

Desmos Tip

Type $y_1 = 2x + 4$ and $y_2 = 10$ in Desmos to confirm the first line stays at or below the second exactly for $x \leq 3$.

Linear Inequalities — Problem 6 (Medium)



How many integer values of x satisfy both $x > 2$ and $3 - x \geq -4$?

Formula / Rule Used: Solve each inequality separately, then find the overlap of both solution sets.

Solution

Step 1: The first inequality is already solved: $x > 2$.

Step 2: For the second, subtract 3 from both sides: $-x \geq -4 - 3$.

Step 3: Simplify the right side: $-x \geq -7$.

Step 4: Divide both sides by -1 , and flip the sign because you're dividing by a negative: $x \leq 7$.

Step 5: Combine both results: the overlap of $x > 2$ and $x \leq 7$ is $2 < x \leq 7$.

Step 6: List every integer in that range: 3, 4, 5, 6, 7.

Step 7: Count them: there are 5 integers.

Answer: 5

Tip

Solve each inequality on its own first, then combine the two ranges to find where they overlap.

Trap

It's easy to forget the flip rule on the second inequality since it starts as $3 - x$, not $-x$, hiding the negative coefficient.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Solved the second inequality correctly to $x \leq 7$ but treated it as strict and left out $x = 7$, undercounting by one.
6	Treated the first inequality $x > 2$ as $x \geq 2$ and mistakenly included $x = 2$ in the count.
0	Miscalculated $-4 - 3$ as -1 instead of -7 , which flips to $x \leq 1$ and leaves no overlap with $x > 2$.

Shortcut

Since both inequalities are already simple, solving them separately and listing integers in the overlap is faster than graphing.

Desmos Tip

Type $y1 = x$, $y2 = 2$, $y3 = 3 - x$, and $y4 = -4$ in Desmos and check the integer x -values where $y1 > y2$ and $y3 \geq y4$ at the same time.

A moving company in Rotterdam charges a flat fee of €25 plus €7 per box. The company also offers free bubble wrap on any order of 20 or more boxes. Lars has a budget of €130 for his move. What is the greatest number of boxes he can order and stay within budget?

Formula / Rule Used: Set up a linear inequality with a flat fee plus a rate, then solve for the variable.

Solution

Step 1: Let b = the number of boxes Lars orders. The free bubble wrap offer doesn't change the price, so it can be ignored.

Step 2: The total cost is the flat fee plus the per-box rate: $25 + 7b$.

Step 3: This total must stay within budget: $25 + 7b \leq 130$.

Step 4: Subtract 25 from both sides: $7b \leq 130 - 25$.

Step 5: Simplify: $7b \leq 105$.

Step 6: Divide both sides by 7, which is positive, so the sign stays the same: $b \leq 15$.

Step 7: Since 15 is already a whole number, the greatest number of boxes he can order is 15.

Answer: 15 boxes

Tip

Pull out the flat fee first, since it's a one-time cost, then divide only the remaining budget by the per-item rate.

Trap

The free bubble wrap detail at 20+ boxes is a distraction — since the real answer (15) never reaches 20, it never actually applies.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
18 boxes	Forgot to subtract the 25 euro flat fee and solved $7b \leq 130$ directly, getting $b \leq 18.57$, rounded down to 18.
22 boxes	Assumed the free bubble wrap offer meant extra budget was available and added boxes past the true limit.
16 boxes	Subtracted $130 - 25$ incorrectly as 115 instead of 105, leading to $b \leq 16.43$ rounded down to 16.

Shortcut

Since 105 divides evenly by 7, you can estimate the budget split quickly before doing the exact division.

Desmos Tip

Type $y_1 = 25 + 7x$ and $y_2 = 130$ in Desmos and find the greatest whole x where the first graph is still at or below the second.

If $5 \leq x \leq 9$, what is the range of possible values of $3x - 4$?

Formula / Rule Used: Apply the same operation to every part of a compound inequality to transform it.

Solution

Step 1: Start with the given compound inequality: $5 \leq x \leq 9$.

Step 2: Multiply all three parts by 3 to build the '3x' from the target expression: $5(3) \leq 3x \leq 9(3)$.

Step 3: Simplify: $15 \leq 3x \leq 27$.

Step 4: Subtract 4 from all three parts to build the '-4' from the target expression: $15 - 4 \leq 3x - 4 \leq 27 - 4$.

Step 5: Simplify both ends: $11 \leq 3x - 4 \leq 23$.

Answer: $11 \leq 3x - 4 \leq 23$

Tip

Rebuild the target expression one operation at a time, applying each step to all three parts of the inequality at once.

Trap

It's tempting to only transform the two outer numbers and forget that the middle part must become $3x - 4$ too, matching the target expression exactly.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$15 \leq 3x - 4 \leq 27$	Multiplied all three parts by 3 correctly but forgot the final step of subtracting 4 from each part.
$1 \leq 3x - 4 \leq 5$	Subtracted 4 from the original bounds (5-4 and 9-4) but forgot to multiply by 3 first.
$11 \leq 3x - 4 \leq 27$	Fully transformed the left bound but only multiplied, not subtracted 4 from, the right bound.

Shortcut

Notice $3x - 4$ is built from x by 'multiply by 3, then subtract 4' — apply those same two moves directly to both bounds instead of solving for x first.

Desmos Tip

Type $y1 = 3x - 4$ in Desmos and trace the graph between $x = 5$ and $x = 9$ to read off the output range directly, from 11 to 23.

4. Absolute Value

Absolute Value — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Absolute value	Απόλυτη τιμή
Distance from zero	Απόσταση από το μηδέν
Two cases (positive/negative)	Δύο περιπτώσεις (θετική/αρνητική)
Extraneous solution	Ξένη (μη αποδεκτή) λύση

Absolute Value — What It Is

What It Is

- Absolute value means 'distance from zero' on the number line.
- Distance is never negative, so $|x|$ is always zero or positive.
- The bars $| |$ just strip away any negative sign once distance is measured.
- The SAT tests this because two different numbers can have the same absolute value.
- That 'two numbers, one distance' idea is why absolute value problems often have two answers.

Absolute Value — Core Rules

Solving $|X| = a$

If $a \geq 0$: $X = a$ or $X = -a$. If $a < 0$: no solution.

- First isolate the absolute value expression completely on one side.
- Once isolated, split into two separate equations: $X = a$ and $X = -a$.
- e.g. $|x| = 5$ means $x = 5$ or $x = -5$, since both are 5 away from zero.
- If a is negative, stop immediately — distance can never equal a negative number.

Absolute Value Inequalities (less than)

$$|X| < a \rightarrow -a < X < a$$

- This only applies when a is positive — 'less than a distance of a .'
- Rewrite it as one compound inequality, X trapped between $-a$ and a .
- e.g. $|x| < 4$ becomes $-4 < x < 4$, all points closer than 4 to zero.
- Think 'less than' = 'between' — the solution is one connected chunk of the line.

Absolute Value Inequalities (greater than)

$|X| > a \rightarrow X < -a \text{ or } X > a$

- This applies when a is positive — 'more than a distance of a .'
- Rewrite it as two separate branches, not one connected range.
- e.g. $|x| > 4$ becomes $x < -4$ or $x > 4$, points farther than 4 from zero.
- Think 'greater than' = 'split apart' — the solution is two separate rays.

$|A| = |B|$

$A = B \text{ or } A = -B$

- Used when absolute value expressions appear on both sides of an equation.
- Set up two equations: the expressions equal, and one negated.
- e.g. $|x - 1| = |2x|$ means $x - 1 = 2x$ or $x - 1 = -2x$.
- Solve each equation separately to get every candidate solution.

Checking for Extraneous Solutions

- Splitting into cases can introduce fake solutions that don't actually work.
- This risk is highest when the variable appears both inside and outside the bars.
- Always plug every candidate answer back into the ORIGINAL equation to test it.
- If a candidate makes the original equation false, throw that solution out.

Distance Interpretation

$|x - a| = \text{distance between } x \text{ and } a$

- $|x - a|$ measures how far the value x is from the number a on the number line.
- This shows up in 'within so many units of a target' word problems.
- e.g. 'within 3 degrees of 70' translates to $|x - 70| \leq 3$.
- Recognizing this pattern lets you skip straight to writing the inequality.

Absolute Value as a Function

$f(x) = |x|$ has a V-shaped graph with its vertex at the origin.

- The graph looks like a V, always touching or above the x -axis.
- Every output (y -value) of an absolute value function is zero or positive.
- If a question asks for the minimum value of $|\text{expression}|$, the answer is often 0.
- Shifting the expression inside or outside the bars moves the V left/right or up/down.

How to Recognize This on the Test

- You see the $| |$ bars directly around a number or expression in the problem.
- Wording like 'distance from,' 'how far,' or 'within ____ units of' signals absolute value.

- The problem mentions two possible values, two solutions, or 'both values of x .'
- A word problem gives a target value plus an allowed margin, tolerance, or error range.
- An equation or inequality has a variable trapped inside bars that must be isolated first.

Common Mistakes — and the Fix

- Forgetting the negative case and writing only one solution instead of two.
- Flipping $<$ and $>$ rules — mixing up when it's a 'between' range vs. a 'split' range.
- Not isolating the absolute value first before splitting into two cases.
- Skipping the check step, so an extraneous (fake) solution slips into the final answer.
- Treating $|X| = -a$ as solvable — distance can never equal a negative number, so it has no solution.

Absolute Value — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Absolute Value — Problem 1 (Easy)

Solve for x : $|x - 3| = 7$.

Formula / Rule Used: Solving $|X| = a$: $X = a$ or $X = -a$.

Solution

Step 1: The absolute value expression is already isolated on the left side.

Step 2: Split the equation into two cases: $x - 3 = 7$ and $x - 3 = -7$.

Step 3: Solve the first case by adding 3 to both sides: $x = 10$.

Step 4: Solve the second case by adding 3 to both sides: $x = -4$.

Step 5: Check case 1: $|10 - 3| = |7| = 7$, which matches.

Step 6: Check case 2: $|-4 - 3| = |-7| = 7$, which also matches.

Answer: $x = 10$ or $x = -4$

Tip

Whenever you see $|X| = a$, immediately isolate the bars, then write both the positive and negative case before solving.

Trap

It's tempting to solve only $x - 3 = 7$ and stop, forgetting the negative case $x - 3 = -7$ is just as valid.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 10$ only	Solved only the positive case ($x - 3 = 7$) and forgot the negative case $x - 3 = -7$ exists.



$x = -4$ only	Solved only the negative case and forgot to also solve $x - 3 = 7$.
$x = 4$ or $x = -10$	Subtracted 3 from both sides instead of adding it when isolating x in both cases.

Shortcut

None needed — for a simple isolated absolute value equation, splitting into the two cases directly is already the fastest method.

Desmos Tip

Graph $y_1 = \text{abs}(x - 3)$ and $y_2 = 7$, then find where the two graphs intersect — the x -coordinates of both intersection points are the answers.

Absolute Value — Problem 2 (Easy)

A bottling plant in Rotterdam fills bottles to a target volume of 500 milliliters. Quality control allows each bottle's volume to differ from the target by at most 4 milliliters. If V is a bottle's volume in milliliters, write this rule as an absolute value inequality, then find the least and greatest acceptable volumes.

Formula / Rule Used: Distance interpretation: $|\text{actual} - \text{target}| \leq \text{tolerance}$.

Solution

Step 1: The distance between the actual volume V and the target 500 must be at most 4 mL.

Step 2: Write this as an absolute value inequality: $|V - 500| \leq 4$.

Step 3: This is a 'less than or equal to' type, so rewrite it as a compound inequality: $-4 \leq V - 500 \leq 4$.

Step 4: Add 500 to all three parts of the inequality to isolate V .

Step 5: $-4 + 500 \leq V \leq 4 + 500$.

Step 6: Simplify both sides: $496 \leq V \leq 504$.

Answer: $496 \text{ mL} \leq V \leq 504 \text{ mL}$ (least acceptable volume 496 mL, greatest 504 mL)

Tip

Start by translating 'differ from the target by at most' directly into $|\text{actual} - \text{target}| \leq \text{tolerance}$.

Trap

A common mistake is writing the inequality backwards, describing the volumes that FAIL instead of the ones that pass.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$V \leq 504$ only	Only solved the upper bound and forgot the inequality also limits how low V can go.
$494 \text{ mL} \leq V \leq 506 \text{ mL}$	Used a tolerance of 6 instead of 4, misreading the number given in the problem.



$500 \text{ mL} \leq V \leq 504 \text{ mL}$	Added 4 to 500 correctly but forgot to also subtract 4, dropping the lower bound entirely.
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Shortcut

Since this is a straightforward tolerance question, just add and subtract the tolerance from the target directly instead of formally rewriting the compound inequality.

Desmos Tip

Graph $y = \text{abs}(x - 500)$ and $y = 4$ on the same screen, then read off the x-values where the V-shaped graph is at or below the horizontal line.

Absolute Value — Problem 3 (Easy)

Solve for x: $|2x + 5| = 11$.

Formula / Rule Used: Solving $|X| = a$: $X = a$ or $X = -a$.

Solution

Step 1: The absolute value is already isolated, so split into two cases.

Step 2: Case 1: $2x + 5 = 11$.

Step 3: Subtract 5 from both sides: $2x = 6$.

Step 4: Divide both sides by 2: $x = 3$.

Step 5: Case 2: $2x + 5 = -11$.

Step 6: Subtract 5 from both sides: $2x = -16$.

Step 7: Divide both sides by 2: $x = -8$.

Answer: $x = 3$ or $x = -8$

Tip

Always set the expression inside the bars equal to both a and -a before doing any more algebra.

Trap

It's tempting to write the negative case incorrectly, mixing up which side of the equation gets the negative sign.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 3$ only	Solved only the positive case and never set up the second equation, $2x + 5 = -11$.
$x = 6$ or $x = -16$	Correctly subtracted 5 in both cases but forgot to divide by 2 at the last step.
$x = 3$ or $x = -3$	Set up the negative case correctly but subtracted 5 in the wrong direction, getting $2x = -6$ instead of $2x = -16$.

**Shortcut**

None needed — since the bars are already isolated, splitting directly into the two linear equations is already the fastest path.

Desmos Tip

Graph $y_1 = \text{abs}(2x + 5)$ and $y_2 = 11$, then find the two x-coordinates where the graphs cross to confirm $x = 3$ and $x = -8$.

Absolute Value — Problem 4 (Easy)

Solve for x: $|4x - 9| = -3$.

Formula / Rule Used: Solving $|X| = a$: if $a < 0$, there is no solution.

Solution

Step 1: Look at the right-hand side of the equation: $|4x - 9| = -3$.

Step 2: Recall that absolute value measures distance from zero, so it must always be zero or positive.

Step 3: The right side, -3, is negative, so no real value of x can make $|4x - 9|$ equal -3.

Step 4: Because of this, do not try to split the equation into two cases — there is nothing valid to solve.

Step 5: Any x-value produced by splitting anyway would be extraneous and would fail the original equation.

Step 6: Conclude that this equation has no solution.

Answer: No solution

Tip

Before splitting into cases, always check the sign of the number on the other side of the equal sign.

Trap

It's tempting to split into $4x - 9 = 3$ and $4x - 9 = -3$ anyway and solve for x, producing fake answers that don't actually work.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 3/2$	Ignored that the right side was negative and solved $4x - 9 = 3$ as if it were a normal, valid case.
$x = 3/2$ or $x = 3$	Split into both cases despite the negative right side, producing two extraneous fake solutions.
$x = 3$	Solved $4x - 9 = -3$ directly, ignoring that an absolute value expression can never equal a negative number.

Shortcut

Recognizing a negative value on the isolated side instantly tells you 'no solution' — no algebra needed at all.

Desmos Tip



Graph $y_1 = \text{abs}(4x - 9)$ and $y_2 = -3$; since the V-shaped graph never dips below the x-axis, it never touches that horizontal line, confirming no solution.

Absolute Value — Problem 5 (Medium)

The solution to the inequality $|x - h| < k$ is the range $5 < x < 17$. What is the value of $h \times k$?

Formula / Rule Used: Reverse of $|X - h| < k$: the range is $h - k < X < h + k$, so h is the midpoint and k is the half-width.

Solution

Step 1: Compare the given compound inequality $5 < x < 17$ to the general form $h - k < x < h + k$.

Step 2: The midpoint of the range gives h , so add the two endpoints: $5 + 17 = 22$.

Step 3: Divide by 2 to find h : $h = 11$.

Step 4: The distance from the midpoint to either endpoint gives k , so subtract: $k = 17 - 11$.

Step 5: Simplify: $k = 6$.

Step 6: Check with the lower endpoint: $h - k = 11 - 6 = 5$, which matches, confirming both values.

Step 7: Multiply to answer the question asked: $h \times k = 11 \times 6 = 66$.

Answer: $h \times k = 66$

Tip

When you're given the solved range, find the midpoint first — that's always h — then measure the distance to either end to get k .

Trap

It's tempting to grab 17 or 5 directly as h without averaging the endpoints — always find the true midpoint first.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
60	Mistook the lower bound 5 for h instead of averaging 5 and 17 to find the true midpoint 11.
132	Used the correct midpoint $h = 11$ but used the full range width $17 - 5 = 12$ as k instead of half the width.
204	Used the upper bound 17 as h and the full range width 12 as k , multiplying two wrong values together.

Shortcut

If only h were needed, averaging the two endpoints would be enough; since this asks for h times k , both values must be found individually first.

Desmos Tip

Type $y = \text{abs}(x - 11)$ and $y = 6$ in Desmos; the graph crosses $y = 6$ exactly at $x = 5$ and $x = 17$, confirming $h = 11$ and $k = 6$.

Absolute Value — Problem 6 (Medium)

Solve for x : $|3x - 1| = |x + 9|$. Give every value of x that satisfies the equation.

Formula / Rule Used: $|A| = |B| \rightarrow A = B \text{ or } A = -B$.

Solution

Step 1: Set up the two cases: $3x - 1 = x + 9$, or $3x - 1 = -(x + 9)$.

Step 2: Case 1: subtract x from both sides: $2x - 1 = 9$.

Step 3: Add 1 to both sides: $2x = 10$, so $x = 5$.

Step 4: Case 2: distribute the negative sign on the right: $3x - 1 = -x - 9$.

Step 5: Add x to both sides: $4x - 1 = -9$.

Step 6: Add 1 to both sides: $4x = -8$, so $x = -2$.

Step 7: Check both in the original equation: $|3(5) - 1| = 14$ and $|5 + 9| = 14$, which match.

Step 8: Check the second: $|3(-2) - 1| = 7$ and $|-2 + 9| = 7$, which also match, so both solutions are valid.

Answer: $x = 5$ or $x = -2$

Tip

Whenever absolute value expressions appear on BOTH sides, set up $A = B$ and $A = -B$, then solve each equation completely.

Trap

The most common slip is forgetting to distribute the negative sign across both terms in the second case.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 5$ only	Solved only the first case ($3x - 1 = x + 9$) and never set up the second case with the negative sign.
$x = 5$ or $x = 2$	Forgot to distribute the negative sign onto the 9, writing $3x - 1 = -x + 9$ instead of $3x - 1 = -x - 9$.
$x = -5$ or $x = -2$	Made a sign error in case 1, writing $2x - 1 = -9$ instead of $2x - 1 = 9$, flipping the first solution's sign.

Shortcut

None genuinely faster here — since the variable appears inside two absolute values, setting up both cases and solving is the most reliable method.

Desmos Tip

Graph $y_1 = \text{abs}(3x - 1)$ and $y_2 = \text{abs}(x + 9)$, then find the two x -coordinates where the graphs intersect to confirm $x = 5$ and $x = -2$.

Absolute Value — Problem 7 (Medium)

A courier company in Vienna, EuroSwift Logistics, accepts packages only if their weight is within 1.5 kilograms of the standard 8-kilogram box. The company also charges 0.40 euros per kilometer for delivery. A customer has four packages weighing 6.2 kg, 6.8 kg, 9.3 kg, and 9.7 kg. How many of these four packages will EuroSwift accept?

Formula / Rule Used: Distance interpretation: $|\text{weight} - 8| \leq 1.5$ defines the acceptable range.

Solution

Step 1: Ignore the delivery cost of 0.40 euros per kilometer — it has nothing to do with the weight rule.

Step 2: Translate the acceptance rule into an inequality: $|w - 8| \leq 1.5$, where w is a package's weight.

Step 3: Rewrite this as a compound inequality: $-1.5 \leq w - 8 \leq 1.5$.

Step 4: Add 8 to all three parts: $6.5 \leq w \leq 9.5$.

Step 5: Compare each given weight to this range: 6.2 kg is below 6.5 kg, so it fails.

Step 6: 6.8 kg is between 6.5 and 9.5, so it passes.

Step 7: 9.3 kg is between 6.5 and 9.5, so it passes.

Step 8: 9.7 kg is above 9.5 kg, so it fails.

Answer: 2 packages will be accepted (the 6.8 kg and 9.3 kg packages).

Tip

First build the acceptable range from the tolerance rule, then just check each given value against it — ignore numbers unrelated to weight.

Trap

The delivery cost of 0.40 euros per kilometer is a distractor — using it anywhere in your inequality is a sign you've misread the question.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3 packages	Miscalculated the upper bound as 9.7 instead of computing $8 + 1.5 = 9.5$, wrongly letting the 9.7 kg package pass.
1 package	Used a tolerance of 1 instead of 1.5, narrowing the range to $7 \leq w \leq 9$ and excluding the 6.8 kg package too.
4 packages	Forgot to apply the tolerance rule at all and assumed every package near 8 kg would be accepted.

Shortcut

Once you have the range 6.5 kg to 9.5 kg, just add/subtract the tolerance from the target directly instead of writing the full inequality each time.

Desmos Tip

Graph $y = \text{abs}(x - 8)$ and $y = 1.5$, then check which of the four weights fall where the V-graph is at or below the horizontal line.

Absolute Value — Problem 8 (Medium)

Let $f(x) = |3x - 12| + 5$. For what value of x is $f(x)$ at its minimum, and what is that minimum value?

Formula / Rule Used: $|expression|$ is minimized (equal to 0) exactly when the expression inside equals 0.

Solution

Step 1: The smallest possible value of an absolute value expression is 0, since distance can't be negative.

Step 2: Find the x that makes the inside expression equal 0: $3x - 12 = 0$.

Step 3: Add 12 to both sides: $3x = 12$.

Step 4: Divide both sides by 3: $x = 4$.

Step 5: At $x = 4$, the absolute value term $|3x - 12|$ becomes $|0| = 0$.

Step 6: Substitute back into $f(x)$: $f(4) = 0 + 5$.

Step 7: Simplify: $f(4) = 5$.

Step 8: So the minimum value of $f(x)$ is 5, occurring at $x = 4$.

Answer: Minimum value is 5, occurring at $x = 4$.

Tip

For $f(x) = |expression| + constant$, always set the inside expression to 0 first — that's where the minimum happens.

Trap

It's tempting to plug $x = 0$ into $f(x)$ instead of solving for the x that zeroes out the inside expression, which gives the wrong minimum.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
17 at $x = 0$	Plugged in $x = 0$ instead of solving $3x - 12 = 0$, computing $f(0) = -12 + 5 = 17$.
5 at $x = 12$	Solved $3x = 12$ correctly but forgot to divide by 3, leaving $x = 12$ instead of $x = 4$.
0 at $x = 4$	Correctly found $x = 4$ but forgot to add the +5 outside the bars, reporting the inside value 0 as the final minimum.

Shortcut

Since this is a straightforward 'V shape plus constant,' the minimum is always just the constant added outside the bars, once the inside can reach 0.

Desmos Tip



Graph $y = \text{abs}(3x - 12) + 5$ and trace to the lowest point of the V, which sits at (4, 5), confirming both the x-value and the minimum.

5. Exponent Rules

Exponent Rules — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Exponent (power)	Εκθέτης (δύναμη)
Base	Βάση
Exponent rules	Ιδιότητες δυνάμεων
Negative exponent	Αρνητικός εκθέτης
Common base	Κοινή βάση

Exponent Rules — What It Is

What It Is

- An exponent is just shorthand for repeated multiplication of the same number.
- means 'multiply a by itself n times,' where a is the base and n is the exponent.
- The SAT tests whether you can simplify exponent expressions using a small set of fixed rules.
- It also tests solving equations where the variable sits up in the exponent itself.
- Every exponent rule on the test can be rebuilt from scratch just by expanding the multiplication.

Exponent Rules — Core Rules

Product Rule (multiplying same base)

- When you multiply two powers with the SAME base, add the exponents.
- This only works if the base is identical on both sides — not .
- e.g. , since that's 2 x's times 5 x's = 7 x's total.

Quotient Rule (dividing same base)

- When you divide two powers with the SAME base, subtract the bottom exponent from the top.
- Think of it as canceling matching factors from top and bottom.
- e.g. , because 3 of the x's on top cancel with the 3 on bottom.

Power Rules (a power raised to a power)

- When a power is raised to another power, MULTIPLY the exponents instead of adding.
- e.g. , since is repeated 4 times.
- Don't confuse this with the product rule — that one adds, this one multiplies.

Power Rules (product or quotient raised to a power)

- An exponent outside parentheses applies to EVERY factor inside, if it's all multiplication or division.
- e.g. — both the 2 and the x get cubed.
- This distributing trick only works for products and quotients, never for a sum or difference.

Zero Exponent

(as long as $a \neq 0$)

- Any nonzero base raised to the power 0 always equals exactly 1.
- This isn't random — it falls out of the quotient rule: , but also .
- e.g. , and even .

Negative Exponents

- A negative exponent does NOT make the answer negative — it means 'take the reciprocal.'
- Flip the base to the other side of a fraction and make the exponent positive.
- e.g. , and .

Solving Exponential Equations (matching bases)

If and a is the same on both sides, then $m = n$.

- If two sides of an equation have the SAME base, you can just set the exponents equal.
- If the bases look different, first rewrite both sides as powers of one common base.
- e.g. becomes (since), so $x = 3$.

How to Recognize This on the Test

- You see a variable with a small raised number attached, like or .
- The problem asks you to 'simplify' an expression full of exponents and the same base repeated.
- The variable itself is stuck up in the exponent, like — that signals an exponential equation.
- Words like 'raised to the power of,' 'squared,' 'cubed,' or 'to the nth power' appear in the wording.
- You see parentheses with an exponent outside them, like or .

**Common Mistakes — and the Fix**

- Adding exponents when the bases are different — the product rule only works for matching bases.
- Treating a negative exponent as a negative number instead of a reciprocal — fix: flip it to .
- Distributing an exponent over addition, writing as — this is never allowed.
- Forgetting that anything to the zero power is 1, and instead writing 0 or leaving it unsimplified.
- Multiplying exponents instead of adding them (or vice versa) — remember: multiply same bases means add, power-of-a-power means multiply.

Exponent Rules — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Exponent Rules — Problem 1 (Easy)

For all nonzero values of x , the expression is equivalent to . What is the value of k ?

Formula / Rule Used: Product rule then quotient rule for exponents with the same base.

Solution

Step 1: First combine the numerator using the product rule: .

Step 2: The expression is now .

Step 3: Apply the quotient rule: subtract the bottom exponent from the top: .

Step 4: So the expression simplifies to , meaning .

Step 5: Therefore $k = 5$.

Answer: $k = 5$

Tip

Handle the multiplication in the numerator first, then apply the quotient rule to the whole fraction.

Trap

The biggest risk is mixing up when to add and when to subtract — multiplying same bases adds, dividing subtracts.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 11$	Added all three exponents ($6+2+3$) instead of subtracting the denominator's exponent.
$k = 1$	Subtracted instead of added in the numerator ($6-2=4$), then subtracted 3 again ($4-3=1$).
$k = 8$	Correctly combined the numerator to but forgot to divide by at all.

**Shortcut**

None needed — combining same-base exponents by adding then subtracting is already the fastest way to simplify this.

Desmos Tip

Type into Desmos and confirm it equals 32, which is , confirming $k = 5$.

Exponent Rules — Problem 2 (Easy)

A tech company based in Athens models its user base with , where x is the number of years after 2020 and U_0 is the number of users in 2020. According to this model, the number of users in 2018 equals U_0 times what fraction?

Formula / Rule Used: Negative exponent rule: .

Solution

Step 1: 2018 is 2 years before 2020, so $x = -2$ in this model.

Step 2: Substitute $x = -2$ into to get .

Step 3: A negative exponent means take the reciprocal of the positive power: .

Step 4: Evaluate the denominator: .

Step 5: So .

Step 6: The number of users in 2018 is , meaning of U_0 .

Answer: (users in 2018 = of U_0)

Tip

Whenever a real-world exponent model asks about a time before the starting point, expect a negative exponent for x .

Trap

A common mistake is treating as -4 , a negative number, instead of the reciprocal .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-4	Treated the negative exponent as making the whole value negative instead of taking a reciprocal.
4	Correctly found but forgot to flip it into a reciprocal for the negative exponent.
	Used $x = -1$ instead of $x = -2$, miscounting how many years before 2020 the year 2018 is.

Shortcut

None needed — plugging $x = -2$ directly into and applying the reciprocal rule is already the fastest path.

Desmos Tip



Type into Desmos and trace the graph to $x = -2$ to confirm the y -value is 0.25, matching .

Exponent Rules — Problem 3 (Easy)

A cube has a side length of centimeters. Which expression, written as a single term with one exponent on y , represents the cube's volume in cubic centimeters?

Formula / Rule Used: Power of a product rule and power of a power rule.

Solution

Step 1: The volume of a cube is , so Volume = .

Step 2: Apply the power of a product rule: distribute the outer exponent to both factors: .

Step 3: Evaluate .

Step 4: Apply the power of a power rule to : multiply the exponents, $3 \times 3 = 9$, giving .

Step 5: Combine both results: Volume = .

Answer: cubic centimeters

Tip

When parentheses raised to a power surround a product, apply the exponent to every factor inside separately.

Trap

It's easy to add the exponents ($3+3=6$) instead of multiplying them when raising to the 3rd power.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Added the exponents 3 and 3 instead of multiplying them for the power-of-a-power step.
	Correctly cubed the term but forgot to also cube the coefficient 2.
	Multiplied the coefficient 2 by 3 instead of raising it to the 3rd power.

Shortcut

None needed — distributing the outer exponent to the coefficient and the variable separately is already the fastest method.

Desmos Tip

Type and separately into Desmos, using $y = 2$ substituted in — both give 4096, confirming the simplified form.

Exponent Rules — Problem 4 (Easy)



What is the value of the expression , where x is any nonzero number?

Formula / Rule Used: Zero exponent rule: for any nonzero a .

Solution

Step 1: In the first term, only x is raised to the 0 power, not the 9: .

Step 2: So the first term becomes $9 \cdot 1 = 9$.

Step 3: In the second term, the entire quantity $(3x)$ is raised to the 0 power: , since $3x$ is nonzero.

Step 4: Add the two results together: $9 + 1$.

Step 5: The total is 10.

Answer: 10

Tip

Check exactly what's inside the parentheses before applying the zero-exponent rule — it changes what the exponent applies to.

Trap

It's tempting to treat the same as , but the parentheses change what the exponent 0 actually applies to.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Treated as , instead of only raising x to the 0 power.
12	Evaluated as 3, applying the 0 power only to x and keeping the 3, giving $9+3=12$.
9	Correctly got 9 for the first term, but mistakenly treated as 0 instead of 1.

Shortcut

None needed — recognizing that anything nonzero to the 0 power is 1 makes this a quick direct evaluation.

Desmos Tip

Type into Desmos using $x = 5$ as a test value and confirm the result is 10.

Exponent Rules — Problem 5 (Medium)

If a is a positive number and $32^a = 1024$, what is the value of a ?

Formula / Rule Used: Power of a power rule, used in reverse.

Solution

Step 1: Notice that 1024 can be rewritten as 2^{10} , since multiplying the exponents 5×2 gives 10 .

Step 2: You're already given that $32^a = 2^{10}$, so substitute that value in directly.

Step 3: This gives $2^{5a} = 2^{10}$.

Step 4: Compute : $32 \times 32 = 1024$.

Step 5: So $a = 2$, without ever needing to find a itself.

Answer: 1024

Tip

Before solving for the base itself, check whether you can rewrite the target expression using the given power directly.

Trap

Many students solve $32^a = 1024$ for a first ($a = 2$), then compute $32 \times 32 = 1024$ — that works but takes longer and risks a slip.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
64	Treated 32 as (doubling) instead of squaring 32 , giving $2 \times 32 = 64$.
20	Solved for $a = 2$ correctly but then multiplied 2×10 instead of computing 2^{10} .
320	Multiplied 32 by 10 instead of recognizing that 1024 is 32 squared, not 10 times 32 .

Shortcut

Recognizing $1024 = 32^2$ and substituting 32 directly is the shortcut — it skips solving for a entirely.

Desmos Tip

Type $32^a = 1024$ into Desmos to confirm it equals 32 (so $a=2$), then type 32^2 to confirm the answer 1024 .

Exponent Rules — Problem 6 (Medium)

If , what is the value of ?

Formula / Rule Used: Solving exponential equations with matching bases, then the power rule.

Solution

Step 1: Rewrite 64 as a power of 2: .

Step 2: Now the equation reads , and since the bases match, set the exponents equal: $3x = 6$.

Step 3: Divide both sides by 3 to solve for x: $x = 2$.

Step 4: The question asks for , not x itself, so substitute $x = 2$: .

Step 5: Evaluate .

Answer: 16

Tip

Always check exactly what the question is asking for at the end — here it's , not x itself.

Trap

It's easy to solve for x, get $x = 2$, and then stop — forgetting the question actually wants .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2	Solved correctly for $x = 2$ but stopped there instead of computing .
8	Found $x = 2$ correctly but then computed $4 \cdot 2$ instead of .
64	Confused with the original and just restated the given value.

Shortcut

You could rewrite as and find from the original equation, but solving for x directly is just as fast here.

Desmos Tip

Type into Desmos, trace to where $y = 64$ to confirm $x = 2$, then type to confirm the final answer 16.

Exponent Rules — Problem 7 (Medium)

A warehouse in Rotterdam models its stock of a spare part with $N(t)$, where t is the number of months since tracking began, and the warehouse operates 6 days a week. According to this model, what is $N(4)$, the number of parts in stock after 4 months?

Formula / Rule Used: Evaluating a negative fractional exponent.

Solution

Step 1: The '6 days a week' detail doesn't affect the formula — it's extra information you don't need.

Step 2: Substitute $t = 4$ into the exponent: $-$.

Step 3: Now the expression is $2916^{-\frac{1}{9}}$.

Step 4: A negative exponent means take the reciprocal: $\frac{1}{2916^{-\frac{1}{9}}}$.

Step 5: Multiply: $\frac{1}{2916^{-\frac{1}{9}}} = 2916^{\frac{1}{9}}$.

Step 6: Divide 2916 by 9: $2916^{\frac{1}{9}} = 324$.

Answer: 324 parts

Tip

Simplify the exponent itself first (here, $-$ at $t=4$), before applying any exponent rules.

Trap

It's tempting to think the '6 days a week' detail belongs in the calculation somewhere — it doesn't, it's just context.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
5832	Multiplied by $2916^{\frac{1}{9}}$ instead of dividing by it, mixing up what a negative exponent means.
972	Computed $2916^{\frac{1}{9}}$ as $2916^{\frac{1}{3}}$ instead of $2916^{\frac{1}{9}}$, forgetting to square the base before taking the reciprocal.
36	Used $t = 4$ directly as the exponent instead of dividing t by 2 first.

Shortcut

None needed — simplifying the exponent first, then applying the negative exponent rule, is already the fastest approach.

Desmos Tip

Type $2916^{\frac{1}{9}}$ into Desmos and trace to $x = 4$ to confirm $y = 324$.

Exponent Rules — Problem 8 (Medium)

What value of x satisfies ?

Formula / Rule Used: Rewriting both sides with a common base, then solving the exponential equation.

Solution

Step 1: Rewrite as a power of 3: , so .

Step 2: Apply the power of a power rule: .

Step 3: Rewrite 27 as a power of 3 as well: .

Step 4: Now the equation reads , and since the bases match, set the exponents equal: $-2x = 3$.

Step 5: Divide both sides by -2 to solve for x : .

Answer:

Tip

Whenever a base is a fraction like , rewrite it as a negative power of the whole number first, before combining exponents.

Trap

It's easy to rewrite as instead of , forgetting the fraction means a negative exponent.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Forgot the negative sign when rewriting as , using instead.
$x = -3$	Forgot to distribute the -2 through the power of a power step, using $-x = 3$ instead of $-2x = 3$.
$x = -6$	Multiplied both sides by -2 instead of dividing, turning $-2x = 3$ into $x = -6$.

Shortcut

None needed — converting both sides to the same base 3 is already the most direct route to solving for x .

Desmos Tip

Type into Desmos and trace to where $y = 27$ to confirm $x = -1.5$.

6. Radicals, Rational Exponents & Scientific Notation

Radicals, Rational Exponents & Scientific Notation — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Radical (root)	Ρίζα (ριζικό)
Rational exponent	Ρητός εκθέτης
Simplify a radical	Απλοποίηση ρίζας
Scientific notation	Επιστημονική σημειογραφία

Radicals, Rational Exponents & Scientific Notation — What It Is

What It Is

- A radical, like a square root or cube root, asks 'what number times itself gives me this?'
- A rational exponent, like $a^{\frac{1}{n}}$, is just another way to write that exact same root.
- Scientific notation is a compact way to write very large or very small numbers using powers of 10.
- The SAT tests these together because they're all really the same exponent rules in different disguises.
- You'll need to convert smoothly between radical form, exponent form, and scientific notation on the same question.

Radicals, Rational Exponents & Scientific Notation — Core Rules

Radical-Exponent Conversion

n -th root of

- The small number outside the radical, the index, becomes the bottom of the fraction.
- The power on the inside, m , becomes the top of the fraction.
- A plain square root with no visible index has an index of 2.
- Example: the cube root of 8 equals 2 .

Simplifying Radicals

- Break the number under the radical into a perfect square times a leftover factor.
- Pull the perfect square out as a whole number, and leave the rest under the root.
- Example: $\sqrt{18} = 3\sqrt{2}$.
- The same idea works for cube roots — look for perfect cubes instead.



Exponent Rules Recap

| |

- Multiplying powers with the same base: add the exponents together.
- Dividing powers with the same base: subtract the bottom exponent from the top one.
- Raising a power to another power: multiply the two exponents.
- These shortcuts only work when both bases are exactly the same number or variable.

Negative and Zero Exponents

and

- A negative exponent means 'take the reciprocal,' not 'make the answer negative.'
- Example: , which is a positive fraction.
- Any nonzero number or variable raised to the power 0 always equals 1.

Combining Radicals with Different Indexes

- First rewrite every radical in the expression as a fractional exponent.
- Give all the fractional exponents a common denominator so they line up.
- Now apply the normal product or quotient exponent rule to combine them.
- Convert the result back into radical form only if the question asks for that.

Solving Equations with Rational Exponents

If , then

- Raise both sides of the equation to the reciprocal power to isolate x.
- Example: if , then .
- Flip the exponent's fraction upside down — don't reuse the same fraction.

Scientific Notation

, where $1 \leq N < 10$

- The coefficient N must always be at least 1 but strictly less than 10.
- The exponent n tells you how many places the decimal point has shifted.
- When multiplying: multiply the coefficients, then add the exponents.
- When dividing: divide the coefficients, then subtract the exponents, and recheck N is still 1 to 10.

How to Recognize This on the Test

- A radical symbol (square root, cube root) or a fractional exponent like signals this skill.
- A number written as 'something times 10 to a power,' like , means scientific notation.
- The instruction 'simplify the expression' next to a root or fractional power points here.



- A variable trapped under a root, or raised to a fraction, needs this skill to solve for it.
- Extremely large or extremely small real-world values (distances, populations, atom sizes) hint at scientific notation.

Common Mistakes — and the Fix

- Treating $\sqrt{\quad}$ as $\sqrt{\quad}$ — a root never distributes over addition or subtraction.
- Reading a negative exponent as making the value negative instead of taking its reciprocal.
- Forgetting to renormalize a scientific-notation coefficient back into the range 1 to 10 after calculating.
- Mixing up which number becomes the top and which becomes the bottom when converting a radical to an exponent.
- Stopping after finding $\sqrt{\quad}$ instead of finishing the job and raising both sides to solve for x .

Radicals, Rational Exponents & Scientific Notation — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Radicals, Rational Exponents & Scientific Notation — Problem 1 (Easy)

Simplify the expression $\sqrt{50}$ so it is written in the form $\sqrt{b}$, where b has no perfect-square factors other than 1.

Formula / Rule Used: Simplifying radicals: $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$.

Solution

Step 1: Look for the largest perfect square that divides evenly into 50.

Step 2: Break 50 into that perfect square times a leftover factor: $50 = 25 \times 2$.

Step 3: Rewrite the radical using the product rule: $\sqrt{50} = \sqrt{25 \times 2}$.

Step 4: Simplify the perfect square part: $\sqrt{25} = 5$.

Step 5: Combine the pieces: the simplified form is $5\sqrt{2}$.

Answer: $5\sqrt{2}$

Tip

Always search for the largest perfect square factor (4, 9, 16, 25, 36...) hiding inside the radicand first.

Trap

It's tempting to pull out a smaller factor like 5 instead of the full perfect square 25, which leaves the answer only partly simplified.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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	Swapped which factor comes out of the root and which stays inside, leaving the perfect square trapped under the radical.
	Forgot to actually take the square root of 25 before pulling it outside, leaving 25 instead of 5.
	Split 50 correctly into 25 x 2 but then multiplied the leftover 2 by the pulled-out 5 under the root by mistake.

Shortcut

None needed — factoring out the largest perfect square right away is already the fastest way to simplify a radical.

Desmos Tip

Type $\sqrt{50}$ on one line and $\sqrt{25 \times 2}$ on the next line in Desmos — matching decimal values confirm the simplification is correct.

Radicals, Rational Exponents & Scientific Notation — Problem 2 (Easy)

What is the value of ?

Formula / Rule Used: Rational exponent rule: .

Solution

Step 1: The denominator of the exponent, 3, tells you to take a cube root.

Step 2: The numerator of the exponent, 2, tells you to then square the result.

Step 3: Find the cube root of 8 first: the cube root of 8 is 2, since .

Step 4: Now square that result: .

Step 5: So .

Answer: 4

Tip

Take the root first using the denominator, then raise to the power using the numerator — the numbers stay small and manageable that way.

Trap

Doing the power first before taking the root makes the numbers much bigger and harder to simplify by hand.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2	Correctly found the cube root of 8 as 2, but forgot to also square it using the numerator 2.
	Treated as a multiplier on 8 instead of using it as a root-and-power exponent.



	Treated the exponent as if it were negative, inverting the correct positive answer into a fraction.
--	---

Shortcut

Recognizing that 8 is a perfect cube lets you jump straight to the small root value without extra work.

Desmos Tip

Type directly into a Desmos line to see it evaluate to 4 and confirm your answer.

Radicals, Rational Exponents & Scientific Notation — Problem 3 (Easy)

A virus particle has a mass of about grams. A vial contains of these particles. What is the total mass of the particles in the vial, in grams, written in scientific notation?

Formula / Rule Used: Scientific notation multiplication: multiply the coefficients, add the exponents.

Solution

Step 1: Set up the multiplication: .

Step 2: Multiply the coefficients together: $5 \times 4 = 20$.

Step 3: Add the exponents together: $-15 + 9 = -6$.

Step 4: Combine the pieces so far: .

Step 5: Check the coefficient: 20 is not between 1 and 10, so it needs to be renormalized.

Step 6: Rewrite 20 as , then combine exponents again: .

Answer: grams

Tip

Multiply the coefficients and add the exponents separately, then always double-check the coefficient ends between 1 and 10.

Trap

Forgetting to renormalize a coefficient like 20 back into proper scientific notation is the single most common slip on these questions.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
grams	Multiplied correctly but stopped without renormalizing the coefficient back into the range 1 to 10.
grams	Renormalized the coefficient to 2 but forgot to adjust the exponent by +1 to compensate.
grams	Subtracted the exponents instead of adding them, getting $-15 - 9 = -24$.

Shortcut



None beyond the direct rule — multiply coefficients, add exponents, then renormalize; there's no faster legitimate path.

Desmos Tip

Type into Desmos to get the decimal value, then compare it to to confirm they match.

Radicals, Rational Exponents & Scientific Notation — Problem 4 (Easy)

Solve for x : .

Formula / Rule Used: Isolate the radical, then square both sides to undo it.

Solution

Step 1: The radical is already alone on the left side of the equation.

Step 2: Square both sides of the equation to undo the square root: .

Step 3: The left side simplifies to just $x + 3$.

Step 4: The right side simplifies to 25, so the equation becomes $x + 3 = 25$.

Step 5: Subtract 3 from both sides to solve for x : $x = 22$.

Step 6: Check by substituting back in: , which matches, so $x = 22$ is valid.

Answer: $x = 22$

Tip

Whenever a variable is trapped under a root that's already alone on one side, square both sides first to free it.

Trap

It's easy to square the left side but forget the right side must also be squared fully, turning 5 into 25.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 2$	Subtracted 3 from 5 directly ($5 - 3 = 2$) instead of squaring both sides before isolating x .
$x = 28$	Added 3 to 25 instead of subtracting it, flipping the sign of the final step.
$x = 25$	Squared both sides correctly to get $x + 3 = 25$ but forgot to subtract the 3 in the last step.

Shortcut

None needed — squaring both sides immediately is already the fastest method for an isolated radical equation.

Desmos Tip

Graph and $y = 5$ as two separate lines in Desmos; the x -value where they cross confirms $x = 22$.

Radicals, Rational Exponents & Scientific Notation — Problem 5 (Medium)

If $x = 8^{1/3}$, what is the value of x ?

Formula / Rule Used: Simplify one side to a number, then raise both sides to the reciprocal exponent.

Solution

Step 1: First simplify the right side: means (cube root of 8) raised to the 4th power.

Step 2: The cube root of 8 is 2, since $2^3 = 8$.

Step 3: Raise that result to the 4th power: $2^4 = 16$.

Step 4: The equation now reads $x = 16$.

Step 5: To isolate x , raise both sides to the reciprocal exponent, $1/4$.

Step 6: means (square root of 16) raised to the 3rd power.

Step 7: The square root of 16 is 4, so $x = 4^3$.

Step 8: So $x = 64$.

Answer: $x = 64$

Tip

Simplify the known side down to a single plain number first, then flip the exponent to solve for the variable.

Trap

Using the same exponent to solve for x instead of correctly flipping it to its reciprocal, $x = 16$, does nothing.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 16$	Correctly simplified the right side to 16 but stopped there, forgetting to actually solve for x .
$x = 12$	Found the square root of 16 correctly as 4, but then multiplied by 3 instead of raising it to the 3rd power.
	Treated the exponent $1/4$ as a multiplier on 8 instead of a root-and-power operation, and stopped there.

Shortcut

Rewriting 8 and 16 as powers of 2, $8 = 2^3$ and $16 = 2^4$, lets you compare exponents directly instead of computing roots.

Desmos Tip

Graph $y = x^{1/4}$ and $y = 16$ in Desmos and find their intersection point — the x -coordinate confirms $x = 64$.

Radicals, Rational Exponents & Scientific Notation — Problem 6 (Medium)

A star system studied by an observatory in Reykjavik is estimated to be kilometers from Earth, a distance the observatory's telescope, upgraded in 2019, took three years to confirm using new sensors. A spacecraft travels at an average speed of kilometers per hour. Approximately how many hours would the spacecraft take to reach the star system, written in scientific notation?

Formula / Rule Used: Scientific notation division: divide coefficients, subtract exponents, then renormalize.

Solution

Step 1: Ignore the detail about the telescope upgrade and the three years — it's not needed to answer the question.

Step 2: Set up the calculation as time = distance / speed: .

Step 3: Divide the coefficients: .

Step 4: Subtract the exponents: $13 - 4 = 9$, giving .

Step 5: Combine the pieces so far: .

Step 6: Check the coefficient: 0.8 is less than 1, so it needs to be renormalized.

Step 7: Rewrite 0.8 as , then combine exponents again: .

Answer: hours

Tip

Set up rate problems as distance divided by speed, then apply the scientific-notation division rule directly.

Trap

The 'upgraded in 2019' and 'three years' details are extra noise meant to distract — the question only needs the distance and speed values.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
hours	Divided correctly but left the coefficient at 0.8 instead of renormalizing it into the range 1 to 10.
hours	Renormalized the coefficient correctly to 8 but forgot to decrease the exponent by 1 to compensate.
hours	Multiplied the two scientific-notation values together instead of dividing distance by speed.

Shortcut

Rounding to the friendlier fraction in your head is faster than long division under time pressure.

Desmos Tip

Type into Desmos to get the decimal value, then compare it to to confirm they match.

If , what is the value of x ?

Formula / Rule Used: Rewrite both sides with a common base, then set the exponents equal.

Solution

Step 1: Notice that 9 and 27 are both powers of the same base, 3: and .

Step 2: Rewrite the left side using that base: .

Step 3: Rewrite the right side the same way: .

Step 4: Since both sides now share the base 3, their exponents must be equal: $2x + 2 = 3x - 3$.

Step 5: Add 3 to both sides: $2x + 5 = 3x$.

Step 6: Subtract $2x$ from both sides: $5 = x$.

Step 7: So $x = 5$.

Answer: $x = 5$

Tip

Whenever two exponential expressions are set equal, rewrite both sides with the same base first, then set exponents equal.

Trap

Trying to solve this without first spotting that 9 and 27 share the common base 3 leads to unnecessary, error-prone work.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = -5$	Made a sign error while moving terms across the equals sign, reversing $2x - 3x = -2 - 3$ by mistake.
$x = 1$	Set 9 equal to 27 directly and solved a simpler but incorrect equation instead of rewriting both with base 3.
$x = -1$	Correctly rewrote the exponents but subtracted 2 from both sides instead of adding 3, mishandling the constant term.

Shortcut

Recognizing 9 and 27 as powers of 3 immediately is much faster than any trial-and-error or guessing approach.

Desmos Tip

Graph and in Desmos and find their intersection point — the x -coordinate confirms $x = 5$.

Radicals, Rational Exponents & Scientific Notation — Problem 8 (Medium)

A radical expression, once fully simplified, equals $\sqrt{36N}$. If the original expression was $\sqrt{N}$ for some positive integer N less than 200, what is the value of N ?

Formula / Rule Used: Reverse of simplifying radicals: $\sqrt{a^2b} = a\sqrt{b}$.

Solution

Step 1: To reverse a simplified radical, the coefficient must be squared and moved back under the root.

Step 2: Square the coefficient: $6^2 = 36$.

Step 3: Multiply that result by the number still under the radical: $36 \times 3 = 108$.

Step 4: So the original expression was $\sqrt{108}$.

Step 5: Check the work: $\sqrt{108} = \sqrt{36 \times 3} = 6\sqrt{3}$, which matches.

Step 6: Confirm 108 is less than 200, satisfying the problem's condition.

Answer: $N = 108$

Tip

To reverse a simplified radical, square the outside coefficient first, then multiply it back under the root.

Trap

Multiplying the coefficient 6 directly by the radicand 3 (getting 18) instead of squaring 6 first is the most common way to get this wrong.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$N = 18$	Multiplied the coefficient 6 by the radicand 3 directly instead of squaring 6 before multiplying.
$N = 9$	Added the coefficient and the radicand ($6 + 3$) instead of squaring the coefficient and multiplying.
$N = 216$	Cubed the coefficient instead of squaring it before combining with the radicand.

Shortcut

None truly faster — reversing a simplified radical always requires squaring the coefficient and multiplying it back in.

Desmos Tip

Type $\sqrt{108}$ and $6\sqrt{3}$ as two separate lines in Desmos — matching decimal values confirm $N = 108$.

PART 2 — PROBLEM-SOLVING AND DATA ANALYSIS

7. Ratios & Proportions

Ratios & Proportions — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Ratio	Λόγος
Proportion	Αναλογία
Cross-multiplication	Εξίσωση γινομένων (χιαστί πολλαπλασιασμός)
Scale factor	Λόγος κλίμακας
Unit rate	Μοναδιαία τιμή

Ratios & Proportions — What It Is

What It Is

- A ratio compares two quantities by division, like comparing boys to girls in a class.
- A proportion is simply two ratios set equal to each other, like $a/b = c/d$.
- The SAT tests ratios because real life is full of scaled comparisons: recipes, maps, mixtures, speeds.
- A ratio only describes a relationship between parts, not the actual sizes of those parts.
- Once you know the relationship, you can scale it up or down to find any real quantity.

Ratios & Proportions — Core Rules

Ratio Notation

$$a : b = a/b$$

- Order matters — the ratio 'boys to girls' lists boys first, girls second.
- Any ratio $a:b$ can be rewritten as the fraction a/b .
- e.g. '3 apples for every 5 oranges' is written 3:5, not 5:3.

The 'x' Method (Common Multiplier)

If $a:b = m:n$, then $a = mx$ and $b = nx$ for some number x .

- A ratio alone only gives a relationship, not actual sizes, so introduce one unknown multiplier x .
- Use another given fact, like a total, to solve for x first.
- Once you know x , plug it back into mx and $n x$ to get the real quantities.

- e.g. ratio 2:3, total 20: $2x + 3x = 20$, so $x = 4$, and the parts are 8 and 12.

Proportions & Cross-Multiplication

$$a/b = c/d \Leftrightarrow ad = bc$$

- A proportion states that two ratios are equal to one another.
- Cross-multiply by multiplying diagonally across the equals sign.
- Always set up both ratios in the same order, such as both as part over whole.
- e.g. $3/4 = x/12$ cross-multiplies to $3(12) = 4x$, so $x = 9$.

Combining Two Ratios (Ratio Chains)

- Sometimes you're given $x:y$ and $y:z$ separately and asked to find $x:z$.
- Scale both ratios so the shared term (y) has the same value in each one.
- Find the LCM of the two y -values, then scale each ratio up to match it.
- e.g. $x:y = 2:3$, $y:z = 4:5$: scale to $y = 12$, giving $8:12$ and $12:15$, so $x:z = 8:15$.

Part-to-Part vs. Part-to-Whole

- A ratio like 2:3 compares one part to another part, not to the total amount.
- The total number of 'shares' is the sum of the ratio's terms: $2 + 3 = 5$ shares.
- To find a part's fraction of the whole, put that part over the total share count, not over the other part.
- e.g. in a 2:3 ratio, the first quantity is $2/5$ of the whole, not $2/3$ of it.

Direct vs. Inverse Proportion

$$\text{Direct: } y = kx. \quad \text{Inverse: } y = k/x.$$

- In a direct proportion, both quantities increase or decrease together.
- In an inverse proportion, as one quantity increases, the other decreases.
- Test which type applies by asking: does doubling one quantity double the other, or cut it in half?
- e.g. more workers finishing a job faster is inverse, since more workers means less time.

Scaling a Ratio (Equivalent Ratios)

$$a:b = ka:kb \text{ for any } k$$

- Multiplying or dividing both terms of a ratio by the same number keeps it equivalent.
- This is exactly how recipes, maps, and scale models get resized without changing the relationship.
- e.g. 2:5 is equivalent to 6:15, since both terms were multiplied by 3.

How to Recognize This on the Test

- The question uses words like 'ratio,' 'for every,' 'per,' or 'proportion.'
- You're given a relationship between two or more quantities plus one extra fact, like a total.
- The question asks you to scale a recipe, mixture, map, or model up or down.
- Two fractions or rates are set equal to each other, often with one unknown value.
- The question describes one quantity changing directly or inversely with another, like speed and time.

Common Mistakes — and the Fix

- Mixing up the order of a ratio, writing $b:a$ when the question actually asked for $a:b$.
- Treating a part-to-part ratio as if it were already a fraction of the whole.
- Guessing numbers that seem close instead of setting up the 'x' method to solve exactly.
- Setting up a proportion with the two ratios in different orders, which breaks the cross-multiplication.
- Confusing direct and inverse proportion, using $y = kx$ in a situation that is actually inverse.

Ratios & Proportions — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Ratios & Proportions — Problem 1 (Easy)

A jar contains only red and green jelly beans in the ratio 3:5. If the jar has 40 jelly beans in total, how many are red?

Formula / Rule Used: The 'x' method: let red = $3x$ and green = $5x$, then use the total.

Solution

Step 1: Let the number of red jelly beans be $3x$ and green be $5x$, based on the ratio 3:5.

Step 2: The total number of jelly beans is red plus green: $3x + 5x = 40$.

Step 3: Combine like terms: $8x = 40$.

Step 4: Divide both sides by 8 to solve for x : $x = 5$.

Step 5: Multiply back to find the actual number of red jelly beans: $3x = 3(5) = 15$.

Answer: 15 red jelly beans

Tip

Whenever a ratio and a total are both given, use the 'x' method — let each part be its ratio number times x , then add them to the total.

Trap

It's tempting to just take $3/5$ of 40 without setting up x , but that skips the step of finding the total number of shares first.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
5	Solved correctly for the multiplier $x = 5$ but reported x itself instead of scaling it back to $3x$ for the actual red count.
24	Used the fraction $3/5$ (red compared to green) instead of $3/8$ (red compared to the total), computing $3/5 \times 40$.
25	Mixed up which color was asked for and calculated the number of green jelly beans ($5x = 25$) instead of red.

Shortcut

Notice $40 \div 8 = 5$ exactly, so each 'share' is worth 5 — just multiply the red share number (3) by 5.

Desmos Tip

Type $y = 8x - 40$ in Desmos and find where the graph crosses zero to confirm $x = 5$, then multiply by 3 for the red count.

Ratios & Proportions — Problem 2 (Easy)

A taverna in Thessaloniki uses 2 cups of olive oil for every 5 servings of moussaka it prepares. If the taverna needs to prepare 35 servings for a large event, how many cups of olive oil are needed?

Formula / Rule Used: Set up a proportion between cups of oil and servings, then cross-multiply.

Solution

Step 1: Set up a proportion comparing oil to servings: $2 \text{ cups} / 5 \text{ servings} = x \text{ cups} / 35 \text{ servings}$.

Step 2: Cross-multiply: $2 \times 35 = 5 \times x$.

Step 3: Simplify the left side: $70 = 5x$.

Step 4: Divide both sides by 5 to solve for x : $x = 14$.

Answer: 14 cups of olive oil

Tip

Set up both ratios in the exact same order — oil over servings on both sides — before cross-multiplying.

Trap

It's easy to flip one side of the proportion (servings over oil) by accident, which gives a much larger, incorrect number.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
87.5	Set up the proportion upside down as $5/2 = x/35$, giving an oil amount larger than the total servings.



70	Cross-multiplied correctly to get $5x = 70$ but forgot the final step of dividing by 5.
17.5	Divided 35 by 2 (the oil number) instead of by 5 (the servings number) when finding the scale factor.

Shortcut

Notice 35 is 7 times 5, so just multiply the oil number 2 by that same scale factor 7 to get 14 instantly.

Desmos Tip

Type $y1 = (2/5)x$ in Desmos and trace the graph to $x = 35$ to confirm the output is 14.

Ratios & Proportions — Problem 3 (Easy)

At a school in Rotterdam, the ratio of students who bike to school to those who don't is 4:3. If there are 210 students in total, how many students bike to school?

Formula / Rule Used: The 'x' method: split the total into ratio shares, then scale up the bikers' share.

Solution

Step 1: Let the number of students who bike be $4x$ and those who don't be $3x$, based on the ratio 4:3.

Step 2: The total number of students is the sum of both groups: $4x + 3x = 210$.

Step 3: Combine like terms: $7x = 210$.

Step 4: Divide both sides by 7 to solve for x : $x = 30$.

Step 5: Multiply back to find the bikers: $4x = 4(30) = 120$.

Answer: 120 students

Tip

Add the two ratio numbers together first to find the total number of 'shares' the whole is split into.

Trap

It's tempting to divide 210 by 4 directly, but 4 is only part of the ratio, not the total number of shares.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
52.5	Divided 210 by 4 directly instead of by the total of 7 shares ($4 + 3$).
90	Solved for the students who bike using the wrong ratio number, computing $3x = 3(30) = 90$ instead of $4x$.
140	Added the ratio parts incorrectly as $4 + 3 = 6$ instead of 7, then computed $210 \times 4/6$.

Shortcut

Since $210 \div 7 = 30$ exactly, find the value of one share first, then multiply it by 4 for the bikers.

Ratios & Proportions — Problem 4 (Easy)

A ferry travels from Piraeus to a nearby island at a constant speed, covering 45 kilometers in 3 hours. At this same speed, how many kilometers will it cover in 7 hours?

Formula / Rule Used: Direct proportion: find the unit rate, then multiply by the new time.

Solution

Step 1: Find the ferry's rate by dividing distance by time: $45 \text{ km} / 3 \text{ hours} = 15 \text{ km per hour}$.

Step 2: Since speed stays constant, distance and time are directly proportional: $\text{distance} = 15 \times \text{time}$.

Step 3: Plug in the new time of 7 hours: $\text{distance} = 15 \times 7$.

Step 4: Multiply: $\text{distance} = 105$.

Answer: 105 kilometers

Tip

Find the unit rate (distance per one hour) first — it turns every other time value into a one-step multiplication.

Trap

Some students set up the proportion upside down as time/distance, which gives a rate less than 1 instead of 15 km/h.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
52	Added the given distance and new time instead of setting up a proportion, computing $45 + 7$.
15	Calculated the unit rate correctly (15 km/h) but reported the rate itself instead of using it to find the new distance.
315	Multiplied 45×7 directly instead of finding the unit rate first, skipping the division by 3.

Shortcut

Since 7 hours isn't a clean multiple of 3, finding the unit rate (15 km/h) first is the fastest path here.

Desmos Tip

Type $y1 = 15x$ in Desmos and read the y-value at $x = 7$ to confirm the distance is 105 kilometers.

Ratios & Proportions — Problem 5 (Medium)

At a summer camp in Crete, the ratio of counselors to junior counselors is 2:5, and the ratio of junior counselors to campers is 3:8. If there are 240 campers at the camp, how many counselors are there?

Formula / Rule Used: Combine two ratios into a chain using their shared term, then apply the 'x' method.

Solution

Step 1: The shared term between the two ratios is junior counselors: 5 in the first ratio, 3 in the second.

Step 2: Find the LCM of 5 and 3, which is 15, so scale each ratio to make its junior counselors term equal 15.

Step 3: Scale the first ratio (2:5) by 3: counselors : junior = 6 : 15.

Step 4: Scale the second ratio (3:8) by 5: junior : campers = 15 : 40.

Step 5: Combine both into a single chain: counselors : junior : campers = 6 : 15 : 40.

Step 6: Let campers = 40x, so $40x = 240$, giving $x = 6$.

Step 7: Find the counselors: $6x = 6(6) = 36$.

Answer: 36 counselors

Tip

When two ratios share a common category, scale both so that shared number matches before combining them into one chain.

Trap

It's tempting to compare counselors to campers directly (2 to 8) without matching the junior counselors term first, which gives a wrong ratio.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
60	Compared counselors to campers directly as 2:8 without scaling through the shared junior counselors term, computing $240 \times 2/8$.
90	Combined the ratios correctly but reported the junior counselors count ($15x = 90$) instead of the counselors count.
30	Found $x = 6$ correctly but multiplied by 5 instead of 6, using the un-scaled ratio number by mistake.

Shortcut

None faster here — combining the ratios into one chain before using the total is necessary since counselors and campers aren't directly compared.

Desmos Tip

Type $y_1 = 6x$, $y_2 = 15x$, and $y_3 = 40x$ in Desmos, then find x where $y_3 = 240$ to instantly read off $y_1 = 36$.

Ratios & Proportions — Problem 6 (Medium)

A shipping warehouse in Rotterdam needs 8 workers to sort a container of packages in 6 hours. If the warehouse wants the same container sorted in only 4 hours, working at the same rate, how many workers are needed?

Formula / Rule Used: Inverse proportion: workers $\times$ time = a constant amount of work.

Solution

Step 1: Since more workers finish the job faster, workers and time are inversely proportional, not directly.

Step 2: Find the total amount of work using the original numbers: 8 workers $\times$ 6 hours = 48 worker-hours.

Step 3: This 48 worker-hours stays the same no matter how the job is split up.

Step 4: Set up the new equation: (number of workers) $\times$ 4 hours = 48 worker-hours.

Step 5: Divide both sides by 4 to solve for the number of workers: $48 / 4 = 12$.

Answer: 12 workers

Tip

Whenever more of one quantity causes less of another, like workers and time, multiply the original pair together to find the constant total work.

Trap

Setting this up as a direct proportion, like $8/6 = x/4$, gives a smaller, incorrect number of workers instead of a larger one.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
5.33	Set up a direct proportion $8/6 = x/4$ instead of recognizing the relationship is inverse, decreasing workers instead of increasing them.
6	Assumed workers and time change by the same amount and simply subtracted 2 from 8 since the time dropped by 2 hours.
48	Correctly found the total work as 48 worker-hours but reported that number instead of dividing by the new time.

Shortcut

Multiply the original workers and hours together first ($8 \times 6 = 48$) — that constant turns every new time value into a one-step division.

Desmos Tip

Type $y1 = 48/x$ in Desmos and read the y-value at $x = 4$ to confirm 12 workers are needed.

Ratios & Proportions — Problem 7 (Medium)

A vineyard in Santorini mixes fertilizer and water in a ratio of 2:15 for its tomato plants, which were first planted back in 2010. If a farmer uses 6 liters of fertilizer, how many total liters of the fertilizer-water mixture will he prepare?

Formula / Rule Used: Scale the ratio to match the given part, then add both parts for the total mixture.

Solution

Step 1: Ignore the planting year — it's background detail that doesn't affect the ratio calculation.

Step 2: Set up a proportion for fertilizer to water: $2/15 = 6/x$.

Step 3: Cross-multiply: $2x = 15 \times 6$.

Step 4: Simplify the right side: $2x = 90$.

Step 5: Divide both sides by 2 to find the water amount: $x = 45$.

Step 6: Add fertilizer and water to get the total mixture: $6 + 45 = 51$.

Answer: 51 liters

Tip

When a question gives one part of a ratio and asks for the total mixture, find the missing part first, then add both parts together.

Trap

The planting year (2010) is an irrelevant detail meant to distract you — it has nothing to do with the fertilizer-to-water ratio.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
45	Correctly solved for the water amount but forgot to add the 6 liters of fertilizer to get the total mixture.
0.8	Mixed up which side of the proportion each value belongs on, computing $2 \times 6 = 15x$ instead of $15 \times 6 = 2x$.
90	Cross-multiplied correctly to $2x = 90$ but forgot to divide by 2, reporting 90 as the water amount.

Shortcut

Find the water amount first using the ratio, then remember the question wants fertilizer plus water together, not water alone.

Desmos Tip

Type $y1 = (15/2)x$ in Desmos, evaluate at $x = 6$ to get 45 liters of water, then add 6 for the total mixture.



The variable y varies directly with x . When $x = 9$, $y = 15$. What is the value of y when $x = 21$?

Formula / Rule Used: Direct variation: $y = kx$. Find k using the given pair, then apply it to the new x .

Solution

Step 1: Direct variation means $y = kx$ for some constant k .

Step 2: Use the given pair to solve for k : $15 = k(9)$.

Step 3: Divide both sides by 9: $k = 15/9$.

Step 4: Simplify the fraction: $k = 5/3$.

Step 5: Use this k to find y when $x = 21$: $y = (5/3)(21)$.

Step 6: Multiply: since $21/3 = 7$, $y = 5 \times 7 = 35$.

Answer: 35

Tip

Whenever a problem says two quantities 'vary directly,' find the constant k first using the given pair, then reuse it for the new value.

Trap

Some students set up k using x/y instead of y/x , which flips the constant and gives a wrong value.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
15	Solved for k correctly but then re-plugged the original $x = 9$ instead of the new $x = 21$, getting back the given y -value.
12.6	Found k as $x/y = 9/15 = 0.6$ instead of y/x , then multiplied $0.6 \times 21 = 12.6$.
27	Used additive thinking instead of proportional scaling, computing $21 - 9 = 12$ and adding it to 15 to get 27.

Shortcut

Notice $21/9 = 7/3$, so multiply $y = 15$ by $7/3$ directly instead of solving for k separately: $15 \times 7/3 = 35$.

Desmos Tip

Type $y1 = (5/3)x$ in Desmos and read the y -value at $x = 21$ to confirm $y = 35$.

8. Probability

Probability — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Probability	Πιθανότητα
Independent events	Ανεξάρτητα ενδεχόμενα
Dependent events	Εξαρτημένα ενδεχόμενα
Mutually exclusive events	Ασυμβίβαστα ενδεχόμενα
Sample space	Δειγματικός χώρος

Probability — What It Is

What It Is

- Probability is a number between 0 and 1 that measures how likely an event is to happen.
- 0 means the event is impossible, 1 means it's certain, and everything else is somewhere in between.
- The SAT tests probability because it shows up in real decisions, like odds, surveys, and games.
- Most SAT probability questions give you counts (how many of each thing) rather than abstract formulas.
- You'll see single events, combined events with 'and'/'or', and probability read straight off a data table.

Probability — Core Rules

Basic Probability

$$P(\text{event}) = \text{favorable outcomes} / \text{total outcomes}$$

- Count how many outcomes match what you want — that's the favorable outcomes.
- Count every possible outcome in the whole situation — that's the total outcomes.
- Divide favorable by total to get a fraction, decimal, or percent between 0 and 1.
- Example: a bag has 3 red and 7 blue marbles, so $P(\text{red}) = 3/10$.

Probability Is Always Between 0 and 1

$$0 \leq P(\text{event}) \leq 1$$

- If your final answer is negative or bigger than 1, you made an arithmetic mistake.
- A probability of 0 means the event never happens; 1 means it always happens.
- You can also write probability as a percent, from 0% to 100%, meaning the same thing.
- Use this as a quick sanity check on every probability answer you calculate.

The Complement Rule

$$P(\text{not } A) = 1 - P(A)$$

- The complement of an event is everything that could happen except that event.
- Every outcome either is event A or is not event A — the two probabilities add to 1.
- This is the fastest way to handle 'at least one' questions.
- Example: if $P(\text{rain}) = 0.3$, then $P(\text{no rain}) = 1 - 0.3 = 0.7$.

'And' — Two Events Both Happening

$$\text{Independent: } P(A \text{ and } B) = P(A) \times P(B)$$

- 'Independent' means the first event doesn't change the odds of the second one.
- Look for the phrase 'with replacement' — that's a strong signal the events are independent.
- Multiply the two individual probabilities together to get the combined probability.
- Example: flipping heads twice in a row is $P(H) \times P(H) = (1/2) \times (1/2) = 1/4$.

'And' With Dependent Events

$$\text{Dependent: } P(A \text{ and } B) = P(A) \times P(B \text{ after } A)$$

- 'Dependent' means the first event changes what's left for the second event.
- Look for the phrase 'without replacement' — the item removed doesn't go back.
- Recalculate the second probability using the new, smaller totals after the first event.
- Example: drawing 2 aces from a deck without replacement, the second draw uses only 51 cards left, with only 3 aces remaining.

'Or' — At Least One of Two Events

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

- 'Or' means you want the event to be A, or B, or possibly both at once.
- Adding $P(A)$ and $P(B)$ alone double-counts any outcomes that satisfy both.
- Subtract $P(A \text{ and } B)$, the overlap, so those outcomes are only counted once.
- If A and B can never happen together (mutually exclusive), just add: $P(A \text{ or } B) = P(A) + P(B)$.

Probability From a Two-Way Table

$$P = (\text{cell value}) / (\text{row total, column total, or grand total})$$

- A two-way table sorts a group by two categories at once, like gender and favorite sport.
- For a simple probability, divide the target cell by the grand total of the whole table.
- For a conditional probability ('given' a certain group), divide by that group's row or column total only, not the grand total.
- Always double check which total the question wants before you divide.

How to Recognize This on the Test

- The word 'probability,' 'likelihood,' or 'chance' directly signals this skill.
- A question asking 'what is the probability that... and...' means you'll combine two events.
- Phrases like 'at random' or 'randomly selected' mean every outcome is equally likely to be chosen.
- A two-way table with row and column totals almost always leads to a probability or conditional probability question.
- 'Without replacement' or 'not returned to the group' is a signal the second event's totals must be recalculated.

Common Mistakes — and the Fix

- Using the original total instead of the reduced total for a 'without replacement' second draw.
- Forgetting to subtract the overlap in an 'or' problem, which double-counts outcomes and inflates the answer.
- Dividing by the grand total in a two-way table when the question actually asked for a conditional probability.
- Multiplying probabilities for an 'or' situation instead of an 'and' situation, mixing up the two rules.
- Reporting a probability greater than 1 or negative without noticing that's a sign of a setup mistake.

Probability — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Probability — Problem 1 (Easy)

A school fair in Athens sells 250 raffle tickets, and 15 of those tickets are winning tickets. One ticket is drawn at random. What is the probability that the drawn ticket is a winning ticket?

Formula / Rule Used: Basic probability: $P = \text{favorable outcomes} / \text{total outcomes}$.

Solution

Step 1: Identify the favorable outcomes: the number of winning tickets is 15.

Step 2: Identify the total outcomes: the total number of tickets is 250.

Step 3: Write the probability as a fraction: $P = 15 / 250$.

Step 4: Find a number that divides evenly into both 15 and 250 — try 5.

Step 5: Divide the numerator by 5: $15 / 5 = 3$.

Step 6: Divide the denominator by 5: $250 / 5 = 50$.

Step 7: The simplified probability is $P = 3/50$.

Answer: 3/50

Tip



Always write favorable outcomes over total outcomes first, then simplify the fraction as a separate last step.

Trap

It's tempting to use the non-winning tickets instead of the winning tickets in the numerator or denominator.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
47/50	Used the number of non-winning tickets ($250 - 15 = 235$) as the favorable count instead of the 15 winning tickets.
15/235	Used the non-winning tickets (235) as the total instead of all 250 tickets sold.
3/5	Misread the total as 25 tickets instead of 250, dropping a zero before simplifying.

Shortcut

No real shortcut needed — writing the fraction and simplifying is already the fastest path for a direct probability like this.

Desmos Tip

Type 15/250 into the Desmos calculator and confirm the decimal matches 3/50 (0.06).

Probability — Problem 2 (Easy)

A weather app in Rotterdam reports a 35% chance of rain tomorrow. What is the probability, written as a decimal, that it does NOT rain tomorrow?

Formula / Rule Used: Complement rule: $P(\text{not } A) = 1 - P(A)$.

Solution

Step 1: Convert the given percent to a decimal: $35\% = 0.35$.

Step 2: This 0.35 represents $P(\text{rain})$, the probability of the event happening.

Step 3: The complement, $P(\text{not rain})$, is everything except that event.

Step 4: Use the complement rule: $P(\text{not rain}) = 1 - P(\text{rain})$.

Step 5: Substitute the value: $P(\text{not rain}) = 1 - 0.35$.

Step 6: Subtract: $P(\text{not rain}) = 0.65$.

Answer: 0.65

Tip

Whenever you're given the probability of one outcome and asked for the opposite outcome, subtract from 1.

Trap

It's tempting to just restate the given 0.35 as the answer instead of subtracting it from 1.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
0.35	Reported the given probability of rain instead of calculating the complement for no rain.
1.35	Added 1 and 0.35 instead of subtracting, reversing the complement rule.
-0.65	Subtracted in the wrong order, computing $0.35 - 1$ instead of $1 - 0.35$.

Shortcut

Once you see 'does not happen,' just subtract the given probability from 1 — no other setup is needed.

Desmos Tip

Type $1 - 0.35$ directly into the Desmos calculator to confirm the complement equals 0.65.

Probability — Problem 3 (Easy)

A game show spinner is divided into 8 equal sections numbered 1 through 8. A contestant spins the spinner, notes the result, and then spins it again under the exact same conditions. What is the probability that both spins land on a number greater than 5?

Formula / Rule Used: Independent events (with replacement): $P(A \text{ and } B) = P(A) \times P(B)$.

Solution

Step 1: List the numbers greater than 5 on the spinner: 6, 7, and 8 — that's 3 favorable outcomes.

Step 2: The total outcomes for one spin is 8, since there are 8 equal sections.

Step 3: Find the probability of one spin landing greater than 5: $P = 3/8$.

Step 4: Since the spinner resets each time, the two spins are independent of each other.

Step 5: For independent events, multiply the individual probabilities: $P(\text{both}) = (3/8) \times (3/8)$.

Step 6: Multiply the numerators: $3 \times 3 = 9$.

Step 7: Multiply the denominators: $8 \times 8 = 64$.

Step 8: The combined probability is $P = 9/64$.

Answer: 9/64

Tip

When an event repeats under identical conditions ('with replacement'), multiply the same probability by itself.

Trap

It's tempting to add the two probabilities together instead of multiplying them for an 'and' situation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3/4	Added the two individual probabilities instead of multiplying them: $3/8 + 3/8 = 6/8 = 3/4$.



1/4	Miscounted 'greater than 5' as including 5 itself, using 4/8 for each spin instead of 3/8.
9/8	Multiplied the numerators correctly but forgot to also multiply the denominators, writing $3/8 \times 3 = 9/8$.

Shortcut

Once you know one spin's probability is $3/8$, just square that fraction — no need to recount outcomes for the second spin.

Desmos Tip

Type $(3/8) * (3/8)$ into the Desmos calculator to confirm the result equals $9/64$.

Probability — Problem 4 (Easy)

A survey of 150 students at a school in Lisbon sorted them by gender and preferred sport. Of the 80 boys, 54 prefer soccer and 26 prefer basketball. Of the 70 girls, 31 prefer soccer and 39 prefer basketball. One student from the entire school is selected at random. What is the probability that the selected student is a girl who prefers basketball?

Formula / Rule Used: Basic probability from a two-way table: target cell / grand total.

Solution

Step 1: Identify the target cell: girls who prefer basketball, which is 39 students.

Step 2: Identify the grand total: all students surveyed, which is $80 + 70 = 150$.

Step 3: Write the probability as a fraction: $P = 39/150$.

Step 4: Find a common factor for 39 and 150 — try 3.

Step 5: Divide the numerator by 3: $39 / 3 = 13$.

Step 6: Divide the denominator by 3: $150 / 3 = 50$.

Step 7: The simplified probability is $P = 13/50$.

Answer: 13/50

Tip

For a plain probability (no 'given'), always divide by the grand total of the whole table, not one row or column.

Trap

It's tempting to divide by the girls' row total (70) instead of the whole school's total (150) since the question mentions girls.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
39/70	Divided by the girls' row total (70) instead of the grand total, accidentally computing a conditional probability.



13/75	Read the boys-and-basketball cell (26) instead of the girls-and-basketball cell, giving $26/150 = 13/75$.
39/85	Divided by the soccer column total (85) instead of the grand total (150).

Shortcut

No shortcut beyond reading the correct cell and dividing by the grand total — that's already the fastest path.

Desmos Tip

Type 39/150 into the Desmos calculator and confirm it matches the simplified fraction 13/50 (0.26).

Probability — Problem 5 (Medium)

A box of USB drives at an electronics store in Seoul contains only red and green drives, 24 drives in total. The probability of randomly selecting a red drive from the box is $3/8$. If two drives are selected at random without replacement, what is the probability that both selected drives are green?

Formula / Rule Used: Dependent events (without replacement): $P(A \text{ and } B) = P(A) \times P(B \text{ after } A)$.

Solution

Step 1: Find the number of red drives: $(3/8) \times 24 = 9$ red drives.

Step 2: Subtract from the total to find the green drives: $24 - 9 = 15$ green drives.

Step 3: Find the probability the first drive selected is green: $15/24$.

Step 4: Since the first drive is not replaced, one green drive and one total drive are now gone.

Step 5: The green drives remaining is $15 - 1 = 14$, and the total remaining is $24 - 1 = 23$.

Step 6: Find the probability the second drive is also green: $14/23$.

Step 7: Multiply the two dependent probabilities: $(15/24) \times (14/23)$.

Step 8: Simplify $15/24$ to $5/8$ first, then multiply: $(5/8) \times (14/23) = 70/184 = 35/92$.

Answer: 35/92

Tip

When a probability and a total are given, first find the actual counts of each group before doing anything else.

Trap

It's tempting to treat this like 'with replacement' and use 24 as the total for both draws, instead of reducing it for the second draw.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
25/64	Treated the two draws as independent (with replacement), computing $(15/24)^2 = 25/64$ instead of reducing the total after the first draw.
3/23	Used the red-drive count (9) as if it were the green-drive count, computing



	$(9/24) \times (8/23)$ instead of using 15 green drives.
35/96	Correctly reduced the number of green drives left (14) but forgot to also reduce the total drives left, using 24 again instead of 23.

Shortcut

None beyond finding the green count first and multiplying the two dependent probabilities directly — that's already the fastest path.

Desmos Tip

Type $(15/24) * (14/23)$ into the Desmos calculator to confirm the decimal matches the simplified fraction $35/92$.

Probability — Problem 6 *(Medium)*

A charity raffle in Dublin sells 60 tickets numbered 1 through 60, and one ticket will be drawn at random to win a prize (the winner's photo is also added to the charity's newsletter). What is the probability that the winning ticket's number is a multiple of 4 or a multiple of 6?

Formula / Rule Used: 'Or' with overlap: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

Solution

Step 1: Ignore the detail about the newsletter photo — it has nothing to do with probability.

Step 2: Count the multiples of 4 from 1 to 60: there are $60/4 = 15$ of them.

Step 3: Count the multiples of 6 from 1 to 60: there are $60/6 = 10$ of them.

Step 4: Find the overlap: numbers that are multiples of BOTH 4 and 6 are multiples of their LCM, 12.

Step 5: Count the multiples of 12 from 1 to 60: there are $60/12 = 5$ of them.

Step 6: Apply the 'or' formula: favorable outcomes = $15 + 10 - 5 = 20$.

Step 7: Write the probability: $P = 20/60$.

Step 8: Simplify by dividing by 20: $P = 1/3$.

Answer: $1/3$

Tip

Whenever you see 'a multiple of ___ or a multiple of ___,' find and subtract the overlap using their LCM.

Trap

It's tempting to just add the two counts (15 and 10) without subtracting the numbers counted twice.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
5/12	Added the multiples of 4 and 6 without subtracting the overlap, double-counting the multiples of 12: $(15 + 10)/60 = 25/60$.
23/60	Subtracted the overlap using multiples of 24 (4×6) instead of the correct overlap, multiples of 12 (the LCM).



1/4	Counted only the multiples of 4 and forgot to include the multiples of 6 in the 'or' at all.
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Shortcut

Recognize this as an inclusive 'or': add the two counts, then subtract the count of multiples of their LCM (12) once.

Desmos Tip

Type $\text{floor}(60/4) + \text{floor}(60/6) - \text{floor}(60/12)$ into the Desmos calculator, then divide the result by 60 to confirm $1/3$.

Probability — Problem 7 (Medium)

A chess club in Warsaw has 120 members total. Of the 50 members under 16, 18 play in tournaments. Of the 70 members who are 16 or older, 22 do not play in tournaments. One member who is 16 or older is selected at random. What is the probability that this member plays in tournaments?

Formula / Rule Used: Conditional probability from a two-way table: target cell / row (or column) total.

Solution

Step 1: The phrase 'who is 16 or older' means the sample space is restricted to that row only, not all 120 members.

Step 2: The 16-or-older row total is given as 70 members.

Step 3: Of those 70, the 22 who do NOT play tournaments are given directly.

Step 4: Find how many 16-or-older members DO play tournaments: $70 - 22 = 48$.

Step 5: Write the conditional probability using only the restricted row: $P = 48/70$.

Step 6: Find a common factor for 48 and 70 — try 2.

Step 7: Divide both by 2: $48/2 = 24$ and $70/2 = 35$.

Step 8: The simplified probability is $P = 24/35$.

Answer: 24/35

Tip

When a question already restricts the group ('given' or 'who is ____'), only use that row or column, never the grand total.

Trap

It's tempting to divide by the grand total of 120 members instead of restricting to the 70 members who are 16 or older.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2/5	Divided by the grand total (120) instead of restricting to the 16-or-older



	row (70), ignoring the 'given' condition: $48/120 = 2/5$.
11/35	Used the non-tournament count (22) as the favorable outcome instead of subtracting to find the tournament count (48): $22/70 = 11/35$.
9/35	Used the under-16 tournament count (18) instead of computing the 16-or-older tournament count: $18/70 = 9/35$.

Shortcut

Since the question already restricts to '16 or older,' you only ever need that one row — skip the grand total entirely.

Desmos Tip

Type $70 - 22$ into the Desmos calculator to confirm there are 48 tournament players before dividing by 70.

Probability — Problem 8 (Medium)

A bag contains only yellow and purple beads. If one bead is drawn at random, the probability it is purple is $2/5$. Then 10 more purple beads are added to the bag (no beads are removed), which changes the probability of drawing a purple bead to $1/2$. How many beads were in the bag originally?

Formula / Rule Used: Reverse setup: build a before/after probability equation and solve for n .

Solution

Step 1: Let n stand for the original total number of beads in the bag.

Step 2: The original number of purple beads is $(2/5) \times n$, since purple started as $2/5$ of the bag.

Step 3: After adding 10 purple beads, the new purple count is $(2n/5) + 10$, and the new total is also larger: $n + 10$.

Step 4: Set up the new probability as an equation: $[(2n/5) + 10] / (n + 10) = 1/2$.

Step 5: Cross-multiply: $2 \times [(2n/5) + 10] = 1 \times (n + 10)$.

Step 6: Distribute the 2 on the left side: $(4n/5) + 20 = n + 10$.

Step 7: Subtract $(4n/5)$ and 10 from both sides: $20 - 10 = n - (4n/5)$, which simplifies to $10 = n/5$.

Step 8: Solve for n by multiplying both sides by 5: $n = 5 \times 10 = 50$.

Answer: 50 beads

Tip

When items are added to only one part of a group, set up 'before' and 'after' expressions using the same variable, then solve.

Trap

It's tempting to add the 10 new beads only to the purple count (numerator) and forget the total (denominator) also grows by 10.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
100	Forgot to add the 10 new beads to the total in the denominator, solving $[(2n/5) + 10] / n = 1/2$ instead.
60	Correctly solved the equation but reported the new total after adding beads ($n + 10 = 60$) instead of the original number of beads asked for.
2	Made an arithmetic slip on the last step, dividing 10 by 5 instead of multiplying, treating $n/5 = 10$ as $n = 10/5$.

Shortcut

No shortcut beyond the before/after equation — setting it up in one variable and solving is already the fastest reliable path.

Desmos Tip

Type $y = ((2x/5) + 10) / (x + 10)$ in Desmos and trace the graph to find where $y = 0.5$, confirming $x = 50$.

9. Unit Conversions & Rates

Unit Conversions & Rates — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Unit conversion	Μετατροπή μονάδων
Conversion factor	Συντελεστής μετατροπής
Rate	Ρυθμός
Dimensional analysis	Ανάλυση διαστάσεων (μονάδων)

Unit Conversions & Rates — What It Is

What It Is

- Unit conversion means rewriting the same real quantity using a different unit, without changing its actual size.
- The SAT tests this by having you multiply by a conversion factor that secretly equals exactly 1.
- A rate compares two different kinds of quantities, like distance and time, or price and quantity.
- These questions check careful, step-by-step proportional thinking, not a memorized single formula.
- You'll meet this inside real-world word problems about distance, time, money, speed, or density.

Unit Conversions & Rates — Core Rules

What a Conversion Factor Is

1 foot = 12 inches, so 12 inches / 1 foot = 1

- A conversion factor is a fraction built from two equal amounts, like 60 minutes = 1 hour.
- Because the top and bottom describe the exact same amount, the whole fraction equals 1.
- Multiplying any quantity by 1 never changes its true value, only how it's labeled.
- e.g. both 60 minutes/1 hour and 1 hour/60 minutes are valid, correct conversion factors.

The Cancel-the-Units Method

Given quantity x (desired unit / given unit) = answer in desired unit

- Write the conversion factor so the unit you want to get rid of sits on the bottom.
- That unit then cancels with the same unit in your starting quantity, like canceling in a fraction.
- Whatever unit is left over on top becomes the unit of your final answer.
- e.g. 5 hours x (60 minutes / 1 hour) = 300 minutes, since 'hours' cancels out completely.

Multi-Step (Chain) Conversions

- Use this when no single conversion factor connects your starting unit to your target unit.
- Convert through a middle unit you already know, one step at a time.
- Multiply all the conversion factors in a row; every in-between unit should cancel out.
- e.g. to go from seconds to hours, convert seconds to minutes first, then minutes to hours.

Squared and Cubed Units (Area and Volume)

$(1 \text{ unit})^2 = (\text{conversion factor})^2$; $(1 \text{ unit})^3 = (\text{conversion factor})^3$

- Area uses squared units, like ft^2 , and volume uses cubed units, like cm^3 .
- You must square or cube the ENTIRE conversion factor, not just the front number.
- e.g. since $1 \text{ ft} = 12 \text{ in}$, then $1 \text{ ft}^2 = 12^2 \text{ in}^2 = 144 \text{ in}^2$, not just 12 in^2 .
- Forgetting to raise the whole factor to that power is the single most common slip here.

Rates as Ratios

Rate = quantity / time (or quantity / another quantity)

- A rate compares two different kinds of quantities, most often something over time.
- Density, speed, and unit price are all just rates wearing different names.
- e.g. density = mass/volume, and speed = distance/time — same ratio idea both times.
- Treat a rate exactly like a conversion factor: it's a fraction you can multiply or flip.

Converting a Rate (Two Units at Once)

- A rate has two units attached, like miles per hour, so both may need converting.
- Multiply by one conversion factor for the top unit, then another for the bottom unit.
- Track which unit cancels at each step so nothing is left unconverted by accident.
- e.g. converting kilometers per hour to meters per second needs two separate conversion factors.

Which Direction to Convert

- Going from a smaller unit to a larger unit means dividing, since fewer larger units are needed.
- Going from a larger unit to a smaller unit means multiplying, since more smaller units are needed.
- Before you calculate, predict whether the answer should come out bigger or smaller.
- e.g. converting 5000 meters to kilometers should shrink the number, giving 5, not 5,000,000.

How to Recognize This on the Test

- The problem gives a measurement in one unit and asks for the same quantity in a different unit.
- You spot words like 'convert,' 'equivalent to,' 'how many,' or two different unit names in one

sentence.

- The wording includes a rate word like 'per,' 'speed,' 'density,' or 'price per item.'
- Area or volume units appear with a small exponent, like square feet or cubic centimeters.
- A word problem hides the real unit-conversion step inside a story about distance, time, or money.

Common Mistakes — and the Fix

- Setting up the conversion factor upside down so the starting unit doesn't actually cancel — check units first.
- Forgetting to square or cube the conversion factor for area or volume — always raise the whole factor to that power.
- Converting in the wrong direction, multiplying when you should divide or vice versa — predict bigger or smaller first.
- Converting only one unit of a rate and leaving the second unit unconverted — convert top and bottom separately.
- Dropping the unit labels midway through the problem and just guessing with raw numbers — always write units next to numbers.

Unit Conversions & Rates — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Unit Conversions & Rates — Problem 1 (Easy)

The distance by road from Vienna to Bratislava is 80 kilometers. A digital road sign displays distances in meters instead. How many meters is 80 kilometers? (1 kilometer = 1000 meters)

Formula / Rule Used: Unit conversion: km to m using $1 \text{ km} = 1000 \text{ m}$.

Solution

Step 1: Write the conversion factor as a fraction: 1 kilometer = 1000 meters, so $1000 \text{ m} / 1 \text{ km}$.

Step 2: Multiply the given distance by this factor: $80 \text{ km} \times (1000 \text{ m} / 1 \text{ km})$.

Step 3: The km unit on top and bottom cancels, leaving only meters.

Step 4: Multiply the remaining numbers: 80×1000 .

Step 5: $80 \times 1000 = 80,000$.

Step 6: So 80 kilometers equals 80,000 meters.

Answer: 80,000 meters

Tip

Write the conversion factor as a fraction with the unit you want to cancel on the bottom, then multiply straight across.

Trap



It's tempting to divide by 1000 instead of multiplying, since meters feels like it should give a 'smaller' number.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
8 meters	Divided by 1000 instead of multiplying, treating a km-to-m conversion as if it shrinks the number.
800 meters	Multiplied by 10 instead of 1000, misremembering how many meters are in a kilometer.
80,000,000 meters	Multiplied by 1000 twice, treating this like a squared (area) conversion instead of a linear one.

Shortcut

Moving the decimal point three places to the right is the fastest way to convert km to m directly.

Desmos Tip

Type 80×1000 into Desmos to instantly confirm the answer is 80000.

Unit Conversions & Rates — Problem 2 (Easy)

A rectangular carpet in an apartment in Lisbon has an area of 4 square meters. A flooring catalog lists carpet sizes in square centimeters. How many square centimeters is 4 square meters? (1 meter = 100 centimeters)

Formula / Rule Used: Squared unit conversion: $(1 \text{ m})^2 = (100 \text{ cm})^2$.

Solution

Step 1: Since the area is in square meters, you must square the whole linear conversion factor, not just multiply by it once.

Step 2: Start with the linear fact: 1 meter = 100 centimeters.

Step 3: Square both sides: $(1 \text{ m})^2 = (100 \text{ cm})^2$, so $1 \text{ m}^2 = 10,000 \text{ cm}^2$.

Step 4: Multiply the given area by this squared factor: $4 \text{ m}^2 \times (10,000 \text{ cm}^2 / 1 \text{ m}^2)$.

Step 5: The m^2 unit cancels, leaving only cm^2 .

Step 6: Multiply the numbers: $4 \times 10,000 = 40,000$.

Answer: 40,000 square centimeters

Tip

Always square the entire linear conversion factor first, then multiply by the given area.

Trap

The most common slip is multiplying by 100 (the linear factor) instead of $100^2 = 10,000$ for an area.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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400 square centimeters	Multiplied by 100 instead of squaring the conversion factor to get 10,000.
4,000,000 square centimeters	Cubed the factor (100^3) as if converting a volume, instead of squaring it for an area.
40 square centimeters	Divided by 100 instead of multiplying by the squared conversion factor.

Shortcut

Squaring the conversion factor once at the start, then multiplying, is already the fastest correct method.

Desmos Tip

Type 4×100^2 into Desmos to confirm the area converts to 40000 square centimeters.

Unit Conversions & Rates — Problem 3 (Easy)

A direct flight from Dublin to Toronto takes 8 hours. How many minutes long is this flight? (1 hour = 60 minutes)

Formula / Rule Used: Unit conversion: hours to minutes using 1 hour = 60 minutes.

Solution

Step 1: Write the conversion factor as a fraction: 1 hour = 60 minutes, so 60 minutes / 1 hour.

Step 2: Multiply the given time by this factor: 8 hours $\times$ (60 minutes / 1 hour).

Step 3: The hours unit cancels, leaving only minutes.

Step 4: Multiply the numbers: 8×60 .

Step 5: $8 \times 60 = 480$.

Step 6: So the flight is 480 minutes long.

Answer: 480 minutes

Tip

When converting to a smaller unit like minutes, multiply by how many of the small unit fit in one of the large unit.

Trap

Some students divide 8 by 60 instead of multiplying, mixing up which direction makes the number bigger.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
0.13 minutes	Divided 8 by 60 instead of multiplying, giving a much smaller number than makes sense.
48 minutes	Multiplied 8 by 6 instead of 8 by 60, dropping a zero from the conversion factor.
68 minutes	Added 60 to 8 instead of multiplying 8 by 60.

**Shortcut**

Multiplying hours by 60 directly is already the fastest way to get minutes — no faster path exists here.

Desmos Tip

Type 8×60 into Desmos to confirm the flight is 480 minutes long.

Unit Conversions & Rates — Problem 4 (Easy)

A block of aluminum has a mass of 540 grams and a volume of 200 cubic centimeters. What is the density of the aluminum, in grams per cubic centimeter?

Formula / Rule Used: Density = mass / volume.

Solution

Step 1: Recall the definition of density: Density = mass / volume.

Step 2: Substitute the given values: Density = 540 grams / 200 cubic centimeters.

Step 3: Divide: $540 / 200$.

Step 4: $540 / 200 = 2.7$.

Step 5: So the density is 2.7 grams per cubic centimeter.

Answer: 2.7 grams per cubic centimeter

Tip

Density is just a rate — always put mass on top and volume on the bottom of the fraction.

Trap

It's easy to flip the fraction and compute volume divided by mass instead of mass divided by volume.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
0.37 grams per cubic centimeter	Flipped the fraction, computing $200/540$ instead of $540/200$.
340 grams per cubic centimeter	Subtracted the volume from the mass ($540 - 200$) instead of dividing.
27 grams per cubic centimeter	Misplaced a decimal, dividing 540 by 20 instead of by 200.

Shortcut

Dividing mass by volume directly is already the fastest way to find density — no shortcut beats it.

Desmos Tip

Type $540/200$ into Desmos to confirm the density is 2.7.

Unit Conversions & Rates — Problem 5 (Medium)



A car travels at a constant speed and covers 108 kilometers in 2 hours. A road sign ahead gives the remaining distance in miles, where 1 mile = 1.6 kilometers. What is the car's speed in miles per hour?

Formula / Rule Used: Find the rate (distance/time), then convert the distance unit.

Solution

Step 1: First find the speed in kilometers per hour: speed = distance / time = 108 km / 2 hours.

Step 2: Divide: $108 / 2 = 54$, so the car travels at 54 kilometers per hour.

Step 3: Now convert kilometers to miles using 1 mile = 1.6 kilometers, so 1 km = $(1/1.6)$ miles.

Step 4: Multiply: 54 km/hour $\times$ (1 mile / 1.6 km).

Step 5: The km unit cancels, leaving miles per hour.

Step 6: Divide: $54 / 1.6 = 33.75$.

Step 7: So the car's speed is 33.75 miles per hour.

Answer: 33.75 miles per hour

Tip

When a problem needs both a rate calculation AND a unit conversion, find the rate first, then convert its unit.

Trap

Since a mile is a bigger distance than a kilometer, converting km to miles should shrink the number — dividing, not multiplying, by 1.6.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
54 miles per hour	Found the correct speed in km/h but forgot to convert kilometers to miles at all.
86.4 miles per hour	Multiplied 54 by 1.6 instead of dividing, going the wrong direction on the conversion.
27 miles per hour	Divided the already-found speed by 2 again, confusing the mile conversion factor with the time value.

Shortcut

You could convert 108 km to miles first ($108/1.6 = 67.5$ miles), then divide by 2 hours — either order reaches 33.75 mph.

Desmos Tip

Type $108/2/1.6$ into Desmos in one line to confirm the speed is 33.75 miles per hour.

A cyclist in Copenhagen rides at a constant speed of 18 kilometers per hour. What is this speed in meters per second? (1 kilometer = 1000 meters, 1 hour = 3600 seconds)

Formula / Rule Used: Rate conversion: convert both the top unit and the bottom unit of the rate.

Solution

Step 1: Start with the rate written as a fraction: 18 kilometers / 1 hour.

Step 2: Convert the top unit first: multiply by (1000 meters / 1 km) to change km into meters.

Step 3: $18 \times 1000 = 18,000$, so the speed is now 18,000 meters per 1 hour.

Step 4: Convert the bottom unit next: multiply by (1 hour / 3600 seconds) to change hours into seconds.

Step 5: Combine both conversions: 18,000 meters / 3600 seconds.

Step 6: Divide: $18,000 / 3600 = 5$.

Step 7: So the cyclist's speed is 5 meters per second.

Answer: 5 meters per second

Tip

Convert the top and bottom units of a rate one at a time, using a separate conversion factor for each.

Trap

A common mistake is converting km to meters but forgetting that hours also still need to become seconds.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
18,000 meters per second	Converted kilometers to meters but never converted hours to seconds at all.
0.005 meters per second	Flipped the final fraction, dividing 3600 by 18,000 instead of 18,000 by 3600.
64.8 meters per second	Multiplied 18 by 3.6 instead of dividing, misremembering the standard km/h-to-m/s shortcut.

Shortcut

There's a well-known shortcut: divide km/h by 3.6 to get m/s directly, so $18/3.6 = 5$, matching the full method.

Desmos Tip

Type $18 \times 1000 / 3600$ into Desmos to confirm the speed converts to exactly 5 meters per second.

Unit Conversions & Rates — Problem 7 (Medium)

A shipping crate built by a company in Rotterdam is a cube with edge length 2 meters, and the empty crate weighs 340 kilograms. What is the volume of the crate in cubic centimeters? (1 meter = 100 centimeters)

Formula / Rule Used: Volume of a cube, then a cubed-unit conversion.

Solution

Step 1: The crate's weight of 340 kilograms is extra information that isn't needed to find volume.

Step 2: Find the volume in cubic meters first: $\text{Volume} = \text{edge}^3 = 2^3 = 8$ cubic meters.

Step 3: To convert to cubic centimeters, cube the whole linear conversion factor: $1 \text{ m} = 100 \text{ cm}$, so $1 \text{ m}^3 = 100^3 \text{ cm}^3$.

Step 4: Compute $100^3 = 1,000,000$, so 1 cubic meter equals 1,000,000 cubic centimeters.

Step 5: Multiply the volume by this factor: $8 \text{ m}^3 \times (1,000,000 \text{ cm}^3 / 1 \text{ m}^3)$.

Step 6: The m^3 unit cancels, leaving only cm^3 .

Step 7: Multiply: $8 \times 1,000,000 = 8,000,000$.

Answer: 8,000,000 cubic centimeters

Tip

Ignore any detail that doesn't help calculate volume, like a weight or mass mentioned in the problem.

Trap

It's tempting to use the crate's weight somewhere in the math, or to square the conversion factor instead of cubing it.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
80,000 cubic centimeters	Squared the conversion factor ($100^2 = 10,000$) instead of cubing it for a volume conversion.
800 cubic centimeters	Used the linear factor 100 with no power at all, multiplying 8×100 instead of 8×100^3 .
2,720,000 cubic centimeters	Mistakenly worked the irrelevant weight of 340 kilograms into the calculation.

Shortcut

Finding the volume in meters first, then cubing the conversion factor a single time, is already the fastest method.

Desmos Tip

Type $2^3 * 100^3$ into Desmos to confirm the crate's volume is 8000000 cubic centimeters.

A water tank in Geneva fills at a constant rate. After 15 minutes, it holds 180 liters. At this same rate, how many liters will the tank hold after 40 minutes?

Formula / Rule Used: Find the unit rate, then apply it to a new amount.

Solution

Step 1: First find the fill rate in liters per minute: rate = 180 liters / 15 minutes.

Step 2: Divide: $180 / 15 = 12$, so the tank fills at 12 liters per minute.

Step 3: Multiply this rate by the new time: 12 liters/minute x 40 minutes.

Step 4: The minutes unit cancels, leaving liters.

Step 5: Multiply: $12 \times 40 = 480$.

Step 6: So the tank holds 480 liters after 40 minutes.

Answer: 480 liters

Tip

When a rate is constant, find the amount 'per one minute' first, then multiply by the new time you need.

Trap

Guessing how the amount scales without ever finding the actual per-minute rate leads to an unreliable answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
195 liters	Added 15 more liters for the extra time instead of finding and using the actual 12 liters/minute rate.
540 liters	Rounded $40/15$ up to 3 and tripled 180, instead of computing the exact rate first.
48 liters	Found the correct rate of 12 liters/minute but multiplied by 4 instead of 40, dropping a zero.

Shortcut

Finding the per-minute rate once and then multiplying by the new time is already the fastest path here.

Desmos Tip

Type $(180/15)*40$ into Desmos in one line to confirm the tank holds 480 liters after 40 minutes.

10. Statistics — Center & Spread

Statistics — Center & Spread — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Mean (average)	Μέση τιμή
Median	Διάμεσος
Mode	Επικρατούσα τιμή
Standard deviation	Τυπική απόκλιση
Range	Εύρος
Outlier	Ακραία τιμή

Statistics — Center & Spread — What It Is

What It Is

- Statistics questions ask you to summarize a whole list of numbers with just one or two key values.
- 'Center' means a typical, middle-ish value for the data — mean, median, or mode.
- 'Spread' means how bunched together or spread out the data is — range or standard deviation.
- The SAT tests this because real data (grades, prices, heights) is easier to compare using summaries than raw lists.
- Most questions just want you to calculate one summary value, or predict how it changes when the data changes.

Statistics — Center & Spread — Core Rules

Mean (Average)

Mean = (sum of all values) / (number of values)

- Add up every value in the data set first.
- Then divide that total by how many values there are, n.
- Example: for 2, 4, 6, sum = 12 and n = 3, so mean = $12/3 = 4$.
- The mean can be pulled far from most of the data by even one extreme value.

Median (Middle Value)

Median = middle value once data is sorted

- First, sort the data from smallest to largest — never skip this step.
- If n is odd, the median is the single middle number.
- If n is even, average the two middle numbers.

- Example: for 3, 7, 9, 20, the two middle values are 7 and 9, so median = $(7+9)/2 = 8$.

Mode and Range

Mode = most frequent value. Range = max - min.

- Mode is whichever value shows up the most times in the list.
- A data set can have one mode, more than one mode, or no mode at all.
- Range measures total spread: just subtract the smallest value from the largest.
- Example: for 5, 5, 8, 12, mode = 5 and range = $12 - 5 = 7$.

Weighted Average

Weighted avg = $(\text{mean1} \times n1 + \text{mean2} \times n2 + \dots) / (n1 + n2 + \dots)$

- Use this whenever you combine two or more groups of different sizes.
- Multiply each group's mean by its own group size first, not by the total.
- Add those products together, then divide by the combined total size.
- Example: 10 students averaging 80 and 20 students averaging 90 gives $(10 \times 80 + 20 \times 90) / 30 = 2600 / 30 \approx 86.7$.

Recovering a Total with Sum = Mean x n

Sum = Mean x n

- Rearranging the mean formula lets you find the total sum instead of the average.
- This is the fastest tool when a value is added, removed, or changed and asks for a new mean.
- Example: if 8 numbers have a mean of 5, their sum must be $8 \times 5 = 40$.
- Once you have the sum, add or remove the changed value, then divide by the new n.

Standard Deviation (compare only, never calculate)

- Standard deviation measures how far, on average, the data spreads out from the mean.
- The SAT never makes you compute it by hand — only compare which set is more spread out.
- A data set with values scattered far from the mean has a bigger standard deviation.
- A data set with values clustered tightly together has a smaller standard deviation.

Effect of Outliers and Skew

Right-skew: mean > median. Left-skew: mean < median. Symmetric: mean $\approx$ median.

- An outlier is one value much bigger or smaller than the rest of the data.
- An outlier drags the mean toward it but barely moves the median at all.
- An outlier also stretches the range and increases the standard deviation.
- In a skewed distribution, the long 'tail' side is where the mean gets pulled, away from the median.

How to Recognize This on the Test

- Words like 'average,' 'typical,' or 'mean' point straight to the mean formula.
- The word 'median' or a request for the 'middle' value means you must sort the data first.
- Combining two groups with different sizes (like two classes) signals a weighted average question.
- 'Standard deviation' or 'more/less spread out' means compare shapes, never calculate by hand.
- A question that adds, removes, or changes one data value and re-asks for the mean signals $\text{Sum} = \text{Mean} \times n$.

Common Mistakes — and the Fix

- Finding the median without sorting the data first, which gives a completely wrong middle value.
- Averaging group means directly instead of weighting each by its own group size.
- Assuming a bigger number always means 'more spread out' instead of comparing actual variability.
- Confusing the effect of an outlier on the mean with its much smaller effect on the median.
- Recomputing an entire mean from scratch instead of using $\text{Sum} = \text{Mean} \times n$ to save time.

Statistics — Center & Spread — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Statistics — Center & Spread — Problem 1 (Easy)

A vending machine in Rotterdam recorded these numbers of energy drinks sold over 5 school days: 18, 22, 15, 20, 25. What is the mean number of energy drinks sold per day?

Formula / Rule Used: $\text{Mean} = \text{sum} / n$

Solution

Step 1: Add up all 5 values: $18 + 22 + 15 + 20 + 25$.

Step 2: Add step by step: $18 + 22 = 40$, then $40 + 15 = 55$.

Step 3: Continue adding: $55 + 20 = 75$, then $75 + 25 = 100$.

Step 4: The total sum of all 5 days is 100.

Step 5: Divide the sum by the number of days, $n = 5$: $\text{mean} = 100 / 5$.

Step 6: $\text{Mean} = 20$.

Answer: 20 energy drinks per day

Tip

Whenever a question asks for the average, add every value first, then divide by how many values there are.

Trap



It's tempting to divide by the wrong count of days instead of the actual $n = 5$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
15	Left out 25 when adding, then divided the incomplete sum of 75 by 5: $75/5 = 15$.
16.7	Miscounted the days as 6 instead of 5, dividing the correct sum 100 by 6 instead of 5.
100	Added all 5 values correctly but forgot the final division step, reporting the total sum instead of the mean.

Shortcut

None needed — adding the 5 values and dividing by 5 is already the fastest method for a set this small.

Desmos Tip

Type `mean(18,22,15,20,25)` into a blank Desmos line to instantly confirm the average equals 20.

Statistics — Center & Spread — Problem 2 (Easy)

On a chemistry quiz in Berlin, seven students scored the following points out of 20: 14, 9, 20, 11, 9, 17, 13. What is the median score?

Formula / Rule Used: Median = middle value once data is sorted

Solution

Step 1: List all seven scores in the order given: 14, 9, 20, 11, 9, 17, 13.

Step 2: Sort them from smallest to largest: 9, 9, 11, 13, 14, 17, 20.

Step 3: Since $n = 7$ is odd, the median is a single middle value, not an average of two.

Step 4: Count inward from both ends to find the middle position: 3 values are on each side of it.

Step 5: The 4th value in the sorted list is 13.

Step 6: The median score is 13.

Answer: 13

Tip

Always sort the data first — the SAT often lists values out of order on purpose to trip you up.

Trap

Picking the 4th value straight from the original, unsorted order gives the wrong middle value.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
11	Took the 4th value from the original unsorted list instead of sorting the



	data first.
9	Confused the median with the mode, since 9 happens to appear twice in the list.
13.5	Treated $n = 7$ as if it were even and averaged the two middle sorted values, 13 and 14.

Shortcut

Cross out the highest and lowest remaining values one pair at a time until only one value is left in the middle.

Desmos Tip

Type median(14,9,20,11,9,17,13) into Desmos to confirm the median is 13.

Statistics — Center & Spread — Problem 3 (Easy)

A shoe store in Lisbon sold these shoe sizes in one afternoon: 38, 40, 42, 40, 39, 40, 41. What is the mode of the sizes sold minus the range of the sizes sold?

Formula / Rule Used: Mode = most frequent value. Range = max - min.

Solution

Step 1: List the shoe sizes sold: 38, 40, 42, 40, 39, 40, 41.

Step 2: Count how many times each size appears: 40 appears three times, more than any other size.

Step 3: So the mode, the most frequent size, is 40.

Step 4: Find the range next: subtract the smallest size from the largest, $42 - 38$.

Step 5: Range = 4.

Step 6: Subtract the range from the mode as the question asks: $40 - 4$.

Step 7: Result = 36.

Answer: 36

Tip

Count frequencies carefully for the mode, and only subtract max minus min for the range — keep the two steps separate.

Trap

It's easy to rush and miss the 42 buried in the middle of the list, undercounting the true maximum.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
40	Found the mode correctly but forgot to subtract the range, reporting the mode alone.
-36	Subtracted in the reverse order, computing range minus mode instead of mode minus range.



37	Overlooked the 42 in the list and used 41 as the maximum, getting a range of 3 instead of 4.
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Shortcut

Scan once for the most-repeated number to get the mode, and once for the highest and lowest to get the range.

Statistics — Center & Spread — Problem 4 (Easy)

A tour company in Vienna ran two guided walking tours today. The morning tour had 15 tourists with an average age of 34, and the afternoon tour had 25 tourists with an average age of 42. What is the average age of all 40 tourists combined?

Formula / Rule Used: Weighted average = $(\text{mean1} \times n1 + \text{mean2} \times n2) / (n1 + n2)$

Solution

Step 1: Identify each group's size and mean: morning has $n1 = 15$ and $\text{mean1} = 34$.

Step 2: The afternoon group has $n2 = 25$ and $\text{mean2} = 42$.

Step 3: Multiply each group's mean by its own size: $15 \times 34 = 510$, and $25 \times 42 = 1050$.

Step 4: Add those two products together: $510 + 1050 = 1560$.

Step 5: Add the group sizes to get the total number of tourists: $15 + 25 = 40$.

Step 6: Divide the combined total by the combined size: $1560 / 40$.

Step 7: Combined average age = 39.

Answer: 39

Tip

Never just average the two means directly — multiply each mean by its own group size first.

Trap

Averaging 34 and 42 straight to get 38 ignores that the afternoon tour had more tourists, which pulls the true average higher.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
38	Averaged the two means directly, $(34+42)/2 = 38$, ignoring that the group sizes were different.
39.5	Divided the correct weighted sum, 1560, by 39 instead of the correct total of 40 tourists.
1560	Computed the weighted sum correctly but forgot the final division by the total number of tourists.

Shortcut



None needed — the weighted average formula is already the fastest way to combine groups of different sizes.

Desmos Tip

Type $(15 \cdot 34 + 25 \cdot 42)/40$ directly into Desmos as one line to get 39 without separate arithmetic steps.

Statistics — Center & Spread — Problem 5 (Medium)

A cycling team based in Lyon has 6 riders, and their mean finishing time in a time trial was 15 minutes. One rider got a flat tire and finished in 33 minutes. If that rider's time is excluded from the data, what is the mean finishing time, in minutes, of the remaining 5 riders?

Formula / Rule Used: $\text{Sum} = \text{Mean} \times n$

Solution

Step 1: Use $\text{Sum} = \text{Mean} \times n$ to turn the original mean back into a total: $\text{sum} = 15 \times 6 = 90$ minutes.

Step 2: This 90 minutes is the combined finishing time of all 6 riders.

Step 3: Remove the excluded rider's time from that total: $90 - 33 = 57$ minutes.

Step 4: There are now only 5 riders left in the data set, so the new n is 5.

Step 5: Divide the new sum by the new count: $\text{mean} = 57 / 5$.

Step 6: New mean = 11.4 minutes.

Answer: 11.4 minutes

Tip

When one value is removed and you need a new mean, first turn the old mean back into a sum with $\text{Sum} = \text{Mean} \times n$.

Trap

It's tempting to subtract 33 from the old mean of 15 directly instead of working through the total sum first.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-18	Subtracted the removed rider's time straight from the mean, $15 - 33$, instead of adjusting the total sum.
9.5	Divided the new sum of 57 by the original count of 6 instead of the new count of 5.
18	Divided the original total of 90 by the new count of 5 but forgot to subtract the removed rider's time first.

Shortcut

$\text{Sum} = \text{Mean} \times n$ turns this into simple arithmetic — you never need to know the other 5 individual times.

Desmos Tip

Type $(15 \cdot 6 - 33)/5$ into Desmos as one line to instantly get the new mean, 11.4.

Statistics — Center & Spread — Problem 6 (Medium)

At a fitness center in Copenhagen, a group of boys and a separate group of 12 girls took the same balance test. The boys averaged 82 points, the girls averaged 90 points, and the combined average for both groups together was 86 points. How many boys were in the group?

Formula / Rule Used: Weighted average equation, solved for an unknown group size

Solution

Step 1: Let x stand for the unknown number of boys.

Step 2: Set up the weighted average equation: $(82x + 90(12)) / (x + 12) = 86$.

Step 3: Multiply out the girls' term: $90(12) = 1080$, so the equation becomes $(82x + 1080)/(x+12) = 86$.

Step 4: Multiply both sides by $(x + 12)$ to clear the fraction: $82x + 1080 = 86(x + 12)$.

Step 5: Distribute the right side: $82x + 1080 = 86x + 1032$.

Step 6: Subtract $82x$ from both sides: $1080 = 4x + 1032$.

Step 7: Subtract 1032 from both sides: $48 = 4x$.

Step 8: Divide both sides by 4: $x = 12$.

Answer: 12 boys

Tip

When a group size is unknown, name it x and build the weighted average equation before solving anything.

Trap

It's tempting to assume the groups must be equal size just because 86 looks roughly in the middle of 82 and 90.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
10	Made an arithmetic slip distributing $86(x+12)$, writing $86x + 1030$ instead of the correct $86x + 1032$.
6	Mixed up which mean belonged to which group, swapping boys' and girls' averages in the equation.
24	Divided 48 by 2 instead of 4 in the last step, an arithmetic slip.

Shortcut

86 is much closer to 90 than to 82, so the 12-person girls' group must outweigh the boys — a quick check that $x = 12$ is reasonable.

Desmos Tip

Type $y = (82x + 90 \cdot 12)/(x+12)$ and $y = 86$ as two separate lines in Desmos, then find their intersection to confirm $x = 12$.

Statistics — Center & Spread — Problem 7 (Medium)

A small startup in Dublin has 5 employees. Four of them earn monthly salaries of 1800, 1900, 2000, and 2100 euros, and the founder earns 8200 euros a month. What is the positive difference between the mean and the median monthly salary at this startup?

Formula / Rule Used: Mean = sum/n , Median = middle sorted value, compared directly

Solution

Step 1: List all 5 salaries: 1800, 1900, 2000, 2100, 8200.

Step 2: Find the mean: add them all up, $1800+1900+2000+2100+8200 = 16000$.

Step 3: Divide by $n = 5$ to get the mean: $16000/5 = 3200$.

Step 4: Now find the median: the salaries are already sorted from smallest to largest.

Step 5: Since $n = 5$ is odd, the median is the single 3rd (middle) value: 2000.

Step 6: Find the positive difference between the mean and median: $3200 - 2000$.

Step 7: Positive difference = 1200.

Answer: 1200 euros

Tip

When one value looks much bigger than the rest, expect the mean and median to disagree — calculate both carefully.

Trap

The founder's high salary pulls the mean way up, but it barely touches the median — don't assume the two stay close.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
0	Assumed the mean and median are always close together and never actually calculated both values.
6200	Subtracted the median from the founder's salary alone instead of from the actual mean, $8200 - 2000$.
1300	Made an addition slip totaling the salaries as 16500 instead of 16000, giving a mean of 3300.

Shortcut

Since 4 of the 5 salaries are already sorted, you only need to insert the founder's salary once before reading off the median.

Desmos Tip

Type `mean(1800,1900,2000,2100,8200)` and `median(1800,1900,2000,2100,8200)` on two Desmos lines, then subtract the results to confirm 1200.

**Statistics — Center & Spread — Problem 8** *(Medium)*

A running club in Madrid timed 7 members in a 5-kilometer race, with finishing times in minutes of 22, 19, 25, 21, 40, 23, and 20. The club has 34 members in total. What is the median finishing time, in minutes, for these 7 runners?

Formula / Rule Used: Median = middle value once data is sorted

Solution

Step 1: Notice that '34 members in total' is extra information not needed to find the median of these 7 runners.

Step 2: List the 7 finishing times: 22, 19, 25, 21, 40, 23, 20.

Step 3: Sort them from smallest to largest: 19, 20, 21, 22, 23, 25, 40.

Step 4: Since $n = 7$ is odd, the median is the single 4th value, with 3 values on each side of it.

Step 5: Counting to the 4th value in the sorted list gives 22.

Step 6: The median finishing time is 22 minutes.

Answer: 22 minutes

Tip

Cross out any numbers the question gives that you don't actually need, like the club's total membership here.

Trap

The 40-minute outlier looks unusually slow and can tempt you to think it drags the median with it, but sorting shows it doesn't.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
24.3	Tried to compute the mean instead of the median, mistakenly averaging all 7 times together.
21	Picked the 4th value from the original unsorted list instead of sorting the times first.
34	Mistakenly used the irrelevant total club membership number as if it were part of the answer.

Shortcut

Cross out the highest and lowest sorted times one pair at a time until only the middle time, 22, is left.

Desmos Tip

Type median(22,19,25,21,40,23,20) into Desmos to instantly confirm the median is 22, ignoring the irrelevant 34.

11. Data Displays — Histograms, Dot Plots & Box Plots

Data Displays — Histograms, Dot Plots & Box Plots — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Histogram	Ιστόγραμμα
Dot plot	Διάγραμμα κουκκίδων
Box plot	Διάγραμμα κουτιού (θηκόγραμμα)
Five-number summary	Σύνοψη πέντε αριθμών
Frequency	Συχνότητα

Data Displays — Histograms, Dot Plots & Box Plots — What It Is

What It Is

- Data displays are pictures that organize a list of numbers so patterns are easy to see at a glance.
- The SAT shows the same kind of data as histograms, dot plots, or box plots and expects you to read each one.
- Each display highlights something different: dot plots show every exact value, histograms show grouped counts, box plots show spread.
- Most questions just ask you to read the picture correctly, not to calculate anything new from scratch.
- You will never need to draw one of these displays yourself, only interpret one that's already given.

Data Displays — Histograms, Dot Plots & Box Plots — Core Rules

Histograms

bar height = frequency (count) in that interval

- A histogram groups numerical data into equal-width intervals, called bins.
- The height of each bar shows how many data values fall inside that bin.
- Bars touch with no gaps, because the data is continuous, not separate labeled categories.
- The overall shape (symmetric, right-skewed, left-skewed) tells you how spread out the data is.

Dot Plots

- A dot plot shows every individual data value as one dot above a number line.
- Stacking several dots at the same spot shows how many times that exact value occurred.
- Because every dot is a real value, you can directly count frequencies, clusters, gaps, and outliers.
- Example: three dots stacked above the number 7 means the value 7 occurred exactly three times.

Box Plots and the Five-Number Summary

Min, Q1, Median, Q3, Max

- A box plot summarizes a data set using five key values: minimum, Q1, median, Q3, and maximum.
- The box itself stretches from Q1 to Q3, and a line inside the box marks the median.
- The two 'whiskers' are lines extending from the box out to the minimum and the maximum.
- The five values split the data into four equal groups, each holding 25% of all the data.

Interquartile Range (IQR)

$$\text{IQR} = \text{Q3} - \text{Q1}$$

- IQR measures the spread of just the middle 50% of the data, ignoring extreme values.
- Subtract Q1 (the 25th percentile mark) from Q3 (the 75th percentile mark) to find it.
- A smaller IQR means the middle half of the data is tightly clustered together.
- Example: if $\text{Q1} = 4$ and $\text{Q3} = 10$, then $\text{IQR} = 10 - 4 = 6$.

Finding the Median from a List or Dot Plot

- First list every value in order from smallest to largest.
- Cross off the smallest and the largest remaining value, one pair at a time.
- If exactly one value is left in the middle, that value is the median.
- If two values are left in the middle, add them together and divide by 2 to average them.

Reading Skew from Shape

Right-skew: mean > median. Left-skew: mean < median.

- A distribution's 'tail' is the side that stretches out further with fewer values.
- Right-skewed means the tail points toward larger values, which pulls the mean above the median.
- Left-skewed means the tail points toward smaller values, which pulls the mean below the median.
- In a box plot, a longer whisker or box section on one side signals skew toward that side.

Comparing Two Data Displays

- The SAT often shows two dot plots, histograms, or box plots side by side to compare groups.
- Compare center (mean or median) to see which group tends to have larger or smaller values.
- Compare spread (range or IQR) to see which group is more consistent or more variable.
- Don't assume taller bars or more dots always mean a 'greater' data value — check the axis labels first.

How to Recognize This on the Test

- Touching bars sitting above a number line labeled with numeric ranges signal a histogram.
- Individual dots stacked in columns above a plain number line signal a dot plot.

- Words like 'median,' 'quartile,' 'IQR,' or 'whisker,' or a box-shaped diagram, signal a box plot.
- Phrases like 'which statement is best supported by the graph' mean you must read values off the picture.
- A question comparing two groups' typical value or consistency usually points you to compare centers and spreads.

Common Mistakes — and the Fix

- Confusing a histogram's bars (grouped number ranges) with a bar graph's bars (separate labeled categories).
- Miscounting a dot plot by reading the number line value instead of counting how many dots sit above it.
- Computing IQR using the min and max instead of only Q1 and Q3.
- Assuming mean equals median for every distribution instead of checking whether the shape is actually skewed.
- Mixing up the median line inside a box plot with the box's edges, which are actually Q1 and Q3.

Data Displays — Histograms, Dot Plots & Box Plots — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 1 (Easy)

A dot plot shows the number of books read last month by 12 students at a school in Thessaloniki. The horizontal axis (number of books) runs from 0 to 6, with one dot above 0, two dots above 1, three dots above 2, four dots above 3, one dot above 4, and one dot above 6 (no dots appear above 5). What is the median number of books read?

Formula / Rule Used: Median of a data set: order all values, then find the middle value(s).

Solution

Step 1: List every value shown by the dots, in order, using how many dots sit above each number.

Step 2: 0 appears once, 1 appears twice, 2 appears three times, 3 appears four times, 4 appears once, and 6 appears once: 0, 1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 6.

Step 3: Count the values: there are 12 data points total, which is an even number.

Step 4: For an even count, the median is the average of the 6th and 7th values in the ordered list.

Step 5: Counting up to the 6th value: 0, 1, 1, 2, 2, 2 — so the 6th value is 2, and the 7th value is 3.

Step 6: Average the two middle values: $(2 + 3) / 2 = 5/2 = 2.5$.

Answer: 2.5 books

Tip

Before finding a median, always write out the full ordered list using the dot counts — don't try to eyeball the middle from the picture.

**Trap**

It's tempting to just pick the number that 'looks like the middle' on the axis, but you must count all 12 individual data points, not the 5 distinct labels.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3	Counted only the 5 distinct values shown on the axis (0,1,2,3,4,6) and picked the middle one, ignoring how many dots sit above each.
2	Correctly found the 6th value but forgot to average it with the 7th value, stopping one step early.
2.75	Averaged the 6th and 8th values instead of the correct 6th and 7th values.

Shortcut

None needed — writing out the ordered list from the dot counts is already the fastest reliable method.

Desmos Tip

Type $L1=\{0,1,1,2,2,2,3,3,3,3,4,6\}$ on the first line, then type $\text{median}(L1)$ on the next line to confirm the median is 2.5.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 2 (Easy)

A histogram shows the scores (out of 100) of 20 students on a chemistry quiz at a school in Lisbon. The horizontal axis shows score intervals of width 10, and the bars (with bar heights) are: 60-69: 3 students, 70-79: 5 students, 80-89: 8 students, 90-99: 4 students. How many students scored 80 or higher on the quiz?

Formula / Rule Used: Total in a range = sum of the bar heights that cover that range.

Solution

Step 1: Identify which bars cover a score of 80 or higher: the 80-89 bar and the 90-99 bar.

Step 2: Read the height of the 80-89 bar: 8 students.

Step 3: Read the height of the 90-99 bar: 4 students.

Step 4: Add these two bar heights together, since both bars fall in the range 80 or higher: $8 + 4$.

Step 5: Simplify: $8 + 4 = 12$.

Answer: 12 students

Tip

When a question asks for 'X or higher,' first identify exactly which bars are included, then just add their heights.

Trap

It's easy to accidentally include the 70-79 bar too, since 79 looks close to 80, but that whole bar is below 80.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
17	Mistakenly included the 70-79 bar (5 students) along with the two correct bars: $5 + 8 + 4 = 17$.
8	Read only the 80-89 bar and forgot to also include the 90-99 bar.
20	Added all four bar heights together ($3+5+8+4=20$), ignoring the 'scored 80 or higher' condition entirely.

Shortcut

Since only two bars matter, skip the others and add just the 80-89 and 90-99 heights directly: $8 + 4$.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 3 (Easy)

A box plot summarizes the daily high temperatures (in degrees Celsius) recorded in Athens over a two-week period. The five-number summary is: minimum = 18, Q1 = 22, median = 25, Q3 = 29, maximum = 34. What is the interquartile range (IQR) of these temperatures?

Formula / Rule Used: $IQR = Q3 - Q1$

Solution

Step 1: Identify Q1 and Q3 directly from the five-number summary: $Q1 = 22$, $Q3 = 29$.

Step 2: Recall the IQR formula: $IQR = Q3 - Q1$.

Step 3: Substitute the values: $IQR = 29 - 22$.

Step 4: Subtract: $29 - 22 = 7$.

Answer: 7 degrees Celsius

Tip

IQR questions only ever need Q1 and Q3 — ignore the minimum, maximum, and median completely.

Trap

It's tempting to use the maximum and minimum ($34 - 18 = 16$) since that also looks like a measure of 'spread,' but that's the range, not the IQR.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
16	Computed the range ($\text{max} - \text{min} = 34 - 18$) instead of the interquartile range.
3	Subtracted median - Q1 ($25 - 22$) instead of Q3 - Q1.
4	Subtracted Q3 - median ($29 - 25$) instead of Q3 - Q1.

Shortcut



None needed — once Q1 and Q3 are read off the plot, one subtraction is already the fastest path.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 4 (Easy)

A dot plot shows the number of goals scored by a football (soccer) club in Rotterdam across 10 matches this season. The horizontal axis (goals per match) runs from 0 to 4, with two dots above 0, three dots above 1, three dots above 2, one dot above 3, and one dot above 4. What is the mean number of goals scored per match?

Formula / Rule Used: Mean = (sum of all values) / (number of values)

Solution

Step 1: Multiply each goal value by how many matches had that many goals: 0×2 , 1×3 , 2×3 , 3×1 , 4×1 .

Step 2: Calculate each product: $0 \times 2 = 0$, $1 \times 3 = 3$, $2 \times 3 = 6$, $3 \times 1 = 3$, $4 \times 1 = 4$.

Step 3: Add all the products together to get the total goals scored: $0 + 3 + 6 + 3 + 4 = 16$.

Step 4: Count the total number of matches by adding the dot counts: $2 + 3 + 3 + 1 + 1 = 10$.

Step 5: Divide total goals by total matches: mean = $16/10$.

Step 6: Simplify: mean = 1.6.

Answer: 1.6 goals per match

Tip

Treat each dot as one data value — multiply value $\times$ frequency for each stack, add everything, then divide by the total count.

Trap

A common mistake is dividing by the number of distinct dot-stacks (5) instead of the actual total number of matches (10).

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3.2	Divided the total goals (16) by 5, the number of distinct values on the axis, instead of by the 10 actual matches.
2	Added only the axis values themselves ($0+1+2+3+4=10$) and divided by 5, ignoring how many dots sat above each value.
1.2	Forgot to include the single dot above 4 goals when summing, using $0+3+6+3=12$ instead of 16, then divided by 10.

Shortcut

Estimate first: most dots cluster around 1 and 2 goals, so the mean should land close to 1.5-2, matching 1.6.

Desmos Tip

Type $L1=\{0,0,1,1,1,2,2,2,3,4\}$ (one entry per match) then type $\text{mean}(L1)$ on the next line to confirm the mean is 1.6.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 5 (Medium)

A dot plot represents the number of hours volunteered last month by 7 students at a community center in Dublin. Six of the values are known: 2, 3, 3, 4, 5, and 5 hours. The seventh student's number of hours, v , is unknown. If the mean number of volunteer hours for all 7 students is 4, what is the value of v ?

Formula / Rule Used: Sum of all values = Mean $\times$ number of values

Solution

Step 1: Recall that Mean = Sum / n , so Sum = Mean $\times$ n .

Step 2: There are 7 students total and the mean is 4, so the sum of all 7 values must be $4 \times 7 = 28$.

Step 3: Add up the six known values: $2 + 3 + 3 + 4 + 5 + 5$.

Step 4: Simplify: $2+3+3+4+5+5 = 22$.

Step 5: The unknown value v must make the total reach 28, so $v = 28 - 22$.

Step 6: Subtract: $v = 6$.

Answer: $v = 6$ hours

Tip

Whenever one value is missing but you know the mean, use Sum = Mean $\times$ n to find the total first, then subtract the known values.

Trap

It's tempting to average the six known values directly to estimate v , but that ignores that the TRUE mean of all seven students must equal 4.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Assumed the missing value must simply equal the given mean, without actually setting up and solving the equation.
2	Used $n = 6$ instead of 7 in Sum = Mean $\times$ n , forgetting the unknown student is also part of the group, getting target sum 24 and $v = 24 - 22 = 2$.
50	Added the known sum to the target total instead of subtracting, computing $28 + 22 = 50$ instead of $28 - 22$.

Shortcut

Sum = Mean $\times$ n is always faster than guessing — it turns any 'missing value' question into a single subtraction.

Desmos Tip

After solving for $v = 6$ by hand, type `mean({2,3,3,4,5,5,6})` in Desmos to confirm it outputs exactly 4.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 6 (Medium)

A box plot summarizes the finishing times (in minutes) of 50 runners in a 10K race in Munich. The five-number summary shows: minimum = 42, $Q1 = 51$, median = 58, maximum = 79, and the interquartile range (IQR) is 20 minutes. First find $Q3$. Then determine whether a runner who finished in 68 minutes falls inside the box (that is, between $Q1$ and $Q3$).

Formula / Rule Used: $IQR = Q3 - Q1$, solved for $Q3$; then compare a value to the box's boundaries.

Solution

Step 1: Start from the IQR formula: $IQR = Q3 - Q1$.

Step 2: Plug in the known values: $20 = Q3 - 51$.

Step 3: Add 51 to both sides to solve for $Q3$: $Q3 = 20 + 51$.

Step 4: Simplify: $Q3 = 71$.

Step 5: The box spans from $Q1 = 51$ to $Q3 = 71$, so any value between 51 and 71 falls inside the box.

Step 6: Check whether 68 minutes falls in that range: $51 \leq 68 \leq 71$ is true.

Step 7: Conclude that the runner's time of 68 minutes does fall inside the box.

Answer: $Q3 = 71$ minutes; yes, 68 minutes falls inside the box (between $Q1 = 51$ and $Q3 = 71$).

Tip

When IQR and one quartile are given, rearrange $IQR = Q3 - Q1$ to solve for the missing quartile before answering the real question.

Trap

It's tempting to compare 68 to the median (58) instead of to the box's actual edges, $Q1$ and $Q3$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$Q3 = 31$; no	Computed $Q1 - IQR$ ($51 - 20 = 31$) instead of $Q1 + IQR$, reversing the direction of the IQR formula.
$Q3 = 79$; yes	Confused $Q3$ with the maximum value (79) already given in the five-number summary.
$Q3 = 71$; no	Correctly solved for $Q3 = 71$ but compared 68 to the median (58) instead of to the box's edges, wrongly concluding it falls outside.

Shortcut

Since the box must contain the median 58, just add IQR directly to $Q1$ — no need to try other combinations.

Desmos Tip

Type $51+20$ in Desmos to instantly confirm $Q3 = 71$ before checking whether 68 falls between 51 and 71.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 7 (Medium)

A histogram shows the number of text messages sent yesterday by 40 students at a school in Valencia. The horizontal axis shows message-count intervals of width 20, and the bars (with bar heights) are: 0-19 messages: 12 students, 20-39: 10 students, 40-59: 8 students, 60-79: 6 students, 80-99: 4 students. Based on the shape of this histogram, which is greater: the mean or the median number of messages sent?

Formula / Rule Used: Right-skew (long tail toward higher values): mean > median.

Solution

Step 1: Look at how the bar heights change across the intervals: 12, 10, 8, 6, 4.

Step 2: The bars are tallest at the low end (0-19) and get shorter and shorter toward the high end (80-99).

Step 3: A long, thin 'tail' stretching toward the higher values means the distribution is right-skewed.

Step 4: Recall the skew rule: in a right-skewed distribution, a few unusually high values pull the mean upward.

Step 5: The median stays anchored near where most of the data actually sits, which is the low-message end.

Step 6: Since the tail pulls the mean up but leaves the median mostly in place, the mean must be greater than the median.

Answer: The mean is greater than the median.

Tip

Before comparing mean and median, always check the histogram's shape first — you almost never need to calculate exact values.

Trap

It's tempting to think 'more students sent fewer messages, so the median must be higher,' but skew pulls the MEAN, not the median.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
The median is greater than the mean	Reversed the skew rule, thinking the long tail pulls the median instead of the mean.
They are equal	Assumed mean always equals median without checking whether the histogram's shape was actually symmetric.
Cannot be determined	Correctly noticed the skewed shape but didn't recall that skew direction alone is enough to compare mean and median.

Shortcut

Just check which side has the longer, thinner tail — you never need to calculate the actual mean or median for a shape-comparison question.

Desmos Tip

This is a shape-reasoning question, not a calculation — Desmos won't help here since no exact data values are given, only bar heights.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 8 (Medium)

Two dot plots compare the number of hours of sleep reported last night by two classes at a school in Krakow, both taught by the same teacher, Mr. Nowak. Class A (12 students): the axis runs from 5 to 9 hours, with one dot above 5, two dots above 6, four dots above 7, four dots above 8, and one dot above 9. Class B (12 students): the axis runs from 4 to 9 hours, with two dots above each of 4, 5, 6, 7, 8, and 9. Which class has the smaller interquartile range (IQR), and what is that IQR?

Formula / Rule Used: $IQR = Q3 - Q1$, found by splitting the ordered data in half at the median.

Solution

Step 1: Ignore the fact that both classes share the same teacher — that detail doesn't affect the data at all.

Step 2: List Class A in order: 5, 6, 6, 7, 7, 7, 8, 8, 8, 8, 9 (12 values).

Step 3: Split Class A into a lower half (first 6 values) and upper half (last 6 values): lower = 5, 6, 6, 7, 7, 7; upper = 7, 8, 8, 8, 8, 9.

Step 4: Class A's $Q1$ is the average of the lower half's middle two values: $(6+7)/2 = 6.5$. $Q3$ is the average of the upper half's middle two: $(8+8)/2 = 8$.

Step 5: Class A's $IQR = Q3 - Q1 = 8 - 6.5 = 1.5$.

Step 6: List Class B in order: 4, 4, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9 (12 values), split the same way: lower = 4, 4, 5, 5, 6, 6; upper = 7, 7, 8, 8, 9, 9.

Step 7: Class B's $Q1 = (5+5)/2 = 5$. $Q3 = (8+8)/2 = 8$. Class B's $IQR = 8 - 5 = 3$.

Step 8: Compare: Class A's IQR (1.5) is smaller than Class B's IQR (3), so Class A has the smaller IQR.

Answer: Class A has the smaller IQR, which equals 1.5 hours.

Tip

For 12 data points, split evenly into a lower 6 and an upper 6 right at the middle — no value is shared between the halves.

Trap

The detail about both classes sharing the same teacher is irrelevant filler — don't waste time trying to use it in the math.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Class B, $IQR = 3$	Correctly computed Class B's IQR but mixed up which class was actually smaller, picking B instead of A.
Class A, $IQR = 4$	Used the range ($\max - \min = 9 - 5 = 4$) for Class A instead of the interquartile range.
Class A, $IQR = 2.5$	Included the median-region value in both halves when splitting Class A's 12 data points instead of splitting evenly into 6 and 6.



Shortcut

Since both classes have 12 values, split each ordered list into a clean lower 6 and upper 6 without worrying about a leftover middle value.

Desmos Tip

Type $L1=\{5,6,6,7,7,7,7,8,8,8,8,9\}$, then $\text{sort}(L1)$ and $\text{median}(L1)$ to double-check the ordering before splitting the list by hand into halves.

12. Scatterplots

Scatterplots — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Scatterplot	Διάγραμμα διασποράς
Line of best fit	Ευθεία βέλτιστης προσαρμογής
Residual	Υπόλοιπο (residual)
Positive/negative association	Θετική/αρνητική συσχέτιση
Nonlinear model	Μη γραμμικό μοντέλο

Scatterplots — What It Is

What It Is

- A scatterplot is a graph with dots showing two number values for each item, one value per axis.
- It's used to see if two things are connected, like hours studied and a test score.
- The SAT tests this because real data is messy, so you must judge the overall trend, not one dot.
- A line of best fit is a single straight line drawn through the cloud of dots to summarize the trend.
- Scatterplot questions combine reading a graph with basic algebra, like slope-intercept substitution.

Scatterplots — Core Rules

Line of Best Fit

$$y = mx + b$$

- The line of best fit is one straight line that best summarizes the trend in the dots.
- It almost never passes through every actual dot — it's only an approximation of the pattern.
- Use this line's equation, not any single dot, whenever you're asked to predict a value.
- Example: if the line is $y = 2x + 10$, then at $x = 5$ the model predicts $y = 20$.

Types of Association

- Positive association: as x increases, y also tends to increase, so the line slants upward.
- Negative association: as x increases, y tends to decrease, so the line slants downward.
- No association: the dots are scattered with no clear upward or downward direction at all.
- Strength: dots hugging tightly around the line show a strong relationship; scattered dots show a weak one.

Interpreting Slope and Intercept in Context**slope = change in y per 1 unit of x**

- Slope tells you the predicted change in y every time x goes up by exactly 1 unit.
- Always attach real units to slope, like 'euros per hour' or 'points per week studied.'
- The y-intercept is the predicted y-value when $x = 0$, where the line crosses the y-axis.
- Check whether $x = 0$ is realistic in context — sometimes it's just a mathematical starting value.

Residuals**residual = actual y - predicted y**

- A residual measures how far one real dot sits above or below the line of best fit.
- Positive residual: the actual point lies above the line, so the line underestimated that value.
- Negative residual: the actual point lies below the line, so the line overestimated that value.
- Example: if actual $y = 15$ and the line predicts $y = 12$ there, the residual is $15 - 12 = 3$.

Points On the Line vs. Actual Data Points

- Data points are the real dots plotted from actual observations, scattered near the line.
- A point described as 'on the line of best fit' is usually not an actual data point.
- To find a point on the line, just pick any x-value and read where the line sits there.
- Don't confuse a value the model predicts with a value that was actually observed.

Nonlinear Models**quadratic: $y = ax^2 + bx + c$; exponential: $y = a \cdot b^x$**

- Not every scatterplot trend is a straight line — some curve upward, downward, or level off.
- A quadratic curve looks like a U or an upside-down U, symmetric around one peak or valley.
- An exponential curve grows or shrinks faster and faster, not at one steady constant rate.
- A clearly curving cloud of dots is a hint you may need a curve equation, not a straight line.

Using the Model's Equation to Predict or Solve**substitute a known x or y into the equation**

- If you're given x, substitute it into the equation and simplify to predict y.
- If you're given y instead, substitute it in and solve algebraically for x.
- Round predictions so they make sense in context, since counts of people can't be decimals.
- This substitution step is usually the very last part of a scatterplot question.

How to Recognize This on the Test

- The words 'scatterplot,' 'line of best fit,' or 'trend line' immediately signal this skill.

- A description or image of scattered dots with one straight line drawn through them is this topic.
- Phrases like 'predicted value,' 'based on the model,' or 'approximately how many' mean use the line's equation.
- A question asking how far one specific point is from the line is asking about a residual.
- Wording like 'best describes the relationship' is asking you to name positive, negative, or no association.

Common Mistakes — and the Fix

- Treating an actual scattered dot as if it must sit exactly on the line of best fit.
- Flipping the sign of a residual by subtracting predicted minus actual instead of actual minus predicted.
- Dropping the real-world units when describing what the slope or intercept means in context.
- Trying to fit a straight-line equation to a scatterplot that clearly curves like a quadratic or exponential.
- Reading two random dots off the graph instead of two points that actually lie on the drawn line.

Scatterplots — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Scatterplots — Problem 1 (Easy)

A scatterplot for a garden center in Valencia plots the number of tomato plants watered per week on the x-axis (ranging from about 2 to 20) against the kilograms of tomatoes harvested that week on the y-axis (ranging from about 5 to 30). The plotted points trend upward and cluster fairly closely around a rising straight line drawn through them. The line of best fit for this data is $y = 1.4x + 3.2$, where x is the number of plants watered and y is the kilograms harvested. Using this line of best fit, predict the kilograms of tomatoes harvested in a week when 10 plants are watered.

Formula / Rule Used: Line of best fit: $y = mx + b$, substitute the given x .

Solution

Step 1: Identify the line of best fit equation: $y = 1.4x + 3.2$.

Step 2: The question wants a prediction, so substitute $x = 10$ into the equation.

Step 3: Write it out: $y = 1.4(10) + 3.2$.

Step 4: Multiply first: $1.4(10) = 14$.

Step 5: Add the intercept: $y = 14 + 3.2$.

Step 6: $y = 17.2$.

Answer: 17.2 kg

**Tip**

When a question asks you to predict using a scatterplot, always use the line of best fit's equation, not a single dot.

Trap

It's tempting to look for an actual dot at $x = 10$ on the graph instead of using the line's equation to predict a value.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
14 kg	Multiplied 1.4 by 10 correctly but forgot to add the y-intercept of 3.2 afterward.
33.4 kg	Swapped the slope and intercept values, computing $3.2(10) + 1.4$ instead of $1.4(10) + 3.2$.
143.2 kg	Misread the slope as 14 instead of 1.4, giving $14(10) + 3.2 = 143.2$.

Shortcut

None needed — plugging $x = 10$ directly into the line of best fit equation is already the fastest path.

Desmos Tip

Type $y = 1.4x + 3.2$ in Desmos, trace along the line to $x = 10$, and confirm y reads 17.2.

Scatterplots — Problem 2 (Easy)

A scatterplot for a high school class plots hours studied on the x-axis (ranging from 0 to 8 hours) against test score out of 100 on the y-axis (ranging from about 50 to 95). The plotted points trend upward and stay fairly close to a straight line drawn through them. The line of best fit is $y = 6x + 54$, where x is hours studied and y is the test score. One specific student in the class studied for 3 hours and actually scored 68 on the test. What is the residual for this student's data point?

Formula / Rule Used: Residual = actual y - predicted y

Solution

Step 1: First find the predicted score using the line of best fit at $x = 3$.

Step 2: Substitute $x = 3$ into $y = 6x + 54$: $y = 6(3) + 54$.

Step 3: Multiply: $6(3) = 18$.

Step 4: Add: predicted $y = 18 + 54 = 72$.

Step 5: The actual score given in the problem is 68.

Step 6: Residual = actual - predicted = $68 - 72$.

Step 7: Residual = -4.

Answer: -4

Tip



Always compute the predicted value from the line first, then subtract it from the actual value to get the residual.

Trap

It's tempting to subtract in the wrong order (predicted minus actual), which flips the sign of the residual.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Subtracted in the wrong order, computing predicted minus actual ($72 - 68$) instead of actual minus predicted.
8	Added the slope and intercept instead of multiplying, using predicted = $6 + 54 = 60$, giving $68 - 60 = 8$.
72	Reported the predicted value itself instead of continuing on to compute the residual.

Shortcut

No real shortcut — computing the predicted value first and subtracting is already the fastest method.

Desmos Tip

Type $y = 6x + 54$ in Desmos, trace to $x = 3$ to see predicted $y = 72$, then subtract from the actual 68 to confirm residual -4.

Scatterplots — Problem 3 (Easy)

A scatterplot for an ice cream shop in Barcelona plots the daily high temperature in degrees Celsius on the x-axis (ranging from about 18 to 34) against the number of ice creams sold that day on the y-axis (ranging from about 40 to 220). As the temperature values increase moving across the graph, the plotted points also rise together, clustering tightly and consistently close to a rising straight line. What type of association does this scatterplot show between temperature and ice cream sales?

Formula / Rule Used: Types of association: positive, negative, or none.

Solution

Step 1: Look at the overall direction of the points as x (temperature) increases.

Step 2: The points rise together with temperature, moving from lower-left to upper-right.

Step 3: When y increases as x increases, that pattern is called a positive association.

Step 4: The tight clustering also shows the association is strong, but the question only asks for its type.

Step 5: The type of association shown is positive.

Answer: Positive association

Tip

Trace the general direction of the cloud of dots from left to right to decide if it's rising, falling, or flat.

Trap



It's tempting to answer with the strength of the clustering instead of naming the direction the question actually asked for.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Negative association	Mixed up the direction, treating a rising graph as if y decreases when x increases.
No association	Focused on individual scattered dots instead of stepping back to see the overall rising pattern.
Strong correlation	Answered with the strength of the relationship instead of naming its direction as asked.

Shortcut

None needed — a quick glance at whether the cloud of dots rises or falls left to right answers this immediately.

Scatterplots — Problem 4 (Easy)

A scatterplot from a phone battery study plots hours of continuous use on the x-axis (ranging from 0 to 10) against the percent of battery charge remaining on the y-axis (ranging from 0 to 100). The plotted points trend downward and lie close to a straight line drawn through them. The line of best fit is $y = -8x + 96$, where x is hours of use and y is percent battery remaining. What does the slope of this line mean in the context of the phone battery?

Formula / Rule Used: Interpreting slope in context: change in y per 1 unit of x .

Solution

Step 1: The slope is the coefficient of x in the equation, so slope = -8.

Step 2: Slope means the predicted change in y for every 1-unit increase in x .

Step 3: Here, x represents hours of use and y represents percent battery remaining.

Step 4: A slope of -8 means the battery charge is predicted to drop by 8 percentage points per extra hour.

Step 5: The negative sign matches the downward trend seen in the graph, since charge is decreasing.

Answer: The battery charge is predicted to decrease by 8 percentage points for each additional hour of use.

Tip

To interpret slope in context, always attach real units to it: 'y-unit per one x-unit.'

Trap

It's tempting to describe the slope as a total drop over all 10 hours instead of the rate for just one hour.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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The battery decreases by 96 percentage points per hour.	Confused the y-intercept, 96, with the slope, -8, when describing the rate.
The battery decreases by 8 percentage points total over 10 hours.	Misread the per-hour rate as a one-time total drop across the whole study.
The battery increases by 8 percentage points per hour.	Dropped the negative sign, reversing the direction of the change.

Shortcut

None needed — reading the coefficient of x directly from the equation and attaching the correct units is already immediate.

Desmos Tip

Type $y = -8x + 96$ in Desmos and trace from $x = 0$ to $x = 1$ to see y drop by 8, confirming the slope's meaning.

Scatterplots — Problem 5 (Medium)

A scatterplot from a sleep study of Belgian students plots hours of sleep the night before on the x -axis (ranging from about 4 to 10) against reaction time in milliseconds on the y -axis (ranging from about 180 to 280). The plotted points trend downward and stay fairly close to a straight line. The line of best fit is $y = -15x + 340$, where x is hours of sleep and y is reaction time in milliseconds. According to this line of best fit, how many hours of sleep does the model predict would produce a reaction time of exactly 190 milliseconds?

Formula / Rule Used: Line of best fit solved in reverse: substitute y and solve for x .

Solution

Step 1: Normally you'd plug in x to find y , but here you're given $y = 190$ and must find x instead.

Step 2: Substitute $y = 190$ into the equation: $190 = -15x + 340$.

Step 3: Subtract 340 from both sides: $190 - 340 = -15x$.

Step 4: Simplify the left side: $-150 = -15x$.

Step 5: Divide both sides by -15: $x = -150 / -15$.

Step 6: $x = 10$.

Answer: 10 hours

Tip

When a question gives you a y -value and asks for x , substitute y into the equation first, then solve for x algebraically.

Trap

It's tempting to keep isolating x the same way you would for a normal x -to- y prediction, forgetting the roles are reversed here.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-10 hours	Made a sign error dividing -150 by -15, forgetting a negative divided by a negative gives a positive result.
-12.67 hours	Divided 190 directly by -15 without first subtracting 340 from both sides.
10.7 hours	Misread the slope in the equation as -14 instead of -15, giving a slightly off value.

Shortcut

Once rearranged, you can isolate x in one clean step: $(340 - 190) / 15 = 150 / 15 = 10$.

Desmos Tip

Type $y = -15x + 340$ in Desmos, then trace the line to find where $y = 190$, confirming $x = 10$.

Scatterplots — Problem 6 (Medium)

A scatterplot for an airline based in Lisbon plots the discount percentage offered on a flight on the x -axis (ranging from 0 to 25) against the number of unsold seats on that flight on the y -axis (ranging from about 10 to 100). The plotted points trend downward and stay close to a straight line. The line of best fit for this data is given by the equation $4x + y = 96$, where x is the discount percentage and y is the number of unsold seats. Using this line of best fit, predict the number of unsold seats when the discount offered is 15%.

Formula / Rule Used: Rearrange to slope-intercept form $y = mx + b$, then substitute x .

Solution

Step 1: The equation is given in a different form, so first solve it for y .

Step 2: Subtract $4x$ from both sides of $4x + y = 96$: $y = 96 - 4x$.

Step 3: Rewrite this in slope-intercept order: $y = -4x + 96$.

Step 4: Now substitute $x = 15$ into the equation: $y = -4(15) + 96$.

Step 5: Multiply first: $-4(15) = -60$.

Step 6: Add: $y = -60 + 96$.

Step 7: $y = 36$.

Answer: 36 unsold seats

Tip

If a line of best fit is given in a form other than $y = mx + b$, rearrange it for y first before substituting anything.

Trap

It's tempting to plug $x = 15$ straight into $4x + y = 96$ and solve for y incorrectly by forgetting to isolate y first.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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24 unsold seats	Substituted $x = 15$ into $4x + y = 96$ and treated $4(15)$ as equal to y instead of isolating $y = 96 - 4x$.
156 unsold seats	Added $4x$ to 96 instead of subtracting, computing $y = 96 + 4(15) = 156$.
-60 unsold seats	Correctly found $-4(15) = -60$ but forgot to add the 96 from the rearranged equation.

Shortcut

You can substitute directly into the original form: $4(15) + y = 96$, so $y = 96 - 60 = 36$, without fully rewriting the equation first.

Desmos Tip

Type $4x + y = 96$ directly into Desmos, trace the line to $x = 15$, and confirm y reads 36.

Scatterplots — Problem 7 (Medium)

A scatterplot from a lab study in Rotterdam plots fish tank size in liters on the x -axis (ranging from about 10 to 60) against the number of fish that thrive in that tank on the y -axis (ranging from about 5 to 35). The plotted points trend upward and stay fairly close to a straight line. The line of best fit is $y = 0.5x + 2$, where x is tank size in liters and y is the number of fish that thrive. While organizing the data, the lab assistant, Marco, relabeled some unrelated storage boxes, which has no effect on the results. One specific tank in the study holds 40 liters and actually had 18 fish thriving in it. What is the residual for this specific tank's data point?

Formula / Rule Used: Residual = actual y - predicted y

Solution

Step 1: Ignore the detail about Marco relabeling boxes — it has no effect on the math.

Step 2: First find the predicted number of fish using the line of best fit at $x = 40$.

Step 3: Substitute $x = 40$ into $y = 0.5x + 2$: $y = 0.5(40) + 2$.

Step 4: Multiply: $0.5(40) = 20$.

Step 5: Add: predicted $y = 20 + 2 = 22$.

Step 6: The actual number of fish given in the problem is 18.

Step 7: Residual = actual - predicted = $18 - 22$.

Step 8: Residual = -4 .

Answer: -4

Tip

Cross out any detail in the problem that doesn't involve numbers connected to x , y , or the equation — it's there to test focus.

Trap

It's tempting to think the mention of Marco somehow affects the data, when it's simply irrelevant information.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
4	Subtracted in the wrong order, computing predicted minus actual ($22 - 18$) instead of actual minus predicted.
-2	Forgot to add the y-intercept of 2 when computing the predicted value, using only $0.5(40) = 20$.
22	Reported the predicted value itself instead of continuing on to compute the residual.

Shortcut

No real shortcut — find the predicted value first, then subtract, ignoring any unrelated details in the story.

Desmos Tip

Type $y = 0.5x + 2$ in Desmos, trace to $x = 40$ to see predicted $y = 22$, then subtract from actual 18 to confirm residual -4.

Scatterplots — Problem 8 (Medium)

A scatterplot for a used car dealership in Munich plots the car's age in years on the x-axis (ranging from 0 to 12) against its price in thousands of euros on the y-axis (ranging from about 6 to 24). The plotted points trend downward and stay close to a straight line. The line of best fit has a slope of -1.5 and is known to pass through the point (2, 18), where 2 is the car's age in years and 18 is its price in thousands of euros. First find the equation of the line of best fit, then use it to predict the price of a car that is 6 years old.

Formula / Rule Used: Point substitution to find b , then $y = mx + b$.

Solution

Step 1: Start with slope-intercept form: $y = mx + b$, using the known slope $m = -1.5$.

Step 2: Substitute the known point (2, 18) to solve for b : $18 = -1.5(2) + b$.

Step 3: Multiply: $-1.5(2) = -3$.

Step 4: So the equation becomes: $18 = -3 + b$.

Step 5: Add 3 to both sides to solve for b : $b = 18 + 3 = 21$.

Step 6: The full line of best fit is $y = -1.5x + 21$.

Step 7: Now substitute $x = 6$ to predict the price: $y = -1.5(6) + 21$.

Step 8: Multiply: $-1.5(6) = -9$.

Step 9: Add: $y = -9 + 21 = 12$.

Answer: 12 (thousand euros, about 12,000 euros)

Tip

When you're only given a slope and one point, first solve for b before you can predict anything else.

**Trap**

It's tempting to substitute $x = 6$ in before ever finding b , using an incomplete equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
9 (thousand euros)	Substituted $x = 6$ using only the slope, computing $-1.5(6) = -9$, and never solved for or added b .
6 (thousand euros)	Made a sign error solving for b , computing $b = 18 - 3 = 15$ instead of $18 + 3 = 21$.
21 (thousand euros)	Reported the y -intercept b itself instead of continuing on to compute the predicted price at $x = 6$.

Shortcut

Once $b = 21$ is found, there's no faster path than plugging $x = 6$ directly into $y = -1.5x + 21$.

Desmos Tip

Type $y = -1.5x + 21$ in Desmos, trace to $x = 6$, and confirm y reads 12.

13. Sampling, Bias & Margin of Error

Sampling, Bias & Margin of Error — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Sample	Δείγμα
Population	Πληθυσμός
Random sampling	Τυχαία δειγματοληψία
Bias	Μεροληψία (σφάλμα)
Margin of error	Περιθώριο σφάλματος

Sampling, Bias & Margin of Error — What It Is

What It Is

- This skill checks whether a survey or study's results can actually be trusted.
- A 'population' is the entire group you want to learn something about.
- A 'sample' is the smaller group that was actually surveyed or tested.
- The SAT tests this because polls, studies, and news reports rely on sampling every day.
- You'll judge whether a sampling method was fair, or work with margin of error numbers.

Sampling, Bias & Margin of Error — Core Rules

Population vs. Sample

- The population is the full group a conclusion is meant to describe.
- The sample is the smaller subset actually measured or asked questions.
- A trustworthy sample must be both randomly chosen AND reasonably large.
- Example: surveying 200 randomly picked students can represent all 3,000 students at a school.

Random Selection Is What Makes a Sample Valid

- Random means every member of the population has an equal chance of being chosen.
- Random selection is exactly what lets results generalize back to the full population.
- A huge sample that wasn't randomly chosen can still give badly misleading results.
- Phrases like 'randomly selected' or 'chosen at random' signal a valid sampling method.

Sampling Bias

- Bias means the method used to pick the sample favors certain outcomes unfairly.

- This happens when the group sampled from isn't representative of the real population.
- Example: surveying only students in the chess club about favorite school activities.
- A bigger sample size never fixes a biased selection method — the method itself must change.

Self-Selection and Nonresponse Bias

- Self-selection bias happens when people choose to volunteer themselves for a study.
- Volunteers tend to have stronger or more extreme opinions than the average person.
- Nonresponse bias happens when some chosen people never actually respond.
- The people who don't respond may differ systematically from the people who do.

Margin of Error

true population value $\approx$ sample statistic $\pm$ margin of error

- Margin of error gives a range where the true population value probably falls.
- A larger, more random sample produces a SMALLER margin of error.
- A smaller sample size produces a LARGER margin of error, meaning less precision.
- The margin of error describes the overall estimate, not any one specific individual.

Building the Interval From Margin of Error

interval = sample estimate $\pm$ margin of error

- Subtract the margin of error from the sample estimate to get the low end.
- Add the margin of error to the sample estimate to get the high end.
- Example: if $62\% \pm 3\%$ support a rule, the true value likely sits between 59% and 65%.

Generalizing Results and Correlation vs. Causation

- A study's results only generalize to the exact population it was randomly sampled from.
- Watch for a mismatch between who was sampled and who a conclusion claims to describe.
- An observational study can show two things are related, but never that one causes the other.
- Only a properly randomized experiment, with random assignment to groups, can support a cause-and-effect claim.

How to Recognize This on the Test

- Words like 'sample,' 'survey,' 'randomly selected,' or 'margin of error' signal this skill directly.
- A question comparing two studies with different sample sizes is testing margin of error logic.
- Phrases like 'volunteers' or 'those who chose to respond' signal self-selection or nonresponse bias.
- A question asking which population a conclusion can be applied to is testing generalization rules.
- The word 'causes' or 'leads to' next to a described study is testing correlation vs. causation.

Common Mistakes — and the Fix

- Assuming a bigger sample size fixes a biased sampling method instead of the method itself.
- Thinking a smaller margin of error means a smaller sample, when it's actually the opposite.
- Applying a study's conclusion to a broader population than the one actually sampled from.
- Treating a strong correlation in an observational study as proof that one thing causes another.
- Forgetting margin of error describes the overall estimate, not what's true for every single person.

Sampling, Bias & Margin of Error — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Sampling, Bias & Margin of Error — Problem 1 (Easy)

A random sample of 400 out of 5,000 factory workers at a manufacturing plant in Stuttgart found that 25% plan to retire within the next 5 years. Based on this sample, about how many of the 5,000 workers are expected to plan to retire within the next 5 years?

Formula / Rule Used: Population estimate = population size x sample percentage (as a decimal).

Solution

Step 1: Convert the sample percentage to a decimal: $25\% = 0.25$.

Step 2: Multiply this decimal by the total population size: $5,000 \times 0.25$.

Step 3: Multiply: $5,000 \times 0.25 = 1,250$.

Step 4: This means about 1,250 of the 5,000 workers are expected to plan to retire within 5 years.

Answer: 1,250 workers

Tip

Convert the sample percent to a decimal first, then multiply it by the full population size, not the sample size.

Trap

It's tempting to multiply by 400 (the sample size) instead of 5,000 (the full population), which gives a meaningless number.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
100 workers	Multiplied 0.25 by the sample size 400 instead of the full population 5,000.
125 workers	Used 25% as 0.025 instead of 0.25, misplacing the decimal point.
20 workers	Divided 5,000 by 25 instead of multiplying 5,000 by 0.25.

Shortcut



Since 25% is the friendly fraction $\frac{1}{4}$, just divide 5,000 by 4 to get 1,250 instantly.

Desmos Tip

Type $5000 * 0.25$ directly into Desmos to instantly confirm the population estimate is 1,250.

Sampling, Bias & Margin of Error — Problem 2 (Easy)

A survey of runners entering the Berlin Marathon estimates that 68% of participants train more than 5 days a week, with a margin of error of 3 percentage points. Based on this estimate, what is the highest percent of participants that could actually train more than 5 days a week?

Formula / Rule Used: Upper bound of interval = sample estimate + margin of error.

Solution

Step 1: The sample estimate is 68%, and the margin of error is 3 percentage points.

Step 2: To find the highest possible true value, add the margin of error to the estimate: $68 + 3$.

Step 3: Add: $68 + 3 = 71$.

Step 4: The highest percent that could actually train more than 5 days a week is 71%.

Answer: 71%

Tip

For the highest possible value in the range, always ADD the margin of error to the sample estimate.

Trap

Subtracting the margin of error instead of adding it gives the lowest bound, not the highest one the question asked for.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
65%	Subtracted the margin of error instead of adding it, finding the lowest bound instead of the highest.
3%	Reported the margin of error itself instead of using it to build the upper bound.
74%	Added the margin of error twice ($68 + 3 + 3$) instead of adding it once.

Shortcut

None needed — adding the margin of error once is already the fastest way to find the upper bound.

Desmos Tip

Type $68+3$ directly into Desmos to instantly confirm the upper bound is 71.

Sampling, Bias & Margin of Error — Problem 3 (Easy)

A study estimates that between 42% and 50% of teenagers in Sweden own a smartwatch. What is the margin of error for this estimate, in percentage points?

Formula / Rule Used: Margin of error = (upper bound - lower bound) / 2.

Solution

Step 1: The interval runs from a lower bound of 42% to an upper bound of 50%.

Step 2: Find the total width of the interval: $50 - 42$.

Step 3: Subtract: $50 - 42 = 8$.

Step 4: The margin of error is half of this width, since the estimate sits exactly in the middle: $8 / 2$.

Step 5: Divide: $8 / 2 = 4$.

Step 6: The margin of error is 4 percentage points.

Answer: 4 percentage points

Tip

Find the full width of the interval first, then cut it in half to get the margin of error.

Trap

Reporting the full width of the interval (8) instead of dividing it by 2 is a common mix-up.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
8 percentage points	Found the width of the interval ($50 - 42$) but forgot to divide by 2.
46 percentage points	Calculated the midpoint of the interval, which is the sample estimate, not the margin of error.
2 percentage points	Divided the width by 4 instead of by 2, splitting it into the wrong number of parts.

Shortcut

Since 42 and 50 are both even, you can quickly see the midpoint is 46 and each side is 4 away from it.

Desmos Tip

Type $(50-42)/2$ directly into Desmos to instantly confirm the margin of error is 4.

Sampling, Bias & Margin of Error — Problem 4 (Easy)



A school newspaper wants to learn how the school's 1,800 students feel about moving the prom date. The editor only posts the survey link in the school's photography club group chat, which has 45 members, and all 45 respond. What percent of the school's total student population actually had a chance to be selected for this sample?

Formula / Rule Used: Percent reached = (number who could respond / total population) x 100.

Solution

Step 1: Only the 45 members of the photography club had any chance of seeing and answering the survey.

Step 2: The total student population is 1,800.

Step 3: Divide the number who could respond by the total population: $45 / 1,800$.

Step 4: Divide: $45 / 1,800 = 0.025$.

Step 5: Multiply by 100 to convert to a percent: $0.025 \times 100 = 2.5$.

Step 6: Only 2.5% of the school's students had any chance of being part of this sample.

Answer: 2.5%

Tip

To check if a sample is representative, ask who actually had a chance to be chosen, not just how many people answered.

Trap

It's tempting to think 45 out of 45 club members responding (100%) makes this a good sample, but compare to the full school instead.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
100%	Compared the 45 respondents to the 45 club members instead of to the full 1,800 students.
40%	Divided 1,800 by 45 and misread the result as the answer instead of dividing 45 by 1,800.
25%	Dropped a zero while converting the decimal 0.025 into a percent.

Shortcut

None needed — dividing the group size by the full population is already the fastest way to see how limited this sample is.

Desmos Tip

Type $(45/1800) * 100$ directly into Desmos to instantly confirm only 2.5% of students had a chance to respond.

Sampling, Bias & Margin of Error — Problem 5 (Medium)

In a city of 12,000 residents, a transportation study estimates that between 3,120 and 3,480 residents bike to work. What is the margin of error for this estimate, expressed as a percentage of the city's population?

Formula / Rule Used: Margin of error (count) = (upper bound - lower bound) / 2, then convert to a percent of the population.

Solution

Step 1: Find the width of the interval in people: $3,480 - 3,120$.

Step 2: Subtract: $3,480 - 3,120 = 360$.

Step 3: The margin of error in people is half the width, since the estimate is the midpoint: $360 / 2$.

Step 4: Divide: $360 / 2 = 180$ people.

Step 5: Convert this to a percent of the population by dividing by 12,000: $180 / 12,000$.

Step 6: Divide: $180 / 12,000 = 0.015$.

Step 7: Multiply by 100 to get a percent: $0.015 \times 100 = 1.5$.

Step 8: The margin of error is 1.5% of the population.

Answer: 1.5%

Tip

When bounds are given as raw counts, find the margin of error in people first, then convert it to a percent of the total population.

Trap

Forgetting to divide the interval's width by 2 gives the full range of people (360), not the actual margin of error.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3%	Used the full width of 360 people without dividing by 2, then converted that full width to a percent.
180%	Found the correct margin of error in people (180) but forgot to convert it into a percent of the population.
2.85%	Divided the margin of 180 by the midpoint estimate of 3,300 instead of by the total population of 12,000.

Shortcut

Notice 360 people halves easily to 180, then divide by 12,000 by simplifying to $18/1,200$ mentally.

Desmos Tip

Type $((3480-3120)/2)/12000*100$ directly into Desmos as one line to get the margin of error percent in a single step.

Study A randomly surveys 150 residents of Lisbon and estimates that 55% support a new park renovation, with a margin of error of 8 percentage points. Study B randomly surveys 600 residents using the same method and gets the same 55% estimate, but with a margin of error of only 4 percentage points. A city planner wants to use whichever study gives the narrower range for the true percentage. What is the total width, in percentage points, of the range from that more precise study?

Formula / Rule Used: Larger random sample = smaller margin of error; interval width = 2 x margin of error.

Solution

Step 1: Compare the two studies: Study B used a larger sample (600 vs. 150) with the same random method.

Step 2: A larger random sample produces a smaller margin of error, so Study B is the more precise study.

Step 3: Study B's margin of error is 4 percentage points.

Step 4: The full range runs from the estimate minus the margin to the estimate plus the margin, so its width is twice the margin: 2×4 .

Step 5: Multiply: $2 \times 4 = 8$.

Step 6: The width of Study B's range is 8 percentage points.

Answer: 8 percentage points

Tip

First decide which study is actually more precise using sample size, then double that study's margin of error for the range width.

Trap

Grabbing Study A's numbers is tempting since it's listed first, but Study A is actually the less precise study here.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
16 percentage points	Doubled Study A's margin of error (8) instead of recognizing Study B is the more precise study.
4 percentage points	Correctly identified Study B as more precise but reported its margin of error instead of doubling it for the width.
55 percentage points	Confused the 55% estimate itself with the width of the range built around it.

Shortcut

A bigger random sample always means a smaller margin of error, so you can pick Study B without extra calculation.

Desmos Tip

Type 4×2 into Desmos to instantly confirm the width of Study B's range is 8 percentage points.

Sampling, Bias & Margin of Error — Problem 7 (Medium)

Delft Polytechnic has 8,400 enrolled students. Researchers randomly sample 350 students and find that 84 of them work a part-time job. That same week, the university also announced 12 new elective courses for next semester. Based on the random sample, about how many of Delft Polytechnic's 8,400 students would be expected to work a part-time job?

Formula / Rule Used: Sample proportion $\times$ total population = population estimate.

Solution

Step 1: The new elective courses have nothing to do with part-time jobs, so ignore that detail completely.

Step 2: Find the sample proportion who work part-time: 84 out of 350.

Step 3: Divide: $84 / 350 = 0.24$.

Step 4: Multiply this proportion by the full population: $8,400 \times 0.24$.

Step 5: Multiply: $8,400 \times 0.24 = 2,016$.

Step 6: About 2,016 of the 8,400 students would be expected to work a part-time job.

Answer: 2,016 students

Tip

Find the sample's proportion as a decimal first, then scale it up to the full population size.

Trap

The detail about 12 new elective courses is irrelevant filler — don't let it pull you into using the number 12 anywhere.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
24 students	Treated 0.24 as if it already meant 24 students, without multiplying by the population of 8,400.
2,000 students	Rounded $84/350$ to 0.2 too early instead of using the exact value 0.24, losing precision.
168 students	Multiplied 8,400 by 0.02 instead of 0.24, misplacing the decimal point.

Shortcut

Simplify $84/350$ down to $12/50$, then $24/100 = 0.24$ mentally before multiplying by 8,400.

Desmos Tip

Type $(84/350) * 8400$ directly into Desmos as one line to get the population estimate immediately.

Sampling, Bias & Margin of Error — Problem 8 (Medium)

A random sample of 500 out of 20,000 employees at Nordholm Logistics found that a certain percentage plan to work remotely at least 3 days a week. Based on this sample, researchers report a confidence interval of 41% to 49% for the true percentage. Using the midpoint of this interval as the best estimate, about how many of the 20,000 employees are expected to plan to work remotely at least 3 days a week?

Formula / Rule Used: Find the midpoint of the interval, then multiply it by the total population.

Solution

Step 1: First find the midpoint of the interval, which is the actual sample estimate: $(41 + 49) / 2$.

Step 2: Add: $41 + 49 = 90$.

Step 3: Divide: $90 / 2 = 45$, so the sample estimate is 45%.

Step 4: Convert 45% to a decimal: 0.45.

Step 5: Multiply this decimal by the total number of employees: $20,000 \times 0.45$.

Step 6: Multiply: $20,000 \times 0.45 = 9,000$.

Step 7: About 9,000 of the 20,000 employees are expected to plan to work remotely at least 3 days a week.

Answer: 9,000 employees

Tip

When you're given a range instead of a single estimate, find the midpoint first — that's the actual sample percentage to use.

Trap

Using one of the endpoints (41% or 49%) instead of the true midpoint estimate throws off the final population count.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
8,200 employees	Used the lower bound 41% instead of the midpoint 45%, computing $20,000 \times 0.41$.
9,800 employees	Used the upper bound 49% instead of the midpoint 45%, computing $20,000 \times 0.49$.
500 employees	Applied the percentage to the sample size of 500 instead of to the full population of 20,000.

Shortcut

Since 41 and 49 are each 4 away from 45, you can spot the midpoint 45% mentally without doing the full addition.

Desmos Tip

Type $((41+49)/2)/100*20000$ directly into Desmos as one line to get the population estimate in a single step.



PART 3 — ADVANCED MATH

14. Functions — Notation, Domain, Range & Graphs

Functions — Notation, Domain, Range & Graphs — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Function	Συνάρτηση
Domain	Πεδίο ορισμού
Range	Πεδίο τιμών
Independent variable (input)	Ανεξάρτητη μεταβλητή
Dependent variable (output)	Εξαρτημένη μεταβλητή
Function notation	Συμβολισμός συνάρτησης
Graph of a function	Γραφική παράσταση συνάρτησης

Functions — Notation, Domain, Range & Graphs — What It Is

What It Is

- A function is a rule that takes an input and gives back exactly one output for it.
- Function notation $f(x)$ is just a labeled way to write 'the output of rule f when the input is x .'
- The SAT uses functions to test whether you can plug in, read graphs, and track inputs vs. outputs.
- You'll meet functions as equations, as graphs, as tables of values, and as word-problem descriptions.
- Domain and range just describe which inputs are allowed and which outputs actually happen.

Functions — Notation, Domain, Range & Graphs — Core Rules

Function Notation

$$y = f(x)$$

- $f(x)$ is read 'f of x' — it does NOT mean f multiplied by x .
- x is the input you plug in; $f(x)$ is the output you get back.
- The letter f just names the rule — $g(x)$ and $h(x)$ work exactly the same way.
- Example: if $f(x) = x + 3$, then f is the rule 'add 3 to whatever goes in.'

Evaluating a Function

$f(a)$ = substitute a for every x

- To find $f(a)$, replace every x in the rule with the number a .
- Simplify using order of operations after substituting, just like any expression.
- Example: if $f(x) = 2x + 1$, then $f(4) = 2(4) + 1 = 8 + 1 = 9$.
- This works the same on a graph: point (a, b) on the graph means $f(a) = b$.

The Vertical Line Test

- This test checks whether a graph actually represents a function.
- Imagine sliding a vertical line left to right across the graph.
- If any vertical line ever crosses the graph more than once, it is NOT a function.
- This makes sense because one input (one x -value) can't give two different outputs.

Domain and Range

Domain = allowed x -values. Range = resulting y -values.

- The domain is every input value the function is allowed to use.
- The range is every output value the function actually produces.
- On a graph, the domain reads along the x -axis and the range along the y -axis.
- To find x when $f(x) = k$, find $y = k$ on the graph and read across to the curve.

Domain Restrictions

Denominator $\neq 0$. Radicand under an even root ≥ 0 .

- A fraction is undefined whenever its denominator equals 0, so exclude those x -values.
- A square root (or any even root) can't have a negative number inside it in real numbers.
- Set the denominator equal to 0, or the radicand less than 0, and solve to find what's excluded.
- If a function has more than one restriction, find each one separately, then combine them all.

Piecewise Functions

$f(x) = \{\text{expression 1 if condition 1; expression 2 if condition 2}\}$

- A piecewise function uses a different rule depending on what x is.
- First check which condition (which range of x) your input value actually falls into.
- Then evaluate using ONLY the matching piece's expression — ignore the other pieces.
- Example: if $f(x) = x+1$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$, then $f(-2) = -1$ but $f(2) = 4$.

Rate of Change Between Two Points

Rate of change = $(f(x_2) - f(x_1)) / (x_2 - x_1)$

- This measures how much the output changes per one-unit increase in the input.
- For a straight-line (linear) function, this rate of change is just the slope.
- In a word problem, this tells you something like dollars per hour or meters per second.
- Compute it the same way as slope: subtract outputs on top, subtract matching inputs on bottom.

How to Recognize This on the Test

- Seeing $f(x)$, $g(x)$, or $h(x)$ notation anywhere is the clearest sign this is a functions question.
- Phrases like 'what is $f(3)$ ' or 'find the value of $g(-2)$ ' mean you just substitute and simplify.
- A question showing a graph and asking for a domain, range, or output value is testing graph-reading.
- The word 'piecewise' or a function written with multiple expressions and conditions signals a piecewise problem.
- 'For which values of x is this function defined' is asking specifically about domain restrictions.

Common Mistakes — and the Fix

- Treating $f(x)$ like multiplication (f times x) instead of 'the output of f at input x .'
- Substituting the input into only part of the expression instead of every occurrence of x .
- Mixing up domain and range — remember domain is x (inputs), range is y (outputs).
- Forgetting to check ALL restrictions when a function has both a root and a denominator.
- Using the wrong piece of a piecewise function because the input wasn't checked against the condition first.

Functions — Notation, Domain, Range & Graphs — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Functions — Notation, Domain, Range & Graphs — Problem 1 (Easy)

A vending machine company in Rotterdam models the number of snacks left in a machine using $N(t) = 200 - 15t$, where t is the number of hours since the machine was filled. How many snacks are left after 6 hours?

Formula / Rule Used: Evaluating a function: substitute the input for t .

Solution

Step 1: You need $N(6)$, so substitute $t = 6$ into the rule everywhere t appears.

Step 2: $N(6) = 200 - 15(6)$.

Step 3: Multiply first, before subtracting: $15(6) = 90$.

Step 4: Now the expression is $N(6) = 200 - 90$.

Step 5: Subtract: $N(6) = 110$.

Answer: 110 snacks

Tip

Substitute the given input for every t in the rule first, then follow order of operations to simplify.

Trap

It's tempting to subtract $200 - 15$ first and then multiply by 6, but multiplication has to happen before the subtraction.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
90	Computed $15(6) = 90$ but forgot to subtract that from 200, reporting only the product.
1110	Did the operations in the wrong order, computing $(200 - 15)(6) = 1110$ instead of $200 - 15(6)$.
185	Subtracted 15 directly from 200 without multiplying it by 6 first, getting $200 - 15 = 185$.

Shortcut

None needed — direct substitution into $N(t)$ is already the fastest way to answer this.

Desmos Tip

Type $y = 200 - 15x$ in Desmos and trace the graph to $x = 6$ to confirm y lands on 110.

If $g(x) = x^2 - 4x + 1$, what is the value of $g(-3)$?

Formula / Rule Used: Evaluating a function: substitute the input for x .

Solution

Step 1: Substitute -3 for every x in the rule: $g(-3) = (-3)^2 - 4(-3) + 1$.

Step 2: Square the negative number first: $(-3)^2 = 9$.

Step 3: Multiply -4 by -3, remembering a negative times a negative is positive: $-4(-3) = 12$.

Step 4: Now the expression reads $g(-3) = 9 + 12 + 1$.

Step 5: Add the terms in order: $9 + 12 = 21$.

Step 6: $g(-3) = 21 + 1 = 22$.

Answer: 22

Tip

When the input is negative, square it and multiply it out carefully before combining anything — sign errors happen fast here.

Trap

The double negative in $-4(-3)$ is exactly where students accidentally drop a sign and get the wrong value.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Treated $(-3)^2$ as -9 instead of 9, then combined $-9 + 12 + 1 = 4$.
-2	Made a sign error on $-4(-3)$, writing it as -12 instead of 12, giving $9 - 12 + 1 = -2$.
21	Correctly computed $9 + 12$ but forgot to add the final +1 term at the end.

Shortcut

None needed — direct substitution is already the fastest route for a single evaluation like this.

Desmos Tip

Type $y = x^2 - 4x + 1$ in Desmos and trace the graph to $x = -3$ to confirm y reads 22.

The function h is defined by $h(x) = 5 / (x - 7)$. For what value of x is $h(x)$ undefined?

Formula / Rule Used: Domain restriction: a denominator can never equal 0.

Solution

Step 1: A fraction is undefined exactly when its denominator equals 0.

Step 2: Set the denominator equal to 0: $x - 7 = 0$.

Step 3: Add 7 to both sides to solve for x : $x = 0 + 7$.

Step 4: $x = 7$.

Answer: $x = 7$

Tip

For any fraction rule, set the denominator equal to 0 and solve — that x -value is always the one to exclude.

Trap

It's easy to accidentally solve $x + 7 = 0$ instead of $x - 7 = 0$, flipping the sign of the answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-7	Solved $x + 7 = 0$ instead of $x - 7 = 0$, a sign error that flips the result.
5	Set the numerator equal to 0 instead of the denominator, confusing which part causes the function to be undefined.
0	Assumed $x = 0$ always makes a fraction undefined instead of actually solving $x - 7 = 0$.

Shortcut

You can skip writing the full equation — just take the denominator's expression and flip its sign: $x - 7$ means $x = 7$ directly.

Desmos Tip

Type $y = 5/(x-7)$ in Desmos and look for the vertical break in the graph at $x = 7$ to confirm it's undefined there.



The table below gives some values of the function p . x : -1 0 2 5 $p(x)$: 6 3 -1 -7 What is the value of $p(2) - p(-1)$?

Formula / Rule Used: Evaluating a function from a table: match each input to its output.

Solution

Step 1: Find $p(2)$ by locating $x = 2$ in the table: $p(2) = -1$.

Step 2: Find $p(-1)$ by locating $x = -1$ in the table: $p(-1) = 6$.

Step 3: Set up the subtraction in the order the question asks: $p(2) - p(-1) = -1 - 6$.

Step 4: Subtract: $-1 - 6 = -7$.

Answer: -7

Tip

Look up both table values first and write them down before subtracting, so you don't mix up which is which.

Trap

It's tempting to subtract in the reverse order, $p(-1) - p(2)$, which gives the opposite sign.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
7	Subtracted in reversed order, computing $p(-1) - p(2) = 6 - (-1) = 7$ instead of what was asked.
-4	Misread the table and used $p(0) = 3$ in place of $p(-1) = 6$, computing $-1 - 3 = -4$.
-5	Misread $p(2)$ as 1 instead of -1 (dropped the negative sign), then computed $1 - 6 = -5$.

Shortcut

This is already the fastest method — just be sure to read straight down each column before subtracting.

Desmos Tip

Enter the table's x and $p(x)$ values into Desmos as a table, then read the y -values at $x = 2$ and $x = -1$ to confirm the subtraction.

Functions — Notation, Domain, Range & Graphs — Problem 5 (Medium)

A function f is defined by $f(x) = 3x + 2$ if $x < 2$, and $f(x) = x^2 - 1$ if $x \geq 2$. For what value of x is $f(x) = 24$?

Formula / Rule Used: Piecewise functions: solve each piece, then check its condition.

Solution

Step 1: Since 24 is a fairly large output, try the $x \geq 2$ piece first: $x^2 - 1 = 24$.

Step 2: Add 1 to both sides: $x^2 = 25$.

Step 3: Take the square root of both sides: $x = 5$ or $x = -5$.

Step 4: Check each against the condition $x \geq 2$: $x = 5$ satisfies it, but $x = -5$ does not.

Step 5: Also check the other piece to be thorough: $3x + 2 = 24$ gives $x = 22/3$.

Step 6: But $22/3$ is about 7.33, which does NOT satisfy that piece's condition $x < 2$, so it's rejected.

Step 7: The only value that works in its own piece is $x = 5$.

Answer: $x = 5$

Tip

Guess which piece the output likely belongs to based on its size, solve that piece, then always verify the condition.

Trap

It's tempting to report both $x = 5$ and $x = -5$ without checking that -5 fails the required condition $x \geq 2$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 5$ or $x = -5$	Solved $x^2 = 25$ correctly but forgot to reject $x = -5$, which fails the piece's condition $x \geq 2$.
$x = 22/3$	Solved the first piece ($3x + 2 = 24$) but never checked that $22/3$ fails that piece's own condition $x < 2$.
$x = -5$	Took only the negative square root of 25 without checking it against the required condition $x \geq 2$.

Shortcut

Starting with the piece whose condition matches the output's size (a large positive output suggests the squared piece) saves time over checking both blindly.

Desmos Tip

Graph $y = 3x + 2$ restricted to $x < 2$ and $y = x^2 - 1$ restricted to $x \geq 2$ in Desmos, then trace to find where $y = 24$.

What is the domain of the function $f(x) = \sqrt{x - 3} / (x - 8)$?

Formula / Rule Used: Combine two domain restrictions: even root and denominator.

Solution

Step 1: This function has two separate restrictions to check: the square root and the denominator.

Step 2: For the square root, the expression inside it cannot be negative: $x - 3 \geq 0$.

Step 3: Solve this: $x \geq 3$.

Step 4: For the denominator, it cannot equal 0: $x - 8 \neq 0$.

Step 5: Solve this: $x \neq 8$.

Step 6: Combine both restrictions into one final domain statement: $x \geq 3$, but x cannot equal 8.

Answer: All real numbers x such that $x \geq 3$ and $x \neq 8$

Tip

Whenever a function has more than one restriction, find each one separately, then combine them at the end.

Trap

It's tempting to stop after handling the square root and forget the denominator also excludes a value.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x \geq 3$	Correctly handled the square root restriction but completely forgot to also exclude $x = 8$ from the denominator.
$x \neq 8$	Only handled the denominator restriction and forgot the square root also restricts the domain to $x \geq 3$.
$x > 3, x \neq 8$	Used a strict inequality $x > 3$ instead of $x \geq 3$, incorrectly excluding $x = 3$ even though $\sqrt{0} = 0$ is a valid, defined value.

Shortcut

No shortcut — a root combined with a denominator always needs both conditions checked separately before combining.

Desmos Tip

Type $y = \sqrt{x-3}/(x-8)$ in Desmos and note the graph starts at $x = 3$ and has a break in it at $x = 8$.

Functions — Notation, Domain, Range & Graphs — Problem 7 (Medium)

A tech startup in Athens tracks its total app downloads, in thousands, using the function $D(t)$, where t is the number of months since launch. The table shows $D(2) = 18$ and $D(7) = 43$. The company also redesigned its logo in month 5. Assuming downloads grew at a constant rate over this period, what was the rate of change of downloads, in thousands per month, between month 2 and month 7?

Formula / Rule Used: Rate of change = $(D(t_2) - D(t_1)) / (t_2 - t_1)$

Solution

Step 1: The logo redesign in month 5 has nothing to do with the download numbers — ignore that detail.

Step 2: Identify the two data points you actually need: (2, 18) and (7, 43).

Step 3: Set up the rate of change formula: $(D(7) - D(2)) / (7 - 2)$.

Step 4: Subtract the outputs on top: $43 - 18 = 25$.

Step 5: Subtract the inputs on the bottom: $7 - 2 = 5$.

Step 6: Divide: $25 / 5 = 5$.

Answer: 5 thousand downloads per month

Tip

List only the two actual data points you need before you calculate — irrelevant details in the problem don't belong in the formula.

Trap

It's tempting to pull the month-5 logo detail into your calculation just because a number is mentioned near it.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-5	Subtracted the outputs in reversed order, computing $(18 - 43) / (7 - 2) = -5$ instead of the correct order.
125	Multiplied the numerator and denominator (25×5) instead of dividing them.
12.5	Mistakenly used month 5 (the logo redesign month) in place of month 2, computing $25 / (7 - 5) = 12.5$.

Shortcut

This is already the fastest method — just be careful to use only the two real download data points, not the rebrand month.

Desmos Tip

Enter the two points (2, 18) and (7, 43) as a table in Desmos and check the slope between them confirms 5.

The function f is defined by $f(x) = ax + 5$, where a is a constant. If $f(4) = 21$, what is the value of $f(10)$?

Formula / Rule Used: Two-step: find the constant first, then evaluate.

Solution

Step 1: You can't evaluate $f(10)$ yet because a is still unknown — find a first.

Step 2: Use the given information $f(4) = 21$: substitute $x = 4$ into the rule: $a(4) + 5 = 21$.

Step 3: Subtract 5 from both sides: $4a = 16$.

Step 4: Divide both sides by 4: $a = 4$.

Step 5: Now the full rule is known: $f(x) = 4x + 5$.

Step 6: Substitute $x = 10$ into this complete rule: $f(10) = 4(10) + 5$.

Step 7: Multiply: $4(10) = 40$.

Step 8: Add: $f(10) = 40 + 5 = 45$.

Answer: 45

Tip

When a constant is unknown, use the given input-output pair to solve for it FIRST, before answering the actual question.

Trap

It's tempting to stop once you've found $a = 4$ and forget that the question actually asked for $f(10)$, not a .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Correctly solved for the constant $a = 4$ but stopped there instead of using it to find $f(10)$.
21	Reused the given value $f(4) = 21$ instead of computing the new output $f(10)$.
125	Solved $4a = 16$ incorrectly by subtracting instead of dividing, getting $a = 12$, then used that wrong constant in $f(10) = 12(10) + 5$.

Shortcut

No shortcut beyond solving for a first — but once you have $a = 4$, plugging into $f(10)$ is quick.

Desmos Tip

Type $y = 4x + 5$ in Desmos, using the constant you solved for, and trace to $x = 10$ to confirm $y = 45$.

15. Composite & Inverse Functions

Composite & Inverse Functions — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Composite function	Σύνθεση συναρτήσεων
Inverse function	Αντίστροφη συνάρτηση
One-to-one function	Συνάρτηση 1-1 (αμφιμονοσήμαντη)
Identity function	Ταυτοτική συνάρτηση
Reflection over $y = x$	Ανάκλαση ως προς την $y = x$

Composite & Inverse Functions — What It Is

What It Is

- A composite function is what you get when you plug one whole function inside another one.
- Writing $f(g(x))$ means: find $g(x)$ first, then use that answer as the new input for f .
- An inverse function is a function that completely undoes what the original function does.
- The SAT tests these to see if you can carefully track which expression substitutes into which.
- Both ideas show up as plain equations, tables of values, or graphs — not just algebra to solve.

Composite & Inverse Functions — Core Rules

Composite Function Notation

$$(f \circ g)(x) = f(g(x))$$

- This is read 'f of g of x,' and it always means g happens first, then f.
- Apply the inner function g to x, then apply the outer function f to that result.
- Order matters a lot: $f(g(x))$ is usually different from $g(f(x))$.
- Example: if $g(x) = x + 1$ and $f(x) = x^2$, then $f(g(3))$ means $g(3) = 4$, then $f(4) = 16$.

Evaluating a Composite at a Number — Work Inside Out

- Start with the innermost function and the given number, and compute that first.
- Take the number you just got and plug it in as the input to the next function out.
- Keep working outward, one function at a time, until nothing is left to apply.
- This method is fastest when you only need ONE specific numeric answer.

Building the Composite Expression First

$f(g(x))$ = substitute $g(x)$ for every x in $f(x)$

- Use this when you need a general formula for $f(g(x))$, not just one number.
- Replace every single x inside $f(x)$ with the entire expression for $g(x)$.
- Wrap the substituted expression in parentheses so you don't lose track of it.
- Simplify fully — expand and combine like terms — before evaluating at any value.

Expanding Multi-Term Substitutions Correctly

$(a + b)^2 = a^2 + 2ab + b^2$

- If $g(x)$ has more than one term, squaring it is NOT the same as squaring each term alone.
- Always expand with FOIL or the perfect-square pattern, never distribute the exponent.
- Example: $(x + 2)^2 = x^2 + 4x + 4$, which is NOT the same as $x^2 + 4$.

What an Inverse Function Is

$f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$

- An inverse function reverses what the original function does, swapping inputs and outputs.
- If f turns 3 into 10, then f^{-1} turns 10 back into 3.
- Composing a function with its own inverse, in either order, always just gives back x .
- $f^{-1}(x)$ is read 'f inverse of x' — it does NOT mean 1 divided by $f(x)$.

Finding an Inverse Algebraically

Steps: 1) $y = f(x)$ 2) swap x and y 3) solve for y 4) $y = f^{-1}(x)$

- Rewrite $f(x)$ as y first — it's much easier to swap two letters than function notation.
- Swap every x and every y in the equation; this reverses the input/output roles.
- Solve the new equation for y using normal algebra steps, one at a time.
- Replace y with $f^{-1}(x)$ in your final answer.

Inverse from a Table, Graph, or Reflection

$f^{-1}(b) = a$ when $f(a) = b$

- From a table, find the value in the OUTPUT column, then read off its matching input.
- On a graph, every point (a, b) on f becomes the point (b, a) on f^{-1} .
- The graphs of f and f^{-1} are always mirror images across the line $y = x$.
- This reflection trick lets you estimate f^{-1} values even without an equation.

How to Recognize This on the Test

- Two function names nested together, like $f(g(x))$ or $g(f(3))$, signal a composite question.
- The notation $f^{-1}(x)$, or the phrase 'inverse function,' always signals an inverse question.

- Being asked to 'undo' a function, or to find the input that produced a given output, is inverse thinking.
- A table with an input column and an output column, used in reverse, hints at an inverse relationship.
- A graph described as a 'reflection across $y = x$ ' of another graph is really about inverses.

Common Mistakes — and the Fix

- Evaluating the outer function first instead of the inner one, which reverses the whole answer.
- Distributing an exponent across a substituted expression instead of expanding it fully.
- Treating $f^{-1}(x)$ as $1/f(x)$ instead of as a completely different, reversed function.
- Forgetting to actually swap x and y before solving when finding an inverse algebraically.
- Computing $g(f(x))$ when the question asked for $f(g(x))$, because the order wasn't checked first.

Composite & Inverse Functions — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Composite & Inverse Functions — Problem 1 (Easy)

If $f(x) = x^2 - 3$ and $g(x) = 2x + 1$, what is the value of $f(g(2))$?

Formula / Rule Used: Composite functions: work inside out.

Solution

Step 1: For $f(g(2))$, work inside out: evaluate the inner function g at $x = 2$ first.

Step 2: $g(2) = 2(2) + 1 = 4 + 1 = 5$.

Step 3: Now use that result, 5, as the new input for the outer function f .

Step 4: $f(5) = 5^2 - 3$.

Step 5: Square first: $5^2 = 25$.

Step 6: Subtract: $25 - 3 = 22$.

Answer: 22

Tip

For $f(g(\text{number}))$, always evaluate the inside function first and treat its output as the new input.

Trap

It's tempting to evaluate f first and g second, which actually computes $g(f(2))$ instead of $f(g(2))$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3	Reversed the order and computed $g(f(2))$ instead: $f(2) = 1$, then $g(1) =$



	$2(1)+1 = 3.$
4	Order-of-operations slip: computed $(5 - 3)^2 = 4$ instead of $5^2 - 3.$
33	Misread $g(2)$ as $2(2+1) = 6$ instead of $2(2)+1 = 5$, then correctly computed $f(6) = 36-3 = 33.$

Shortcut

No shortcut needed — working strictly inside out is already the fastest, cleanest method here.

Desmos Tip

Define $f(x)=x^2-3$ and $g(x)=2x+1$ on two lines in Desmos, then type $f(g(2))$ on a third line to confirm 22.

Composite & Inverse Functions — Problem 2 (Easy)

The table below gives values of functions f and g for several inputs. x : 1, 2, 3, 4. $f(x)$: 4, 7, 10, 13. $g(x)$: 3, 1, 4, 2. What is the value of $g(f(1))$?

Formula / Rule Used: Composite functions from a table: look up the inner output first.

Solution

Step 1: Work inside out: first find $f(1)$ using the table's $f(x)$ row.

Step 2: Reading the $x = 1$ column, $f(1) = 4$.

Step 3: Now use that output, 4, as the new input to look up in g .

Step 4: Find $g(4)$ using the table's $g(x)$ row, reading the $x = 4$ column.

Step 5: $g(4) = 2$.

Answer: 2

Tip

With a table, treat the inner function's output as the new x -value you look up in the outer function's row.

Trap

It's easy to accidentally look up $g(1)$ directly instead of using $f(1)$'s result as the new input for g .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3	Skipped the inner step and read $g(1)$ straight from the table instead of first finding $f(1)$.
4	Stopped after finding $f(1) = 4$ and reported it, forgetting to apply g afterward.
10	Reversed the composite order and computed $f(g(1)) = f(3) = 10$ instead.

Shortcut



No shortcut needed with a table — just read one row's output, then look that number up as the input in the other row.

Composite & Inverse Functions — Problem 3 (Easy)

The function h is defined by $h(x) = (x - 5)/2$. What is $h^{-1}(x)$?

Formula / Rule Used: Finding an inverse: swap x and y , then solve for y .

Solution

Step 1: Rewrite $h(x)$ as y to make the swap easier to see: $y = (x - 5)/2$.

Step 2: Swap x and y : $x = (y - 5)/2$.

Step 3: Multiply both sides by 2 to clear the fraction: $2x = y - 5$.

Step 4: Add 5 to both sides to isolate y : $2x + 5 = y$.

Step 5: Rewrite using inverse notation: $h^{-1}(x) = 2x + 5$.

Answer: $h^{-1}(x) = 2x + 5$

Tip

To find an inverse function, always swap x and y first before doing any other algebra.

Trap

It's tempting to just rearrange the equation for x without truly swapping x and y , which skips the key step.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$h^{-1}(x) = 2x - 5$	Sign error isolating y : subtracted 5 instead of adding it to both sides.
$h^{-1}(x) = 2/(x-5)$	Confused the inverse function with the reciprocal $1/h(x)$ instead of reversing the operations.
$h^{-1}(x) = (x+5)/2$	Undid the operations in the wrong order, adding 5 before clearing the division by 2.

Shortcut

Check any inverse fast: plug your $h^{-1}(x)$ back into h and confirm the x 's cancel out to leave just x .

Desmos Tip

Graph $y = (x-5)/2$ and $y = 2x+5$ in Desmos and confirm the two lines are mirror images across the line $y = x$.

Composite & Inverse Functions — Problem 4 (Easy)

A clothing store in Athens applies a 20 euro discount to a jacket's price, then charges 24% sales tax on the discounted price. The discount is modeled by $d(p) = p - 20$ and the tax step is modeled by $t(p) = 1.24p$, where p is a price in euros. What is the final price of a jacket that originally cost 80 euros?

Formula / Rule Used: Composite functions in context: apply the steps in the order described.

Solution

Step 1: The discount happens before the tax, so apply d first, then t : this is $t(d(80))$.

Step 2: Apply the discount function to the original price: $d(80) = 80 - 20 = 60$.

Step 3: Now apply the tax function to that discounted price: $t(60) = 1.24(60)$.

Step 4: Multiply: $1.24 \times 60 = 74.4$.

Answer: 74.40 euros

Tip

When a word problem describes steps happening in order, apply the matching functions in that same order.

Trap

It's tempting to apply the tax function first and the discount second, which changes the final price since order matters.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
79.2 euros	Applied tax before discount, reversing the order the problem describes: $1.24(80) = 99.2$, then $99.2 - 20 = 79.2$.
60 euros	Correctly computed the discount, $d(80) = 60$, but forgot to apply the tax step at all.
62.4 euros	Treated the 24% tax as adding 2.40 instead of multiplying by 1.24, computing $60 + 2.4$.

Shortcut

Once you know discount comes first, you can combine both into $(80 - 20) \times 1.24$ in one line.

Desmos Tip

Type $y = 1.24(x - 20)$ in Desmos and trace to $x = 80$ to confirm the final price reads 74.4.

Composite & Inverse Functions — Problem 5 (Medium)



If $h(x - 3) = 2x^2 + x - 4$, what is $h(5)$?

Formula / Rule Used: Disguised composite: match the inside expression to the target input.

Solution

Step 1: h is defined in terms of $(x - 3)$, not x directly, so $h(5)$ needs $x - 3$ to equal 5.

Step 2: Solve $x - 3 = 5$ for x : $x = 8$.

Step 3: Substitute $x = 8$ into the given right-hand expression: $2(8)^2 + 8 - 4$.

Step 4: Square first: $8^2 = 64$, so $2(64) = 128$.

Step 5: Add and subtract left to right: $128 + 8 = 136$, then $136 - 4 = 132$.

Answer: 132

Tip

When you see $h(\text{something} - \text{number})$, set that something equal to the target input and solve for x first.

Trap

It's tempting to plug $x = 5$ straight into the expression, skipping the step of solving $x - 3 = 5$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
51	Plugged $x = 5$ directly into $2x^2 + x - 4$ instead of first solving $x - 3 = 5$ for $x = 8$.
124	Found $x = 8$ correctly, but mis-sequenced the last step as $128 - 8 + 4$ instead of $128 + 8 - 4$.
6	Solved $x - 3 = 5$ with a sign error, getting $x = 2$ instead of $x = 8$, then used that wrong x .

Shortcut

None beyond solving for x first — once $x = 8$ is found, direct substitution is already fastest.

Desmos Tip

Type $y1 = 2x^2 + x - 4$ in Desmos and trace to $x = 8$ to confirm $y1 = 132$, matching $h(5)$.

Composite & Inverse Functions — Problem 6 (Medium)

The functions f and g are defined by $f(x) = 2x + 3$ and $g(x) = x^2 - 4$. What is the value of $f^{-1}(g(3))$?

Formula / Rule Used: Combine an inverse with a composite: evaluate inside out.

Solution

Step 1: Work inside out: first evaluate $g(3)$.

Step 2: $g(3) = 3^2 - 4 = 9 - 4 = 5$.

Step 3: Now find the inverse of f algebraically, starting from $y = 2x + 3$.

Step 4: Swap x and y : $x = 2y + 3$.

Step 5: Solve for y : $x - 3 = 2y$, so $y = (x - 3)/2$, meaning $f^{-1}(x) = (x - 3)/2$.

Step 6: Substitute $x = 5$ into f^{-1} : $f^{-1}(5) = (5 - 3)/2 = 2/2 = 1$.

Answer: 1

Tip

When a problem combines an inverse with a composite, evaluate the plain function part first, then apply the inverse.

Trap

It's tempting to apply f itself instead of f^{-1} to $g(3)$, since f is the function that's given directly.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
13	Computed $f(g(3))$ instead of $f^{-1}(g(3))$, using f directly instead of its inverse: $f(5) = 2(5) + 3 = 13$.
2.5	Used $f^{-1}(x) = x/2$ instead of the correct $(x - 3)/2$, skipping the subtract-3 step entirely.
-1	Miscomputed $g(3)$ as $(3 - 4)^2 = 1$ instead of $3^2 - 4 = 5$, then found $f^{-1}(1) = (1 - 3)/2 = -1$.

Shortcut

No real shortcut — evaluate $g(3)$ first, then build f^{-1} algebraically; that order is already fastest.

Desmos Tip

Graph $y_1 = 2x + 3$ and $y_2 = (x - 3)/2$ in Desmos to confirm they're reflections across $y = x$, then trace y_2 to $x = 5$ for $y = 1$.

Composite & Inverse Functions — Problem 7 (Medium)

The functions f and g are defined by $f(x) = ax + 5$ and $g(x) = x - 1$, where a is a constant. If $f(g(4)) = 17$, what is $f(10)$?

Formula / Rule Used: Find the constant first, then use the completed function.

Solution

Step 1: First evaluate the inner function: $g(4) = 4 - 1 = 3$.

Step 2: Substitute that result into f : $f(3) = a(3) + 5 = 3a + 5$.

Step 3: Set this equal to the given value: $3a + 5 = 17$.

Step 4: Subtract 5 from both sides: $3a = 12$.

Step 5: Divide by 3 to solve for a : $a = 4$.

Step 6: Now the full function is $f(x) = 4x + 5$.

Step 7: Substitute $x = 10$: $f(10) = 4(10) + 5 = 40 + 5 = 45$.

Answer: 45

Tip

When a problem gives you $f(g(\text{number})) = \text{value}$ to find an unknown constant, solve for the constant first, then answer the actual question asked.

Trap

It's tempting to stop as soon as you find $a = 4$, but the question actually asks for $f(10)$, not the value of a .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Stopped after solving for a and reported that value instead of continuing to compute $f(10)$.
40	Forgot to add the constant 5 when computing $f(10)$, calculating only $4(10)$.
35	Skipped evaluating $g(4)$ and set $f(4) = 17$ directly, giving $a = 3$ instead of the correct $a = 4$.

Shortcut

None beyond solving for a first — this two-step 'find the constant, then use it' structure needs both steps in order.

Desmos Tip

Once you solve for a , type $y = 4x + 5$ in Desmos and trace to $x = 10$ to confirm $y = 45$.

A conversion function models temperature: $F(c) = (9/5)c + 32$, which converts a Celsius reading c to Fahrenheit. The formula was popularized in a physics course taught by Mr. Papadopoulos in Athens. Using the inverse of this function, what Celsius temperature corresponds to a Fahrenheit reading of 98.6?

Formula / Rule Used: Inverse of a linear function, applied to a real-world value.

Solution

Step 1: The detail about Mr. Papadopoulos is background only and doesn't affect the math — ignore it.

Step 2: Start from $F = (9/5)c + 32$ and solve directly for c .

Step 3: Subtract 32 from both sides: $F - 32 = (9/5)c$.

Step 4: Multiply both sides by $5/9$ to isolate c : $c = (5/9)(F - 32)$.

Step 5: Substitute $F = 98.6$: $c = (5/9)(98.6 - 32)$.

Step 6: Subtract inside the parentheses: $98.6 - 32 = 66.6$.

Step 7: Multiply: $c = (5/9)(66.6) = 37$.

Answer: 37 degrees Celsius

Tip

For 'convert back' questions, solve the original equation for the OTHER variable first, then substitute the given value.

Trap

It's tempting to get distracted by the irrelevant detail about Mr. Papadopoulos, or to multiply by $9/5$ instead of its reciprocal.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
approximately 54.8	Forgot to subtract 32 first, computing $(5/9)(98.6)$ instead of $(5/9)(98.6 - 32)$.
approximately 119.9	Used $9/5$ instead of the reciprocal $5/9$ when isolating c , multiplying by the wrong factor.
66.6	Correctly subtracted 32 but forgot to multiply by $5/9$, stopping one step early.

Shortcut

98.6 F is the well-known value for human body temperature, which is exactly 37 C — worth recognizing on sight.

Desmos Tip

Type $y = (5/9)(x - 32)$ in Desmos and trace to $x = 98.6$ to confirm $y = 37$.

16. Function Transformations

Function Transformations — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Transformation	Μετασχηματισμός
Translation (shift)	Μεταφορά
Reflection	Ανάκλαση
Vertical/horizontal stretch	Κατακόρυφη/οριζόντια διαστολή
Vertical/horizontal compression	Κατακόρυφη/οριζόντια συστολή
Parent function	Βασική (μητρική) συνάρτηση

Function Transformations — What It Is

What It Is

- A transformation takes a function's graph and shifts, flips, stretches, or squishes it, without changing its basic shape.
- You start from a 'parent' graph, like $y = f(x)$, and a small change to the equation moves the whole picture.
- The SAT tests whether you can predict exactly how a graph moves just by reading numbers added or multiplied in the equation.
- Every transformation question is really asking: does this change move the graph up/down, left/right, flip it, or resize it?
- You'll see this tested with equations, with graphs you compare by eye, and with word problems describing a shifted model.

Function Transformations — Core Rules

Vertical Shift (moves up or down)

$$y = f(x) + k$$

- Adding a number k OUTSIDE the function, after $f(x)$, moves the whole graph up or down.
- If k is positive, the graph shifts up by k units; if k is negative, it shifts down.
- Every single y -value increases or decreases by k — the x -values never change.
- Example: $y = f(x) + 3$ takes every point on $f(x)$ and moves it up 3 spaces.

Horizontal Shift (moves left or right)

$$y = f(x + h)$$

- Adding a number h INSIDE the parentheses, next to x , moves the graph sideways.
- This direction feels backwards: $f(x + h)$ shifts LEFT when h is positive.
- $f(x - h)$ shifts RIGHT when h is positive — the sign flips your expectation.
- Example: $y = f(x - 4)$ shifts the graph 4 units to the right, not left.

Why Horizontal Shifts Flip Direction

- To get the SAME output as the original graph had at $x = 0$, the new x must cancel out h .
- In $f(x + 2)$, plugging in $x = -2$ gives $f(-2 + 2) = f(0)$, so the 'start' moved left to -2 .
- This is the single most missed idea in this entire topic, so double-check the direction every time.
- A quick trick: set the inside expression equal to 0 and solve for x to find the key point.

Reflections (flips the graph)

$y = -f(x)$ flips over the x-axis. $y = f(-x)$ flips over the y-axis.

- A negative sign OUTSIDE the function (in front of f) flips the graph upside down.
- A negative sign INSIDE the function (in front of x) flips the graph left-to-right.
- Outside changes affect y -values (up/down effects); inside changes affect x -values (left/right effects).
- Example: if $f(2) = 5$, then $-f(2) = -5$, but $f(-2)$ needs the graph's value at $x = -2$ instead.

Vertical Stretch and Compression

$y = a \cdot f(x)$

- Multiplying the WHOLE function by a number a changes how tall or short the graph looks.
- If $|a|$ is greater than 1 (like 2 or 3), the graph stretches vertically and looks taller/narrower.
- If $|a|$ is between 0 and 1 (like $1/2$), the graph compresses vertically and looks shorter/flatter.
- If a is negative, it stretches or compresses AND reflects over the x -axis at the same time.

Horizontal Stretch and Compression

$y = f(bx)$

- Multiplying x itself (inside the function) by a number b changes the graph's width sideways.
- If $|b|$ is greater than 1, the graph compresses horizontally and looks squeezed narrower.
- If $|b|$ is between 0 and 1, the graph stretches horizontally and looks wider.
- This one is less common on the SAT than vertical stretch, but the inside/outside logic still applies.

Combining Several Transformations at Once

$y = a \cdot f(x - h) + k$

- SAT problems often combine two or three transformations into a single equation like this.
- Read from the inside out: first the horizontal shift ($x - h$), then the stretch/reflection (a), then the



vertical shift (+k).

- Handle one transformation at a time instead of trying to picture all of them at once.
- Example: $y = 2f(x - 1) + 3$ shifts right 1, stretches vertically by 2, then shifts up 3.

How to Recognize This on the Test

- Seeing $f(x)$ with something added, subtracted, or multiplied around it signals a transformation question.
- Phrases like 'shifted,' 'reflected,' 'stretched,' or 'compressed' are direct signals for this skill.
- A question showing two graphs and asking how one relates to the other is testing transformations.
- A word problem where a model is 'increased by a constant' or 'delayed by a few units' is a transformation in disguise.
- Seeing $g(x)$ written in terms of $f(x)$ means you're comparing a transformed graph to its parent.

Common Mistakes — and the Fix

- Assuming $f(x + 3)$ shifts right — it actually shifts LEFT; always check the sign rule for horizontal shifts.
- Mixing up inside changes (affect x , do the opposite of what you'd guess) with outside changes (affect y , match your guess).
- Applying transformations in the wrong order instead of working from the inside out, one step at a time.
- Forgetting that a negative value of a both flips AND stretches/compresses the graph, not just one or the other.
- Reading a stretch/compression backwards — forgetting that a fraction like $1/2$ compresses, while a number over 1 stretches.

Function Transformations — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Function Transformations — Problem 1 (Easy)

A weather station in Copenhagen models the temperature in degrees Celsius t hours after midnight using the function $C(t)$. After recalibrating a sensor, the station reports temperatures using a new function $D(t) = C(t) + 4$ for the same day. If $C(6) = 12$, what is $D(6)$?

Formula / Rule Used: Vertical shift: $y = f(x) + k$

Solution

Step 1: $D(t) = C(t) + 4$ means every output of C gets increased by 4.

Step 2: We need $D(6)$, and we're told $C(6) = 12$.

Step 3: Substitute the known value into the new function: $D(6) = C(6) + 4$.

Step 4: Replace $C(6)$ with 12: $D(6) = 12 + 4$.

Step 5: Add: $D(6) = 16$.

Answer: 16

Tip

When a function is defined as another function plus a constant, just add that constant to the known output.

Trap

It's tempting to subtract the 4 instead of adding it, treating the shift as if it were downward.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
8	Subtracted 4 instead of adding it: $12 - 4 = 8$.
12	Forgot to apply the vertical shift at all and just restated $C(6)$.
4	Reported only the constant k instead of adding it to $C(6)$.

Shortcut

None needed — this is already the fastest path, just add the shift directly to the known value.

Function Transformations — Problem 2 (Easy)



The graph of $y = f(x)$ contains the point $(5, 9)$. What point must lie on the graph of $y = f(x - 3)$?

Formula / Rule Used: Horizontal shift: $y = f(x - h)$

Solution

Step 1: $y = f(x - 3)$ shifts the graph of f to the RIGHT by 3 units.

Step 2: Since $(5, 9)$ is on $f(x)$, we know $f(5) = 9$.

Step 3: We need the new x so that the inside expression equals 5: $x - 3 = 5$.

Step 4: Add 3 to both sides to solve for x : $x = 5 + 3$.

Step 5: $x = 8$, and the y -value stays 9 since the outside of the function didn't change.

Step 6: So the point $(8, 9)$ lies on $y = f(x - 3)$.

Answer: $(8, 9)$

Tip

To move a known point through a horizontal shift, set the inside expression equal to the original x -value and solve.

Trap

It's easy to subtract 3 from the x -coordinate instead of adding it, since $'-3'$ looks like it should shift left.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(2, 9)$	Subtracted 3 from the x -coordinate instead of adding: $5 - 3 = 2$.
$(5, 6)$	Shifted the y -coordinate instead of the x -coordinate: $9 - 3 = 6$.
$(8, 12)$	Correctly shifted x but also added 3 to the y -coordinate: $9 + 3 = 12$.

Shortcut

For $f(x - h)$, just add h to the original x -coordinate — no need to resolve the whole function.

Desmos Tip

Define $f(x)$ as any simple curve through $(5, 9)$, then type $g(x) = f(x - 3)$ and trace g to confirm the point lands at $(8, 9)$.

Function Transformations — Problem 3 (Easy)



The graph of $y = f(x)$ passes through the point $(-2, 7)$. What point must lie on the graph of $y = -f(x)$?

Formula / Rule Used: Reflection over the x-axis: $y = -f(x)$

Solution

Step 1: $y = -f(x)$ reflects the graph of f over the x-axis.

Step 2: This flips the SIGN of every y-value, while every x-value stays exactly the same.

Step 3: Since $(-2, 7)$ is on $f(x)$, we know $f(-2) = 7$.

Step 4: Compute the new y-value: $-f(-2) = -7$.

Step 5: So the point $(-2, -7)$ lies on $y = -f(x)$.

Answer: $(-2, -7)$

Tip

For $y = -f(x)$, keep the x-coordinate exactly the same and just flip the sign of the y-coordinate.

Trap

It's easy to flip the sign of the x-coordinate instead, which is actually what $y = f(-x)$ would do.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(2, 7)$	Flipped the sign of the x-coordinate instead of the y-coordinate, confusing it with $y = f(-x)$.
$(-2, 7)$	Forgot to flip any sign and just restated the original point.
$(2, -7)$	Flipped the signs of BOTH coordinates instead of only the y-coordinate.

Shortcut

None needed — flipping the sign of the given y-value is already the fastest way.

Desmos Tip

Plot the point $(-2, 7)$, define any curve $f(x)$ through it, graph $y = -f(x)$, and confirm the reflected point $(-2, -7)$ appears directly below.

Function Transformations — Problem 4 (Easy)

The graph of $y = f(x)$ contains the point $(4, 6)$. What point must lie on the graph of $y = 3f(x)$?

Formula / Rule Used: Vertical stretch: $y = a \cdot f(x)$

Solution

Step 1: $y = 3f(x)$ stretches the graph vertically by multiplying every output by 3.

Step 2: The x-coordinate never changes in a vertical stretch.

Step 3: Since $(4, 6)$ is on $f(x)$, we know $f(4) = 6$.

Step 4: Multiply the y-value by the stretch factor: $3 \cdot f(4) = 3 \cdot 6$.

Step 5: $3 \cdot 6 = 18$.

Step 6: So the point $(4, 18)$ lies on $y = 3f(x)$.

Answer: $(4, 18)$

Tip

For $y = a \cdot f(x)$, keep the x-coordinate the same and multiply only the y-coordinate by a .

Trap

It's tempting to multiply the x-coordinate by 3 instead, since the '3' sits right in front of $f(x)$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(12, 6)$	Multiplied the x-coordinate by 3 instead of the y-coordinate: $4 \cdot 3 = 12$.
$(4, 9)$	Added 3 to the y-value instead of multiplying by it: $6 + 3 = 9$.
$(4, 2)$	Divided the y-value by 3 instead of multiplying: $6 / 3 = 2$.

Shortcut

None needed — multiplying the known y-value by the stretch factor is already the fastest path.

Desmos Tip

Define $f(x)$ as any curve through $(4, 6)$, graph $g(x) = 3 \cdot f(x)$, and trace g at $x = 4$ to confirm $y = 18$.

Function Transformations — Problem 5 (Medium)

The graph of $y = f(x)$ contains the point $(3, 5)$. The function $g(x) = 2f(x - 4) + 1$ is a transformation of f . What point corresponds to $(3, 5)$ on the graph of g ?

Formula / Rule Used: Combined transformation: $y = a \cdot f(x - h) + k$

Solution

Step 1: $g(x) = 2f(x - 4) + 1$ combines a shift right 4, a vertical stretch by 2, and a shift up 1.

Step 2: First find the new x-coordinate: g uses f at input $(x - 4)$, so set $x - 4 = 3$.

Step 3: Add 4 to both sides: $x = 3 + 4 = 7$.

Step 4: This tells us the original point moves to $x = 7$ on the graph of g .

Step 5: Next find the new y-coordinate by applying the outside operations to $y = 5$.

Step 6: Multiply first: $2 \cdot 5 = 10$.

Step 7: Then add the vertical shift: $10 + 1 = 11$.

Step 8: So the corresponding point on g is $(7, 11)$.

Answer: $(7, 11)$

Tip

Handle the inside (solve $x - 4 = 3$ for the new x) and the outside (compute $2(5) + 1$ for the new y) as two separate steps.

Trap

A common mistake is doing the addition before the multiplication on the y-value, or solving $x - 4 = 3$ backwards.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(7, 10)$	Forgot to add the final $+1$, stopping after $2(5) = 10$.
$(-1, 11)$	Solved $x - 4 = 3$ incorrectly as $x = 3 - 4 = -1$ instead of $x = 3 + 4 = 7$.
$(7, 12)$	Added 1 before multiplying by 2, computing $2(5 + 1) = 12$ instead of $2(5) + 1$.

Shortcut

None needed — solving the inside and outside separately is already the fastest method for a combined transformation.

Desmos Tip

Define $f(x)$ as any curve through $(3, 5)$, type $g(x) = 2f(x-4)+1$, and trace g at $x = 7$ to confirm $y = 11$.

Function Transformations — Problem 6 (Medium)



The graph of $h(x) = f(x + 2) - 5$ passes through the point (1, 3). What is the value of $f(3)$?

Formula / Rule Used: Reverse transformation: solve for the original function's value.

Solution

Step 1: $h(x) = f(x + 2) - 5$ means $h(1) = f(1 + 2) - 5 = f(3) - 5$.

Step 2: We're told $h(1) = 3$, so set up the equation: $f(3) - 5 = 3$.

Step 3: Add 5 to both sides to isolate $f(3)$: $f(3) = 3 + 5$.

Step 4: $f(3) = 8$.

Answer: $f(3) = 8$

Tip

Substitute the given x-value into the inside expression first, so you see exactly which input of f you're solving for.

Trap

It's easy to forget to undo the outside '-5' shift and just report $h(1)$'s value as if it were $f(3)$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3	Reported $h(1)$ directly, forgetting to undo the vertical shift of -5.
-2	Subtracted 5 instead of adding it: $3 - 5 = -2$.
1	Confused the input value $x = 1$ with the value being solved for, reporting x instead of $f(3)$.

Shortcut

Since $1 + 2$ already equals the 3 you're asked about, you can skip extra work and jump straight to solving $f(3) - 5 = 3$.

Desmos Tip

Graph any $h(x)$ through (1, 3), define $f(x) = h(x - 2) + 5$ to undo the transformation, and confirm $f(3) = 8$.

Function Transformations — Problem 7 (Medium)

A ride-sharing company in Berlin models its weekly revenue, in thousands of euros, with the function $R(w)$, where w is the number of weeks since launch. After a marketing campaign, the company's new weekly revenue is modeled by $N(w) = R(w - 2) + 15$, since results from the campaign appear two weeks later than expected and add a flat bonus. The company also reports that average weekly signups rose by 12% over the same period. If $R(10) = 40$, what is $N(12)$?

Formula / Rule Used: Horizontal shift + vertical shift: $N(w) = R(w - 2) + 15$

Solution

Step 1: $N(w) = R(w - 2) + 15$ shifts the graph right 2 weeks, then adds 15.

Step 2: We want $N(12)$, so substitute $w = 12$: $N(12) = R(12 - 2) + 15$.

Step 3: Simplify inside: $12 - 2 = 10$, so $N(12) = R(10) + 15$.

Step 4: We're given $R(10) = 40$.

Step 5: Substitute: $N(12) = 40 + 15$.

Step 6: Add: $N(12) = 55$.

Step 7: The 12% signup increase is extra information that isn't needed for this calculation.

Answer: 55 (thousand euros)

Tip

When you see $w - 2$ inside the function, plug in the given w first to figure out exactly which R -value you actually need.

Trap

The 12% signup statistic is designed to tempt you into using it, but the revenue calculation never needs it.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
40	Found $R(10)$ correctly but forgot to add the +15 vertical shift afterward.
44.8	Mistakenly applied the irrelevant 12% signup growth to the revenue: $40 \times 1.12 = 44.8$.
57	Added the shift amount 2 into the final sum instead of only using it to find the right R -value: $40 + 2 + 15 = 57$.

Shortcut

Since $R(10)$ is handed to you directly, just add 15 once you confirm $12 - 2 = 10$ — no need to re-derive anything else.

The graph of $y = f(x)$ passes through the points $(6, 4)$ and $(9, -2)$. The transformed function $g(x) = a \cdot f(x)$ passes through the point $(6, 10)$. What is $g(9)$?

Formula / Rule Used: Vertical stretch with an unknown constant: $g(x) = a \cdot f(x)$

Solution

Step 1: First use the known point on g to solve for a : $g(6) = a \cdot f(6)$.

Step 2: We know $f(6) = 4$ and $g(6) = 10$, so $10 = a \cdot 4$.

Step 3: Divide both sides by 4 to solve for a : $a = 10 / 4$.

Step 4: $a = 2.5$.

Step 5: Now use this same a to find $g(9)$: $g(9) = a \cdot f(9)$.

Step 6: We know $f(9) = -2$ from the second given point.

Step 7: Substitute: $g(9) = 2.5 \cdot (-2)$.

Step 8: Multiply: $g(9) = -5$.

Answer: -5

Tip

When the stretch factor isn't given directly, use one known point to solve for a first, then apply that same a to the other point.

Trap

It's easy to solve for a using the wrong pair of values, or to drop the negative sign when multiplying by $f(9)$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-2	Forgot to multiply by a at all and just reported $f(9)$ as the final answer.
-0.8	Solved for a using the reversed ratio, $a = 4/10 = 0.4$, then computed $0.4 \times -2 = -0.8$.
5	Solved for a correctly as 2.5, but dropped the negative sign from $f(9) = -2$, computing $2.5 \times 2 = 5$.

Shortcut

None needed — solving for a once, then reusing it for the second point, is already the fastest method.

Desmos Tip

Define $f(x)$ as any curve through $(6, 4)$ and $(9, -2)$, type $g(x) = 2.5 \cdot f(x)$, and trace g at $x = 9$ to confirm $g(9) = -5$.

17. Polynomial Operations & Special Products

Polynomial Operations & Special Products — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Polynomial	Πολυώνυμο
Term / coefficient	Όρος / συντελεστής
Degree of a polynomial	Βαθμός πολυωνύμου
Like terms	Όμοιοι όροι
Special products (identities)	Αξιοσημείωτες ταυτότητες
Binomial	Διώνυμο

Polynomial Operations & Special Products — What It Is

What It Is

- A polynomial is an expression made of terms with whole-number exponents, like $3x^2 + 5x - 7$.
- Polynomial operations means adding, subtracting, or multiplying these expressions together.
- Special products are multiplication patterns, like $(a+b)^2$, that appear so often they're worth memorizing.
- Memorizing the patterns lets you expand instantly instead of multiplying every term by every term.
- The SAT tests this because these patterns hide inside harder algebra, geometry, and factoring problems.

Polynomial Operations & Special Products — Core Rules

Polynomial Vocabulary

$$a_n x^n + \dots + a_1 x + a_0$$

- The degree of a polynomial is its highest exponent on the variable.
- The leading coefficient is the number multiplying that highest-degree term.
- Like terms have the exact same variable raised to the exact same exponent.
- Example: $3x^2$ and $-5x^2$ are like terms, but $3x^2$ and $3x$ are not.

Adding Polynomials

- Line up or group the like terms from both polynomials together.
- Add only the coefficients of matching like terms; the exponent never changes.
- Example: $(2x^2 + 3x) + (x^2 + 5) = 3x^2 + 3x + 5$.

- Any term with no matching like term just carries down unchanged.

Subtracting Polynomials

- Subtracting a polynomial means distributing a negative sign to every term inside it.
- Flip the sign of EVERY term in the second polynomial before combining anything.
- Example: $(5x^2 + 2x) - (3x^2 - 4) = 5x^2 + 2x - 3x^2 + 4$.
- After flipping signs, combine like terms exactly like an addition problem.

Multiplying Polynomials / FOIL

$$(a+b)(c+d) = ac + ad + bc + bd$$

- FOIL stands for First, Outer, Inner, Last — the four pairs to multiply for two binomials.
- First: multiply the first terms. Outer: the two outside terms. Inner: the two inside terms. Last: the last terms.
- For anything bigger than two binomials, multiply every term in the first factor by every term in the second.
- Always finish by combining any resulting like terms into one simplified expression.

Square of a Sum

$$(a + b)^2 = a^2 + 2ab + b^2$$

- Square the first term to get a^2 .
- Double the product of the two terms to get the middle term, $2ab$.
- Square the last term to get b^2 .
- Example: $(x + 4)^2 = x^2 + 2(x)(4) + 4^2 = x^2 + 8x + 16$.

Square of a Difference

$$(a - b)^2 = a^2 - 2ab + b^2$$

- This follows the exact same pattern as the square of a sum.
- The only change is the middle term becomes negative instead of positive.
- Example: $(x - 4)^2 = x^2 - 2(x)(4) + 4^2 = x^2 - 8x + 16$.
- The squared terms, a^2 and b^2 , are always positive either way.

Difference of Squares

$$(a + b)(a - b) = a^2 - b^2$$

- This pattern appears when one factor has a plus sign and the other has a minus sign.
- The two middle terms (the outer and inner products) are opposites, so they cancel out.
- The result only ever has two terms: a squared minus b squared, never a middle term.
- Example: $(x + 3)(x - 3) = x^2 - 9$.

How to Recognize This on the Test

- Two binomials multiplied together, written as $(\quad)(\quad)$, signal FOIL or a special product.
- A binomial being squared, like $(x + 7)^2$, is a direct signal to use the square-of-a-sum or -difference pattern.
- Seeing one plus sign and one minus sign between otherwise identical terms, like $(x+2)(x-2)$, means difference of squares.
- The instruction 'expand' or 'write in standard form' means you need to multiply everything out and combine like terms.
- The instruction 'simplify' next to added or subtracted polynomials means combine like terms carefully.

Common Mistakes — and the Fix

- Writing $(a + b)^2$ as $a^2 + b^2$ — the middle term $2ab$ is real and cannot be skipped.
- Forgetting to flip the sign of every term when subtracting one polynomial from another.
- Combining terms that look similar but aren't, like adding x^2 and x as if they were like terms.
- Multiplying only the first and last terms of two binomials and skipping the outer and inner products.
- Mixing up the difference-of-squares pattern with the square-of-a-difference pattern, which have different results.

Polynomial Operations & Special Products — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Polynomial Operations & Special Products — Problem 1 (Easy)

A gardener in Thessaloniki is building two rectangular flower beds side by side. The first bed has an area of $(x^2 + 3x + 2)$ square meters, and the second has an area of $(2x^2 - x + 5)$ square meters. Write a simplified polynomial for the total area of both beds combined.

Formula / Rule Used: Adding polynomials: combine like terms.

Solution

Step 1: Write the sum of the two areas: $(x^2 + 3x + 2) + (2x^2 - x + 5)$.

Step 2: Group the terms by type: x^2 terms, x terms, and constant terms.

Step 3: Combine the x^2 terms: $x^2 + 2x^2 = 3x^2$.

Step 4: Combine the x terms: $3x + (-x) = 2x$.

Step 5: Combine the constant terms: $2 + 5 = 7$.

Step 6: Put the combined terms together: $3x^2 + 2x + 7$.

Answer: $3x^2 + 2x + 7$ (square meters)

**Tip**

Group same-degree terms together in your head first so you don't accidentally combine unlike terms.

Trap

It's easy to mix the x^2 terms and the x terms together into one term since they both contain the letter x .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$3x^2 + 4x + 7$	Combined $3x$ and $-x$ by adding their sizes instead of subtracting, treating $-x$ as if it were $+x$.
$3x^2 + 2x + 3$	Subtracted the constant terms ($5 - 2 = 3$) instead of adding them ($2 + 5 = 7$).
$3x^4 + 2x + 7$	Added the exponents on the x^2 terms like a multiplication rule, getting x^4 instead of keeping x^2 .

Shortcut

None needed here — combining like terms directly is already the fastest way to add two polynomials.

Desmos Tip

Type $y1 = (x^2 + 3x + 2) + (2x^2 - x + 5)$ and $y2 = 3x^2 + 2x + 7$ — if the two graphs lie exactly on top of each other, the simplification is correct.

Polynomial Operations & Special Products — Problem 2 (Easy)

A bakery in Vienna models its weekly revenue as $(4x^2 + 6x + 20)$ euros and its weekly cost as $(x^2 + 2x + 15)$ euros, where x is the number of hundreds of pastries sold. Write a simplified polynomial for the bakery's weekly profit (profit = revenue - cost).

Formula / Rule Used: Subtracting polynomials: distribute the negative sign, then combine like terms.

Solution

Step 1: Set up the subtraction: $(4x^2 + 6x + 20) - (x^2 + 2x + 15)$.

Step 2: Distribute the negative sign to every term in the cost polynomial: $4x^2 + 6x + 20 - x^2 - 2x - 15$.

Step 3: Combine the x^2 terms: $4x^2 - x^2 = 3x^2$.

Step 4: Combine the x terms: $6x - 2x = 4x$.

Step 5: Combine the constant terms: $20 - 15 = 5$.

Step 6: Put the combined terms together: $3x^2 + 4x + 5$.

Answer: $3x^2 + 4x + 5$ (euros)

Tip

Before combining anything, flip the sign of every single term inside the polynomial being subtracted.

**Trap**

Only flipping the sign of the first term inside the parentheses is the classic error in a subtraction like this.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$3x^2 + 8x + 5$	Only flipped the sign of the first term in the cost polynomial, so added $6x$ and $2x$ instead of subtracting.
$5x^2 + 8x + 35$	Added the two polynomials together instead of subtracting the second from the first.
$3x^2 + 4x + 35$	Correctly combined the x^2 and x terms but added the constants ($20 + 15$) instead of subtracting them.

Shortcut

None needed here — distributing the negative sign before combining terms is already the safest, fastest method.

Desmos Tip

Type $y_1 = (4x^2 + 6x + 20) - (x^2 + 2x + 15)$ and $y_2 = 3x^2 + 4x + 5$ — matching graphs confirm the subtraction was done correctly.

Polynomial Operations & Special Products — Problem 3 (Easy)

Expand and simplify the expression $(x + 6)^2$.

Formula / Rule Used: Square of a sum: $(a+b)^2 = a^2 + 2ab + b^2$.

Solution

Step 1: Match the expression to the pattern $(a+b)^2$ with $a = x$ and $b = 6$.

Step 2: Square the first term: $a^2 = x^2$.

Step 3: Double the product of the two terms: $2ab = 2(x)(6) = 12x$.

Step 4: Square the last term: $b^2 = 6^2 = 36$.

Step 5: Combine all three pieces in order: $x^2 + 12x + 36$.

Answer: $x^2 + 12x + 36$

Tip

Write out what a and b are first, then apply the three-piece pattern a^2 , $2ab$, b^2 in that order.

Trap

Skipping the middle term entirely and writing $x^2 + 36$ is the single most common mistake with this pattern.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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$x^2 + 36$	Forgot the middle term $2ab$ entirely, squaring only the first and last terms of the binomial.
$x^2 + 6x + 36$	Forgot to double the product of a and b , writing $6x$ instead of $12x$.
$x^2 + 12x + 12$	Doubled the product ab correctly but forgot to square b , writing 12 instead of 36 .

Shortcut

None needed — applying $a^2 + 2ab + b^2$ directly is faster than multiplying $(x+6)(x+6)$ term by term with FOIL.

Desmos Tip

Type $y_1 = (x+6)^2$ and $y_2 = x^2 + 12x + 36$ in Desmos — the two graphs should lie perfectly on top of each other.

Polynomial Operations & Special Products — Problem 4 (Easy)

A rug maker in Toronto sells a rectangular rug with a length of $(x + 5)$ feet and a width of $(x + 2)$ feet. Write a simplified polynomial for the area of the rug.

Formula / Rule Used: Multiplying binomials (FOIL): $(a+b)(c+d) = ac + ad + bc + bd$.

Solution

Step 1: Set up the area as length times width: $(x + 5)(x + 2)$.

Step 2: Multiply the First terms: x times $x = x^2$.

Step 3: Multiply the Outer terms: x times $2 = 2x$.

Step 4: Multiply the Inner terms: 5 times $x = 5x$.

Step 5: Multiply the Last terms: 5 times $2 = 10$.

Step 6: Add all four results together: $x^2 + 2x + 5x + 10$.

Step 7: Combine the two like x -terms: $2x + 5x = 7x$.

Step 8: Final simplified area: $x^2 + 7x + 10$.

Answer: $x^2 + 7x + 10$ (square feet)

Tip

Work through FOIL in order — First, Outer, Inner, Last — so you never miss one of the four products.

Trap

It's tempting to multiply only the First and Last terms and skip the Outer and Inner cross terms.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x^2 + 10$	Multiplied only the First and Last terms, skipping the Outer and Inner cross products entirely.
$x^2 + 3x + 10$	Subtracted the Outer and Inner terms ($5x - 2x$) instead of adding them ($2x$



	$+ 5x$).
$2x^2 + 7x + 10$	Made an error multiplying the First terms, writing $2x^2$ instead of x^2 .

Shortcut

None needed — FOIL is already the fastest way to expand the product of two binomials.

Desmos Tip

Type $y_1 = (x+5)(x+2)$ and $y_2 = x^2 + 7x + 10$ — overlapping graphs confirm the expansion is correct.

Polynomial Operations & Special Products — Problem 5 (Medium)

The expression $x^2 + 14x + k$ is a perfect square trinomial, equal to $(x + h)^2$ for some positive constants h and k . What is the value of k ?

Formula / Rule Used: Perfect square trinomial pattern: $(x+h)^2 = x^2 + 2hx + h^2$.

Solution

Step 1: Compare $x^2 + 14x + k$ to the pattern $x^2 + 2hx + h^2$.

Step 2: Match the middle terms: the coefficient of x must satisfy $2h = 14$.

Step 3: Solve for h by dividing both sides by 2: $h = 7$.

Step 4: The constant term of a perfect square trinomial is always h^2 , so $k = h^2$.

Step 5: Substitute $h = 7$: $k = 7^2 = 49$.

Answer: $k = 49$

Tip

Match the middle-term coefficient to $2h$ first — once you know h , k is simply h squared.

Trap

A common mistake is reporting k as equal to h itself (7) instead of squaring h to get 49.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 7$	Correctly found $h = 7$ but reported it directly as k instead of squaring it.
$k = 14$	Confused the middle-term coefficient 14 with the value of h^2 and reported it unchanged.
$k = 196$	Squared the middle-term coefficient 14 directly instead of first solving for h and then squaring h .

Shortcut

Since the trinomial is guaranteed to be a perfect square, jump straight to $k = (\text{coefficient of } x / 2)^2$ without expanding anything.

**Desmos Tip**

Type $y = x^2 + 14x + 49$ and confirm it touches the x-axis at exactly one point, showing the trinomial is a perfect square when $k = 49$.

Polynomial Operations & Special Products — Problem 6 (Medium)

Expand and simplify the expression $(x + 4)^2 - (x - 4)^2$. Write your final answer as a simplified expression with no exponent higher than 1.

Formula / Rule Used: Combine the square-of-a-sum and square-of-a-difference patterns, then subtract.

Solution

Step 1: Expand $(x + 4)^2$ using $a^2 + 2ab + b^2$: $x^2 + 8x + 16$.

Step 2: Expand $(x - 4)^2$ using $a^2 - 2ab + b^2$: $x^2 - 8x + 16$.

Step 3: Set up the subtraction of the two expansions: $(x^2 + 8x + 16) - (x^2 - 8x + 16)$.

Step 4: Distribute the negative sign to every term in the second expansion: $x^2 + 8x + 16 - x^2 + 8x - 16$.

Step 5: Combine the x^2 terms: $x^2 - x^2 = 0$.

Step 6: Combine the constant terms: $16 - 16 = 0$.

Step 7: Combine the remaining x terms: $8x + 8x = 16x$.

Answer: $16x$

Tip

Expand each square completely first, then subtract term by term — don't try to shortcut the squaring step.

Trap

Forgetting to flip the sign of every term in the second expansion when subtracting is exactly what this problem is built to catch.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
0	Expanded $(x-4)^2$ with the wrong sign pattern as $x^2 + 8x + 16$, making both squares identical and canceling everything.
$16x + 32$	Flipped the sign of the x^2 and x terms when subtracting but forgot to also flip the sign of the constant 16.
$-16x$	Subtracted the expressions in reverse order, computing $(x-4)^2 - (x+4)^2$ instead of the order given.

Shortcut

Recognize this as a difference of squares of the two binomials: $[(x+4)+(x-4)] \times [(x+4)-(x-4)] = (2x)(8) = 16x$.

Desmos Tip

Type $y_1 = (x+4)^2 - (x-4)^2$ and $y_2 = 16x$ on separate lines — the graphs overlapping confirms the simplification is correct.

Polynomial Operations & Special Products — Problem 7 (Medium)

A tech startup in Athens, which employs 12 people, models its monthly revenue as $(5x^2 + 40x + 300)$ euros and its monthly cost as $(2x^2 + 25x + 180)$ euros, where x is the number of new clients signed that month. Write the startup's monthly profit as a simplified polynomial, then find the profit when $x = 10$.

Formula / Rule Used: Subtract polynomials to model profit, then substitute a value.

Solution

Step 1: The number of employees, 12, is not needed to find profit — ignore it.

Step 2: Set up profit as revenue minus cost: $(5x^2 + 40x + 300) - (2x^2 + 25x + 180)$.

Step 3: Distribute the negative sign to every term in the cost polynomial: $5x^2 + 40x + 300 - 2x^2 - 25x - 180$.

Step 4: Combine the x^2 terms: $5x^2 - 2x^2 = 3x^2$.

Step 5: Combine the x terms: $40x - 25x = 15x$.

Step 6: Combine the constant terms: $300 - 180 = 120$.

Step 7: The profit polynomial is $3x^2 + 15x + 120$; substitute $x = 10$: $3(10)^2 + 15(10) + 120$.

Step 8: Evaluate: $3(100) + 150 + 120 = 300 + 150 + 120 = 570$.

Answer: Profit = $3x^2 + 15x + 120$; at $x = 10$, profit = 570 euros

Tip

Cross out any detail that isn't part of the two polynomials, like the employee count here, before you start combining terms.

Trap

Trying to work the '12 employees' figure into the profit calculation is a trap — it has nothing to do with the model.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
330	Subtracted the constant terms in the wrong order ($180 - 300$) instead of ($300 - 180$), turning $+120$ into -120 .
1070	Flipped the sign of the x^2 and constant terms when subtracting but forgot to flip the sign of the x term.
582	Added the irrelevant detail of 12 employees onto the final profit value.

Shortcut

None needed here — fully simplifying the profit polynomial before substituting $x=10$ is faster than substituting into both original polynomials separately.

Desmos Tip



Type $y_1 = (5x^2 + 40x + 300) - (2x^2 + 25x + 180)$ and check the table of values at $x=10$ to confirm the profit is 570.

Polynomial Operations & Special Products — Problem 8 (Medium)

Multiply and simplify: $(-x + 9)(x + 9)$.

Formula / Rule Used: Difference of squares hidden behind a disguised negative sign.

Solution

Step 1: Notice the first factor is $-x + 9$, which can be rewritten as $-(x - 9)$.

Step 2: Rewrite the whole product using this form: $-(x - 9)(x + 9)$.

Step 3: Recognize $(x - 9)(x + 9)$ as a difference of squares: $x^2 - 81$.

Step 4: Apply the outside negative sign to that entire result: $-(x^2 - 81)$.

Step 5: Distribute the negative sign to both terms: $-x^2 + 81$.

Answer: $81 - x^2$ (equivalently $-x^2 + 81$)

Tip

When a binomial starts with a negative variable term, factor out the negative sign first to reveal the difference-of-squares pattern underneath.

Trap

Treating $(-x+9)(x+9)$ as an ordinary $(a+b)(a-b)$ difference of squares without adjusting for the sign flips the final answer's sign.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x^2 - 81$	Ignored the negative sign on the x term and applied the difference-of-squares pattern directly, flipping the overall sign.
$-x^2 - 81$	Distributed the outside negative sign onto the x^2 term but forgot to also flip the sign of 81.
$-x^2 - 9x + 81$	Found the Outer and Inner cross terms ($-9x$ and $+9x$) correctly but forgot they cancel, leaving an extra $-9x$.

Shortcut

Factor -1 out of the binomial with the negative variable term first, then it's an ordinary difference of squares to finish.

Desmos Tip

Type $y_1 = (-x+9)(x+9)$ and $y_2 = 81 - x^2$ in Desmos — overlapping graphs confirm the simplification.

18. Factoring Polynomials

Factoring Polynomials — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Factoring	Παραγοντοποίηση
Common factor	Κοινός παράγοντας
Difference of squares	Διαφορά τετραγώνων
Perfect square trinomial	Τέλειο τετράγωνο τριωνύμου
Grouping	Ομαδοποίηση όρων
Root / zero of a polynomial	Ρίζα πολυωνύμου

Factoring Polynomials — What It Is

What It Is

- Factoring takes an expanded expression, like $x^2 + 5x + 6$, and rewrites it as a product of factors.
- It's the exact reverse of multiplying (distributing) two expressions together.
- The SAT tests factoring because it's usually the fastest way to solve a quadratic or cubic equation.
- Recognizing the right pattern instantly (GCF, trinomial, difference of squares) saves real time on the test.
- Factoring also hides inside other topics, like simplifying rational expressions and reading a graph's roots.

Factoring Polynomials — Core Rules

Factor Out the GCF First

GCF $\times$ (remaining terms)

- Always start here, before trying any other method, on any polynomial.
- Find the largest number that divides every term's coefficient evenly.
- For variables, use the SMALLEST exponent of that variable appearing in every term.
- e.g. $6x^3 + 9x^2$ has GCF $3x^2$, so it becomes $3x^2(2x + 3)$.

Factoring $x^2 + bx + c$ (leading coefficient 1)

Find r, s where $r \times s = c$ and $r + s = b$, then $(x + r)(x + s)$

- List pairs of numbers that multiply together to give c .
- From that list, find the one pair that also adds up to give b .
- Write the answer as $(x + r)(x + s)$ using those exact two numbers.

- e.g. $x^2 + 5x + 6$: the pair 2 and 3 multiplies to 6 and adds to 5, so it's $(x+2)(x+3)$.

The AC Method for $ax^2 + bx + c$ (a is not 1)

Find two numbers that multiply to $a \times c$ and add to b

- Multiply the leading coefficient a by the constant c to get one target number.
- Find two numbers that multiply to that target and add to b , the middle coefficient.
- Rewrite the middle term bx as the sum of those two numbers, each times x .
- Group the resulting four terms into two pairs and factor each pair to finish.

Difference of Squares

$$a^2 - b^2 = (a - b)(a + b)$$

- Spot it when a subtraction has exactly two terms, and both are perfect squares.
- Take the square root of each term to identify what a and b are.
- Write the factored form as $(a - b)(a + b)$ right away.
- e.g. $x^2 - 49 = (x - 7)(x + 7)$ since $\sqrt{x^2} = x$ and $\sqrt{49} = 7$.

Perfect Square Trinomial

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

- Check whether the first term of the trinomial is a perfect square.
- Check whether the last term of the trinomial is also a perfect square.
- Check whether the middle term equals exactly twice the product of those two square roots.
- If all three checks pass, write the answer instantly as $(a \pm b)^2$, matching the sign.

Factoring by Grouping (four-term polynomials, including cubics)

Group the first two terms and the last two terms separately

- Split the polynomial into two pairs: the first two terms, and the last two terms.
- Factor out the GCF from each pair on its own.
- If both pairs now share the exact same binomial factor, pull that binomial out front.
- e.g. $x^3 + 3x^2 + 2x + 6$ groups into $x^2(x+3) + 2(x+3)$, which becomes $(x+3)(x^2+2)$.

Multiplicity of Roots

The exponent on a factor = that root's multiplicity

- Once a polynomial is fully factored, look at the exponent on each factor.
- That exponent tells you the multiplicity, meaning how many times that root repeats.
- Odd multiplicity (1, 3, 5...) means the graph crosses straight through the x -axis there.
- Even multiplicity (2, 4, 6...) means the graph just touches the x -axis and turns back.

How to Recognize This on the Test

- The instruction says 'factor completely' or 'write in factored form' next to a polynomial.
- An equation is set equal to zero and needs to be solved for x .
- The question asks about x -intercepts, zeros, or roots of a polynomial or its graph.
- You see a subtraction between two terms that are each perfect squares, like $x^2 - 16$.
- A three-term expression (trinomial) has a squared variable and clean, small-looking coefficients.

Common Mistakes — and the Fix

- Skipping the GCF check first, which makes every later factoring step messier than it needs to be.
- Stopping after one factoring step instead of checking whether a resulting factor can be factored again.
- Mixing up the signs when listing factor pairs of c , especially when c itself is negative.
- Trying to factor a sum of squares like $x^2 + 9$, which cannot be factored over real numbers.
- Dropping the $\pm$ sign or losing one whole factor when writing the final factored answer.

Factoring Polynomials — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Factoring Polynomials — Problem 1 (Easy)

A packaging company in Thessaloniki prints banners whose area, in square centimeters, is given by $12x^2 + 18x$, where x is the banner's width in centimeters. Factor this expression completely by pulling out the greatest common factor.

Formula / Rule Used: Factor out the GCF: GCF x (remaining terms).

Solution

Step 1: Look at the coefficients 12 and 18 and find their greatest common factor: 6.

Step 2: Look at the variable part of each term, x^2 and x — the smallest power present is x^1 .

Step 3: Combine both parts: the GCF of the whole expression is $6x$.

Step 4: Divide each term by $6x$: $12x^2 / 6x = 2x$, and $18x / 6x = 3$.

Step 5: Write the GCF outside a parenthesis and the leftover terms inside: $6x(2x + 3)$.

Step 6: Check by distributing: $6x(2x) + 6x(3) = 12x^2 + 18x$, which matches the original expression.

Answer: $6x(2x + 3)$

Tip

Before doing anything else to a polynomial, always check for a GCF in both the numbers and the variables.

Trap

It's tempting to pull out only the number 6 or only the variable x , instead of combining both into the full GCF, $6x$.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
$6(2x^2 + 3x)$	Pulled out only the numeric GCF, 6, and forgot to also factor out the shared variable factor, x .
$3x(4x + 6)$	Used $3x$ as the GCF instead of the largest possible GCF, $6x$, so the terms left inside can still be simplified further.
$2x(6x + 9)$	Divided by $2x$ instead of the full GCF $6x$, leaving a parenthesis $(6x + 9)$ that still shares a common factor of 3.

Shortcut

None faster here — pulling out the full GCF directly is already the quickest path; just double-check you grabbed the largest one.

Desmos Tip

Type $y_1 = 12x^2 + 18x$ and $y_2 = 6x(2x + 3)$ into Desmos — the two graphs overlap exactly, confirming the factored form is equivalent.

Factoring Polynomials — Problem 2 (Easy)

If $x^2 - 3x - 10 = 0$ and x is positive, what is the value of x ?

Formula / Rule Used: Factoring $x^2 + bx + c$: find r, s with $r \times s = c$ and $r + s = b$, then $(x+r)(x+s) = 0$.

Solution

Step 1: List pairs of numbers that multiply together to give -10 : $(1, -10)$, $(2, -5)$, $(5, -2)$, $(10, -1)$.

Step 2: Check which pair also adds up to give -3 : $2 + (-5) = -3$, so use 2 and -5 .

Step 3: Write the factored form using that pair: $(x + 2)(x - 5) = 0$.

Step 4: Set each factor equal to 0 separately: $x + 2 = 0$ or $x - 5 = 0$.

Step 5: Solve each small equation: $x = -2$ or $x = 5$.

Step 6: The question asks for the positive value of x , so the answer is $x = 5$.

Answer: $x = 5$

Tip

Once a quadratic is set equal to zero, factor it and set EACH factor equal to zero separately to find both solutions.

Trap

It's easy to correctly find both x -values but then report the wrong one when the question asks for just the positive (or negative) solution.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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-2	Correctly found both solutions, -2 and 5, but reported the negative one instead of the positive one the question asked for.
3	Mixed up the sign pattern, factoring as $(x-2)(x+5)$ instead of $(x+2)(x-5)$, which doesn't actually multiply back to the original expression.
10	Mistook the constant term, -10, itself for a solution instead of actually finding the correct factor pair.

Shortcut

None needed — factoring directly is already the fastest way to solve a clean quadratic like this one.

Desmos Tip

Type $y = x^2 - 3x - 10$ into Desmos and use the zero/root feature to see both x-intercepts, confirming $x = -2$ and $x = 5$.

Factoring Polynomials — Problem 3 (Easy)

What is the factored form of $25x^2 - 64$?

Formula / Rule Used: Difference of squares: $a^2 - b^2 = (a - b)(a + b)$.

Solution

Step 1: Check both terms: $25x^2$ is a perfect square, $(5x)^2$, and 64 is a perfect square, 8^2 .

Step 2: Since it's a subtraction of two perfect squares, this expression fits the difference of squares pattern.

Step 3: Take the square root of each term: $\sqrt{25x^2} = 5x$, and $\sqrt{64} = 8$.

Step 4: Write the factored form using one subtraction factor and one addition factor: $(5x - 8)(5x + 8)$.

Step 5: Check by distributing: $(5x)(5x) + (5x)(8) - (8)(5x) - (8)(8) = 25x^2 - 64$, which matches.

Answer: $(5x - 8)(5x + 8)$

Tip

Whenever you see exactly two terms connected by subtraction, check if both are perfect squares before trying anything else.

Trap

It's tempting to write $(5x - 8)^2$ instead of two separate binomial factors, mixing this up with a perfect square trinomial.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(5x - 8)^2$	Confused difference of squares with a perfect square trinomial pattern, which doesn't apply since there's no middle term here.
$(5x - 8)(5x - 8)$	Used the same subtraction sign in both factors instead of one subtraction



	and one addition.
$(5x - 4)(5x + 4)$	Miscalculated the square root of 64 as 4 instead of 8, mixing it up with the square root of 16.

Shortcut

None needed — spotting the difference-of-squares pattern lets you write the factored form in one step, with no expanding required.

Desmos Tip

Type $y1 = 25x^2 - 64$ and $y2 = (5x - 8)(5x + 8)$ into Desmos — the two graphs overlap perfectly, confirming the factoring is correct.

Factoring Polynomials — Problem 4 (Easy)

A framing company in Vienna cuts square glass tiles whose area, in square centimeters, is given by $9x^2 + 30x + 25$. Factor this expression to find an expression for the tile's side length.

Formula / Rule Used: Perfect square trinomial: $a^2 + 2ab + b^2 = (a + b)^2$.

Solution

Step 1: Check if the first term, $9x^2$, is a perfect square: yes, it equals $(3x)^2$.

Step 2: Check if the last term, 25, is a perfect square: yes, it equals 5^2 .

Step 3: Check if the middle term, $30x$, equals twice the product of $3x$ and 5 : $2(3x)(5) = 30x$, which matches.

Step 4: Since all three checks pass, this expression is a perfect square trinomial.

Step 5: Write the factored form as $(3x + 5)^2$, which means $(3x + 5)(3x + 5)$.

Step 6: Since area equals side times side, the tile's side length is represented by $3x + 5$.

Answer: $3x + 5$ (centimeters)

Tip

Before grinding through the full AC method, quickly check if the first and last terms are perfect squares — it might be a faster perfect-square-trinomial pattern.

Trap

It's tempting to check only the first and last terms and skip verifying the middle term, which can lead to a wrong $(a+b)^2$ guess.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$9x + 25$	Added the square roots of the first and last terms directly instead of recognizing the full binomial, $3x + 5$.
$3x - 5$	Used a subtraction sign instead of addition, without checking that the middle term, $30x$, is positive, which requires the $+$ version.



$(3x + 5)(3x - 5)$	Mistakenly treated the trinomial like a difference of squares, ignoring the actual positive middle term, $30x$.
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Shortcut

Recognizing the perfect-square pattern instantly gives the side length as $\sqrt{\text{first term}} + \sqrt{\text{last term}}$ — no need for the full AC method.

Desmos Tip

Type $y1 = 9x^2 + 30x + 25$ and $y2 = (3x + 5)^2$ into Desmos to confirm the two graphs overlap exactly.

Factoring Polynomials — Problem 5 (Medium)

What is the complete factored form of $x^4 - 81$?

Formula / Rule Used: Difference of squares, applied twice (a disguised form).

Solution

Step 1: Recognize that $x^4 - 81$ is itself a difference of squares: $(x^2)^2 - 9^2$.

Step 2: Apply the difference of squares pattern once: $x^4 - 81 = (x^2 - 9)(x^2 + 9)$.

Step 3: Look at each new factor separately to check whether either can be factored again.

Step 4: The factor $x^2 + 9$ is a SUM of squares, which cannot be factored over the real numbers.

Step 5: The factor $x^2 - 9$ is itself another difference of squares: $(x - 3)(x + 3)$.

Step 6: Replace $x^2 - 9$ with its factored form, keeping $x^2 + 9$ exactly as it is.

Step 7: The fully factored form is $(x - 3)(x + 3)(x^2 + 9)$.

Answer: $(x - 3)(x + 3)(x^2 + 9)$

Tip

After one round of difference of squares, always check whether either new factor is itself a difference of squares.

Trap

It's tempting to also try to factor $x^2 + 9$ — but a SUM of squares never factors over the real numbers, so stop there.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(x - 3)(x + 3)(x - 3)(x + 3)$	Mistakenly treated $x^2 + 9$ as another difference of squares and force-factored it, even though a sum of squares can't be factored.
$(x^2 - 9)(x^2 + 9)$	Stopped after the first factoring step without checking whether the factor $x^2 - 9$ could be factored again.
$(x - 9)(x + 9)$	Incorrectly took the square root of x^4 as x instead of x^2 , misapplying the pattern to the wrong power.

**Shortcut**

None faster here — the two-step process of factoring, then checking each new piece again, is the reliable way to fully factor a disguised form.

Desmos Tip

Type $y1 = x^4 - 81$ and $y2 = (x-3)(x+3)(x^2+9)$ into Desmos to confirm the two graphs match exactly.

Factoring Polynomials — Problem 6 (Medium)

The expression $x^2 + bx + 24$ factors as $(x + 4)(x + c)$, where b and c are positive integers. What is the value of b ?

Formula / Rule Used: Reverse trinomial factoring: expand $(x+4)(x+c)$ and match coefficients.

Solution

Step 1: Multiply out $(x + 4)(x + c)$ using FOIL: $x^2 + cx + 4x + 4c$.

Step 2: Combine the like x -terms: $x^2 + (c + 4)x + 4c$.

Step 3: Compare this to the original expression, $x^2 + bx + 24$ — the constant terms must be equal, so $4c = 24$.

Step 4: Divide both sides by 4 to solve for c : $c = 6$.

Step 5: Now that c is known, the coefficient of x must also match: $b = c + 4$.

Step 6: Substitute $c = 6$: $b = 6 + 4 = 10$.

Answer: $b = 10$

Tip

When you're given a partial factored form, expand it and match coefficients to the original expression one at a time.

Trap

It's tempting to correctly solve for c and stop there, forgetting the question actually asks for b , not c .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
6	Correctly solved for $c = 6$ but reported it instead of b , the value the question actually asked for.
28	Multiplied 4 and 6 together by mistake instead of adding them, computing 4×6 instead of $6 + 4$.
2	Made a division slip solving $4c = 24$, getting $c = 2$ instead of $c = 6$, which then threw off the final value of b .

Shortcut

Since one factor, $(x + 4)$, is already known, just divide the constant term by 4 to find the other factor's number instantly.

**Desmos Tip**

Type $y_1 = x^2 + 10x + 24$ and $y_2 = (x+4)(x+6)$ into Desmos to confirm the graphs overlap, checking your value of b .

Factoring Polynomials — Problem 7 (Medium)

A sheet-metal company in Seoul cuts rectangular plates whose area, in square inches, is given by $3x^3 - 48x$. Factor this expression completely.

Formula / Rule Used: Factor out the GCF first, then check the leftover factor for a difference of squares.

Solution

Step 1: Look for a GCF among all terms: both $3x^3$ and $48x$ share a common factor of $3x$.

Step 2: Divide each term by $3x$: $3x^3 / 3x = x^2$, and $48x / 3x = 16$.

Step 3: Write the GCF outside a parenthesis: $3x(x^2 - 16)$.

Step 4: Look at the remaining factor, $x^2 - 16$, and notice it is a subtraction of two perfect squares.

Step 5: Apply the difference of squares pattern: $x^2 - 16 = (x - 4)(x + 4)$.

Step 6: Replace $x^2 - 16$ inside the expression with its factored form.

Step 7: The fully factored expression is $3x(x - 4)(x + 4)$.

Answer: $3x(x - 4)(x + 4)$

Tip

After factoring out a GCF, always look at what's left inside the parentheses to see if it factors again.

Trap

It's tempting to stop right after pulling out the GCF, without noticing the leftover binomial can still be factored as a difference of squares.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$3x(x^2 - 16)$	Correctly factored out the GCF, $3x$, but stopped without noticing $x^2 - 16$ is a difference of squares that factors further.
$3x(x - 16)(x + 1)$	Tried to factor $x^2 - 16$ as if it were a regular trinomial instead of recognizing the difference of squares pattern.
$x(3x^2 - 48)$	Pulled out only x instead of the full GCF, $3x$, leaving a factor that still shares a common factor of 3.

Shortcut

None faster here — factoring out the GCF first, then checking the leftover factor for a pattern, is already the quickest reliable route.

Desmos Tip

Type $y_1 = 3x^3 - 48x$ and $y_2 = 3x(x-4)(x+4)$ into Desmos to confirm the two graphs overlap exactly.

Factoring Polynomials — Problem 8 (Medium)

A polynomial function is given by $h(x) = (x - k)^2(x + 3)$, and h has a double root at $x = 5$. How many distinct x -values are there where the graph of h actually crosses the x -axis, rather than just touching it?

Formula / Rule Used: Multiplicity of roots: odd multiplicity crosses, even multiplicity touches.

Solution

Step 1: A double root means that factor's exponent is 2, so it appears in the function as $(x - k)^2$.

Step 2: Since the double root occurs at $x = 5$, this tells us $k = 5$, giving the factor $(x - 5)^2$.

Step 3: So the fully identified function is $h(x) = (x - 5)^2(x + 3)$.

Step 4: The factor $(x - 5)^2$ has an even exponent, 2, so at $x = 5$ the graph only touches the x -axis and turns back.

Step 5: The factor $(x + 3)$ has an exponent of 1, which is odd, so at $x = -3$ the graph crosses straight through the x -axis.

Step 6: Counting only the x -values where the graph actually crosses through, there is exactly one such x -value.

Answer: 1 (the graph only crosses the x -axis at $x = -3$)

Tip

Find the missing constant first, then use the exponent on each factor to decide whether the graph crosses or just touches at each root.

Trap

It's tempting to count both roots as crossing points, forgetting that an even multiplicity means the graph only touches and bounces back.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2	Counted both $x = 5$ and $x = -3$ as crossing points, forgetting that the double root at $x = 5$ only touches the x -axis instead of crossing it.
0	Mistakenly believed a double root means the graph never touches the x -axis at all, rather than touching without crossing.
5	Correctly found $k = 5$ but reported the value of k instead of answering the actual question about how many crossing points exist.

Shortcut

Once you know each factor's exponent, you can immediately tell odd from even without sketching or expanding the whole function.

Desmos Tip

Type $y = (x-5)^2(x+3)$ into Desmos and zoom into both x -intercepts — the curve bounces off $x=5$ but cuts straight through at $x=-3$.



19. Dividing Polynomials & Rational Expressions

Dividing Polynomials & Rational Expressions — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Polynomial long division	Ευκλείδεια διαίρεση πολυωνύμων
Quotient / remainder	Πηλίκο / υπόλοιπο
Rational expression	Ρητή παράσταση
Excluded value (undefined)	Εξαιρούμενη τιμή (μη ορισμένη)
Simplify (reduce)	Απλοποίηση
Remainder theorem	Θεώρημα υπολοίπου

Dividing Polynomials & Rational Expressions — What It Is

What It Is

- Dividing polynomials means splitting one polynomial expression by another, much like long division with numbers.
- A rational expression is simply a fraction where the top, the bottom, or both are polynomials.
- This skill combines factoring with fraction rules, so both must be solid before you start.
- Any x-value that would make a denominator zero must always be excluded from the answer.
- These ideas show up together: simplifying, combining, dividing, and solving equations built from rational expressions.

Dividing Polynomials & Rational Expressions — Core Rules

Dividing a Polynomial by a Monomial

$$(a+b+c)/d = a/d + b/d + c/d$$

- Split the numerator into separate fractions, one for each term, all over the same denominator.
- Simplify each small fraction on its own by canceling common factors.
- Example: $(6x^2 + 9x)/3x$ splits into $6x^2/3x + 9x/3x$, which simplifies to $2x + 3$.
- This shortcut only works when the divisor is a single term, not a full polynomial.

Polynomial Long Division

- Divide the leading term of the dividend by the leading term of the divisor to start the quotient.
- Multiply that new quotient term by the entire divisor, then subtract the result from the dividend.
- Bring down the next term and repeat the divide-multiply-subtract cycle.
- Stop once the remainder's degree is smaller than the divisor's degree.

Simplifying Rational Expressions

- Factor the numerator and the denominator completely before doing anything else.
- Cancel any whole factor that appears in both the top and the bottom.
- The simplified expression still excludes any x-value that made the ORIGINAL denominator zero.
- Example: $(x^2-9)/(x+3)$ factors to $[(x-3)(x+3)]/(x+3)$, which simplifies to $x-3$, but x cannot equal -3 .

Multiplying and Dividing Rational Expressions

$$(a/b) \times (c/d) = ac/bd$$

- Factor every numerator and denominator first, before multiplying anything out.
- Cancel matching factors across the whole product, not just inside one single fraction.
- To divide by a fraction, flip that second fraction upside down and multiply instead.
- Example: dividing by $(x+1)/(x-2)$ is the same as multiplying by $(x-2)/(x+1)$.

Adding and Subtracting Rational Expressions

$$a/b + c/d = (ad + bc)/bd$$

- Find the least common denominator (LCD) shared by all the fractions in the problem.
- Rewrite every fraction as an equivalent fraction using that same LCD on the bottom.
- Combine the numerators over the one shared denominator, then simplify if possible.
- Factoring each denominator first often reveals a smaller, simpler LCD than multiplying them all together.

Solving Rational Equations

- Multiply every single term in the equation by the LCD to clear all the denominators at once.
- Solve the resulting equation using normal algebra, since it's no longer a fraction equation.
- Plug each solution back into the ORIGINAL equation's denominators, not the cleared version.
- Throw out any solution that makes an original denominator equal zero — it's extraneous, not a real answer.

Complex Fractions

- A complex fraction has a smaller fraction stacked inside its numerator, denominator, or both.
- One method: turn the top into one single fraction, turn the bottom into one single fraction, then divide.
- Another method: multiply the entire complex fraction by the LCD of every small fraction hiding inside it.
- Either method should leave you with one clean, simplified rational expression at the end.

How to Recognize This on the Test

- A fraction bar with a variable expression on the top, the bottom, or both signals a rational

expression.

- The instruction 'simplify the expression' next to a fraction full of polynomials points straight to this skill.
- One polynomial written divided by another, like (x^2+5x+6) over $(x+2)$, calls for long division.
- An equation with x sitting inside a denominator means you must clear denominators and check for extraneous roots.
- A fraction stacked inside another fraction's numerator or denominator is a complex fraction problem.

Common Mistakes — and the Fix

- Canceling a single term instead of a whole factor, like crossing out the x in $(x+3)/x$.
- Simplifying a rational expression correctly but forgetting to state the x -value it excludes.
- Adding or subtracting rational expressions by combining numerators before finding a common denominator.
- Accepting every algebraic solution to a rational equation without checking it against the original denominators.
- Forgetting to flip the second fraction upside down when dividing rational expressions.

Dividing Polynomials & Rational Expressions — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Dividing Polynomials & Rational Expressions — Problem 1 (Easy)

A rectangular solar panel array has an area, in square meters, represented by the expression $8x^3 + 12x^2 - 4x$. If the width of the array is $4x$ meters, what is an expression for its length, in meters?

Formula / Rule Used: Dividing a polynomial by a monomial: $(a+b+c)/d = a/d + b/d + c/d$.

Solution

Step 1: Length = Area divided by width, so set up $(8x^3 + 12x^2 - 4x) / (4x)$.

Step 2: Split the numerator into three separate fractions over the same denominator: $8x^3/4x + 12x^2/4x - 4x/4x$.

Step 3: Simplify the first piece: $8x^3/4x = 2x^2$.

Step 4: Simplify the second piece: $12x^2/4x = 3x$.

Step 5: Simplify the third piece: $-4x/4x = -1$.

Step 6: Combine the simplified pieces to get the length: $2x^2 + 3x - 1$.

Answer: $2x^2 + 3x - 1$ meters

Tip



When dividing a polynomial by a single monomial, split it into separate fractions and simplify each piece one at a time.

Trap

It's easy to divide only the first term by $4x$ and forget the other two terms need the same treatment.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$2x^2 + 3x$	Divided the first two terms correctly but forgot the last term entirely, dropping it instead of turning it into -1 .
$2x^2 + 3x + 1$	Divided the last term correctly in size but lost the negative sign, writing $+1$ instead of -1 .
$8x^2 + 12x - 4$	Divided every term by x only, forgetting to also divide each coefficient by 4 .

Shortcut

None faster — splitting into three simple fractions and dividing each term is already the quickest method here.

Desmos Tip

Type $y_1 = (8x^3 + 12x^2 - 4x)/(4x)$ and $y_2 = 2x^2 + 3x - 1$ in Desmos; identical graphs confirm the division is correct.

Dividing Polynomials & Rational Expressions — Problem 2 (Easy)

Divide the polynomial $x^2 - 5x - 24$ by $x - 8$.

Formula / Rule Used: Polynomial long division: divide leading terms, multiply, subtract, repeat.

Solution

Step 1: Divide the leading term of the dividend by the leading term of the divisor: $x^2 \div x = x$.

Step 2: Multiply this quotient term by the whole divisor: $x(x-8) = x^2 - 8x$.

Step 3: Subtract this from the dividend: $(x^2 - 5x - 24) - (x^2 - 8x) = 3x - 24$.

Step 4: Divide the new leading term by x : $3x \div x = 3$.

Step 5: Multiply this quotient term by the divisor: $3(x-8) = 3x - 24$.

Step 6: Subtract again: $(3x - 24) - (3x - 24) = 0$, so there is no remainder.

Step 7: Combine the two quotient terms found along the way: $x + 3$.

Answer: $x + 3$

Tip

Treat polynomial long division exactly like numeric long division: divide, multiply, subtract, then bring down and repeat.

Trap



Forgetting to distribute the subtraction across every term of the divisor's product can flip a sign and derail the whole problem.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x - 3$	Made a sign error while subtracting, treating $-8x$ as $+8x$ when combining like terms.
$x + 3$, remainder -24	Stopped after the first subtraction and reported the leftover $3x - 24$ as a remainder instead of continuing the division.
$x + 8$	Copied the divisor's constant term directly into the answer instead of carrying out the division.

Shortcut

Since $x-8$ divides evenly, you could also look for factor pairs of -24 that sum to -5 and factor directly: $(x-8)(x+3)$.

Desmos Tip

Type $y_1 = (x^2 - 5x - 24)/(x - 8)$ and $y_2 = x + 3$ in Desmos; the graphs overlap everywhere except a hole at $x = 8$, confirming the division.

Dividing Polynomials & Rational Expressions — Problem 3 (Easy)

Simplify the expression $(x^2 - 25)/(3x + 15)$ fully, and state any value of x that must be excluded.

Formula / Rule Used: Simplifying rational expressions: factor completely, then cancel common factors.

Solution

Step 1: Factor the numerator as a difference of squares: $x^2 - 25 = (x-5)(x+5)$.

Step 2: Factor the denominator by pulling out the common factor 3: $3x + 15 = 3(x+5)$.

Step 3: Rewrite the expression using both factored forms: $[(x-5)(x+5)] / [3(x+5)]$.

Step 4: Cancel the common factor $(x+5)$ from the numerator and denominator.

Step 5: The simplified expression is $(x-5)/3$.

Step 6: Since $(x+5)$ was canceled, x cannot equal -5 , because that made the original denominator zero.

Answer: $(x-5)/3$, where $x \neq -5$

Tip

Always factor the top and bottom completely before canceling anything — never cancel pieces of a sum.

Trap

It's tempting to cancel x^2 and 25 straight against $3x$ and 15, but you can only cancel whole matching factors.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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$(x-5)/3$, no restriction stated	Simplified correctly but forgot to note that x cannot equal -5 , since that value zeroed the original denominator.
$x - 5$	Canceled the leftover 3 in the denominator along with the $(x+5)$ factor instead of leaving it in place.
$(x-5)/(3x+5)$	Canceled just one x from inside $(x+5)$ instead of canceling the entire matching factor $(x+5)$.

Shortcut

Recognizing x^2-25 as a difference of squares immediately, without expanding anything, saves factoring time.

Desmos Tip

Type $y1=(x^2-25)/(3x+15)$ and $y2=(x-5)/3$ in Desmos; the graphs match everywhere except for a hole at $x=-5$.

Dividing Polynomials & Rational Expressions — Problem 4 (Easy)

Simplify $(x^2 - 9)/(x+1)$ divided by $(x-3)/(2x+2)$, and write the result in simplest form.

Formula / Rule Used: Dividing rational expressions: multiply by the reciprocal of the second fraction.

Solution

Step 1: Rewrite the division as multiplication by the reciprocal: $[(x^2-9)/(x+1)] \times [(2x+2)/(x-3)]$.

Step 2: Factor every part first: $x^2 - 9 = (x-3)(x+3)$, and $2x+2 = 2(x+1)$.

Step 3: Rewrite the expression showing all factors: $[(x-3)(x+3)/(x+1)] \times [2(x+1)/(x-3)]$.

Step 4: Cancel the common factor $(x+1)$ from one numerator and the other denominator.

Step 5: Cancel the common factor $(x-3)$ from the other numerator and denominator.

Step 6: What remains is $(x+3) \times 2$, which multiplies out to $2x + 6$.

Answer: $2x + 6$, where $x \neq -1$ and $x \neq 3$

Tip

Before multiplying or dividing rational expressions, factor everything first — canceling is much easier after factoring.

Trap

Forgetting to flip the second fraction upside down before multiplying is the most common way to miss this type of problem.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(x+3)/2$	Divided by the leftover factor of 2 instead of multiplying by it after canceling.
$x + 3$	Canceled $(x+1)$ and $(x-3)$ correctly but dropped the leftover factor of 2 instead of multiplying by it.



$2x - 6$	Made a sign error after canceling, treating the surviving factor as $(x-3)$ instead of $(x+3)$.
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Shortcut

Factoring first and canceling matching factors before multiplying anything out avoids working with large expanded expressions.

Desmos Tip

Type $y1=((x^2-9)/(x+1))/((x-3)/(2x+2))$ and $y2=2x+6$ in Desmos; matching graphs (minus two holes) confirm the simplification.

Dividing Polynomials & Rational Expressions — Problem 5 (Medium)

Write $2/(x+3) + 1/(x^2-9)$ as a single fraction in simplest form.

Formula / Rule Used: Adding rational expressions: factor first, find the LCD, then combine numerators.

Solution

Step 1: Factor the second denominator: $x^2 - 9 = (x-3)(x+3)$.

Step 2: Compare the two denominators, $(x+3)$ and $(x-3)(x+3)$ — the LCD is $(x-3)(x+3)$.

Step 3: Rewrite the first fraction using the LCD: $2/(x+3) = [2(x-3)] / [(x-3)(x+3)]$.

Step 4: Multiply out that new numerator: $2(x-3) = 2x - 6$.

Step 5: The second fraction already uses the LCD, so add the numerators: $(2x-6) + 1 = 2x - 5$.

Step 6: Combine everything over the shared denominator: $(2x-5) / [(x-3)(x+3)]$.

Step 7: Multiply the denominator back out if needed: $(2x-5)/(x^2-9)$.

Answer: $(2x-5)/(x^2-9)$, where $x \neq 3$ and $x \neq -3$

Tip

Whenever a denominator can be factored, factor it first — it usually reveals the true LCD hiding underneath.

Trap

Treating $(x+3)$ and (x^2-9) as unrelated and multiplying them together as the LCD creates unnecessary extra factors.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$3/(x^2-9)$	Added the numerators 2 and 1 directly to get 3 without first rewriting $2/(x+3)$ over the LCD.
$(2x-5)/[(x+3)^2(x-3)]$	Multiplied the two original denominators together as if they shared no common factor, creating an oversized LCD.
$(2x-6)/(x^2-9)$	Rewrote the first fraction correctly but forgot to add the second numerator's 1 to the top.

**Shortcut**

None faster — spotting the factored relationship between the two denominators is itself the shortcut here.

Desmos Tip

Type $y1=2/(x+3)+1/(x^2-9)$ and $y2=(2x-5)/(x^2-9)$ in Desmos; identical graphs confirm the combined fraction is correct.

Dividing Polynomials & Rational Expressions — Problem 6 (Medium)

Solve for x: $1/(x-3) + 1/(x+3) = 6/(x^2-9)$.

Formula / Rule Used: Solving rational equations: multiply every term by the LCD, solve, then check for extraneous solutions.

Solution

Step 1: Factor the right-side denominator: $x^2 - 9 = (x-3)(x+3)$, which is also the LCD of all three terms.

Step 2: Multiply every term in the equation by the LCD, $(x-3)(x+3)$.

Step 3: The first term becomes $(x+3)$, since the $(x-3)$ factors cancel.

Step 4: The second term becomes $(x-3)$, since the $(x+3)$ factors cancel.

Step 5: The right side becomes just 6, since its whole denominator cancels.

Step 6: The equation is now: $(x+3) + (x-3) = 6$.

Step 7: Combine like terms on the left: $2x = 6$, so $x = 3$.

Step 8: Check this solution against the ORIGINAL denominators: $x=3$ makes $(x-3)$ equal zero, which is not allowed.

Step 9: Since the only algebraic solution is extraneous, this equation has no valid solution.

Answer: No solution ($x = 3$ is extraneous)

Tip

After clearing denominators and solving, always plug your answer back into the ORIGINAL equation's denominators before accepting it.

Trap

This equation is built so the only algebraic solution is exactly the value that breaks a denominator — skipping the check misses it.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 3$	Solved the cleared equation correctly but never checked the result against the original denominators.
$x = 6$	Divided incorrectly, treating $2x=6$ as $x=6-2$ instead of dividing both sides by 2.
$x = 0$	Mishandled the constants when combining $(x+3)+(x-3)$, incorrectly canceling the x terms along with the 3's.



Shortcut

Noticing that x^2-9 factors into the same two denominators already in the equation instantly reveals the LCD.

Desmos Tip

Graph $y_1=1/(x-3)+1/(x+3)$ and $y_2=6/(x^2-9)$ in Desmos; the graphs never actually touch, confirming there is no real solution.

Dividing Polynomials & Rational Expressions — Problem 7 (Medium)

Simplify the complex fraction $(1 - 1/x) / (1 - 1/x^2)$.

Formula / Rule Used: Complex fractions: combine numerator and denominator into single fractions, then divide.

Solution

Step 1: Rewrite the top as a single fraction: $1 - 1/x = (x-1)/x$.

Step 2: Rewrite the bottom as a single fraction: $1 - 1/x^2 = (x^2-1)/x^2$.

Step 3: Factor the new bottom's numerator: $x^2 - 1 = (x-1)(x+1)$.

Step 4: The complex fraction is now $[(x-1)/x]$ divided by $[(x-1)(x+1)/x^2]$.

Step 5: Dividing by a fraction means multiplying by its reciprocal: $(x-1)/x$ times $x^2/[(x-1)(x+1)]$.

Step 6: Cancel the common factor $(x-1)$ from the numerator and denominator.

Step 7: Cancel one factor of x , leaving a single x in the numerator.

Step 8: The simplified result is $x/(x+1)$.

Answer: $x/(x+1)$, where $x \neq 0, 1, -1$

Tip

Turn the top and the bottom of a complex fraction into single fractions first, then divide like any other fraction division.

Trap

Multiplying everything by x^2 too early without tracking each piece separately can easily drop or duplicate a factor.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x - 1$	Canceled the leftover x^2 from the denominator completely instead of leaving one factor of x in the answer.
$(x-1)/(x+1)$	Forgot to bring down the extra factor of x from x^2 after canceling one x , dropping it from the final numerator.
$x/(x-1)$	Mixed up which factor, $(x-1)$ or $(x+1)$, actually survived the cancellation step.

**Shortcut**

Multiplying the whole complex fraction, top and bottom, by x^2 (the LCD of all inner fractions) skips the two-step rewriting.

Desmos Tip

Type $y1=(1-1/x)/(1-1/x^2)$ and $y2=x/(x+1)$ in Desmos; matching graphs (minus a few holes) confirm the simplification.

Dividing Polynomials & Rational Expressions — Problem 8 (Medium)

The expression $(x^2 + kx - 12)/(x-3)$ is a polynomial with no remainder, for some constant k . What is the value of k ?

Formula / Rule Used: Polynomial long division with an unknown coefficient; set the remainder equal to 0.

Solution

Step 1: Begin dividing $x^2 + kx - 12$ by $x - 3$: divide the leading terms, $x^2 \div x = x$.

Step 2: Multiply this quotient term by the divisor: $x(x-3) = x^2 - 3x$.

Step 3: Subtract from the original expression: $(x^2+kx-12) - (x^2-3x) = (k+3)x - 12$.

Step 4: Divide the new leading term by x : $(k+3)x \div x = (k+3)$.

Step 5: Multiply this quotient term by the divisor: $(k+3)(x-3) = (k+3)x - 3(k+3)$.

Step 6: Subtract to find the remainder: $-12 - [-3(k+3)] = -12 + 3k + 9 = 3k - 3$.

Step 7: Since there is no remainder, set this leftover expression equal to 0: $3k - 3 = 0$.

Step 8: Add 3 to both sides, then divide by 3: $k = 1$.

Answer: $k = 1$

Tip

When a division must leave no remainder, set the leftover expression from the final subtraction step equal to 0 and solve.

Trap

It's easy to stop the long division too early and solve using the wrong leftover expression before it's fully simplified.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = -3$	Set the coefficient $(k+3)$ itself equal to 0 instead of finishing the division to find the true remainder.
$k = 3$	Made a sign error simplifying $-12 + 3(k+3)$, writing $-12 + 3k - 9$ instead of $-12 + 3k + 9$.
$k = -1$	Made a sign error in the last step, writing $3k = -3$ instead of $3k = 3$, before dividing by 3.



Shortcut

Since $x-3$ must divide evenly, plugging $x=3$ directly into the numerator and setting it equal to 0 finds k faster than full long division.

Desmos Tip

Type $y=(x^2+x-12)/(x-3)$ with $k=1$ substituted into Desmos; the graph shows a straight line with a single hole at $x=3$, confirming no remainder.

20. Solving Quadratic Equations & the Discriminant

Solving Quadratic Equations & the Discriminant — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Quadratic equation	Δευτεροβάθμια εξίσωση
Discriminant	Διακρίνουσα
Quadratic formula	Τύπος επίλυσης δευτεροβάθμιας εξίσωσης
Completing the square	Συμπλήρωση τετραγώνου
Real / complex roots	Πραγματικές / μιγαδικές ρίζες
Sum and product of roots	Άθροισμα και γινόμενο ριζών

Solving Quadratic Equations & the Discriminant — What It Is

What It Is

- A quadratic equation is any equation where the highest power of x is x^2 , like $ax^2 + bx + c = 0$.
- Solving it means finding every x -value that makes the equation true.
- Most quadratics have two solutions, but some have exactly one, and some have none at all in real numbers.
- The SAT tests four different solving methods, plus a shortcut, the discriminant, that predicts how many solutions exist.
- These skills reappear constantly, since many word problems and later topics ultimately reduce to solving a quadratic.

Solving Quadratic Equations & the Discriminant — Core Rules

Solving by Factoring

$ax^2 + bx + c = 0 \rightarrow$ factor $\rightarrow$ set each factor to 0

- This method only works once the equation is set equal to 0, so rearrange first if it isn't.
- Factor the left side into two binomials, the same way you would in a factoring problem.
- Use the zero product property: if two things multiply to give 0, at least one of them must be 0.
- e.g. $(x-2)(x+5) = 0$ means $x-2 = 0$ or $x+5 = 0$, so $x = 2$ or $x = -5$.

Solving by Taking Square Roots

If $x^2 = k$, then $x = \pm\sqrt{k}$

- Use this when the equation is already isolated as something squared equal to a plain number.
- Taking the square root of both sides always creates TWO possible answers, positive and negative.

- e.g. if $x^2 = 49$, then $x = \pm\sqrt{49} = \pm 7$, giving $x = 7$ or $x = -7$.
- If k turns out negative, there is no real solution, since a negative number has no real square root.

Solving by Completing the Square

$$x^2 + bx \rightarrow (x + b/2)^2 - (b/2)^2$$

- Move the constant term by itself to the other side of the equation first.
- Take half of the x -coefficient, then square that result.
- Add that squared value to BOTH sides of the equation to keep it balanced.
- The left side is now a perfect square trinomial, so rewrite it as one squared binomial and take square roots.

The Quadratic Formula

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

- This formula solves ANY quadratic equation, even ones that don't factor nicely.
- First identify a , b , and c carefully from the standard form $ax^2 + bx + c = 0$.
- Plug those three numbers into the formula one at a time, watching every negative sign closely.
- e.g. for $x^2 + 3x - 4 = 0$: $a=1$, $b=3$, $c=-4$, so $x = (-3 \pm \sqrt{9+16}) / 2$.

The Discriminant

$$D = b^2 - 4ac$$

- The discriminant is just the part of the quadratic formula that sits under the square root sign.
- It tells you how many real solutions exist before you ever solve the full equation.
- If D is positive, there are two different real solutions; if $D = 0$, there is exactly one repeated real solution.
- If D is negative, there are no real solutions, since you can't take the square root of a negative number.

Choosing the Best Method

- If the numbers look small and clean, try factoring first, since it's usually the fastest route.
- If the equation looks like $x^2 = k$ with no middle x -term at all, take square roots directly.
- If factoring looks messy or you're not sure it factors, the quadratic formula always works as a backup.
- Completing the square is mostly used for finding a vertex, but it can also solve any quadratic.

How to Recognize This on the Test

- The equation contains an x^2 term and is set equal to a number, most often 0.
- The problem says 'solve for x ,' 'find the roots,' or 'how many real solutions' does an equation have.
- You're asked how many times a parabola's graph crosses the x -axis, without being asked for the actual points.

- The word 'discriminant' appears directly, or you're asked to find the number of solutions without solving.
- A word problem's setup naturally builds an equation with a squared variable, like area or projectile height.

Common Mistakes — and the Fix

- Forgetting the $\pm$ sign when taking a square root, which throws away one of the two valid solutions.
- Plugging numbers into the discriminant or quadratic formula before rearranging the equation into standard form.
- Making a sign error when substituting a negative b or c into the quadratic formula.
- Stopping after solving only ONE factor equal to zero instead of setting both factors equal to zero.
- Misreading what the discriminant means, like calling $D = 0$ 'no solutions' instead of 'exactly one solution.'

Solving Quadratic Equations & the Discriminant — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Solving Quadratic Equations & the Discriminant — Problem 1 (Easy)

A garden design company in Rotterdam is building a rectangular flower bed whose length, in meters, satisfies $x^2 - 5x - 24 = 0$. If the length x must be a positive number, what is the length of the flower bed, in meters?

Formula / Rule Used: Solving by factoring: find r, s where $r \times s = c$ and $r + s = b$, then $(x+r)(x+s) = 0$.

Solution

Step 1: The equation is already set equal to 0, so you can go straight to factoring.

Step 2: List pairs of numbers that multiply together to give -24: $(-8, 3)$, $(8, -3)$, $(-6, 4)$, $(6, -4)$, $(-12, 2)$, $(12, -2)$.

Step 3: Find the pair that also adds up to give -5: -8 and 3, since $-8 + 3 = -5$.

Step 4: Write the factored form using that pair: $(x - 8)(x + 3) = 0$.

Step 5: Set each factor equal to 0 separately: $x - 8 = 0$ or $x + 3 = 0$.

Step 6: Solve each small equation: $x = 8$ or $x = -3$.

Step 7: Since the length must be positive, the length is $x = 8$ meters.

Answer: 8 meters

Tip

Whenever an equation is already set equal to 0 with clean integer coefficients, try factoring before anything else.

Trap



It's tempting to report both solutions or grab the wrong one, forgetting the problem restricts x to a positive length.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-3 meters	Correctly found both solutions, 8 and -3, but reported the negative one, which can't be a real length.
5 meters	Mistook the coefficient of the middle term, -5, for the actual answer instead of solving the equation.
24 meters	Mistook the constant term, -24, for the length instead of finding the correct factor pair.

Shortcut

None needed — factoring directly is already the fastest way to solve a clean quadratic like this one.

Desmos Tip

Type $y = x^2 - 5x - 24$ into Desmos and use the zero/root feature to see both x -intercepts, confirming $x=8$ and $x=-3$.

Solving Quadratic Equations & the Discriminant — Problem 2 (Easy)

Solve for all real values of x : $3x^2 - 27 = 0$.

Formula / Rule Used: Solving by taking square roots: if $x^2 = k$, then $x = \pm\sqrt{k}$.

Solution

Step 1: Add 27 to both sides of the equation: $3x^2 = 27$.

Step 2: Divide both sides by 3 to isolate x^2 by itself: $x^2 = 9$.

Step 3: Take the square root of both sides, remembering both signs: $x = \pm\sqrt{9}$.

Step 4: Simplify the square root: $\sqrt{9} = 3$.

Step 5: So the two solutions are $x = 3$ and $x = -3$.

Answer: $x = 3$ or $x = -3$

Tip

Before taking a square root, always fully isolate the squared term by itself on one side first.

Trap

It's easy to take the square root too early, before dividing by the 3 in front of x^2 , which puts the wrong number under the root.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 9$	Correctly isolated $x^2 = 9$ but forgot to actually take the square root to solve for x .



$x = 3$	Took the square root correctly but dropped the negative solution, forgetting the $\pm$ sign.
$x = \pm\sqrt{27}$	Took the square root before dividing both sides by 3, using the wrong number under the root.

Shortcut

None needed — isolating x^2 and taking the square root is already the fastest method for an equation with no middle x -term.

Desmos Tip

Type $y = 3x^2 - 27$ into Desmos and check where the graph crosses the x -axis to confirm $x=3$ and $x=-3$.

Solving Quadratic Equations & the Discriminant — Problem 3 (Easy)

What is the positive solution to $x^2 + 4x - 7 = 0$? Round your answer to the nearest tenth.

Formula / Rule Used: The quadratic formula: $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$.

Solution

Step 1: Identify a , b , and c from the standard form: $a = 1$, $b = 4$, $c = -7$.

Step 2: Plug these into the quadratic formula: $x = (-4 \pm \sqrt{4^2 - 4(1)(-7)}) / 2(1)$.

Step 3: Compute inside the square root: $4^2 = 16$, and $4(1)(-7) = -28$, so $16 - (-28) = 16 + 28 = 44$.

Step 4: So the formula becomes $x = (-4 \pm \sqrt{44}) / 2$.

Step 5: Approximate the square root: $\sqrt{44}$ is about 6.6.

Step 6: Take the positive branch, using $+$: $x = (-4 + 6.6) / 2 = 2.6 / 2$.

Step 7: Divide to finish: $x \approx 1.3$.

Answer: $x \approx 1.3$

Tip

When a quadratic doesn't factor with nice integers, go straight to the quadratic formula instead of guessing factor pairs.

Trap

A common slip is computing $4(1)(-7)$ as -28 and then subtracting it as $16 - 28$, forgetting that subtracting a negative adds.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x \approx -5.3$	Correctly computed both branches, 1.3 and -5.3, but reported the negative one instead of the positive solution asked for.
$x \approx -0.3$	Computed $16 - (-28)$ as $16 - 28 = -12$ instead of 44, mishandling the double negative, then used the wrong number under the root.



$x \approx 4.6$

Divided only the -4 by 2a but forgot to also divide the $\sqrt{44}$ term by 2, instead of dividing the whole numerator by 2a.

Shortcut

None faster here — since the numbers don't factor nicely, the quadratic formula is already the most direct method.

Desmos Tip

Type $y = x^2 + 4x - 7$ into Desmos and use the zero/root feature near the positive intercept to confirm $x \approx 1.3$.

Solving Quadratic Equations & the Discriminant — Problem 4 (Easy)

Without solving the equation, how many real solutions does $2x^2 - 3x + 5 = 0$ have?

Formula / Rule Used: The discriminant: $D = b^2 - 4ac$.

Solution

Step 1: Identify a, b, and c from the equation: $a = 2$, $b = -3$, $c = 5$.

Step 2: Compute b^2 : $(-3)^2 = 9$.

Step 3: Compute $4ac$: $4(2)(5) = 40$.

Step 4: Subtract to find the discriminant: $D = 9 - 40 = -31$.

Step 5: Since D is negative, the equation has 0 real solutions.

Answer: 0 real solutions

Tip

When a question asks 'how many solutions' without asking you to actually solve, compute just the discriminant.

Trap

It's tempting to only look at b^2 being positive and assume that alone means real solutions exist.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2 real solutions	Used $D = b^2 + 4ac$ instead of $b^2 - 4ac$, incorrectly getting a positive discriminant of 49.
1 real solution	Treated the negative discriminant as 'close enough' to 0 and assumed that meant exactly one repeated root.
-31 real solutions	Correctly computed the discriminant as -31 but reported that number directly instead of using its sign to count solutions.

Shortcut

The discriminant alone answers this — there's no need to ever run the full quadratic formula.

Desmos Tip



Type $y = 2x^2 - 3x + 5$ into Desmos — the graph never touches the x-axis, confirming there are 0 real solutions.

Solving Quadratic Equations & the Discriminant — Problem 5 (Medium)

One solution of $x^2 - 4x + k = 0$ is $x = 6$. What is the other solution?

Formula / Rule Used: Substitute a known root to find a missing constant, then factor to find the other root.

Solution

Step 1: Since $x = 6$ is a solution, substitute it into the equation to solve for the missing constant, k : $6^2 - 4(6) + k = 0$.

Step 2: Compute the numbers: $36 - 24 + k = 0$.

Step 3: Simplify: $12 + k = 0$.

Step 4: Solve for k : $k = -12$.

Step 5: Now the full equation is $x^2 - 4x - 12 = 0$.

Step 6: Factor this: find two numbers that multiply to -12 and add to -4 , which are -6 and 2 .

Step 7: Write the factored form: $(x - 6)(x + 2) = 0$, so $x = 6$ or $x = -2$.

Step 8: Since 6 is already the given solution, the other solution is $x = -2$.

Answer: $x = -2$

Tip

When one solution is given, plug it in first to solve for any unknown constant before finding the other solution.

Trap

It's tempting to stop once you've solved for k , forgetting the question actually asks for the OTHER value of x .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = -12$	Correctly solved for the missing constant but reported k instead of the other value of x the question asked for.
$x = 2$	Made a sign error factoring -12 and -4 , using the pair 6 and -2 in the wrong factor, flipping the sign of the answer.
$x = -6$	Reused the given root, 6 , with a flipped sign as the 'other' solution instead of actually factoring the equation.

Shortcut

None faster here — substituting the known root to find k , then factoring, is the reliable two-step path.

Desmos Tip

Type $y = x^2 - 4x - 12$ into Desmos and use the zero/root feature to confirm the two x-intercepts are 6 and -2 .

Solving Quadratic Equations & the Discriminant — Problem 6 (Medium)

For what value of x does the equation $4x^2 + 9 = 12x$ have its one real solution?

Formula / Rule Used: Rearrange to standard form, confirm $D=0$, then use $x = -b/2a$ for the repeated root.

Solution

Step 1: The equation isn't in standard form yet, so move every term to one side first.

Step 2: Subtract $12x$ from both sides: $4x^2 - 12x + 9 = 0$.

Step 3: Identify a , b , and c from this standard form: $a = 4$, $b = -12$, $c = 9$.

Step 4: Check the discriminant to confirm there's exactly one solution: $D = (-12)^2 - 4(4)(9) = 144 - 144 = 0$.

Step 5: Since $D = 0$, the repeated solution comes from $x = -b / 2a$.

Step 6: Substitute: $x = -(-12) / (2 \times 4) = 12 / 8$.

Step 7: Simplify the fraction: $x = 3/2$.

Answer: $x = 3/2$ (or 1.5)

Tip

'One real solution' is your cue to check that $D=0$, then jump straight to $x=-b/2a$ instead of the full $\pm$ formula.

Trap

It's tempting to plug the equation into the formula before rearranging it, using 12 as b instead of -12 from the correct standard form.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = -3/2$	Used $b=12$ instead of $b=-12$ when applying $x=-b/2a$, flipping the sign of the final answer.
$x = 3$	Used $x=-b/a$ instead of $x=-b/(2a)$, forgetting to divide by the extra factor of 2 in the denominator.
$x = 9/4$	Simplified the fraction $12/8$ incorrectly down to $9/4$ instead of correctly reducing it to $3/2$.

Shortcut

Once you confirm $D=0$, you can skip the $\pm$ part of the quadratic formula entirely and just compute $-b/2a$.

Desmos Tip

Type $y = 4x^2 - 12x + 9$ into Desmos — the graph just touches the x -axis at one point, $x=1.5$, confirming the repeated root.

Solving Quadratic Equations & the Discriminant — Problem 7 (Medium)

A rectangular banner in Lisbon has a length that is 3 meters more than its width, and the banner also has a decorative border painted along its edge. If the banner's area is 88 square meters, what is its width, in meters?

Formula / Rule Used: Translate the scenario into a quadratic, ignore irrelevant details, then factor.

Solution

Step 1: Let w stand for the width, so the length is $w + 3$, based on the problem's wording.

Step 2: Set up the area equation, since area = width $\times$ length: $w(w + 3) = 88$.

Step 3: Expand the left side: $w^2 + 3w = 88$.

Step 4: Move 88 to the left side to set the equation equal to 0: $w^2 + 3w - 88 = 0$.

Step 5: The detail about the decorative border doesn't affect the area at all, so ignore it.

Step 6: Find two numbers that multiply to -88 and add to 3: 11 and -8, since $11 \times (-8) = -88$ and $11 + (-8) = 3$.

Step 7: Write the factored form: $(w + 11)(w - 8) = 0$, so $w = -11$ or $w = 8$.

Step 8: Since width must be positive, the width is $w = 8$ meters.

Answer: 8 meters

Tip

When a word problem includes extra descriptive details, focus only on the numbers actually needed to build the equation.

Trap

It's tempting to think the decorative border somehow affects the area equation, when it's really just irrelevant description.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-11 meters	Correctly solved both values, 8 and -11, but reported the negative one, which isn't a valid width.
11 meters	Picked the wrong number from the factor pair (11, -8) instead of correctly matching $w=8$ to the positive solution.
42.5 meters	Added the width and length instead of multiplying them for area, solving $w + (w+3) = 88$ instead of $w(w+3) = 88$.

Shortcut

Setting up $w(w+3)=88$ directly and factoring is already the fastest path — no need for the quadratic formula on numbers this clean.

Desmos Tip

Type $y = x^2 + 3x - 88$ into Desmos and use the zero/root feature to confirm the positive x -intercept is 8.

Solving Quadratic Equations & the Discriminant — Problem 8 (Medium)

For what positive value of k does the equation $x^2 + kx + 16 = 0$ have exactly one real solution?

Formula / Rule Used: Set the discriminant equal to 0 for exactly one real solution, then solve for the unknown constant.

Solution

Step 1: For an equation to have exactly one real solution, its discriminant must equal 0: $D = b^2 - 4ac = 0$.

Step 2: Identify a , b , and c from the equation: $a = 1$, $b = k$, $c = 16$.

Step 3: Set up the discriminant equation: $k^2 - 4(1)(16) = 0$.

Step 4: Simplify: $k^2 - 64 = 0$.

Step 5: Add 64 to both sides: $k^2 = 64$.

Step 6: Take the square root of both sides: $k = \pm 8$.

Step 7: Since the problem asks for the positive value, $k = 8$.

Answer: $k = 8$

Tip

'Exactly one real solution' is a direct signal to set the discriminant equal to 0 and solve for the unknown.

Trap

It's tempting to stop at $k^2 = 64$ and report 64, or forget to check which sign the question actually wants.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 64$	Solved as far as $k^2 = 64$ but reported 64 itself instead of taking the square root to find k .
$k = -8$	Correctly solved $k = \pm 8$ but reported the negative value instead of the positive one the question requested.
$k = 4$	Computed $\sqrt{64}$ but then divided it by 2 out of habit, computing $8/2=4$ instead of just using $\sqrt{64}=8$.

Shortcut

Recognizing 'exactly one real solution' means $D=0$ skips straight to a simple equation, with no need for the full quadratic formula.

Desmos Tip

Type $y = x^2 + 8x + 16$ into Desmos and confirm the parabola just touches the x -axis at a single point, verifying $k=8$ works.

21. Quadratic Graphs, Roots & Word Problems

Quadratic Graphs, Roots & Word Problems — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Parabola	Παραβολή
Vertex	Κορυφή
Axis of symmetry	Άξονας συμμετρίας
x-intercepts (roots)	Σημεία τομής με τον άξονα x (ρίζες)
Vertex form	Μορφή κορυφής
Maximum / minimum value	Μέγιστη / ελάχιστη τιμή

Quadratic Graphs, Roots & Word Problems — What It Is

What It Is

- A quadratic equation always contains an x^2 term, and its graph is a curved shape called a parabola.
- A parabola always opens either upward (like a smile) or downward (like a frown), never sideways.
- The SAT writes the same parabola in three different equation forms, and each form reveals a different feature instantly.
- Roots, also called x-intercepts or zeros, are the x-values where the parabola crosses the x-axis.
- This skill tests whether you can read a parabola's vertex, intercepts, and roots straight from its equation, without graphing it.

Quadratic Graphs, Roots & Word Problems — Core Rules

Standard Form

$$y = ax^2 + bx + c$$

- The constant term c is always the y-intercept, the point where the graph crosses the y-axis.
- The vertex's x-coordinate is always $x = -b/(2a)$, no matter what a , b , and c are.
- e.g. for $y = x^2 - 4x + 3$, the vertex x-coordinate is $x = -(-4)/(2(1)) = 2$.
- This form is fastest for reading the y-intercept, but it hides the roots and the vertex height.

Factored Form

$$y = a(x - p)(x - q)$$

- p and q are the roots (x-intercepts) — the graph crosses the x-axis exactly at those two x-values.
- Set each factor equal to zero to find them: $x - p = 0$ gives $x = p$.

- The vertex's x-coordinate is always the midpoint of the two roots: $x_v = (p + q) / 2$.
- e.g. for $y = (x - 2)(x - 6)$, the roots are 2 and 6, so the vertex sits at $x = 4$.

Vertex Form

$$y = a(x - h)^2 + k$$

- (h, k) is the vertex itself — you can read it straight off the equation with no calculation.
- If $a > 0$, the vertex is the lowest point (a minimum); if $a < 0$, it's the highest point (a maximum).
- e.g. $y = 2(x - 3)^2 + 1$ has vertex $(3, 1)$, and since $a = 2$ is positive, that's a minimum.
- This form is the fastest way to answer any 'maximum value' or 'minimum value' question.

Completing the Square

Converts standard form into vertex form

- Step 1: factor a out of only the x^2 and x terms, leaving the constant c outside.
- Step 2: inside the parentheses, add and subtract (half the x -coefficient) squared.
- Step 3: the first three terms now form a perfect square, so rewrite them as $(x + \text{something})^2$.
- Step 4: distribute a back through, then combine the leftover numbers to get the final k .

Vieta's Formulas (Sum and Product of Roots)

$$\text{Sum of roots} = -b/a. \quad \text{Product of roots} = c/a.$$

- These give you a combination of the roots directly from a , b , and c , with no solving required.
- Use the sum formula when a question asks for $p + q$, and the product formula for p times q .
- e.g. for $x^2 - 7x + 10 = 0$, the sum of roots is 7 and the product is 10.

The Discriminant and the Graph

$$\text{Discriminant} = b^2 - 4ac$$

- A positive discriminant means the parabola crosses the x -axis at two separate points (two real roots).
- A discriminant of exactly zero means the parabola just touches the x -axis at its vertex (one repeated root).
- A negative discriminant means the parabola never touches the x -axis at all (no real roots).

Quadratic Word Problems

- Translate the real-world scenario into one of the three quadratic forms above, then solve normally.
- Common setups: area = length times width, or height over time for a thrown or launched object.
- After solving, check both solutions against the real-world context described in the problem.
- Throw out any solution that isn't physically possible, such as negative time, negative length, or negative width.

How to Recognize This on the Test

- The question gives an equation with an x^2 term and asks about its graph, vertex, roots, or intercepts.
- Phrases like 'minimum value' or 'maximum value' signal you should switch to (or build) vertex form.
- Phrases like 'x-intercepts,' 'zeros,' or 'roots' signal you need factored form or the discriminant.
- A word problem describing area, a thrown object, or a value that rises then falls is a quadratic setup.
- A question asking for a sum or product of the solutions, without asking you to fully solve, signals Vieta's formulas.

Common Mistakes — and the Fix

- Forgetting that a downward parabola ($a < 0$) has a maximum, not a minimum — always check the sign of a first.
- Reading the vertex's x-coordinate as $-b/a$ instead of the correct $-b/(2a)$, forgetting to double the denominator.
- Misreading a factor like $(x + 7)$ as giving root $+7$ instead of the correct root -7 .
- In word problems, keeping a negative length or negative time solution instead of discarding it as impossible.
- Mixing up sum and product of roots, such as using c/a when the question actually asks for the sum.

Quadratic Graphs, Roots & Word Problems — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Quadratic Graphs, Roots & Word Problems — Problem 1 (Easy)

The graph of $y = 2x^2 - 5x + 9$ in the xy -plane has a y-intercept at $(0, k)$. What is the value of k ?

Formula / Rule Used: Standard form $y = ax^2 + bx + c$: the y-intercept is c .

Solution

Step 1: In standard form $y = ax^2 + bx + c$, the constant term c is always the y-intercept.

Step 2: Compare $y = 2x^2 - 5x + 9$ to $y = ax^2 + bx + c$.

Step 3: Matching terms gives $a = 2$, $b = -5$, and $c = 9$.

Step 4: The y-intercept occurs where $x = 0$, so plug in $x = 0$: $y = 2(0)^2 - 5(0) + 9$.

Step 5: That simplifies to $y = 0 - 0 + 9 = 9$.

Step 6: So $k = 9$.

Answer: $k = 9$

**Tip**

Whenever a question asks for a y-intercept from standard form, just read off the constant term c — no computation needed.

Trap

It's tempting to plug in $x = 1$ out of habit instead of $x = 0$, which changes the answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = -5$	Reported the b-coefficient instead of the constant term c .
$k = 2$	Reported the a-coefficient instead of the constant term c .
$k = 6$	Plugged in $x = 1$ instead of $x = 0$, computing $2(1)^2 - 5(1) + 9 = 6$.

Shortcut

None needed — the y-intercept in standard form is always just the constant term, read directly.

Desmos Tip

Type $y = 2x^2 - 5x + 9$ into Desmos and trace the curve to $x = 0$ to confirm it crosses the y-axis at 9.

Quadratic Graphs, Roots & Word Problems — Problem 2 (Easy)

The graph of $y = (x - 3)(x + 7)$ is a parabola in the xy -plane. What is the sum of its two x-intercepts?

Formula / Rule Used: Factored form $y = a(x - p)(x - q)$: roots are p and q .

Solution

Step 1: In factored form $y = a(x - p)(x - q)$, the x-intercepts (roots) are $x = p$ and $x = q$.

Step 2: Rewrite $(x + 7)$ as $(x - (-7))$ so it matches the $(x - q)$ pattern.

Step 3: Comparing to $(x - p)(x - q)$, the roots are $p = 3$ and $q = -7$.

Step 4: The sum of the roots is $p + q = 3 + (-7)$.

Step 5: Adding those gives $3 + (-7) = -4$.

Answer: -4

Tip

Rewrite any $(x + \text{number})$ factor as $(x - (-\text{number}))$ first, so you don't lose track of the root's sign.

Trap

It's tempting to read the root from $(x + 7)$ as $+7$ instead of the correct -7 .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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10	Added 3 and 7 directly, ignoring that the second root is actually -7.
4	Subtracted 7 - 3 instead of correctly adding 3 + (-7).
-10	Misread the first factor's root sign too, using -3 and -7 instead of 3 and -7.

Shortcut

Since the equation is already factored, you can identify and add the roots directly without expanding to standard form.

Desmos Tip

Graph $y = (x-3)(x+7)$ in Desmos and check where the curve crosses the x-axis to confirm the roots are 3 and -7.

Quadratic Graphs, Roots & Word Problems — Problem 3 (Easy)

The graph of $y = -3(x - 4)^2 + 10$ is a parabola in the xy-plane. What is the maximum value of y?

Formula / Rule Used: Vertex form $y = a(x - h)^2 + k$: vertex is (h, k).

Solution

Step 1: In vertex form $y = a(x - h)^2 + k$, the point (h, k) is the vertex of the parabola.

Step 2: Compare $y = -3(x - 4)^2 + 10$ to $y = a(x - h)^2 + k$: $a = -3$, $h = 4$, $k = 10$.

Step 3: Since $a = -3$ is negative, the parabola opens downward, so the vertex is the highest point.

Step 4: The maximum value of y is the k-coordinate of the vertex, not the h-coordinate.

Step 5: So the maximum value of y is 10.

Answer: 10

Tip

Always check the sign of a first in vertex form — negative means the vertex is a maximum, positive means a minimum.

Trap

It's tempting to report $h = 4$ as the maximum value instead of the actual output value $k = 10$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4	Reported the h-coordinate (the x-value of the vertex) instead of the k-coordinate.
-10	Assumed the negative sign on a also flips the sign of k, giving -10 instead of 10.
7	Incorrectly subtracted a from k ($10 - 3$) instead of simply reading k directly.

Shortcut



None needed — vertex form hands you the vertex directly, so there's nothing left to calculate once you check the sign of a .

Desmos Tip

Graph $y = -3(x-4)^2+10$ in Desmos; the highest point on the curve confirms a maximum value of 10 at $x = 4$.

Quadratic Graphs, Roots & Word Problems — Problem 4 (Easy)

In the equation $x^2 - 9x + 20 = 0$, what is the product of the two solutions?

Formula / Rule Used: Vieta's formula: product of roots = c/a .

Solution

Step 1: Compare $x^2 - 9x + 20 = 0$ to standard form $ax^2 + bx + c = 0$.

Step 2: This gives $a = 1$, $b = -9$, and $c = 20$.

Step 3: Vieta's formula says the product of the roots equals c/a .

Step 4: Divide: product = $20 / 1$.

Step 5: So the product of the two solutions is 20.

Answer: 20

Tip

Whenever a question asks for a product (or sum) of solutions, use Vieta's formulas instead of factoring first.

Trap

It's tempting to compute the sum of the roots instead of the product, since both use similar-looking formulas.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
9	Calculated the sum of the roots ($-b/a = 9$) instead of the product the question asked for.
-20	Applied an unnecessary sign flip to c/a , even though a and c are both positive here.
-9	Used the sum formula $-b/a$ but mismanaged the sign of b .

Shortcut

None needed — Vieta's formula gives the product instantly, without factoring or finding the actual roots.

Desmos Tip

Graph $y = x^2 - 9x + 20$ in Desmos, read the two x -intercepts (4 and 5), and multiply them to confirm the product is 20.

Quadratic Graphs, Roots & Word Problems — Problem 5 (Medium)

A parabola has vertex (2, -3) and passes through the point (5, 15). The equation can be written as $y = a(x - 2)^2 - 3$. What is the value of a ?

Formula / Rule Used: Vertex form $y = a(x - h)^2 + k$, solved for a using a known point.

Solution

Step 1: Since the vertex is (2, -3), the equation has the form $y = a(x - 2)^2 - 3$.

Step 2: The parabola also passes through (5, 15), so substitute $x = 5$ and $y = 15$.

Step 3: This gives $15 = a(5 - 2)^2 - 3$.

Step 4: Simplify inside the parentheses first: $15 = a(3)^2 - 3$, which is $15 = 9a - 3$.

Step 5: Add 3 to both sides: $18 = 9a$.

Step 6: Divide both sides by 9: $a = 2$.

Answer: $a = 2$

Tip

When you're given a vertex and one more point, plug that point straight into vertex form and solve for the leftover constant.

Trap

It's easy to forget to square (5 - 2) before multiplying by a , treating the equation as $15 = 3a - 3$ instead of $15 = 9a - 3$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$a = 6$	Skipped squaring (5-2), using $15 = 3a - 3$ instead of $15 = 9a - 3$, giving $18 = 3a$.
$a = -2$	Made a sign error moving the -3, writing $15 = 9a + 3$ instead of $15 = 9a - 3$.
$a = 1.8$	Divided 18 by 10 instead of by 9, a simple arithmetic slip.

Shortcut

None faster here — substituting the known point into vertex form is already the most direct route to a .

Desmos Tip

In Desmos, graph $y = a(x-2)^2-3$ with a slider for a , and adjust it until the curve passes through (5,15) to confirm $a = 2$.

Quadratic Graphs, Roots & Word Problems — Problem 6 (Medium)

A rectangular garden in Rotterdam has a length that is 3 meters longer than its width. The garden's area is 108 square meters. What is the width of the garden, in meters?

Formula / Rule Used: Area = length x width, translated into a quadratic equation.

Solution

Step 1: Let w stand for the width in meters; then the length is $w + 3$.

Step 2: Area = length times width, so $w(w + 3) = 108$.

Step 3: Expand the left side: $w^2 + 3w = 108$.

Step 4: Move 108 to the left so the equation equals zero: $w^2 + 3w - 108 = 0$.

Step 5: Factor: find two numbers that multiply to -108 and add to 3, which are 12 and -9.

Step 6: So $(w + 12)(w - 9) = 0$, giving $w = -12$ or $w = 9$.

Step 7: Discard $w = -12$ since a width can't be negative, leaving $w = 9$.

Answer: 9 meters

Tip

Translate the word problem into a quadratic equation set equal to zero, then check which root actually makes sense in context.

Trap

It's tempting to report both roots, or the negative root, instead of discarding the one that's physically impossible.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-12 meters	Reported the negative root instead of discarding it as an impossible width.
12 meters	Gave the length ($w + 3 = 12$) instead of the width the question actually asked for.
10.4 meters	Ignored the 'longer by 3' condition and solved $w^2 = 108$ alone, treating the garden as a square.

Shortcut

None truly faster — setting up the quadratic and factoring (or using the quadratic formula) is the direct path here.

Desmos Tip

Graph $y = x(x+3)$ and $y = 108$ in Desmos; the intersection at positive x confirms the width is 9.

For what value of k does the equation $x^2 + 6x + k = 0$ have exactly one real solution?

Formula / Rule Used: Discriminant $b^2 - 4ac = 0$ gives exactly one real solution.

Solution

Step 1: A quadratic has exactly one real solution exactly when its discriminant equals zero.

Step 2: The discriminant is $b^2 - 4ac$, and here $a = 1$, $b = 6$, $c = k$.

Step 3: Set the discriminant equal to zero: $6^2 - 4(1)(k) = 0$.

Step 4: Simplify 6^2 to get $36 - 4k = 0$.

Step 5: Add $4k$ to both sides: $36 = 4k$.

Step 6: Divide both sides by 4: $k = 9$.

Answer: $k = 9$

Tip

The phrase 'exactly one real solution' is a strong signal to set the discriminant $b^2 - 4ac$ equal to zero and solve.

Trap

It's tempting to set the discriminant greater than or less than zero instead of exactly equal to zero.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 36$	Stopped after $4k = 36$ without dividing by 4, reporting 36 instead of 9.
$k = -9$	Made a sign error simplifying $-4k$, ending up with $4k = -36$ and $k = -9$.
$k = 3$	Confused the problem with a vertex calculation and reported $b/2 = 3$ instead of using the discriminant.

Shortcut

Recognizing that 'one real solution' always means $b^2 = 4ac$ skips writing out the full discriminant formula.

Desmos Tip

Graph $y = x^2 + 6x + 9$ in Desmos and confirm the parabola just touches the x -axis at one point, verifying $k = 9$.

Quadratic Graphs, Roots & Word Problems — Problem 8 (Medium)



The sum of two numbers is 11 and their product is 24. What is the positive difference between the two numbers?

Formula / Rule Used: Vieta's formulas in reverse: build $x^2 - (\text{sum})x + (\text{product}) = 0$.

Solution

Step 1: Vieta's formulas say that for $x^2 - (\text{sum})x + (\text{product}) = 0$, the roots are exactly the two numbers.

Step 2: Using $\text{sum} = 11$ and $\text{product} = 24$, the equation is $x^2 - 11x + 24 = 0$.

Step 3: Factor by finding two numbers that multiply to 24 and add to 11: those are 8 and 3.

Step 4: So $x^2 - 11x + 24 = (x - 8)(x - 3) = 0$.

Step 5: The two numbers are 8 and 3.

Step 6: The positive difference is $8 - 3 = 5$.

Answer: 5

Tip

When a problem gives you a sum and a product of two unknown numbers, build the quadratic $x^2 - (\text{sum})x + (\text{product}) = 0$ right away.

Trap

It's tempting to stop once you find the two numbers and report one of them, instead of finishing to find their difference.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
8	Reported one of the two numbers instead of computing the difference the question asked for.
11	Confused the difference with the given sum of the two numbers.
24	Reported the given product value instead of finishing to find the difference.

Shortcut

Since the numbers are nice integers, guessing factor pairs of 24 that add to 11 can be faster than formally factoring.

Desmos Tip

Graph $y = x^2 - 11x + 24$ in Desmos, read the two x-intercepts (3 and 8), and subtract to confirm the difference is 5.

22. Radicals & Rational Exponents

Radicals & Rational Exponents — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Radical (root)	Ρίζα (ριζικό)
Rational exponent	Ρητός εκθέτης
Exponent rules	Ιδιότητες δυνάμεων
Simplify a radical	Απλοποίηση ρίζας
Rationalize the denominator	Ρητοποίηση παρονομαστή

Radicals & Rational Exponents — What It Is

What It Is

- A radical, like a square root, and a rational exponent, like $x^{(1/2)}$, are two ways to write the exact same thing.
- This skill is about simplifying and combining these expressions once a variable is involved, not just plain numbers.
- The SAT hides radicals and fractional exponents inside bigger algebra and function questions to test this fluency.
- You need to convert between the two forms, simplify, combine, and clear radicals from denominators confidently.

Radicals & Rational Exponents — Core Rules

Radical-Exponent Conversion (with Variables)

$$n\text{-th root of } (x^m) = x^{(m/n)}$$

- The index, the small root number, becomes the denominator of the fractional exponent.
- The power inside the radical becomes the numerator of the fractional exponent.
- A plain square root has an invisible index of 2, so it becomes an exponent of $1/2$.
- Example: the cube root of x^5 equals $x^{(5/3)}$.

Properties of Radicals

$$\sqrt{ab} = \sqrt{a} * \sqrt{b} \quad \text{and} \quad \sqrt{a/b} = \sqrt{a} / \sqrt{b}$$

- A radical over a product can be split into two separate radicals multiplied together.
- A radical over a quotient can be split into a radical on top and a radical on bottom.
- These rules also work in reverse, letting you combine two radicals into one.
- This splitting-and-combining idea is the main tool behind simplifying and rationalizing.

Simplifying Radicals with Variables

- Split the number under the root into a perfect-power factor times a leftover factor.
- Split any variable power into the largest matching perfect power times a leftover power.
- Pull each perfect-power piece outside the radical, leaving only the leftovers inside.
- Example: $\sqrt{x^5} = \sqrt{x^4 \cdot x} = x^2 \cdot \sqrt{x}$.

Adding and Subtracting Radicals

$$a\sqrt{c} + b\sqrt{c} = (a + b)\sqrt{c}$$

- Only radicals with the exact same index AND the exact same radicand can combine.
- Treat matching radicals like like terms, adding or subtracting only the outside coefficients.
- Always simplify every radical first, since matches often stay hidden until you do.
- Example: $\sqrt{8} + \sqrt{2}$ becomes $2\sqrt{2} + \sqrt{2} = 3\sqrt{2}$ once simplified.

Rationalizing Denominators

Single radical: multiply by that radical. Binomial: multiply by the conjugate.

- A fraction isn't fully simplified if a radical is still sitting in the denominator.
- For a single-term radical denominator, multiply top and bottom by that exact radical.
- For a binomial denominator like $a + \sqrt{b}$, multiply top and bottom by its conjugate, $a - \sqrt{b}$.
- The conjugate works because it turns the denominator into a difference of squares, erasing the radical.

Exponent Rules with Rational Exponents

$$x^{(m/n)} \cdot x^{(p/n)} = x^{((m + p)/n)}$$

- The same product, quotient, and power rules from whole-number exponents still apply.
- The bases must match exactly before you can add, subtract, or multiply exponents.
- Example: $x^{(1/4)} \cdot x^{(3/4)} = x^{(4/4)} = x^1 = x$.
- A power raised to another power still means you multiply the two exponents together.

How to Recognize This on the Test

- A variable sitting directly under a radical sign, like $\sqrt{x}$ or the cube root of x^2 .
- A variable raised to a visible fraction, like $x^{(2/3)}$ or $x^{(1/2)}$.
- A fraction with a radical left in the denominator that clearly needs to be cleared.
- Phrasing like 'simplify the expression' or 'which expression is equivalent to' next to a root or fractional power.

Common Mistakes — and the Fix

- Treating $\sqrt{x + y}$ as $\sqrt{x} + \sqrt{y}$ — a radical never distributes over addition or subtraction.
- Trying to add or subtract radicals before simplifying them, missing that they actually match.
- Multiplying only the denominator by the conjugate and forgetting to multiply the numerator too.
- Mixing up which number becomes the exponent's numerator and which becomes its denominator.
- Adding or multiplying exponents on expressions that don't actually share the same base.

Radicals & Rational Exponents — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Radicals & Rational Exponents — Problem 1 (Easy)

Simplify $\sqrt{50x^3}$ for $x > 0$, writing your answer as $a\sqrt{b}$ with the variable pulled outside the radical as far as possible.

Formula / Rule Used: Simplifying radicals with variables: split into perfect-power factors times leftovers.

Solution

Step 1: Break 50 into a perfect square times a leftover factor: $50 = 25 \cdot 2$.

Step 2: Break x^3 into an even power times a leftover factor: $x^3 = x^2 \cdot x$.

Step 3: Rewrite the radical using these pieces: $\sqrt{25 \cdot 2 \cdot x^2 \cdot x}$.

Step 4: Separate the perfect-power factors from the leftover factors: $\sqrt{25} \cdot \sqrt{x^2} \cdot \sqrt{2x}$.

Step 5: Take the square roots of the perfect-power pieces: $\sqrt{25} = 5$ and $\sqrt{x^2} = x$.

Step 6: Multiply the pieces that came outside the radical together: $5 \cdot x = 5x$.

Step 7: Write the final simplified form: $5x \cdot \sqrt{2x}$.

Answer: $5x \cdot \sqrt{2x}$

Tip

Split the number and the variable power separately into perfect-power times leftover before pulling anything out.

Trap

It's easy to pull the entire x^3 out as x , forgetting that only the even part, x^2 , can fully leave the radical.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$5x^3 \cdot \sqrt{2}$	Pulled the entire x^3 out from under the radical without checking that only x^2 is a perfect square.
$25x \cdot \sqrt{2x}$	Forgot to actually take the square root of 25, leaving it as 25 instead of simplifying it to 5.



$5x \cdot \sqrt{2x^3}$

Never separated x^2 from x^3 , leaving the entire variable power stuck inside the radical.

Shortcut

None faster — splitting the number and the variable power separately is already the quickest way to simplify.

Desmos Tip

Substitute $x = 2$ into $\sqrt{50x^3}$ and into $5x \cdot \sqrt{2x}$ as two separate Desmos lines and confirm the outputs match.

Radicals & Rational Exponents — Problem 2 (Easy)

Which expression is equivalent to $x^{(3/4)} \cdot x^{(1/4)}$ for $x > 0$?

Formula / Rule Used: Product rule for exponents: $x^m \cdot x^n = x^{(m+n)}$.

Solution

Step 1: Both terms share the same base, x , so the product rule for exponents applies.

Step 2: Add the two exponents together: $3/4 + 1/4$.

Step 3: The fractions already share a denominator, so add the numerators: $3 + 1 = 4$.

Step 4: Write the sum as a fraction: $4/4$.

Step 5: Simplify the fraction: $4/4 = 1$.

Step 6: So the expression simplifies to x^1 , which is just x .

Answer: x

Tip

When multiplying powers of the same base, add the exponents first, then simplify the resulting fraction.

Trap

It's tempting to multiply the exponents instead of adding them, since the original expression involves multiplication.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x^{(3/16)}$	Multiplied the two exponents, $3/4$ times $1/4$, instead of adding them together.
x^2	Combined the numerators and denominators incorrectly, getting $4/2$ instead of the correct $4/4$.
$x^{(1/2)}$	Averaged the two exponents together instead of adding them as the product rule requires.

Shortcut



Recognizing the denominators already match lets you add the numerators directly: $3 + 1 = 4$.

Desmos Tip

Type $x^{(3/4)} * x^{(1/4)}$ and x on two separate lines in Desmos and confirm the two graphs overlap exactly.

Radicals & Rational Exponents — Problem 3 (Easy)

Rationalize the denominator of $6/\sqrt{2}$, writing your answer in the form $a*\sqrt{b}$.

Formula / Rule Used: Rationalizing a single-term radical denominator: multiply top and bottom by that radical.

Solution

Step 1: Identify the radical sitting in the denominator: $\sqrt{2}$.

Step 2: Multiply both the numerator and the denominator by $\sqrt{2}$: $(6*\sqrt{2}) / (\sqrt{2}*\sqrt{2})$.

Step 3: Simplify the denominator: $\sqrt{2} * \sqrt{2} = 2$.

Step 4: Keep the numerator as is for now: $6*\sqrt{2}$.

Step 5: Write the fraction so far: $6*\sqrt{2} / 2$.

Step 6: Reduce the fraction by dividing 6 and 2 by their common factor, 2: $3*\sqrt{2}$.

Answer: $3*\sqrt{2}$

Tip

Multiply the top and bottom by the exact radical in the denominator to turn it into a whole number.

Trap

Forgetting to also multiply the numerator by $\sqrt{2}$, and only changing the denominator, breaks the equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$6*\sqrt{2}$	Multiplied top and bottom by $\sqrt{2}$ correctly but forgot to divide by the resulting denominator of 2.
3	Simplified the denominator correctly to 2 but dropped the $\sqrt{2}$ from the numerator entirely.
$6*\sqrt{4}$	Multiplied $\sqrt{2}$ by $\sqrt{2}$ inside the numerator instead of leaving the numerator as $6*\sqrt{2}$.

Shortcut

None needed — multiplying by the denominator's own radical is already the fastest legitimate method.

Desmos Tip

Type $6/\sqrt{2}$ and $3*\sqrt{2}$ on two separate lines in Desmos and confirm they give the same decimal value.

Radicals & Rational Exponents — Problem 4 (Easy)

An engineering lab in Lyon simplifies the expression $\sqrt{18x^5}$ for a positive value of x . What is the fully simplified form?

Formula / Rule Used: Simplifying radicals with variables: split into perfect-power factors times leftovers.

Solution

Step 1: Break 18 into a perfect square times a leftover factor: $18 = 9 \cdot 2$.

Step 2: Break x^5 into the largest even power times a leftover factor: $x^5 = x^4 \cdot x$.

Step 3: Rewrite the radical using these pieces: $\sqrt{9 \cdot 2 \cdot x^4 \cdot x}$.

Step 4: Separate the perfect-power factors from the leftover factors: $\sqrt{9} \cdot \sqrt{x^4} \cdot \sqrt{2x}$.

Step 5: Take the roots of the perfect-power pieces: $\sqrt{9} = 3$ and $\sqrt{x^4} = x^2$.

Step 6: Multiply the pieces that come outside the radical: $3 \cdot x^2 = 3x^2$.

Step 7: Write the final simplified form: $3x^2 \cdot \sqrt{2x}$.

Answer: $3x^2 \cdot \sqrt{2x}$

Tip

Handle the number and the variable power as two separate perfect-square searches, then combine what comes out.

Trap

A common slip is writing $\sqrt{x^4}$ as x instead of x^2 , undercounting how many x 's actually leave the radical.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$3x \cdot \sqrt{2x^4}$	Took the square root of x^4 as x instead of x^2 , leaving too many x 's trapped inside the radical.
$9x^2 \cdot \sqrt{2x}$	Forgot to actually take the square root of 9, leaving it as 9 instead of simplifying it to 3.
$3x^2 \cdot \sqrt{2x^5}$	Never split x^5 into x^4 times x , leaving the entire variable power stuck under the radical.

Shortcut

None faster — splitting the number and the variable power separately is already the quickest method here.

Desmos Tip

Substitute $x = 2$ into $\sqrt{18x^5}$ and into $3x^2 \cdot \sqrt{2x}$ as two Desmos lines and confirm the outputs match.

Radicals & Rational Exponents — Problem 5 (Medium)

Simplify and combine the expression $\sqrt{45x} + \sqrt{20x}$ for $x > 0$.

Formula / Rule Used: Simplify each radical first, then combine any resulting like radicals.

Solution

Step 1: Simplify the first radical: $45x = 9 \cdot 5x$, so $\sqrt{45x} = \sqrt{9} \cdot \sqrt{5x} = 3\sqrt{5x}$.

Step 2: Simplify the second radical: $20x = 4 \cdot 5x$, so $\sqrt{20x} = \sqrt{4} \cdot \sqrt{5x} = 2\sqrt{5x}$.

Step 3: Rewrite the original expression using these simplified forms: $3\sqrt{5x} + 2\sqrt{5x}$.

Step 4: Notice both terms now share the exact same radicand, $5x$, so they can be combined.

Step 5: Add the outside coefficients only, leaving the radical untouched: $3 + 2 = 5$.

Step 6: Write the combined result: $5\sqrt{5x}$.

Answer: $5\sqrt{5x}$

Tip

Whenever two radicals look unrelated, simplify each one first — matching radicands often only appear afterward.

Trap

Trying to add $\sqrt{45x}$ and $\sqrt{20x}$ directly, since they look different, without simplifying first leads nowhere.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$\sqrt{65x}$	Added the numbers under the two radicals directly, $45 + 20$, instead of simplifying and combining like radicals.
$3\sqrt{5x}$	Simplified only the first radical correctly and forgot to also simplify and add in the second term.
$6\sqrt{5x}$	Simplified both radicals correctly but multiplied the coefficients, 3 times 2 , instead of adding them.

Shortcut

Spotting that 45 and 20 are both multiples of 5 hints early that both radicals will simplify to a multiple of $\sqrt{5x}$.

Desmos Tip

Substitute $x = 1$ into $\sqrt{45x} + \sqrt{20x}$ and into $5\sqrt{5x}$ as two Desmos lines and confirm the outputs match.

Radicals & Rational Exponents — Problem 6 (Medium)

Simplify $4 / (3 - \sqrt{5})$ by rationalizing the denominator.

Formula / Rule Used: Rationalizing a binomial denominator: multiply top and bottom by the conjugate.

Solution

Step 1: Identify the conjugate of the denominator $3 - \sqrt{5}$, which is $3 + \sqrt{5}$.

Step 2: Multiply both the numerator and denominator by that conjugate: $4(3+\sqrt{5}) / [(3-\sqrt{5})(3+\sqrt{5})]$.

Step 3: Expand the denominator using the difference-of-squares pattern: $(3)^2 - (\sqrt{5})^2 = 9 - 5$.

Step 4: Simplify the denominator: $9 - 5 = 4$.

Step 5: Expand the numerator: $4(3 + \sqrt{5}) = 12 + 4\sqrt{5}$.

Step 6: Write the fraction so far: $(12 + 4\sqrt{5}) / 4$.

Step 7: Divide every term in the numerator by 4: $3 + \sqrt{5}$.

Answer: $3 + \sqrt{5}$

Tip

Multiply by the conjugate, the same two terms with the middle sign flipped, so the radical cancels out.

Trap

Multiplying by $(3 - \sqrt{5})$ again instead of flipping the sign to the true conjugate leaves the radical behind.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$12 + 4\sqrt{5}$	Correctly expanded the numerator but forgot to divide every term by the simplified denominator of 4.
$3 - \sqrt{5}$	Used the original denominator again as the multiplier instead of the true conjugate, keeping the wrong sign.
$(12 + 4\sqrt{5})/14$	Made an arithmetic slip computing $9 - 5$ and used 14 instead of the correct value, 4.

Shortcut

None faster — multiplying by the conjugate is the only reliable way to clear a binomial radical denominator.

Desmos Tip

Type $4/(3-\sqrt{5})$ and $3+\sqrt{5}$ on two separate lines in Desmos and confirm they produce the same decimal value.



If $x^{2/3} = 25$ and $x > 0$, what is the value of $x^{4/3}$?

Formula / Rule Used: Power rule for exponents: $(x^m)^n = x^{mn}$.

Solution

Step 1: Notice $x^{4/3}$ can be rewritten as $(x^{2/3})$ raised to the power 2, since $(2/3) * 2 = 4/3$.

Step 2: You are already given that $x^{2/3} = 25$.

Step 3: Substitute that value in: $x^{4/3} = 25^2$.

Step 4: Calculate 25^2 by multiplying 25 by itself: $25 * 25$.

Step 5: Multiply: $25 * 25 = 625$.

Step 6: So $x^{4/3} = 625$.

Answer: 625

Tip

Look for a way to rewrite the target exponent as a power of the exponent you were already given, instead of solving for x first.

Trap

Fully solving for x first, getting $x = 125$, before computing $x^{4/3}$ works but wastes time and invites arithmetic errors.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
50	Doubled 25 instead of squaring it, treating the exponent relationship as multiplication instead of a power.
125	Correctly solved for x itself but stopped there without going on to compute x raised to the $4/3$ power.
156.25	Solved for x as 125, then mistakenly took its square root instead of raising it to the $4/3$ power.

Shortcut

Rewriting $x^{4/3}$ as $(x^{2/3})^2$ skips solving for x entirely and jumps straight to 625.

Desmos Tip

Graph $y = x^{2/3}$ in Desmos, trace to the x -value where $y = 25$, then evaluate $x^{4/3}$ there to confirm 625.

Radicals & Rational Exponents — Problem 8 (Medium)

A materials lab in Rotterdam, which also logs each cable's weight for shipping records, models cable strength S using $S = k * d^{(3/2)}$, where d is the cable's diameter in centimeters. A cable with diameter 4 cm has a strength of 80. What is the value of k ?

Formula / Rule Used: Evaluate the rational exponent term, then solve for the constant k .

Solution

Step 1: Ignore the detail about logging cable weight for shipping records — it isn't needed to find k .

Step 2: Substitute the known values into the model: $80 = k * 4^{(3/2)}$.

Step 3: Evaluate $4^{(3/2)}$: the denominator 2 means take the square root of 4 first.

Step 4: The square root of 4 is 2.

Step 5: Now raise that result to the numerator's power: $2^3 = 8$.

Step 6: The equation becomes $80 = k * 8$.

Step 7: Divide both sides by 8 to solve for k : $k = 10$.

Answer: $k = 10$

Tip

Evaluate the rational exponent term completely first, turning the equation into a simple one-step division.

Trap

The detail about logging cable weight is irrelevant noise — don't let it distract you from the actual d and S values.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 60$	Computed $4^{(3/2)}$ incorrectly as 4 times $3/2$ instead of taking the square root first, then cubing.
$k = 26.67$	Divided 80 by 3 instead of by the correctly evaluated value of $4^{(3/2)}$, which is 8.
$k = 640$	Multiplied 80 by 8 instead of dividing by it when isolating k in the final step.

Shortcut

Recognizing $4^{(3/2)}$ as $(\sqrt{4})^3$ keeps the numbers small: $2^3 = 8$, then divide 80 by 8 directly.

Desmos Tip

Type $80/(4^{(3/2)})$ directly into Desmos to see it evaluate to 10 and confirm the value of k .

23. Exponential Functions — Growth & Decay

Exponential Functions — Growth & Decay — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Exponential function	Εκθετική συνάρτηση
Growth / decay rate	Ρυθμός αύξησης / μείωσης
Growth / decay factor	Συντελεστής αύξησης / μείωσης
Compound interest	Ανατοκισμός
Asymptote	Ασύμπτωτη
Initial value	Αρχική τιμή

Exponential Functions — Growth & Decay — What It Is

What It Is

- Exponential functions describe a quantity that grows or shrinks by the same PERCENT every time period.
- The general form is $y = a \cdot b^x$, where x is usually time and y is the changing amount.
- This is different from a linear function, which adds the same fixed amount every step instead of multiplying.
- The SAT tests this both as pure algebra (solving equations with x in the exponent) and as real-world stories.
- Deciding whether a situation is growth or decay is always the first move before writing an equation.

Exponential Functions — Growth & Decay — Core Rules

General Exponential Form

$$y = a \cdot b^x$$

- a is the starting value — the value of y when $x = 0$.
- b is the growth or decay factor — what y gets multiplied by each time x increases by 1.
- If $b > 1$, the function grows; if $0 < b < 1$, the function decays.
- e.g. $y = 5 \cdot 2^x$ starts at 5 and doubles every time x goes up by 1.

Turning a Percent Rate into the Base

Growth: $A = P(1+r)^t$. **Decay:** $A = P(1-r)^t$.

- P is the starting amount, r is the percent rate written as a decimal, and t is the number of time periods.

- For growth (increasing by $r\%$), the base becomes $(1 + r)$.
- For decay (decreasing by $r\%$), the base becomes $(1 - r)$, which is a positive number less than 1.
- e.g. growing 8% a year means $r = 0.08$, so the base is 1.08; decaying 8% a year means the base is 0.92.

Reading the Basic Graph

y-intercept: $(0, a)$. Horizontal asymptote: $y = 0$.

- The graph always crosses the y-axis at $(0, a)$, the starting value.
- The x-axis ($y = 0$) is a horizontal asymptote — the curve gets closer and closer but never actually touches it.
- The domain is all real numbers, but the range is only $y > 0$ when a is positive.
- Because of the asymptote, an exponential function's y-value is never exactly zero.

Solving by Matching Bases

If $a^m = a^n$ (same base a), then $m = n$.

- Rewrite both sides of the equation as powers of the exact same base number.
- Once the bases match exactly, you can drop them and just set the exponents equal to each other.
- e.g. to solve $4^x = 32$, rewrite both sides as powers of 2: $2^{2x} = 2^5$, so $2x = 5$.
- This shortcut only works once both sides truly share the same base — not just similar-looking numbers.

Vertical Shifts Move the Asymptote

$y = b^x + k$

- Adding k at the end of the equation slides the whole graph up (if $k > 0$) or down (if $k < 0$).
- The horizontal asymptote shifts too — from $y = 0$ to $y = k$.
- e.g. $y = 2^x + 3$ has an asymptote at $y = 3$, not the usual $y = 0$.
- This is different from most other shifts you've seen, where the graph's 'floor' doesn't move.

Horizontal Shifts

$y = b^{(x - h)}$

- Subtracting h inside the exponent slides the graph h units to the RIGHT.
- Writing it as $(x + h)$ instead slides the graph h units to the LEFT.
- e.g. $y = 2^{(x-3)}$ looks exactly like $y = 2^x$, just moved 3 units to the right.
- A horizontal shift changes where a given y-value happens, but it does not move the asymptote.

How to Recognize This on the Test

- The problem describes a quantity growing or shrinking by a PERCENTAGE each year, month, hour, or period.

- Keywords like 'doubles,' 'triples,' 'increases by r%,' 'half-life,' or 'decays' show up in the wording.
- A variable sits up in the exponent itself, like 2^x or $5^{(x+1)}$, instead of being squared or cubed.
- You're given an equation shaped like $y = a \cdot b^x$ or $A = P(1+r)^t$ and asked about its behavior.
- A table shows values being MULTIPLIED by the same factor each step, not increased by the same fixed amount.

Common Mistakes — and the Fix

- Writing the decay base as a negative number instead of $(1 - r)$ — a decaying base is always positive and less than 1.
- Forgetting a vertical shift moves the asymptote too, and still assuming the asymptote is $y = 0$ after a shift.
- Adding exponents instead of multiplying them when raising a power to another power while matching bases.
- Mixing up growth and decay from a decimal base — remember $b > 1$ means growth, and $0 < b < 1$ means decay.
- Treating exponential change (multiplying by a constant factor) like linear change (adding a constant amount).

Exponential Functions — Growth & Decay — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Exponential Functions — Growth & Decay — Problem 1 (Easy)

A biology class is tracking a bacteria culture that starts with 400 cells and doubles every hour. The population after t hours is modeled by $P(t) = 400 \cdot 2^t$. How many cells are in the culture after 3 hours?

Formula / Rule Used: Exponential growth model: evaluate $y = a \cdot b^x$ at a given x .

Solution

Step 1: Identify what's given: $a = 400$ (starting cells), $b = 2$ (doubles each hour), and you want $t = 3$.

Step 2: Substitute $t = 3$ into the model: $P(3) = 400 \cdot 2^3$.

Step 3: Evaluate the power first: $2^3 = 2 \times 2 \times 2 = 8$.

Step 4: Multiply the starting value by that result: $P(3) = 400 \cdot 8$.

Step 5: Compute the final product: $400 \cdot 8 = 3200$.

Answer: 3200 cells

Tip

Always evaluate the exponent part first (2^3), then multiply by the starting value — order matters.

Trap



It's tempting to multiply 400 by 3 (the hours) instead of raising 2 to the power of 3 — that turns exponential growth into linear growth.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
1200	Multiplied 400 by the number of hours (3) directly, treating the growth as linear instead of exponential.
800	Doubled the population only once instead of applying the doubling three separate times for $t=3$.
2400	Computed 2^3 as $2 \times 3 = 6$ instead of $2 \times 2 \times 2 = 8$, then multiplied 400×6 .

Shortcut

None needed — plugging $t=3$ directly into the model and evaluating is already the fastest path.

Desmos Tip

Type $y = 400 \cdot 2^x$ into Desmos and trace the graph to $x = 3$ to confirm $y = 3200$.

Exponential Functions — Growth & Decay — Problem 2 (Easy)

A used car in Lisbon is purchased for €18,000. Its value decreases by 10% every year, modeled by $V(t) = 18000(0.9)^t$, where t is the number of years since purchase. What is the car's value after 2 years?

Formula / Rule Used: Percent decay model: $A = P(1-r)^t$.

Solution

Step 1: The base 0.9 already reflects a 10% decrease each year ($1 - 0.10 = 0.9$), so use the formula as given.

Step 2: Substitute $t = 2$ into the model: $V(2) = 18000 \cdot (0.9)^2$.

Step 3: Evaluate the power first: $(0.9)^2 = 0.9 \times 0.9 = 0.81$.

Step 4: Multiply the starting value by that result: $V(2) = 18000 \cdot 0.81$.

Step 5: Compute the final product: $18000 \cdot 0.81 = 14580$.

Answer: €14,580

Tip

Evaluate the decay factor raised to the power first, then multiply by the starting value — never multiply the percent by t directly.

Trap

A common mistake is subtracting 10% of the original price twice in a row instead of applying the compounding decay factor $(0.9)^2$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
€14,400	Subtracted 10% of the original price twice in a row ($18000 - 1800 - 1800$) instead of compounding with $(0.9)^2$.
€16,200	Applied the 10% decrease only once (18000×0.9) instead of squaring the decay factor for 2 years.
€32,400	Added $0.9 + 0.9$ to get 1.8 instead of multiplying 0.9×0.9 for the squared decay factor.

Shortcut

None needed — squaring the decay factor and multiplying by the starting price is already the fastest method.

Desmos Tip

Type $y = 18000 \cdot (0.9)^x$ into Desmos and trace to $x = 2$ to confirm $y = 14580$.

Exponential Functions — Growth & Decay — Problem 3 (Easy)

What value of x satisfies $27^x = 81$?

Formula / Rule Used: Solving exponential equations by matching bases.

Solution

Step 1: Rewrite both sides as powers of the same base. Both 27 and 81 are powers of 3.

Step 2: $27 = 3^3$, so $27^x = (3^3)^x$.

Step 3: Apply the power of a power rule: $(3^3)^x = 3^{(3x)}$.

Step 4: Rewrite 81 as a power of 3 as well: $81 = 3^4$.

Step 5: Now the equation reads $3^{(3x)} = 3^4$. Since the bases match, set the exponents equal: $3x = 4$.

Step 6: Divide both sides by 3 to solve for x : $x = 4/3$.

Answer: $x = 4/3$

Tip

When the bases aren't already the same, look for a smaller common base that both numbers share, like 3 here.

Trap

A common mistake is forgetting that 27 itself equals 3^3 , and setting the exponent x equal to 4 directly.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 4$	Forgot that $27 = 3^3$ and treated 27^x as if it were already 3^x , setting x equal to 4 directly.
$x = 5/4$	Distributed incorrectly, writing the exponent as $4x+1$ instead of $4x+4$ while combining bases.



$x = 3/4$	Mixed up which exponent belongs on top, dividing 3 by 4 instead of 4 by 3.
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Shortcut

None needed — converting both numbers to base 3 is already the most direct way to solve this equation.

Desmos Tip

Type $y = 27^x$ into Desmos and trace to where $y = 81$ to confirm $x = 1.333...$, which matches $4/3$.

Exponential Functions — Growth & Decay — Problem 4 (Easy)

The graph of $y = 3(0.5)^x - 7$ has a horizontal asymptote at $y = k$. What is the value of k ?

Formula / Rule Used: Vertical shift rule: $y = b^x + k$ shifts the asymptote to $y = k$.

Solution

Step 1: The basic graph $y = 3(0.5)^x$, with no addition or subtraction, has a horizontal asymptote at $y = 0$.

Step 2: The '-7' at the end of the equation is a vertical shift applied to the entire graph.

Step 3: A vertical shift moves the asymptote by the same amount as the shift, not just the curve itself.

Step 4: Since the graph is shifted down by 7, the asymptote moves from $y = 0$ to $y = 0 - 7$.

Step 5: So the horizontal asymptote is at $y = -7$.

Answer: $k = -7$

Tip

Ignore the coefficient (3) and the base (0.5) when finding the asymptote — only the added or subtracted constant at the end matters.

Trap

A common mistake is assuming the asymptote is always $y = 0$ no matter what, ignoring the '-7' shift entirely.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 0$	Assumed every exponential graph has asymptote $y = 0$, ignoring the vertical shift of -7.
$k = 7$	Shifted the asymptote in the wrong direction, treating -7 as an upward shift instead of downward.
$k = 3$	Mistook the coefficient 3 for the shift value instead of using the actual constant -7.

Shortcut



None needed — reading off the added or subtracted constant directly gives the asymptote without any graphing.

Desmos Tip

Type $y = 3 \cdot (0.5)^x - 7$ into Desmos and zoom out to watch the curve flatten out and approach $y = -7$.

Exponential Functions — Growth & Decay — Problem 5 (Medium)

A population of fruit flies is modeled by an exponential function $N(t) = N_0 \cdot b^t$, where t is measured in days. At $t = 0$ there are 50 flies, and at $t = 2$ there are 200 flies. Assuming this exponential pattern continues, how many flies will there be at $t = 4$?

Formula / Rule Used: Building an exponential model from two points, then evaluating forward.

Solution

Step 1: At $t = 0$, N_0 is just the starting value: $N(0) = 50$, so $N_0 = 50$.

Step 2: Use the second point to find b : $N(2) = 50 \cdot b^2 = 200$.

Step 3: Divide both sides by 50 to isolate b^2 : $b^2 = 200/50 = 4$.

Step 4: Take the square root of both sides (b must be positive for a real population): $b = 2$.

Step 5: The full model is now $N(t) = 50 \cdot 2^t$. Substitute $t = 4$: $N(4) = 50 \cdot 2^4$.

Step 6: Evaluate the power: $2^4 = 16$.

Step 7: Multiply: $N(4) = 50 \cdot 16 = 800$.

Answer: 800 flies

Tip

When you're given two points instead of the growth factor, find b first by dividing, then build the full equation before evaluating.

Trap

A common mistake is assuming the population keeps adding the same fixed amount (150 flies) instead of continuing to multiply.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
350	Assumed linear growth (adding the same 150-fly increase again) instead of continuing the exponential pattern.
400	Found $b^2 = 4$ correctly but only multiplied 200 by $b = 2$ instead of by $b^2 = 4$ for the jump from $t=2$ to $t=4$.
3200	Used the value at $t=2$ (200) as N_0 by mistake, then computed $200 \cdot 2^4$ instead of $50 \cdot 2^4$.

Shortcut



Once you know $b=2$, jump straight from $N(2)=200$ to $N(4)$ by multiplying by $b^2=4$, without rewriting the full equation.

Desmos Tip

Type $y = 50 \cdot 2^x$ into Desmos, confirm the curve passes through (0,50) and (2,200), then trace to $x=4$ to confirm $y=800$.

Exponential Functions — Growth & Decay — Problem 6 (Medium)

If $16^{(x+1)} = 64$, what is the value of x ?

Formula / Rule Used: Matching bases combined with the power of a power rule.

Solution

Step 1: Rewrite both sides as powers of 2: $16 = 2^4$ and $64 = 2^6$.

Step 2: Substitute these in: $(2^4)^{(x+1)} = 2^6$.

Step 3: Apply the power of a power rule on the left side by multiplying the exponents: $4(x+1) = 4x + 4$.

Step 4: So the equation becomes $2^{(4x+4)} = 2^6$, and since the bases now match, set the exponents equal: $4x + 4 = 6$.

Step 5: Subtract 4 from both sides: $4x = 2$.

Step 6: Divide both sides by 4 to solve for x : $x = 2/4 = 1/2$.

Answer: $x = 1/2$

Tip

Rewrite every number as a power of the smallest common base first (here, 2), before touching the exponents.

Trap

It's easy to distribute the exponent incorrectly, writing $4x+1$ instead of correctly multiplying 4 through both x and 1 to get $4x+4$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 5/4$	Distributed incorrectly, writing the left exponent as $4x+1$ instead of $4x+4$, giving $4x+1=6$.
$x = 3$	Divided 64 by 16 to get 4 and set $x+1 = 4$ directly, instead of rewriting both sides as powers of 2.
$x = 3/2$	Forgot to subtract 4 from both sides before dividing, solving $4x=6$ instead of $4x=6-4=2$.

Shortcut

None needed — converting both sides to base 2 and matching exponents is already the fastest way to solve this type of equation.

Desmos Tip



Type $y = 16^{(x+1)}$ into Desmos and trace to where $y = 64$ to confirm $x = 0.5$.

Exponential Functions — Growth & Decay — Problem 7 (Medium)

A radioactive isotope has a half-life of 5 years, meaning half of it decays away every 5 years. A lab starts with 80 grams of the isotope, and the lab is closed every weekend. Using the model $A(t) = 80 \cdot (1/2)^{(t/5)}$, where t is the number of years since the start, how many grams remain after 15 years?

Formula / Rule Used: Exponential decay using a fractional exponent (half-life model).

Solution

Step 1: The detail about the lab being closed on weekends doesn't affect the model — it's extra information you can ignore.

Step 2: Substitute $t = 15$ into the exponent: $t/5 = 15/5 = 3$.

Step 3: Now the expression is $A(15) = 80 \cdot (1/2)^3$.

Step 4: Evaluate the power: $(1/2)^3 = 1/8$.

Step 5: Multiply: $A(15) = 80 \cdot (1/8)$.

Step 6: Divide 80 by 8: $80/8 = 10$.

Answer: 10 grams

Tip

For half-life problems, first find how many full half-life periods have passed (t divided by 5 here), then use that as the exponent.

Trap

It's tempting to use $t=15$ directly as the exponent instead of dividing by the half-life length of 5 first.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
about 0.0024 grams ($80/2^{15}$)	Used $t=15$ directly as the exponent instead of dividing by the half-life of 5 first.
640 grams	Multiplied 80 by 8 instead of dividing by 8, treating $(1/2)^3$ like a growth factor instead of decay.
about 26.67 grams ($80/3$)	Divided the starting amount by the number of half-lives (3) instead of raising $1/2$ to the 3rd power.

Shortcut

You can also just halve 80 three times in a row (80 to 40 to 20 to 10) since 15 years is exactly 3 half-lives.

Desmos Tip

Type $y = 80 \cdot (0.5)^{(x/5)}$ into Desmos and trace to $x = 15$ to confirm $y = 10$.

Exponential Functions — Growth & Decay — Problem 8 (Medium)

A rare stamp's value is modeled by $V(t) = V_0 \cdot (1.12)^t$, where t is the number of years since it was purchased. Based on this model, by what percent does the stamp's value increase every 2 years?

Formula / Rule Used: Converting a yearly growth base into a combined percent over multiple periods.

Solution

Step 1: The base 1.12 means the value is multiplied by 1.12 every single year (a 12% increase per year).

Step 2: To find the growth over 2 years, apply the yearly factor twice: multiply 1.12 by itself, $(1.12)^2$.

Step 3: Compute $(1.12)^2 = 1.12 \times 1.12 = 1.2544$.

Step 4: This means every 2 years, the value is multiplied by 1.2544.

Step 5: Subtract 1 to isolate just the increase part: $1.2544 - 1 = 0.2544$.

Step 6: Convert the decimal to a percent: $0.2544 = 25.44\%$.

Answer: 25.44%

Tip

To find growth over multiple periods, raise the yearly factor to that many periods first, then subtract 1 to isolate the percent increase.

Trap

A common mistake is just doubling the yearly percent ($12\% \times 2 = 24\%$) instead of properly compounding it with $(1.12)^2$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
24%	Doubled the yearly percent rate ($12\% \times 2$) instead of compounding the growth factor with $(1.12)^2$.
125.44%	Forgot to subtract 1 from $(1.12)^2 = 1.2544$ before converting to a percent, reporting the whole multiplier as the increase.
13.44%	Squared only the percent part ($0.12^2 = 0.0144$) and added it to 12%, instead of squaring the full growth factor 1.12.

Shortcut

None needed — squaring the yearly growth factor and subtracting 1 is already the fastest way to find a multi-year percent change.

Desmos Tip

Type $y = (1.12)^2$ into Desmos to confirm it equals 1.2544, confirming a 25.44% increase over 2 years.

24. Radical & Rational Equations

Radical & Rational Equations — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Radical equation	Ριζική εξίσωση
Rational equation	Ρητή εξίσωση
Extraneous solution	Ξένη (μη αποδεκτή) λύση
Isolate the radical	Απομόνωση ριζικού
Square both sides	Ύψωση στο τετράγωνο και των δύο μελών
Verify / check solutions	Επαλήθευση λύσεων

Radical & Rational Equations — What It Is

What It Is

- A radical equation has the variable trapped under a square root, cube root, or other root symbol.
- A rational equation has the variable sitting inside a fraction, specifically in the denominator.
- The SAT groups these together because both require the same first move: clear the root or the fraction.
- That clearing step can accidentally create fake solutions, so checking your work matters more here than usual.
- Once the root or fraction is gone, what's left is just a normal equation you already know how to solve.

Radical & Rational Equations — Core Rules

Isolate the Radical First

- Before doing anything else, get the radical completely alone on one side of the equation.
- Move every other term to the opposite side using addition or subtraction.
- Example: in $\sqrt{x} + 4 = 10$, subtract 4 from both sides to get $\sqrt{x} = 6$.
- Squaring too early, before isolating, creates a much messier equation to untangle.

Raise Both Sides to Match the Index

If n -th root of $x = a$, then $x = a^n$

- Once the radical is alone, raise both sides to the power matching the root's index.
- A square root needs both sides squared; a cube root needs both sides cubed.
- This power undoes the root, turning it back into a normal, root-free equation.



- Example: if $\sqrt{x} = 6$, squaring both sides gives $x = 36$.

Always Check for Extraneous Solutions

- Raising both sides to an EVEN power can create a solution that doesn't actually work.
- This happens because squaring turns both a negative and a positive result into the same number.
- Plug every solution back into the ORIGINAL equation, before any squaring, to check it.
- Cube roots (an odd index) never cause this problem, so checking matters most for square roots.

Equations with Two Radicals

- Isolate one radical on one side of the equation, leaving the other radical on the other side.
- Square both sides once to eliminate that first radical.
- If a radical still remains afterward, isolate it too and square both sides again.
- Always finish by checking the solution back in the full original two-radical equation.

Clear a Rational Equation's Denominators

Multiply every term by the LCD

- Find the least common denominator, or LCD, of every fraction in the equation.
- Multiply every single term, on both sides of the equation, by that LCD.
- This cancels all the denominators at once, leaving a normal polynomial equation.
- Example: for $x/2 + 1 = 3/4$, the LCD is 4, so multiply every term by 4.

Extraneous Solutions in Rational Equations

- A solution is extraneous if it makes ANY original denominator equal to zero.
- Dividing by zero is undefined, so that value can never be a valid final answer.
- Check your solution against every denominator in the original equation, not just the obvious one.
- If a solution zeroes out a denominator, throw it out and keep only the remaining solution(s).

Complex Fractions

- A complex fraction has a smaller fraction stacked inside its own numerator or denominator.
- Multiply the entire complex fraction, top and bottom, by the LCD of all its inner fractions.
- This clears every inner fraction in one single step, instead of simplifying piece by piece.
- What remains is one ordinary, simplified fraction with no fractions left inside it.

How to Recognize This on the Test

- The variable sits directly under a square root, cube root, or other radical symbol.

- The variable appears in the bottom (denominator) of a fraction somewhere in the equation.
- The problem says 'solve for x' next to an equation containing a root or a fraction with x below the line.
- The question explicitly asks you to identify or explain an 'extraneous solution.'
- Two separate fractions with variable denominators are set equal to each other.

Common Mistakes — and the Fix

- Squaring both sides before fully isolating the radical, which creates messy extra terms.
- Forgetting to check solutions in the original equation, missing a hidden extraneous solution.
- Multiplying only some terms by the LCD instead of every single term on both sides.
- Checking only the most obvious denominator instead of every denominator in the equation.
- Treating $\sqrt{x + y}$ as $\sqrt{x} + \sqrt{y}$ — a root never splits apart over addition like that.

Radical & Rational Equations — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Radical & Rational Equations — Problem 1 (Easy)

Solve for x: $\sqrt{2x + 7} = 9$.

Formula / Rule Used: Isolate the radical, then square both sides to undo it.

Solution

Step 1: The radical is already alone on the left side of the equation.

Step 2: Square both sides to undo the square root: $(\sqrt{2x + 7})^2 = 9^2$.

Step 3: The left side simplifies to just $2x + 7$.

Step 4: The right side simplifies to 81, so the equation becomes $2x + 7 = 81$.

Step 5: Subtract 7 from both sides: $2x = 74$.

Step 6: Divide both sides by 2 to solve for x: $x = 37$.

Step 7: Check by substituting back in: $\sqrt{2(37) + 7} = \sqrt{81} = 9$, which matches, so $x = 37$ is valid.

Answer: $x = 37$

Tip

When the radical is already alone on one side, square both sides immediately to remove it.

Trap

It's easy to square the 9 correctly but forget the entire left side becomes $2x + 7$, not just x.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



$x = 1$	Set $2x + 7$ equal to 9 directly instead of squaring both sides first, skipping the key step entirely.
$x = 44$	Added 7 to 81 instead of subtracting, getting $2x = 88$ instead of $2x = 74$.
$x = 74$	Correctly solved $2x = 74$ but forgot the final step of dividing both sides by 2.

Shortcut

None needed — squaring both sides right away is already the fastest way to solve an isolated radical equation.

Desmos Tip

Graph $y = \sqrt{2x + 7}$ and $y = 9$ in Desmos; the x-value where the two graphs cross confirms $x = 37$.

Radical & Rational Equations — Problem 2 (Easy)

A ceramics studio in Thessaloniki cuts square floor tiles. The side length s , in centimeters, of a tile with area A is given by $s = \sqrt{A}$. A tile has area $A = x + 9$, and its measured side length is 6 centimeters. What is the value of x ?

Formula / Rule Used: Substitute into the radical formula, then square both sides to undo the root.

Solution

Step 1: Substitute the given values into the formula: $6 = \sqrt{x + 9}$.

Step 2: The radical is already alone on the right side of the equation.

Step 3: Square both sides to undo the square root: $6^2 = (\sqrt{x + 9})^2$.

Step 4: The left side simplifies to 36, and the right side simplifies to $x + 9$.

Step 5: So the equation becomes $36 = x + 9$.

Step 6: Subtract 9 from both sides to solve for x : $x = 27$.

Step 7: Check: $\sqrt{27 + 9} = \sqrt{36} = 6$, which matches the given side length, so $x = 27$ is valid.

Answer: $x = 27$

Tip

Substitute the known values into the formula first, then treat it like any other radical equation.

Trap

It's tempting to square 6 and 9 separately and combine them, instead of squaring 6 and then subtracting 9 in order.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 45$	Added 9 to 36 instead of subtracting it, reversing the final step.
$x = 3$	Took the square root of 9 instead of substituting and squaring the given side length 6.



$x = 21$	Miscalculated 6^2 as 30 instead of 36 through a multiplication slip, then subtracted 9.
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Shortcut

None needed — substituting into the formula and squaring immediately is already the fastest method.

Desmos Tip

Type $\sqrt{x + 9}$ in Desmos and trace the graph to where $x = 27$ to confirm the output reaches 6.

Radical & Rational Equations — Problem 3 (Easy)

Solve for x : $6/x + 2 = 5$.

Formula / Rule Used: Isolate the fraction, then multiply both sides by x to clear the denominator.

Solution

Step 1: Subtract 2 from both sides to isolate the fraction: $6/x = 3$.

Step 2: Multiply both sides by x to clear the denominator: $6 = 3x$.

Step 3: Divide both sides by 3 to solve for x : $x = 2$.

Step 4: Check that $x = 2$ doesn't make the original denominator zero — it doesn't, so it's allowed.

Step 5: Check by substituting back in: $6/2 + 2 = 3 + 2 = 5$, which matches, so $x = 2$ is valid.

Answer: $x = 2$

Tip

Isolate the fraction by itself first, then multiply both sides by the denominator to clear it.

Trap

Multiplying by x before isolating the fraction forces you to distribute across every term, inviting mistakes.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 18$	Correctly found $6/x = 3$ but then multiplied both sides by 3 instead of dividing, turning $6 = 3x$ into $x = 18$.
$x = 6/7$	Added 2 to 5 instead of subtracting, getting $6/x = 7$ instead of $6/x = 3$.
$x = 3$	Matched x directly to the isolated value 3 instead of multiplying both sides by x first.

Shortcut

None needed — isolating the fraction before clearing the denominator is already the fastest approach here.

Desmos Tip

Type $6/x + 2$ in Desmos and trace the graph to where it equals 5 to confirm $x = 2$.

Radical & Rational Equations — Problem 4 (Easy)

A bakery in Vienna tracks its average cost per pastry using the formula $c = 180/n$, where c is the cost in euros and n is the number of pastries baked. On a day when the average cost was 3.60 euros per pastry, how many pastries did the bakery bake?

Formula / Rule Used: Substitute the known value, then multiply both sides by n to clear the denominator.

Solution

Step 1: Substitute the known cost into the formula: $3.60 = 180/n$.

Step 2: Multiply both sides by n to clear the denominator: $3.60n = 180$.

Step 3: Divide both sides by 3.60 to solve for n : $n = 180 / 3.60$.

Step 4: Carry out the division: $n = 50$.

Step 5: Check that $n = 50$ doesn't make the original denominator zero — it doesn't, so it's allowed.

Step 6: Check by substituting back in: $180/50 = 3.60$, which matches, so $n = 50$ is valid.

Answer: $n = 50$ pastries

Tip

Substitute the known value into the rational formula first, then multiply both sides by the variable denominator.

Trap

Dividing 180 by n before clearing the denominator can tempt you to divide the wrong two numbers together.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$n = 648$	Multiplied 180 by 3.60 instead of dividing, reversing the final operation.
$n = 176.4$	Subtracted 3.60 from 180 instead of dividing to isolate n .
$n = 5$	Misplaced the decimal point while dividing 180 by 3.60, getting a tenth of the correct value.

Shortcut

Recognizing $180 / 3.6$ is the same as $1800 / 36 = 50$ makes the division friendlier without a calculator.

Desmos Tip

Type $180/3.6$ directly into Desmos to confirm it evaluates to 50.

Radical & Rational Equations — Problem 5 (Medium)

Solve for x : $\sqrt{x + 6} = x$.

Formula / Rule Used: Square both sides to eliminate the radical, then check every solution in the original equation.

Solution

Step 1: The radical is already alone on the left side of the equation.

Step 2: Square both sides: $(\sqrt{x + 6})^2 = x^2$.

Step 3: The left side simplifies to $x + 6$, so the equation becomes $x + 6 = x^2$.

Step 4: Rearrange into standard form by subtracting x and 6 from both sides: $0 = x^2 - x - 6$.

Step 5: Factor the quadratic: $x^2 - x - 6 = (x - 3)(x + 2)$.

Step 6: Set each factor equal to zero: $x - 3 = 0$ or $x + 2 = 0$, giving $x = 3$ or $x = -2$.

Step 7: Check $x = 3$ in the original equation: $\sqrt{3 + 6} = \sqrt{9} = 3$, which matches.

Step 8: Check $x = -2$ in the original equation: $\sqrt{-2 + 6} = \sqrt{4} = 2$, but 2 does not equal -2 , so $x = -2$ is extraneous and must be rejected.

Answer: $x = 3$ ($x = -2$ is extraneous and rejected)

Tip

Whenever squaring a radical equation produces a quadratic, expect two candidate answers and check both in the original equation.

Trap

It's tempting to report both factored solutions without checking that one of them fails in the original equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 3$ and $x = -2$	Solved the quadratic correctly but skipped checking each solution in the original equation, keeping the extraneous root.
$x = -2$	Checked the solutions but mixed up which one satisfies the original equation, keeping the wrong root.
$x = 2$ or $x = -3$	Made a sign error while factoring $x^2 - x - 6$, reversing the signs inside the two factors.

Shortcut

None needed — squaring, factoring, and checking both roots is already the fastest reliable path for this equation type.

Desmos Tip

Graph $y = \sqrt{x + 6}$ and $y = x$ in Desmos; the graphs only cross once, at $x = 3$, confirming the other algebraic root is extraneous.

Solve for x : $\sqrt{x+9} - \sqrt{x} = 1$.

Formula / Rule Used: Isolate one radical, square both sides, then isolate and square again if needed.

Solution

Step 1: Isolate one radical by adding $\sqrt{x}$ to both sides: $\sqrt{x+9} = 1 + \sqrt{x}$.

Step 2: Square both sides: $(\sqrt{x+9})^2 = (1 + \sqrt{x})^2$.

Step 3: The left side simplifies to $x + 9$, and the right side expands to $1 + 2\sqrt{x} + x$.

Step 4: So the equation becomes $x + 9 = 1 + 2\sqrt{x} + x$.

Step 5: Subtract x and 1 from both sides: $8 = 2\sqrt{x}$.

Step 6: Divide both sides by 2: $\sqrt{x} = 4$.

Step 7: Square both sides again to finish solving: $x = 16$.

Step 8: Check: $\sqrt{16+9} - \sqrt{16} = \sqrt{25} - 4 = 5 - 4 = 1$, which matches, so $x = 16$ is valid.

Answer: $x = 16$

Tip

When two radicals are subtracted, isolate one first, square once, then isolate and square the remaining radical.

Trap

Forgetting the middle term when expanding $(1 + \sqrt{x})^2$ drops the $2\sqrt{x}$ piece and breaks the rest of the equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 4$	Stopped after finding $\sqrt{x} = 4$ and reported that value instead of squaring again to solve for x .
$x = 64$	Expanded $(1 + \sqrt{x})^2$ as $1 + x$ only, skipping the $2\sqrt{x}$ term, which shifted every later step.
$x = 25$	Mistakenly used the 5 from the check step as the final answer instead of continuing from $\sqrt{x} = 4$.

Shortcut

None truly faster — isolating one radical at a time and squaring twice is the reliable path whenever two radicals appear.

Desmos Tip

Graph $y = \sqrt{x+9} - \sqrt{x}$ and $y = 1$ in Desmos; the x -coordinate where they cross confirms $x = 16$.

A shipping company in Rotterdam, founded in 1998, uses the equation $120/x + 3 = 120/(x - 2)$ to model its staffing needs, where x is the number of workers assigned to Warehouse A. What is the value of x ?

Formula / Rule Used: Clear both denominators using the LCD, then solve the resulting quadratic.

Solution

Step 1: Ignore the detail about the company being founded in 1998 — it isn't needed to solve the equation.

Step 2: Multiply every term by the LCD, $x(x - 2)$, to clear both denominators: $120(x - 2) + 3x(x - 2) = 120x$.

Step 3: Expand each product: $120x - 240 + 3x^2 - 6x = 120x$.

Step 4: Combine like terms on the left side: $3x^2 + 114x - 240 = 120x$.

Step 5: Subtract $120x$ from both sides and simplify: $3x^2 - 6x - 240 = 0$.

Step 6: Divide every term by 3 to simplify: $x^2 - 2x - 80 = 0$.

Step 7: Factor the quadratic: $(x - 10)(x + 8) = 0$, giving $x = 10$ or $x = -8$.

Step 8: Since x counts a number of workers, it cannot be negative, so reject $x = -8$ and keep $x = 10$.

Answer: $x = 10$ workers

Tip

Multiply every term by the LCD right away to turn a two-denominator rational equation into a normal quadratic.

Trap

Ignoring the real-world constraint and reporting both $x = 10$ and $x = -8$ gives an impossible negative number of workers.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 10$ or $x = -8$	Solved the quadratic correctly but forgot to reject the negative root using the real-world context.
$x = -8$	Picked the wrong factor's root instead of checking which value makes sense for a number of workers.
$x = 8$	Combined the $120x$ and $-6x$ terms incorrectly while simplifying, leading to a shifted quadratic and a wrong root.

Shortcut

None faster here — clearing both denominators with the LCD is required before this quadratic can be solved.

Desmos Tip

Type $120/x + 3$ and $120/(x - 2)$ as two separate lines in Desmos; the crossing point at a positive x confirms $x = 10$.

If $(3 + 1/x) / (1 - 1/x) = 2$, what is the value of x ?

Formula / Rule Used: Multiply the top and bottom of the complex fraction by the LCD of its inner fractions.

Solution

Step 1: The inner fractions each have denominator x , so the LCD of the complex fraction is x .

Step 2: Multiply every term in the numerator and denominator by x : $(3x + 1) / (x - 1) = 2$.

Step 3: Cross-multiply to clear the remaining fraction: $3x + 1 = 2(x - 1)$.

Step 4: Distribute the 2 on the right side: $3x + 1 = 2x - 2$.

Step 5: Subtract $2x$ from both sides: $x + 1 = -2$.

Step 6: Subtract 1 from both sides: $x = -3$.

Step 7: Check that $x = -3$ doesn't make any original denominator zero — it doesn't, so it's allowed.

Step 8: Check by substituting back in: $(3 + 1/(-3)) / (1 - 1/(-3)) = (8/3) / (4/3) = 2$, which matches, so $x = -3$ is valid.

Answer: $x = -3$

Tip

Clear a complex fraction first by multiplying every piece by the LCD of its small inner fractions.

Trap

Cross-multiplying before clearing the inner fractions mixes up the small fractions with the outer equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 3$	Made a sign error while solving $x + 1 = -2$, treating it as $x = 2 - 1$ instead of $x = -2 - 1$.
$x = -1/3$	Solved for $1/x$ directly from the simplified equation instead of continuing to solve for x itself.
$x = 1$	Cross-multiplied the original complex fraction directly without first clearing the inner $1/x$ terms.

Shortcut

None faster — clearing the inner fractions with the LCD before cross-multiplying is the reliable method here.

Desmos Tip

Type $(3 + 1/x)/(1 - 1/x)$ in Desmos and trace the graph to where $x = -3$ to confirm the output equals 2.

PART 4 — GEOMETRY AND TRIGONOMETRY

25. Angles, Parallel Lines & Triangle Fundamentals

Angles, Parallel Lines & Triangle Fundamentals — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Angle	Γωνία
Complementary / supplementary angles	Συμπληρωματικές / παραπληρωματικές γωνίες
Vertical angles	Κατακορυφήν γωνίες
Parallel lines / transversal	Παράλληλες ευθείες / τέμνουσα
Triangle	Τρίγωνο
Isosceles / equilateral / scalene	Ισοσκελές / ισόπλευρο / σκαληνό
Exterior angle	Εξωτερική γωνία

Angles, Parallel Lines & Triangle Fundamentals — What It Is

What It Is

- An angle measures how much two rays turn away from each other at a shared starting point.
- Angles are measured in degrees, where a full circle around a point is 360 degrees.
- A triangle is the simplest closed shape, and its angle and side rules never change, no matter the size.
- The SAT tests these rules because they let you find unknown angles and sides without ever measuring anything.
- Parallel lines crossed by a third line create a repeating, predictable pattern of equal and supplementary angles.

Angles, Parallel Lines & Triangle Fundamentals — Core Rules

Vertical, Complementary & Supplementary Angles

Vertical angles: equal. Complementary: sum = 90°. Supplementary: sum = 180°.

- Vertical angles are the pair of opposite angles formed when two straight lines cross.
- Vertical angles are always exactly equal, no matter how large or small the angle is.
- Complementary angles are two angles that add up to exactly 90 degrees.
- Supplementary angles are two angles that add up to exactly 180 degrees, like angles along a straight

line.

- e.g. if one angle on a line measures 65° , the angle next to it is $180 - 65 = 115^\circ$.

Parallel Lines Cut by a Transversal

- A transversal is a single line that crosses two parallel lines, creating eight angles total.
- Corresponding angles sit in the matching corner position at each crossing, and they are always equal.
- Alternate interior angles sit between the two parallel lines, on opposite sides of the transversal, and are equal.
- Alternate exterior angles sit outside the two parallel lines, on opposite sides, and are also equal.
- Same-side interior angles are the one pair that are supplementary (sum to 180°), not equal.

Triangle Angle Sum

$$A + B + C = 180^\circ$$

- Every triangle's three interior angles add up to exactly 180 degrees, with no exceptions.
- Knowing any two angles instantly gives you the third one just by subtracting from 180.
- e.g. if two angles are 50° and 70° , the third angle is $180 - 50 - 70 = 60^\circ$.

Classifying Triangles

- By angle: acute (all angles under 90°), right (exactly one 90° angle), obtuse (one angle over 90°).
- By side: equilateral (all three sides equal), isosceles (exactly two sides equal), scalene (no sides equal).
- In an isosceles triangle, the two base angles (opposite the equal sides) are equal to each other too.
- An equilateral triangle is always equiangular as well, with all three angles equal to 60° .

Triangle Inequality Theorem

$$a + b > c \text{ (true for every pair of sides)}$$

- The two shorter sides added together must be greater than the longest side.
- If this fails for any pair of sides, those three lengths cannot actually form a triangle.
- Use this rule to find the possible RANGE for a missing side, not just one value.
- e.g. sides of 4 and 9 mean the third side must be more than 5 and less than 13.

Side-Angle Relationship

- The largest angle in a triangle is always opposite the longest side.
- The smallest angle in a triangle is always opposite the shortest side.
- This lets you compare or order sides and angles without calculating any exact values.

Exterior Angle Theorem

Exterior angle = sum of the two remote interior angles

- An exterior angle is formed by extending one side of a triangle past a vertex.
- It always equals the sum of the two interior angles that are NOT next to it.
- e.g. if the two remote interior angles are 40° and 65° , the exterior angle is 105° .
- This is a shortcut that skips finding the third interior angle first.

How to Recognize This on the Test

- Two parallel lines drawn with a third line crossing them signal the transversal angle rules.
- Tick marks on sides or arc marks on angles in a figure signal equal sides or equal angles.
- A side of a triangle extended past a vertex points straight to the Exterior Angle Theorem.
- A question asking for a 'possible value' or 'range' of a side length points to the Triangle Inequality.
- Any question giving degree measures inside a triangle almost always needs the 180° angle sum somewhere.

Common Mistakes — and the Fix

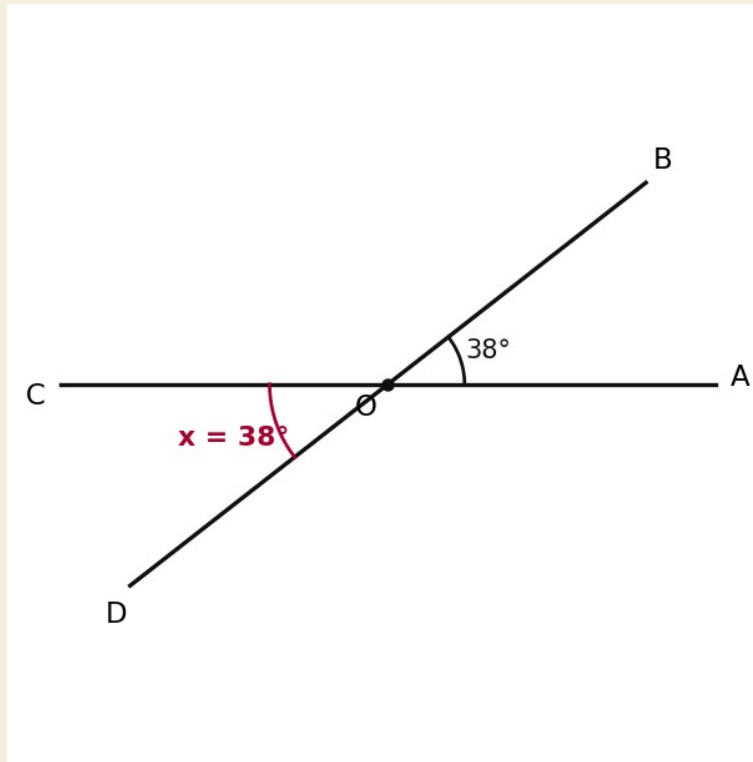
- Assuming all eight angles at a transversal crossing are equal, instead of checking which pairs are supplementary.
- Confusing corresponding angles (equal) with same-side interior angles (supplementary) — they look similar but aren't the same.
- Using only one bound of the Triangle Inequality Theorem instead of checking both the upper and lower bound.
- Mixing up which side is opposite which angle when comparing triangle side and angle sizes.
- Stopping after finding two angles of a triangle without subtracting from 180° to get the third.

Angles, Parallel Lines & Triangle Fundamentals — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Angles, Parallel Lines & Triangle Fundamentals — Problem 1 (Easy)

Two straight lines cross at point O, forming four angles. One of the four angles measures 38 degrees. Directly across from it, on the opposite side of point O, is angle x, formed by the other two rays. What is the measure of angle x?



Formula / Rule Used: Vertical angles are equal.

Solution

Step 1: When two straight lines cross, they form two pairs of angles directly across from each other at the crossing point.

Step 2: These opposite pairs are called vertical angles.

Step 3: Vertical angles are always exactly equal to each other, no matter how the lines are tilted.

Step 4: Angle x is directly across from the 38 degree angle, so it is a vertical angle to it.

Step 5: Since vertical angles are equal, $x = 38$ degrees.

Answer: 38 degrees

Tip

Whenever two lines cross, spot the angle directly opposite the one you're given — it's automatically equal, no calculation needed.

Trap

It's tempting to assume the angle next to the 38 degree angle (not across from it) is also 38 degrees, but adjacent angles here are supplementary, not equal.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong

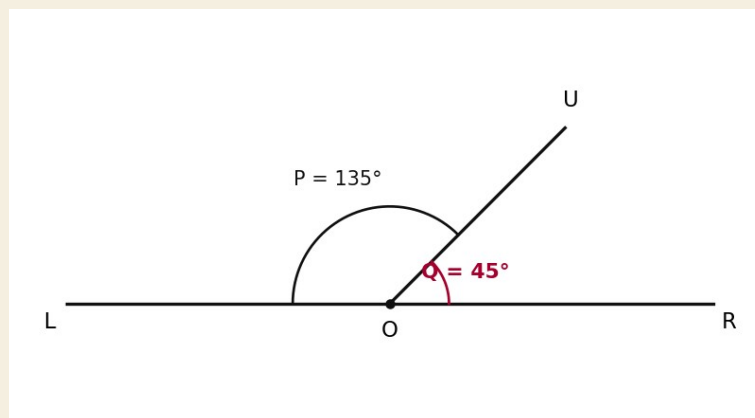
142 degrees	Found the supplementary angle next to the 38 degree angle ($180 - 38$) instead of the vertical angle across from it.
90 degrees	Assumed the two crossing lines must be perpendicular without any evidence in the problem.
52 degrees	Tried to use complementary angles ($90 - 38$) even though nothing in the figure suggests a 90 degree total.

Shortcut

None needed — vertical angles are simply equal by definition, so $x = 38$ degrees immediately.

Angles, Parallel Lines & Triangle Fundamentals — Problem 2 (Easy)

Angle P and angle Q are supplementary. Angle P measures 3 times as many degrees as angle Q. What is the measure of angle Q?



Formula / Rule Used: Supplementary angles: sum = 180 degrees.

Solution

Step 1: Supplementary angles are two angles whose measures add up to exactly 180 degrees.

Step 2: Let angle Q = q degrees. Since angle P is 3 times as large, angle P = $3q$ degrees.

Step 3: Write the supplementary equation: $q + 3q = 180$.

Step 4: Combine the like terms on the left: $4q = 180$.

Step 5: Divide both sides by 4 to solve for q : $q = 180 / 4$.

Step 6: $q = 45$.

Answer: 45 degrees

Tip

When a problem describes angles with words like 'supplementary' or 'complementary,' translate it into an equation first, then use the given ratio to write one variable in terms of the other.

Trap



A common mistake is writing angle P as $3q$ and forgetting to include the original q term when combining, i.e. treating the equation as just $3q = 180$.

Process of Elimination

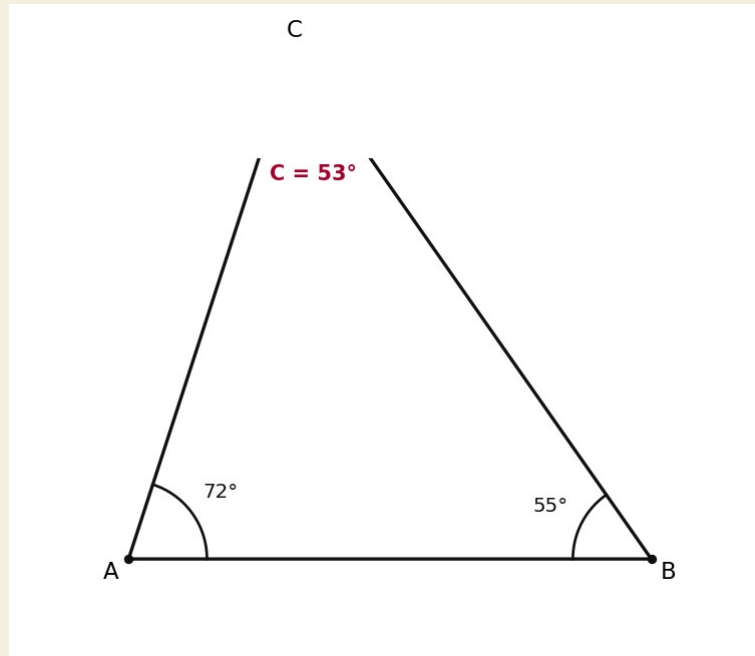
Tempting Wrong Answer	Exactly What Went Wrong
60 degrees	Set up $3q = 180$ by forgetting angle Q's own share, solving $q = 60$ instead of $q = 45$.
135 degrees	Solved correctly for $q = 45$, then reported angle P ($3 \times 45 = 135$) instead of the angle the question asked for, angle Q.
22.5 degrees	Used 90 degrees (complementary) instead of 180 degrees (supplementary) as the total, solving $4q = 90$.

Shortcut

Since P is 3 times Q, Q is 1 out of 4 total 'parts' of 180 degrees, so $Q = 180 / 4 = 45$ directly.

Angles, Parallel Lines & Triangle Fundamentals — Problem 3 (Easy)

In triangle ABC, angle A measures 72 degrees and angle B measures 55 degrees. What is the measure of angle C?



Formula / Rule Used: Triangle angle sum: $A + B + C = 180$ degrees.

Solution

Step 1: Every triangle's three interior angles add up to exactly 180 degrees.

Step 2: Write the equation: $72 + 55 + C = 180$.

Step 3: Add the two known angles first: $72 + 55 = 127$.

Step 4: So the equation becomes $127 + C = 180$.

Step 5: Subtract 127 from both sides to solve for C: $C = 180 - 127$.

Step 6: $C = 53$.

Answer: 53 degrees

Tip

Whenever you know two angles of a triangle, add them together and subtract from 180 to get the third instantly.

Trap

It's easy to subtract only one of the two given angles from 180 and forget to subtract the second one too.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
108 degrees	Subtracted only angle A from 180 and forgot angle B: $180 - 72 = 108$.
125 degrees	Subtracted only angle B from 180 and forgot angle A: $180 - 55 = 125$.
37 degrees	Added 72 and 55 incorrectly as 143 instead of 127, then subtracted from

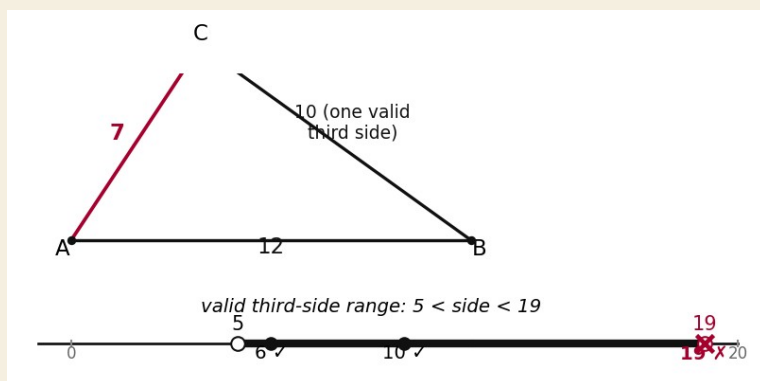
180.

Shortcut

No shortcut needed — adding the two given angles and subtracting from 180 is already the fastest method.

Angles, Parallel Lines & Triangle Fundamentals — Problem 4 (Easy)

A triangle has sides of length 7 and 12. Which of the following could NOT be the length of the third side: 6, 10, or 19?



Formula / Rule Used: Triangle Inequality Theorem: the sum of any two sides must exceed the third side.

Solution

Step 1: The Triangle Inequality Theorem says the sum of the two shorter sides must be greater than the longest side.

Step 2: The third side must satisfy: greater than $12 - 7 = 5$, and less than $12 + 7 = 19$.

Step 3: So the valid range for the third side is between 5 and 19, not including 5 or 19 themselves.

Step 4: Check 6: it is between 5 and 19, so 6 works.

Step 5: Check 10: it is between 5 and 19, so 10 works.

Step 6: Check 19: it is NOT less than 19 (it equals 19), so 19 does not work.

Answer: 19

Tip

Build the full range first (larger side minus smaller side, up to larger side plus smaller side), then test each choice against both ends.

Trap

It's tempting to only check whether a value is small enough and forget that a value can also fail by being too large or exactly equal to the upper bound.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong

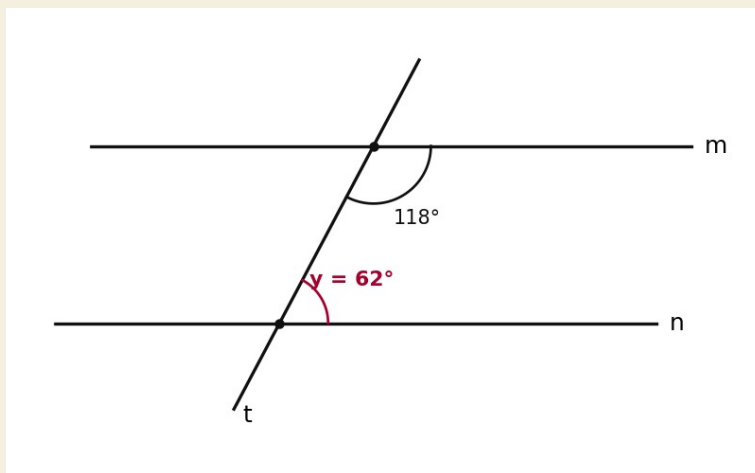
6	Miscalculated the lower bound as $12 - 7 = 5$ but then treated 6 as too close and excluded it anyway, without checking that $6 > 5$ is true.
10	Assumed only very small values fail the theorem and mistakenly flagged a valid middle value.
Both 6 and 19	Forgot that the range is strict but open only at the exact boundary, and mistakenly excluded 6 along with 19.

Shortcut

Just compute $12 - 7 = 5$ and $12 + 7 = 19$, then instantly spot that 19 sits exactly on the excluded upper boundary.

Angles, Parallel Lines & Triangle Fundamentals — Problem 5 (Medium)

Line m and line n are parallel, and both are crossed by a transversal line t . At the crossing with line m , one of the angles formed measures 118 degrees. Angle y is the same-side interior angle to it at the crossing with line n (angle y is between lines m and n , on the same side of the transversal as the 118 degree angle). What is the measure of angle y ?



Formula / Rule Used: Same-side interior angles are supplementary (sum to 180 degrees).

Solution

Step 1: A transversal crossing two parallel lines creates several angle pairs with predictable relationships.

Step 2: Same-side interior angles are the pair that lies between the two parallel lines, on the same side of the transversal.

Step 3: Unlike corresponding or alternate angles, same-side interior angles are NOT equal — they are supplementary.

Step 4: Set up the supplementary equation: $118 + y = 180$.

Step 5: Subtract 118 from both sides to solve for y : $y = 180 - 118$.

Step 6: $y = 62$ degrees.

Answer: 62 degrees

**Tip**

Before you solve, always identify exactly which of the eight transversal angles you're comparing — same-side interior angles are the one pair that's supplementary, not equal.

Trap

This problem is built to tempt you into treating angle y as equal to 118 degrees, the way corresponding or alternate interior angles would be.

Process of Elimination

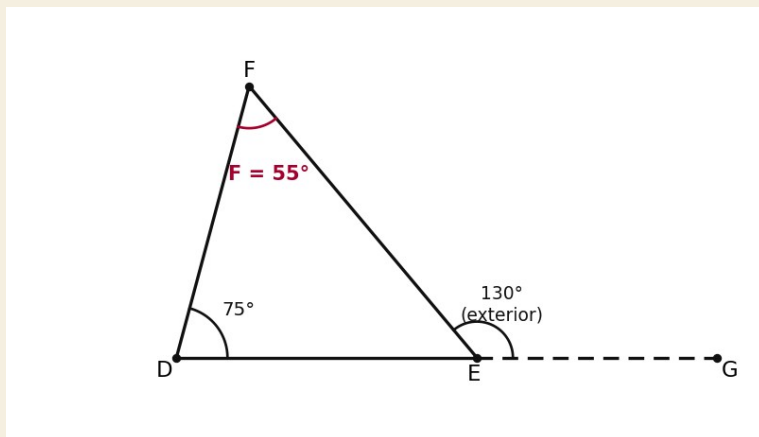
Tempting Wrong Answer	Exactly What Went Wrong
118 degrees	Treated the same-side interior pair as equal, like corresponding angles, instead of supplementary.
72 degrees	Used complementary angles ($90 - 118$ mishandled) instead of the correct supplementary relationship.
242 degrees	Added $118 + 180 - 56$ style arithmetic error instead of correctly subtracting 118 from 180.

Shortcut

Since same-side interior angles always sum to 180, subtract the given angle from 180 in one step: $180 - 118 = 62$.

Angles, Parallel Lines & Triangle Fundamentals — Problem 6 (Medium)

In triangle DEF, side DE is extended past point E to a new point G, creating exterior angle DEG which measures 130 degrees. Inside the triangle, angle D measures 75 degrees. What is the measure of angle F?



Formula / Rule Used: Exterior Angle Theorem: exterior angle = sum of the two remote interior angles.

Solution

Step 1: The exterior angle DEG is formed by extending side DE past vertex E.

Step 2: The Exterior Angle Theorem says this exterior angle equals the sum of the two interior angles that are NOT next to it, which are angle D and angle F.

Step 3: Write the equation: angle D + angle F = 130.

Step 4: Substitute the known angle D = 75: $75 + \text{angle F} = 130$.

Step 5: Subtract 75 from both sides: angle F = $130 - 75$.

Step 6: angle F = 55 degrees.

Answer: 55 degrees

Tip

When you see a side extended past a vertex, jump straight to the Exterior Angle Theorem instead of first finding the triangle's interior angle at that vertex.

Trap

It's tempting to treat the 130 degree exterior angle as the third interior angle of the triangle and just subtract it along with angle D from 180, which skips a step incorrectly.

Process of Elimination

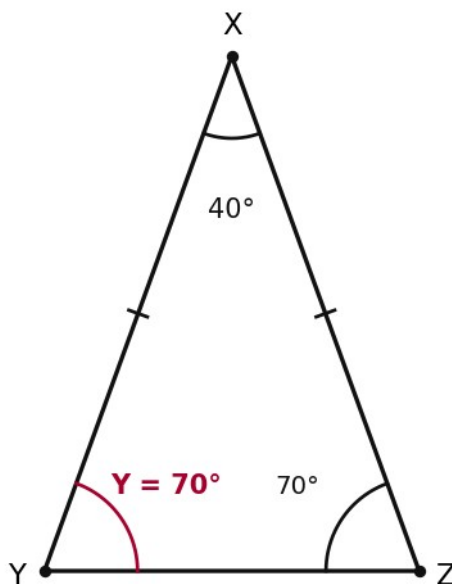
Tempting Wrong Answer	Exactly What Went Wrong
5 degrees	Wrongly treated 130 as an interior angle and computed $180 - 75 - 130$ - (extra adjustment), landing far off after a sign mistake.
50 degrees	Used the interior angle adjacent to the exterior angle ($180 - 130 = 50$) as if it were angle F, instead of applying the exterior angle theorem correctly.
205 degrees	Added 130 and 75 instead of subtracting, reversing the theorem's setup.

Shortcut

Skip finding the interior angle at E first — exterior angle minus the one given remote angle gives angle F directly:
 $130 - 75 = 55$.

Angles, Parallel Lines & Triangle Fundamentals — Problem 7 (Medium)

Triangle XYZ is isosceles, with side XY equal in length to side XZ. Angle X, the angle between the two equal sides, measures 40 degrees. What is the measure of angle Y, one of the two base angles?



Formula / Rule Used: Isosceles triangle base angles are equal, plus the triangle angle sum.

Solution

Step 1: In an isosceles triangle, the two base angles (the angles opposite the two equal sides) are equal to each other.

Step 2: Since $XY = XZ$, the equal sides are X's two sides, so the base angles are angle Y and angle Z, and angle Y = angle Z.

Step 3: Let angle Y = angle Z = k . The triangle angle sum gives: $40 + k + k = 180$.

Step 4: Combine the like terms: $40 + 2k = 180$.

Step 5: Subtract 40 from both sides: $2k = 140$.

Step 6: Divide both sides by 2: $k = 70$.

Step 7: So angle Y = 70 degrees.

Answer: 70 degrees

**Tip**

First identify which two angles are the base angles (opposite the equal sides), then let both equal the same variable before using the 180 degree sum.

Trap

This problem tempts you to assume angle X itself is a base angle, when it's actually the angle between the two equal sides, so it's the odd one out.

Process of Elimination

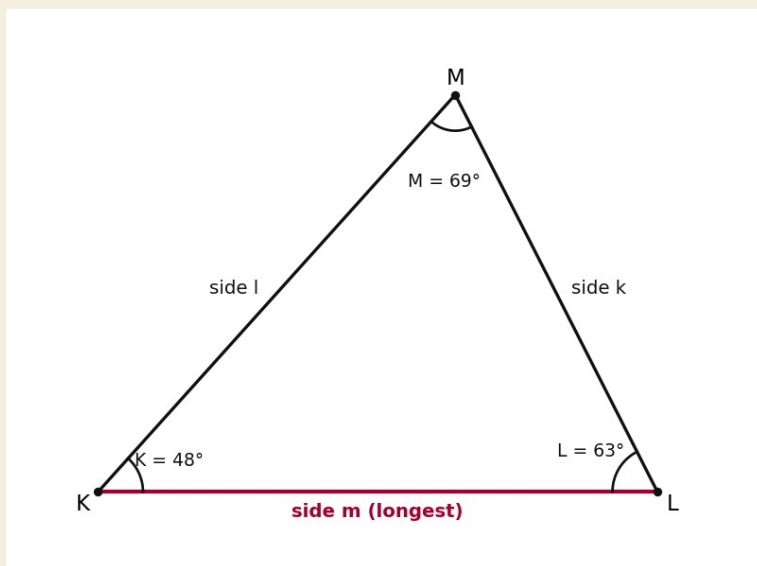
Tempting Wrong Answer	Exactly What Went Wrong
40 degrees	Mistakenly assumed angle Y equals angle X because both looked like 'the given angle,' without checking which sides are equal.
140 degrees	Solved $2k = 140$ correctly but forgot to divide by 2, reporting the combined value of both base angles instead of one.
80 degrees	Used 90 instead of 180 as the angle sum total, solving $2k = 40...$ with an arithmetic slip that overshoot to 80.

Shortcut

Since the two base angles split the remaining degrees evenly, compute $(180 - 40) / 2 = 70$ in one line.

Angles, Parallel Lines & Triangle Fundamentals — Problem 8 (Medium)

Triangle KLM has angle K = 48 degrees and angle L = 63 degrees. Side k is opposite angle K, side l is opposite angle L, and side m is opposite angle M. Which of the three sides, k, l, or m, is the longest?



Formula / Rule Used: Triangle angle sum, then the Side-Angle Relationship.

Solution

Step 1: First find angle M using the triangle angle sum, since it isn't given directly.

Step 2: Set up the equation: $48 + 63 + M = 180$.

Step 3: Add the two known angles: $48 + 63 = 111$.

Step 4: Subtract from 180 to solve for M: $M = 180 - 111 = 69$.

Step 5: Now compare all three angles: $K = 48$, $L = 63$, $M = 69$ degrees.

Step 6: The Side-Angle Relationship says the longest side is always opposite the largest angle.

Step 7: Angle M is the largest angle, and side m is opposite angle M, so side m is the longest side.

Answer: side m

Tip

When a question compares side lengths but only gives angles, first find any missing angle, then match the largest angle to its opposite side.

Trap

It's tempting to assume side k is longest just because angle K is named first or listed first, without actually comparing all three angle values.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
side k	Assumed the first-named angle and side must be the largest, without comparing the actual degree measures.
side l	Correctly found angle $L = 63$ but forgot to compute angle M, wrongly



	assuming L was the biggest of the three.
side m, but from comparing angle M to only angle K	Compared M only to K (69 vs 48) and skipped checking angle L (63) before concluding, arriving at the right answer through an incomplete comparison that could fail on similar problems.

Shortcut

You don't need exact side lengths at all — just find the missing angle, then instantly pair the largest angle with its opposite side.

26. Right Triangles & Trigonometry

Right Triangles & Trigonometry — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Right triangle	Ορθογώνιο τρίγωνο
Hypotenuse	Υποτείνουσα
Leg (of a right triangle)	Κάθετη πλευρά
Pythagorean theorem	Πυθαγόρειο θεώρημα
Sine / cosine / tangent	Ημίτονο / συνημίτονο / εφαπτομένη
Special right triangles (30-60-90, 45-45-90)	Ειδικά ορθογώνια τρίγωνα
Angle of elevation / depression	Γωνία ανύψωσης / υποβιβασμού

Right Triangles & Trigonometry — What It Is

What It Is

- A right triangle is any triangle with exactly one 90-degree angle, marked by a small square in diagrams.
- The Pythagorean theorem links the two shorter sides to the longest side in every right triangle.
- Trigonometric ratios (sine, cosine, tangent) connect an angle to the ratio of two of the triangle's sides.
- The SAT tests this because right triangles model real heights, distances, ramps, and shadows.
- Two special right triangles let you find every side instantly, without ever squaring or using a calculator.

Right Triangles & Trigonometry — Core Rules

The Pythagorean Theorem

$$a^2 + b^2 = c^2$$

- a and b are the two legs — the shorter sides that form the right angle.
- c is always the hypotenuse, the longest side, sitting directly across from the right angle.
- This only works on right triangles, so confirm the right-angle mark before using it.
- Example: legs 3 and 4 give $c^2 = 9 + 16 = 25$, so $c = \sqrt{25} = 5$.

Pythagorean Triples

3-4-5, 5-12-13, 8-15-17, 7-24-25

- These are whole-number side sets that satisfy $a^2 + b^2 = c^2$ exactly, with no decimals.



- Any whole-number multiple of a triple is also a valid triple, like 6-8-10 or 15-20-25.
- Recognizing a scaled triple lets you skip squaring and adding completely.
- Example: legs 10 and 24 are $2 \times (5-12-13)$, so the hypotenuse is $2 \times 13 = 26$.

45-45-90 Triangle

legs: x, x . hypotenuse: $x\sqrt{2}$

- This is an isosceles right triangle, so its two legs are always equal.
- Both acute angles measure 45 degrees, which matches the two equal legs.
- Multiply either leg by $\sqrt{2}$ to get the hypotenuse.
- Example: if each leg is 5, the hypotenuse is $5\sqrt{2}$, roughly 7.07.

30-60-90 Triangle

sides: $x, x\sqrt{3}, 2x$ (opposite 30, 60, 90)

- The side opposite the 30-degree angle is always the shortest side, called x .
- The hypotenuse, opposite the 90-degree angle, is always exactly double that shortest side.
- The side opposite the 60-degree angle equals the shortest side times $\sqrt{3}$.
- Example: if the shortest side is 4, the other sides are $4\sqrt{3}$ and 8.

SOH-CAH-TOA

$\sin\theta = \text{Opp/Hyp}$, $\cos\theta = \text{Adj/Hyp}$, $\tan\theta = \text{Opp/Adj}$

- Opposite means the side directly across from the angle θ you're working with.
- Adjacent means the side touching angle θ that is NOT the hypotenuse.
- The hypotenuse is always the side across from the right angle, no matter which angle θ is.
- Choose the one ratio that connects the side(s) you know to the side or angle you want.

Similar Right Triangles

AA, SSS, or SAS similarity

- Similar triangles have identical angles but can be scaled up or down in size.
- Because the angles match, corresponding sides are always in the same fixed ratio.
- Set up a proportion between two pairs of matching sides to solve for an unknown length.
- Match sides that sit opposite equal angles — never pair sides just because they look close in length.

Complementary Angles & the Pythagorean Identity

$\sin(90^\circ - \theta) = \cos\theta$. $\sin^2\theta + \cos^2\theta = 1$

- The two non-right angles in any right triangle always add up to exactly 90 degrees.
- Because those two angles are complementary, sine of one equals cosine of the other.



- This means if you're given $\sin(90^\circ - \theta)$, you can instantly write it as $\cos\theta$.
- $\sin^2\theta + \cos^2\theta = 1$ is always true for any single angle θ , right triangle or not.

How to Recognize This on the Test

- A small square drawn in the corner of a triangle marks the right angle — that's your cue to use these rules.
- Words like 'ladder,' 'ramp,' 'height of a pole,' 'shadow,' or 'diagonal' usually describe a right triangle.
- An angle of exactly 45 degrees, or a pair of 30 and 60 degrees, signals a special-triangle shortcut applies.
- The words 'sine,' 'cosine,' 'tangent,' or 'angle of elevation/depression' mean you need SOH-CAH-TOA.
- Two triangles drawn as scaled-looking copies, sharing angles, point to a similar-triangles proportion.

Common Mistakes — and the Fix

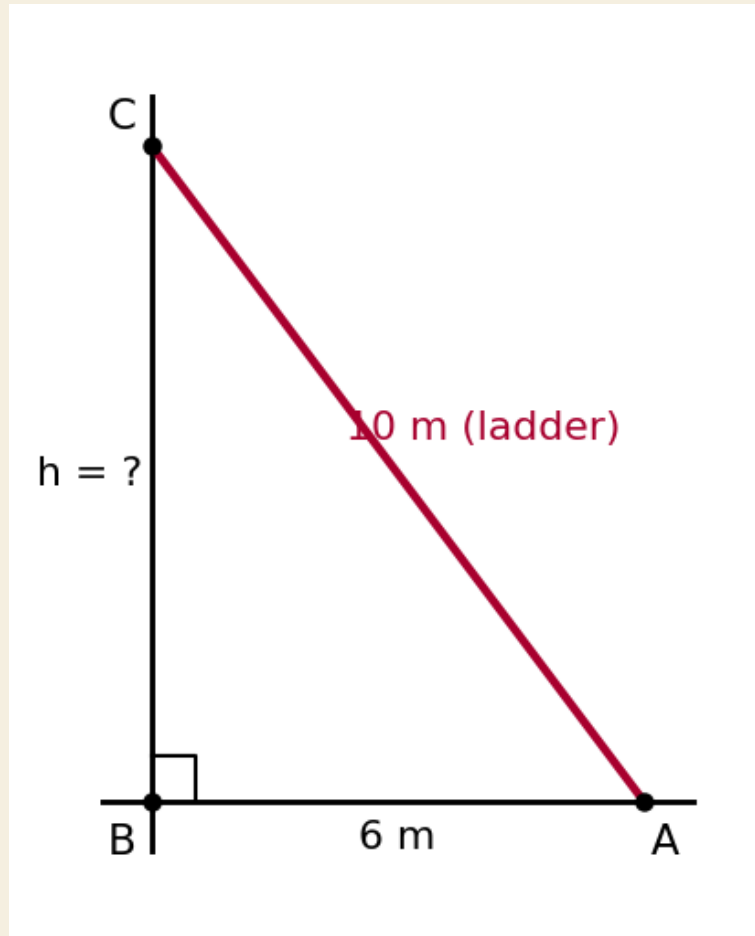
- Mixing up opposite and adjacent — relabel both sides fresh for whichever angle the problem names.
- Assuming a triangle is right just because it looks that way — always confirm the marked right angle.
- Putting $x\sqrt{3}$ on the wrong side of a 30-60-90 triangle — check which angle that side faces first.
- Squaring the legs correctly but forgetting the final square root step when solving for c .
- Pairing up mismatched sides in a similar-triangles proportion instead of sides opposite equal angles.

Right Triangles & Trigonometry — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Right Triangles & Trigonometry — Problem 1 (Easy)

A ladder leans against a vertical wall in Vienna. The foot of the ladder sits 6 meters from the base of the wall, and the ladder itself is 10 meters long. The wall, the ground, and the ladder form a right triangle: label the point where the wall meets the ground as B (the right angle), the foot of the ladder as A, and the top of the ladder (touching the wall) as C. Leg AB = 6 meters, hypotenuse AC = 10 meters (the ladder), and leg BC = the unknown height h . Find h , how high up the wall the ladder reaches.



Formula / Rule Used: Pythagorean theorem: $a^2 + b^2 = c^2$

Solution

Step 1: The right angle is at B (wall meets ground), so the ladder AC is the hypotenuse, and AB and BC are the two legs.

Step 2: Apply the Pythagorean theorem: $AB^2 + BC^2 = AC^2$, so $6^2 + h^2 = 10^2$.

Step 3: Square the known numbers: $36 + h^2 = 100$.

Step 4: Subtract 36 from both sides to isolate h^2 : $h^2 = 64$.

Step 5: Take the square root of both sides: $h = \sqrt{64} = 8$.

Step 6: The ladder reaches 8 meters up the wall.

Answer: 8 meters



Before plugging into $a^2 + b^2 = c^2$, always identify the right angle first — the side across from it is the hypotenuse, never a leg.

Trap

It's tempting to treat the ladder (10) as a leg since it's drawn on a slant, but the slanted side opposite the right angle is always the hypotenuse.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$h = \sqrt{136}$, about 11.7	Added $6^2 + 10^2$ instead of subtracting, mistakenly treating 10 as a leg rather than the hypotenuse.
$h = 4$	Subtracted the side lengths directly ($10 - 6 = 4$) instead of squaring both, subtracting the squares, and then taking a square root.
$h = 64$	Solved correctly up through $h^2 = 64$ but forgot the final square-root step, leaving h^2 as the answer instead of h .

Shortcut

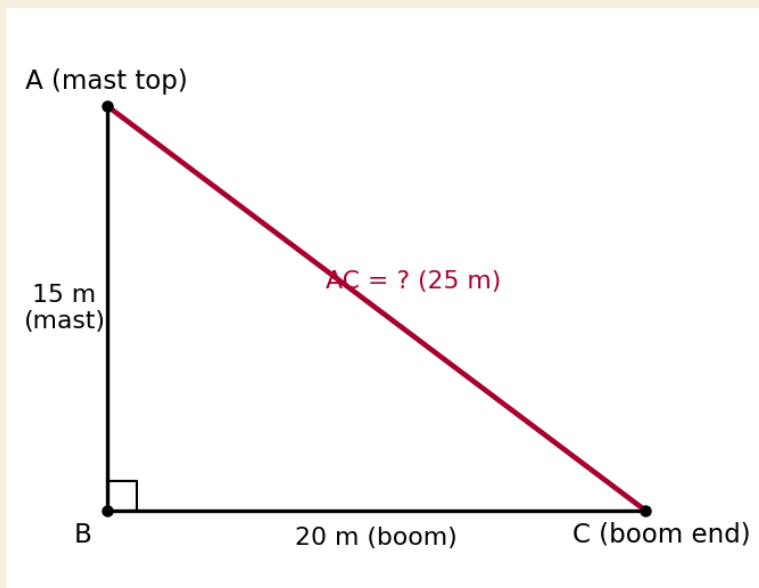
Notice 6 and 10 are 2×3 and 2×5 — this is a 3-4-5 triple scaled by 2 (6-8-10), so the missing leg is $2 \times 4 = 8$ with no squaring needed.

Desmos Tip

Type $\sqrt{10^2 - 6^2}$ as a line in the calculator to instantly confirm the missing leg equals 8.

Right Triangles & Trigonometry — Problem 2 (Easy)

A triangular sail on a boat docked in Split, Croatia, has a right angle where the mast meets the boom. Label that corner B, the top of the mast A, and the far end of the boom C. Leg AB (the mast) = 15 meters, leg BC (the boom) = 20 meters, and the slanted edge AC (from the top of the mast to the far end of the boom) is the hypotenuse. Find the length of AC.



Formula / Rule Used: Pythagorean theorem: $a^2 + b^2 = c^2$

Solution

Step 1: The right angle is at B, so AB and BC are the two legs, and AC is the hypotenuse.

Step 2: Apply the Pythagorean theorem: $AB^2 + BC^2 = AC^2$, so $15^2 + 20^2 = AC^2$.

Step 3: Square each leg: $225 + 400 = AC^2$.

Step 4: Add the two results: $625 = AC^2$.

Step 5: Take the square root of both sides: $AC = \sqrt{625} = 25$.

Step 6: The slanted edge of the sail (AC) is 25 meters.

Answer: 25 meters

Tip

Before squaring two big numbers, check whether they share a common factor with a known triple like 3-4-5 — it can save you the arithmetic.

Trap

Squaring 15 and 20 correctly and adding them to get 625 but then forgetting to take the square root, leaving 625 as the final answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
625	Correctly reached $AC^2 = 625$ but forgot the final square-root step.
35	Added the two legs directly ($15 + 20 = 35$) instead of applying the



	Pythagorean theorem.
5	Subtracted the legs ($20 - 15 = 5$) instead of using $a^2 + b^2 = c^2$, confusing a difference with the hypotenuse.

Shortcut

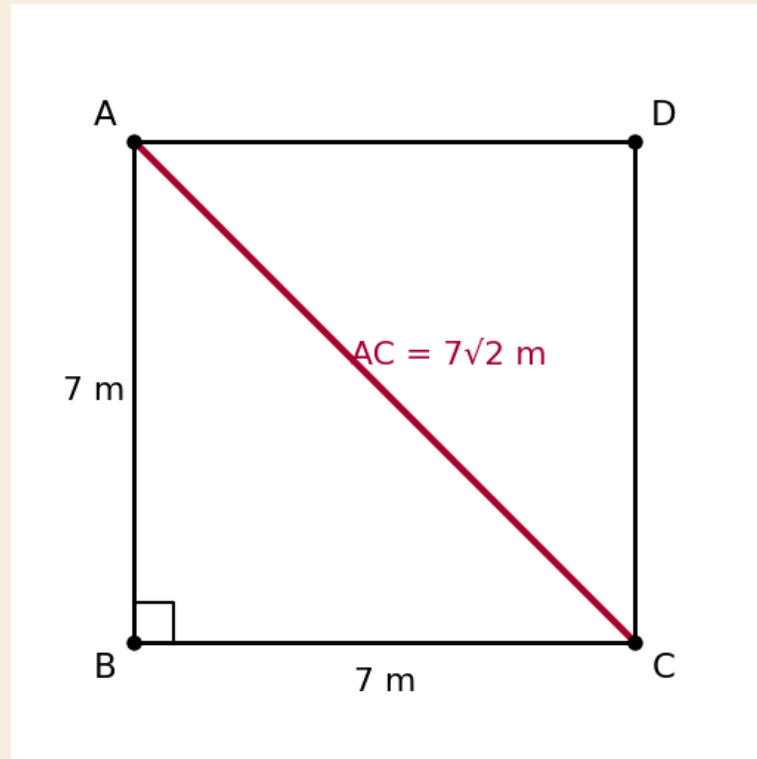
Recognize 15 and 20 as 5×3 and 5×4 — this is a 3-4-5 triple scaled by 5, so the hypotenuse is $5 \times 5 = 25$ with no squaring needed.

Desmos Tip

Type $\text{sqrt}(15^2 + 20^2)$ into a calculator line to instantly confirm the hypotenuse equals 25.

Right Triangles & Trigonometry — Problem 3 (Easy)

A square courtyard in Athens has each side measuring 7 meters. A student walks in a straight line from one corner of the courtyard, point A, to the opposite corner, point C, crossing the diagonal. This diagonal AC, together with two sides of the square (AB and BC, each 7 meters, meeting at right angle B), forms a 45-45-90 right triangle. Find the length of diagonal AC.



Formula / Rule Used: 45-45-90 triangle: hypotenuse = leg * $\sqrt{2}$

Solution

Step 1: Since the courtyard is a square, legs AB and BC of the right triangle are equal: both 7 meters — this makes it a 45-45-90 triangle.

Step 2: In a 45-45-90 triangle, the hypotenuse always equals a leg times $\sqrt{2}$.

Step 3: Multiply: $AC = 7 * \sqrt{2}$.

Step 4: This exact form, $7 * \sqrt{2}$, is the precise answer.

Step 5: As a decimal, $\sqrt{2}$ is about 1.414, so $7 * 1.414$ is about 9.9.

Step 6: The diagonal path AC is $7 * \sqrt{2}$ meters, about 9.9 meters.

Answer: $7 * \sqrt{2}$ meters (about 9.9 meters)

Tip

Whenever a square's diagonal shows up, recognize it right away as a 45-45-90 triangle with both legs equal to the side length.

Trap

Multiplying the leg by 2 instead of $\sqrt{2}$, which confuses the 45-45-90 rule with simply doubling the side.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
14	Multiplied the leg by 2 instead of $\sqrt{2}$, doubling the side length instead of applying the 45-45-90 rule.
$7\sqrt{3}$, about 12.1	Used $\sqrt{3}$ instead of $\sqrt{2}$, confusing this with the 30-60-90 side ratio.
49	Squared the leg ($7^2 = 49$) instead of multiplying by $\sqrt{2}$, confusing this with a Pythagorean-theorem squaring step.

Shortcut

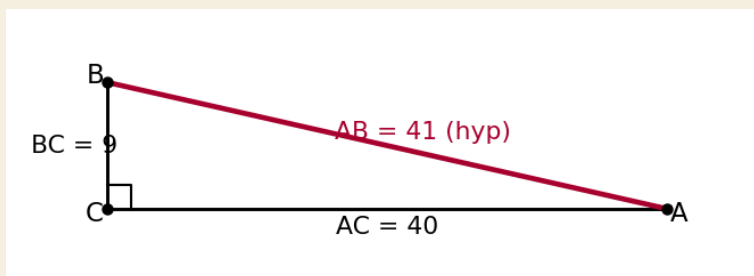
For a square's diagonal, skip the Pythagorean theorem entirely — just multiply the side length by $\sqrt{2}$.

Desmos Tip

Type $7\sqrt{2}$ into the calculator to instantly get the decimal value, about 9.9.

Right Triangles & Trigonometry — Problem 4 (Easy)

In right triangle ABC, the right angle is at C. Side BC = 9, side AC = 40, and the hypotenuse AB = 41. What is the value of $\sin A$ (the sine of angle A)?



Formula / Rule Used: SOH: $\sin\theta = \text{Opposite} / \text{Hypotenuse}$

Solution

Step 1: The right angle is at C, so AB (across from C) is the hypotenuse.

Step 2: For angle A, find the side that does NOT touch A: that side is BC, so BC is opposite angle A.

Step 3: SOH says $\sin\theta = \text{Opposite} / \text{Hypotenuse}$, so $\sin A = BC / AB$.

Step 4: Substitute the given lengths: $\sin A = 9 / 41$.

Step 5: Check that the fraction is fully simplified: 9 and 41 share no common factors, so $9/41$ is the final form.

Answer: $9/41$

Tip

Before plugging into SOH-CAH-TOA, physically point to the side directly across from the angle to confirm which one is 'opposite.'

Trap



Using AC (the side adjacent to A) instead of BC in the numerator, which mixes up opposite and adjacent for angle A.

Process of Elimination

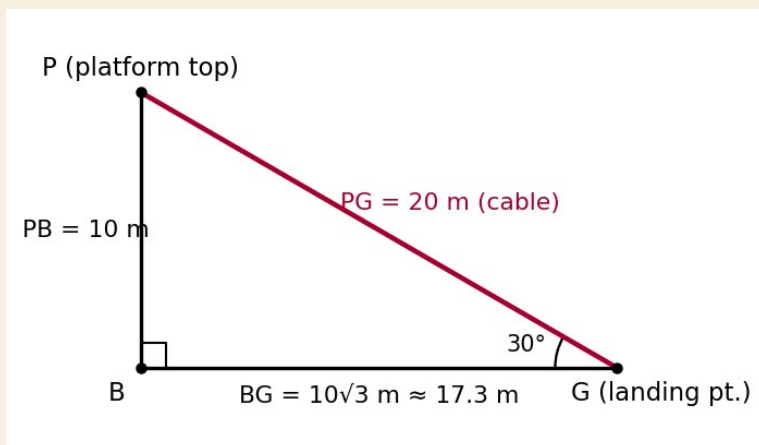
Tempting Wrong Answer	Exactly What Went Wrong
40/41	Used AC (adjacent to A) instead of BC (opposite A) in the numerator — that ratio is actually $\cos A$, not $\sin A$.
9/40	Used AC (the adjacent side) as the denominator instead of the hypotenuse AB — that ratio is actually $\tan A$.
41/9	Flipped the ratio, putting the hypotenuse over the opposite side instead of opposite over hypotenuse.

Shortcut

None needed — reading $\sin A = \text{opposite/hypotenuse}$ directly off the three given side lengths is already the fastest path.

Right Triangles & Trigonometry — Problem 5 (Medium)

A zip-line cable in Salzburg runs from a launch platform down to a landing point on the ground, making a 30-degree angle with the ground at the landing point. Label the landing point G, the point on the ground directly below the platform B (right angle here), and the top of the platform P. The cable PG is the hypotenuse and is 20 meters long. Side PB (vertical, opposite the 30-degree angle at G) is the platform's height. Side BG (horizontal, opposite the 60-degree angle at P) is the horizontal distance a rider covers. Find BG, the horizontal distance covered.



Formula / Rule Used: 30-60-90 side ratios: x (opp. 30 deg), $x\sqrt{3}$ (opp. 60 deg), $2x$ (opp. 90 deg, hypotenuse)

Solution

Step 1: In a 30-60-90 triangle, the shortest side (opposite the 30-degree angle) is always half the hypotenuse — call it x .

Step 2: The hypotenuse (cable PG) is 20, so $x = 20 / 2 = 10$. This x is the platform's height, PB — not yet what's asked.

Step 3: The side opposite the 60-degree angle (BG, the horizontal distance) always equals $x * \sqrt{3}$.

Step 4: Substitute $x = 10$: $BG = 10 * \sqrt{3}$.

Step 5: As a decimal, $\sqrt{3}$ is about 1.732, so $10 * 1.732$ is about 17.3.

Step 6: The horizontal distance covered is $10\sqrt{3}$ meters, about 17.3 meters.

Answer: $10\sqrt{3}$ meters (about 17.3 meters)

Tip

In a 30-60-90 word problem, first find the shortest side (half the hypotenuse) — every other side is built from it.

Trap

Stopping after finding the shortest side (10, the platform's height) and reporting that as the final answer, instead of continuing on to find the side opposite 60 degrees.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
10	Stopped after finding the 30-degree-opposite side (the platform height) without continuing to the 60-degree-opposite side that the question actually asks for.



40	Doubled the hypotenuse instead of halving it to find x, throwing off every side that follows.
$20\sqrt{3}$, about 34.6	Used the full hypotenuse (20) times $\sqrt{3}$ instead of the shortest side (10) times $\sqrt{3}$.

Shortcut

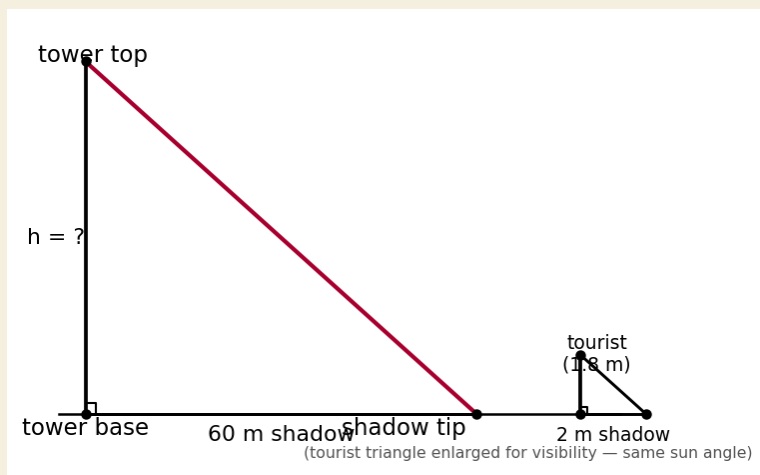
Once you know one side of a 30-60-90 triangle, every other side is a fixed multiple away — no need to set up sine or cosine at all.

Desmos Tip

Type $10\sqrt{3}$ into the calculator to confirm the horizontal distance is about 17.3 meters.

Right Triangles & Trigonometry — Problem 6 *(Medium)*

At a certain time of day in Paris, the Eiffel Tower casts a shadow 60 meters long on the ground. At that same moment, a tourist who is 1.8 meters tall, standing nearby, casts a shadow 2 meters long. A light breeze of 10 km/h is blowing at the time. The tower and its shadow form a right triangle (right angle where the tower meets the ground), and the tourist and their shadow form a smaller, similar right triangle (right angle where the tourist's feet meet the ground) — both triangles share the same sun angle. Find the height of the tower, h .



Formula / Rule Used: Similar triangles: corresponding sides are in equal ratio

Solution

Step 1: The wind speed (10 km/h) is extra information that has no effect on shadow length or height — ignore it.

Step 2: Because the two triangles are similar, height-to-shadow is the same ratio for both the tourist and the tower.

Step 3: Set up the proportion: $(\text{tourist height})/(\text{tourist shadow}) = (\text{tower height})/(\text{tower shadow})$, so $1.8/2 = h/60$.

Step 4: Cross-multiply: $1.8 \cdot 60 = 2 \cdot h$.

Step 5: Multiply the left side: $108 = 2h$.

Step 6: Divide both sides by 2 to solve for h : $h = 54$.

Step 7: The Eiffel Tower's height in this triangle is 54 meters.

Answer: 54 meters

Tip

In a shadow problem, match height-to-shadow for the SAME object first, then set that ratio equal to the other object's — and ignore any detail that isn't a length.

Trap

Trying to work the wind speed into the calculation somehow, when only the two heights and two shadow lengths matter for the similar-triangles proportion.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong

66.7	Set up the proportion upside down, as (tourist shadow)/(tourist height) = (tower height)/(tower shadow), giving $2/1.8 = h/60$.
108	Correctly cross-multiplied to $108 = 2h$ but forgot to divide by 2, leaving 108 as the final answer.
5.4	Divided 108 by 2 but misplaced the decimal point, writing 5.4 instead of 54.

Shortcut

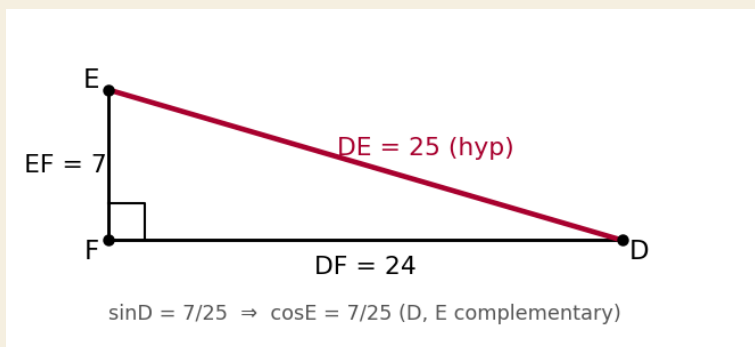
The tourist's ratio simplifies to $1.8/2 = 0.9$, so just multiply the tower's shadow by 0.9 to get its height directly: $60 * 0.9 = 54$.

Desmos Tip

Type $h = 1.8/2 * 60$ as a single line in the calculator to compute the tower's height in one step.

Right Triangles & Trigonometry — Problem 7 (Medium)

In right triangle DEF, the right angle is at F, so angle D and angle E are the two acute angles. If $\sin D = 7/25$, what is the value of $\cos E$?



Formula / Rule Used: Complementary angle identity: $\sin(90^\circ - \theta) = \cos(\theta)$

Solution

Step 1: Angles D and E are the two acute angles in a right triangle, so $D + E = 90^\circ$ — they're complementary.

Step 2: The complementary angle identity says $\sin(90^\circ - \theta) = \cos(\theta)$, which means sine of one acute angle always equals cosine of the other.

Step 3: Since D and E are complementary, $\sin D = \cos E$ automatically, with no side-length computation needed.

Step 4: Substitute the given value: $\cos E = 7/25$.

Step 5: No triangle side lengths are actually needed here — the identity alone gives the answer.

Answer: $7/25$

Tip



Whenever two angles in a right triangle are described as 'the two acute angles,' check whether the complementary angle identity lets you skip straight to the answer.

Trap

Trying to build a 7-24-25 triangle and compute an actual side ratio for E, when the identity means $\cos E$ is already equal to $\sin D$ with zero extra work.

Process of Elimination

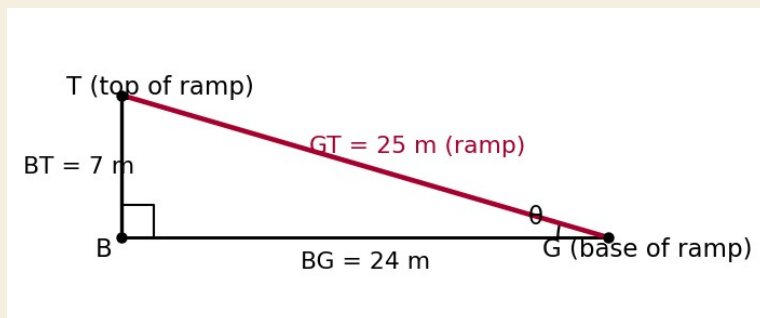
Tempting Wrong Answer	Exactly What Went Wrong
25/7	Flipped the fraction, treating $\cos E$ as the reciprocal of $\sin D$ instead of equal to it.
18/25	Tried to compute $25 - 7 = 18$ and used it as a numerator instead of applying the complementary angle identity.
24/25	Assumed $\sin D = 7/25$ meant a 7-24-25 triangle and calculated $\cos D$ (24/25, adjacent-to-D over hypotenuse) instead of $\cos E$.

Shortcut

This is already the fastest path — the complementary identity turns what looks like a two-step side-length problem into a direct substitution.

Right Triangles & Trigonometry — Problem 8 (Medium)

A skate ramp being built in Rotterdam rises 7 meters vertically over a horizontal base of 24 meters. Label the bottom corner of the vertical support B (the right angle), the top of the ramp T, and the base of the ramp where it meets the ground G. Leg BT (vertical) = 7 meters, leg BG (horizontal) = 24 meters, and the sloped ramp surface GT is the hypotenuse. Angle theta is at G, where the ramp meets the ground. Find $\cos(\theta)$.



Formula / Rule Used: Pythagorean theorem to find the missing side, then CAH: $\cos(\theta) = \text{Adjacent} / \text{Hypotenuse}$

Solution

Step 1: $\cos(\theta)$ needs the hypotenuse, GT, which isn't given yet — find it first using the Pythagorean theorem: $BG^2 + BT^2 = GT^2$.

Step 2: Substitute the legs: $24^2 + 7^2 = GT^2$.

Step 3: Square each leg: $576 + 49 = GT^2$.

Step 4: Add: $625 = GT^2$.

Step 5: Take the square root of both sides: $GT = \sqrt{625} = 25$.

Step 6: Now apply CAH: theta is at G, and the side touching G that is NOT the hypotenuse is BG (24), so BG is adjacent to theta.

Step 7: $\cos(\theta) = \text{Adjacent} / \text{Hypotenuse} = BG / GT = 24/25$.

Answer: 24/25

Tip

When a trig ratio needs a side you don't have yet, find that missing side with the Pythagorean theorem first, then apply SOH-CAH-TOA.

Trap

Jumping straight into a ratio using only 7 and 24 without ever finding the hypotenuse, when cosine specifically requires the hypotenuse in the denominator.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
7/24	Computed $\tan(\theta)$ (opposite/adjacent) instead of $\cos(\theta)$, skipping the need for the hypotenuse entirely.
24/7	Computed the reciprocal of the correct ratio, putting the wrong side on top.

**7/25**

Used the vertical rise (7) as if it were adjacent to theta instead of the horizontal base (24), computing $\sin(\theta)$ instead of $\cos(\theta)$.

Shortcut

Recognize 7 and 24 as a 7-24-25 triple right away, so the hypotenuse is 25 with no squaring needed — then apply $\cos(\theta) = 24/25$ directly.

Desmos Tip

Type $\sqrt{7^2 + 24^2}$ into the calculator first to confirm the hypotenuse is 25, then compute $24/25$ to check $\cos(\theta)$.

27. Radians & the Unit Circle

Radians & the Unit Circle — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Radian	Ακτίνιο (ραδιάνιο)
Unit circle	Μοναδιαίος κύκλος
Reference angle	Γωνία αναφοράς
Coterminal angles	Συνοριακές (ταυτόσημες) γωνίες
Quadrant	Τεταρτημόριο

Radians & the Unit Circle — What It Is

What It Is

- A radian is just another way to measure an angle, like degrees, but based on the circle's own radius.
- The unit circle is a circle with radius 1, centered exactly at the origin (0, 0).
- Any point on the unit circle can be found using just the angle from the positive x-axis.
- The SAT tests this because it connects angles, coordinates, and the trig functions sine and cosine in one picture.
- You'll see radians in equations, and unit-circle points as coordinates you must identify or use.

Radians & the Unit Circle — Core Rules

Degree-Radian Conversion

π radians = 180 degrees

- Degrees and radians both measure the same angle, just with different units.
- To turn degrees into radians, multiply the degree value by $\pi/180$.
- To turn radians into degrees, multiply the radian value by $180/\pi$.
- Example: $90 \text{ degrees} \times (\pi/180) = \pi/2 \text{ radians}$.

Common Angles Table

$30=\pi/6$, $45=\pi/4$, $60=\pi/3$, $90=\pi/2$, $180=\pi$, $270=3\pi/2$, $360=2\pi$

- These are the angles that show up over and over on the SAT, so memorize this table.
- Knowing it by heart saves time versus converting with the formula every single time.
- Notice the pattern: the denominator (6, 4, 3, 2) shrinks as the angle grows toward 180.
- A full circle is 360 degrees, which equals 2π radians.

The Unit Circle Itself

$x^2 + y^2 = 1$, centered at the origin, radius 1

- Every point on this circle is exactly 1 unit away from the origin (0, 0).
- An angle theta is measured starting at the positive x-axis and sweeping counterclockwise.
- The unit circle is the picture that links every angle to one specific (x, y) point.
- A radius of exactly 1 is what makes the coordinates line up perfectly with cosine and sine.

Coordinates from an Angle

point at angle theta = (cos(theta), sin(theta))

- The x-coordinate of the point at angle theta is always cos(theta).
- The y-coordinate of the point at angle theta is always sin(theta).
- This means cosine and sine are not separate ideas, they're just the coordinates of one point.
- Example: at theta = 0, the point is (1, 0), so cos(0) = 1 and sin(0) = 0.

Key Reference Points

0 deg=(1,0), 90 deg=(0,1), 180 deg=(-1,0), 270 deg=(0,-1)

- These four points sit exactly on the x-axis or y-axis, so they're the easiest to remember.
- Use them as anchors, then reason about angles that fall between them.
- At 90 degrees the point is straight up, so cos(90) = 0 and sin(90) = 1.
- At 180 degrees the point is straight left, so cos(180) = -1 and sin(180) = 0.

Quadrant Signs

QI (+,+), QII (-,+), QIII (-,-), QIV (+,-)

- The circle is split into 4 quadrants by the x-axis and y-axis.
- The sign of the x-coordinate and y-coordinate tells you the sign of cosine and sine.
- Find the reference angle (the acute angle to the nearest x-axis) for the size of the value.
- Then attach the correct + or - sign based on which quadrant the angle lands in.

Coterminal Angles

add or subtract 360 degrees (or 2pi radians)

- Coterminal angles end up pointing in the exact same direction on the circle.
- A negative angle or one bigger than 360 degrees is not wrong, just not in standard range yet.
- Add or subtract full rotations of 360 degrees (or 2pi) until the angle lands between 0 and 360.
- Example: 400 degrees is coterminal with 400 - 360 = 40 degrees.

How to Recognize This on the Test

- The word 'radians,' or an angle written with π in it, signals this skill immediately.
- A question asking for coordinates of a point 'on the unit circle' at a given angle.
- A question giving an angle outside 0 to 360 degrees (or 0 to 2π) and asking for its equivalent angle.
- A question asking for the sign (positive or negative) of sine or cosine without needing the exact value.
- Any question that mixes degrees and radians and asks you to convert between them.

Common Mistakes — and the Fix

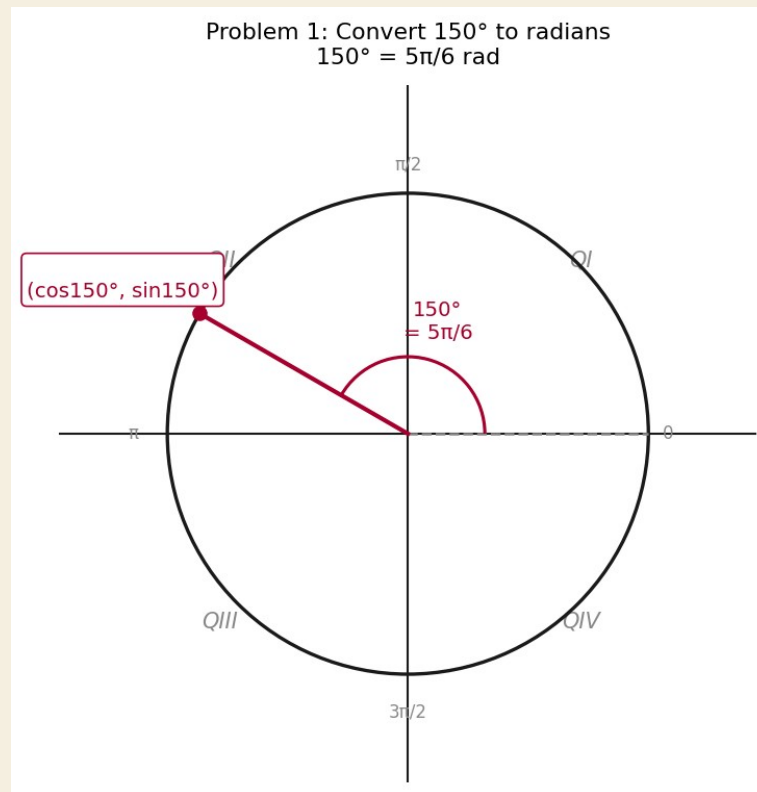
- Forgetting to flip the conversion fraction, multiplying by $180/\pi$ when going from degrees to radians instead of $\pi/180$.
- Mixing up which coordinate is cosine and which is sine, since x always goes with cosine and y always goes with sine.
- Assigning the wrong sign to a trig value because the quadrant wasn't identified before evaluating.
- Treating an angle over 360 degrees as unsolvable instead of just subtracting 360 to find its coterminal angle.
- Misreading the common-angles table and mixing up $\pi/6$ (30 degrees) with $\pi/3$ (60 degrees).

Radians & the Unit Circle — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Radians & the Unit Circle — Problem 1 (Easy)

A robotic arm on an assembly line in Stuttgart, Germany rotates through an angle of 150 degrees to complete one motion cycle. What is 150 degrees in radians? Express your answer as a fraction times pi.



Formula / Rule Used: Degree-to-radian conversion: radians = degrees $\times$ ($\pi/180$).

Solution

Step 1: To convert degrees to radians, multiply the degree measure by $\pi/180$.

Step 2: Set up the multiplication: $150 \times (\pi/180)$.

Step 3: Write it as a single fraction: $(150/180) \times \pi$.

Step 4: Simplify the fraction $150/180$ by dividing both the top and bottom by their greatest common factor, 30.

Step 5: $150 \div 30 = 5$ and $180 \div 30 = 6$, so the fraction becomes $5/6$.

Step 6: The angle in radians is $(5/6) \times \pi$, written as $5\pi/6$.

Answer: $5\pi/6$ radians

Tip

Start by writing degrees $\times (\pi/180)$ exactly as it's given, then simplify the fraction at the end instead of trying to reduce early.

Trap

A common mistake is multiplying by $180/\pi$ instead of $\pi/180$, which is the conversion for going the other direction (radians to degrees).

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$6\pi/5$	Flipped the fraction $150/180$ upside down to $180/150$ before simplifying, which is the reciprocal error.
$5/6$	Simplified $150/180$ correctly but forgot to attach π to the final answer.
150π	Multiplied 150 by π but forgot to divide by 180 at all.

Shortcut

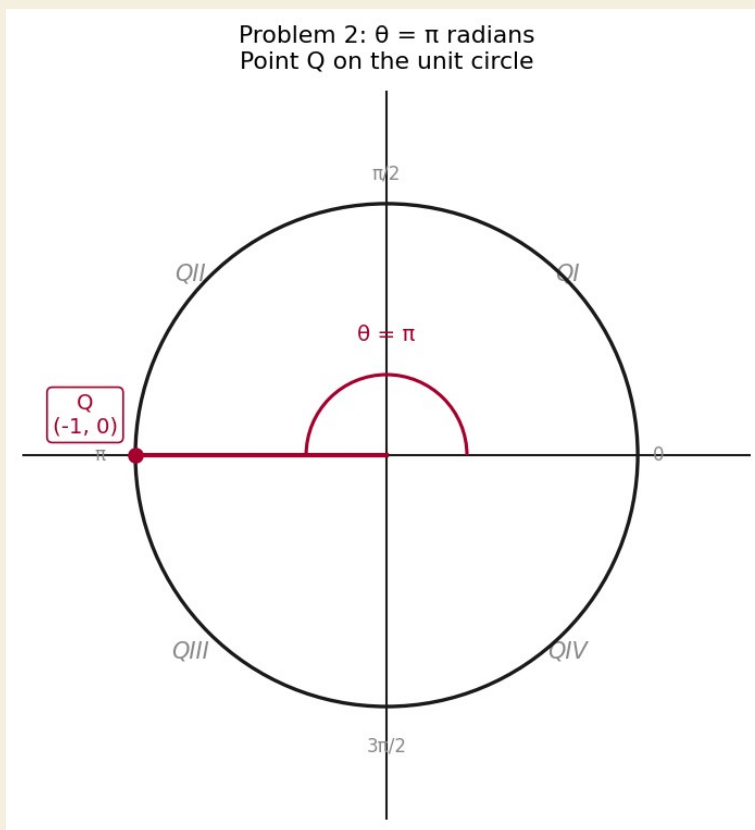
If you've memorized that $30^\circ = \pi/6$, you can also think of 150 as 5×30 , so the answer is immediately $5 \times (\pi/6) = 5\pi/6$.

Desmos Tip

Type $150 \cdot (\pi/180)$ directly into a Desmos line and press enter to see the decimal value, then type $5 \cdot \pi/6$ on the next line to confirm the two match.

Radians & the Unit Circle — Problem 2 (Easy)

Point Q lies on the unit circle $x^2 + y^2 = 1$, which is centered at the origin O with radius 1. The terminal side of angle theta, measured counterclockwise from the positive x-axis, meets the circle exactly at point Q. If theta = pi radians, what are the coordinates of point Q?



Formula / Rule Used: Point at angle theta = (cos theta, sin theta).

Solution

Step 1: Recall that any point on the unit circle at angle theta has coordinates (cos theta, sin theta).

Step 2: theta = pi radians is the same as 180 degrees, since pi radians = 180 degrees.

Step 3: At 180 degrees, the terminal side points straight along the negative x-axis.

Step 4: This means point Q is exactly 1 unit to the left of the origin, at (-1, 0).

Step 5: Check: $\cos(180 \text{ degrees}) = -1$ and $\sin(180 \text{ degrees}) = 0$, which matches (-1, 0).

Answer: (-1, 0)

Tip

Whenever the angle is one of the four axis angles (0, $\pi/2$, π , $3\pi/2$), skip straight to the key reference points instead of doing any calculation.

Trap

A common mistake is swapping the coordinates and writing (0, -1), which is actually the point for theta = $3\pi/2$, not theta = π .

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$(0, -1)$	Mixed up $\theta = \pi$ with $\theta = 3\pi/2$ and used the bottom-of-circle point instead of the left-of-circle point.
$(1, 0)$	Forgot the negative sign and used the point for $\theta = 0$ instead of $\theta = \pi$.
$(-1, 1)$	Assumed both coordinates change at 180 degrees instead of remembering the y-coordinate stays 0 on the x-axis.

Shortcut

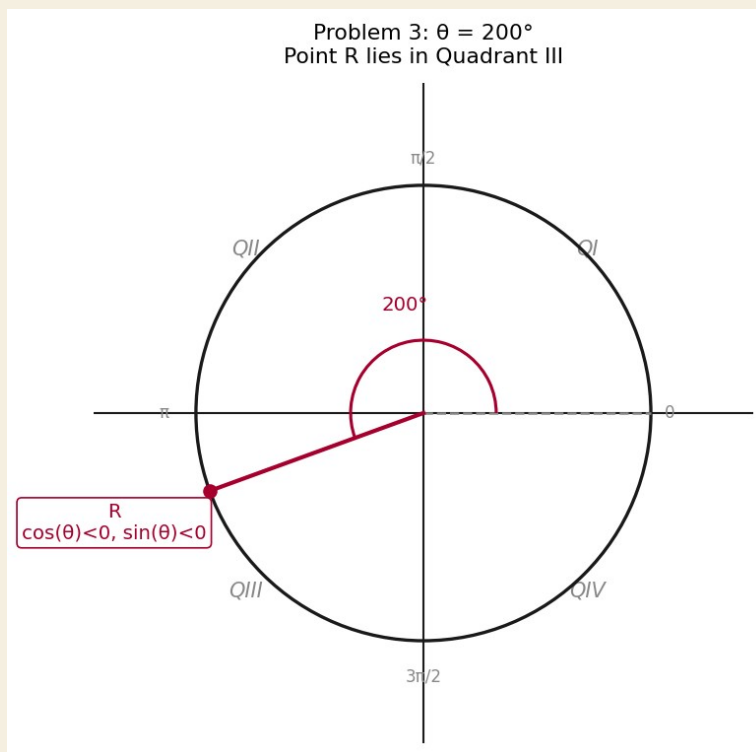
Since π is one of the four key axis angles, you can name the point instantly from memory without computing $\cos$ or $\sin$.

Desmos Tip

Type $x^2 + y^2 = 1$ in Desmos to draw the unit circle, then type the point $(-1, 0)$ on the next line to see it land exactly on the circle at the leftmost edge.

Radians & the Unit Circle — Problem 3 (Easy)

Angle theta measures 200 degrees, measured counterclockwise from the positive x-axis to point R on the unit circle $x^2 + y^2 = 1$. In which quadrant does point R lie, and what are the signs of $\cos(\theta)$ and $\sin(\theta)$?



Formula / Rule Used: Quadrant signs: QI(+,+), QII(-,+), QIII(-,-), QIV(+,-).

Solution

Step 1: Recall the quadrant boundaries: QI is 0-90 degrees, QII is 90-180, QIII is 180-270, QIV is 270-360.

Step 2: 200 degrees is greater than 180 but less than 270, so it falls in Quadrant III.

Step 3: In Quadrant III, both the x-coordinate and y-coordinate of a point on the unit circle are negative.

Step 4: Since $\cos(\theta)$ is the x-coordinate, $\cos(\theta)$ is negative.

Step 5: Since $\sin(\theta)$ is the y-coordinate, $\sin(\theta)$ is negative.

Answer: Quadrant III; $\cos(\theta)$ is negative and $\sin(\theta)$ is negative.

Tip

Always locate the quadrant first by comparing the angle to 90, 180, 270, and 360 before worrying about any exact value.

Trap

A common mistake is assuming an angle past 180 degrees must be in Quadrant IV, when it's actually still in Quadrant III until it passes 270 degrees.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



Quadrant IV; cos positive, sin negative	Assumed any angle over 180 degrees jumps straight to Quadrant IV instead of checking it's still under 270.
Quadrant III; cos positive, sin negative	Correctly found the quadrant but only remembered that y is negative there and mixed up which sign goes with x.
Quadrant II; cos negative, sin positive	Mistook 200 degrees for just under 180 instead of just over 180, misplacing it in Quadrant II.

Shortcut

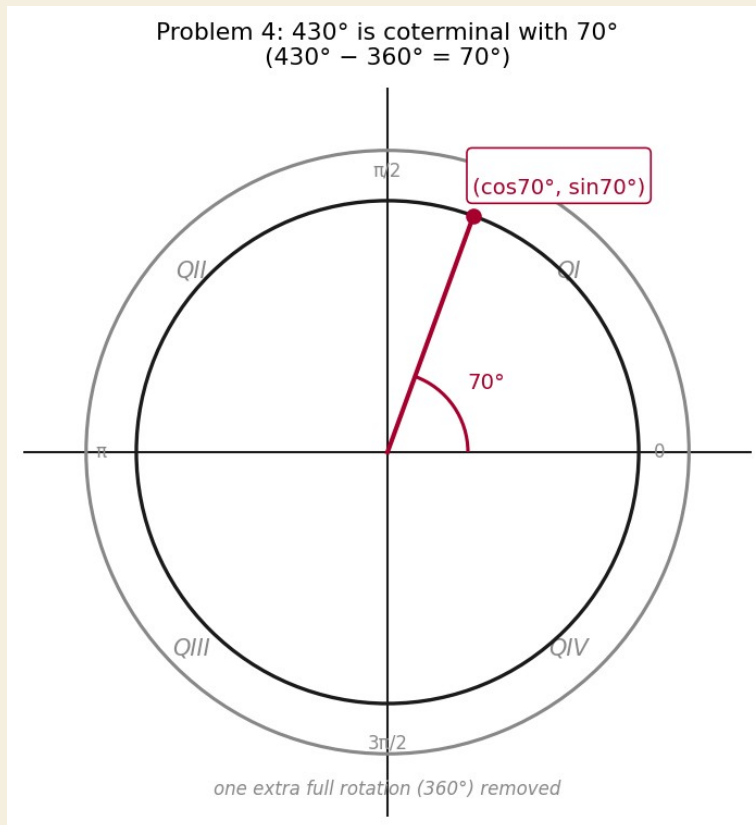
No calculation needed here — just place the angle on a quick mental clock face of the four quadrants to read off the quadrant and signs.

Desmos Tip

Type $(\cos(200^\circ), \sin(200^\circ))$ into Desmos using the degree symbol to see the actual decimal coordinates and confirm both are negative.

Radians & the Unit Circle — Problem 4 (Easy)

An angle measures 430 degrees. What is its coterminal angle that lies between 0 degrees and 360 degrees?



Formula / Rule Used: Coterminal angles: add or subtract 360 degrees (or 2π radians).

Solution

Step 1: A coterminal angle points in the same direction but differs by a full rotation of 360 degrees.

Step 2: Since 430 degrees is more than 360, subtract one full rotation: $430 - 360$.

Step 3: $430 - 360 = 70$.

Step 4: Check that 70 degrees is between 0 and 360 degrees, which it is, so no further subtraction is needed.

Answer: 70 degrees

Tip

Always subtract (or add) 360 in full chunks until the result lands between 0 and 360, checking after each step.

Trap

A common mistake is subtracting 180 instead of a full 360, which gives an angle that points in the opposite direction, not the same one.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
250 degrees	Subtracted 180 instead of 360, landing on the wrong angle entirely.



430 degrees	Forgot that a coterminal angle must be brought into the 0-360 range and left the original angle unchanged.
-70 degrees	Subtracted correctly to get 70 but then wrote it as negative instead of a positive angle in the required range.

Shortcut

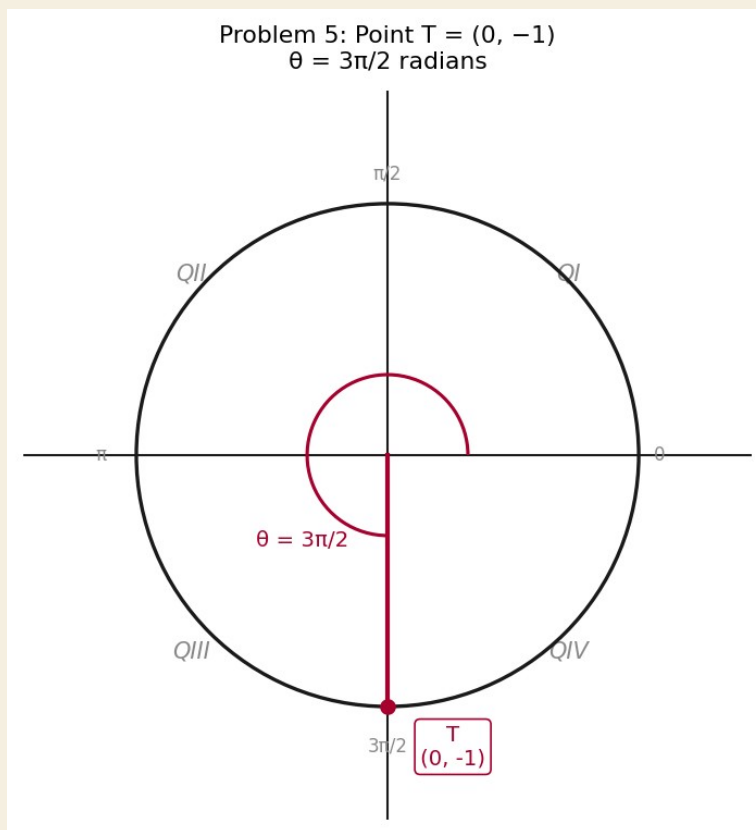
If the angle is only slightly over 360, subtracting one 360 is always enough — no need to check for multiple rotations unless the number is much larger.

Desmos Tip

Type $\cos(430^\circ)$ and $\cos(70^\circ)$ on two separate lines in Desmos to confirm they give the same decimal value, proving the angles are coterminal.

Radians & the Unit Circle — Problem 5 *(Medium)*

Point T lies on the unit circle $x^2 + y^2 = 1$, centered at the origin O. The terminal side of angle theta, measured counterclockwise from the positive x-axis, meets the circle exactly at point T with coordinates (0, -1). If $0 \leq \theta < 2\pi$, what is the value of theta in radians?



Formula / Rule Used: Point at angle theta = (cos theta, sin theta); key reference points.

Solution

Step 1: The coordinates (0, -1) mean $\cos(\theta) = 0$ and $\sin(\theta) = -1$.

Step 2: $\cos(\theta) = 0$ only happens at the top or bottom of the circle, so theta is $\pi/2$ or $3\pi/2$.

Step 3: $\sin(\theta) = -1$ means the point is at the very bottom of the circle, not the top.

Step 4: The point straight down from the origin is reached at $\theta = 3\pi/2$ radians (270 degrees).

Step 5: Check: $\cos(3\pi/2) = 0$ and $\sin(3\pi/2) = -1$, which matches point T exactly.

Answer: $\theta = 3\pi/2$ radians

Tip

When you're given coordinates instead of an angle, first check which axis the point sits on, then use the key reference points to name the angle.

Trap

A common mistake is picking $\theta = \pi/2$ because $\cos(\theta) = 0$ there too, without checking that $\sin(\theta) = 1$ (not -1) at that angle.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$\pi/2$	Matched $\cos(\theta) = 0$ correctly but ignored that $\sin(\pi/2) = 1$, not -1 , so the sign didn't check out.
π	Confused the bottom point $(0,-1)$ with the left point $(-1,0)$, mixing up which axis the coordinates lie on.
270	Found the correct angle but reported it in degrees instead of the radians the question asked for.

Shortcut

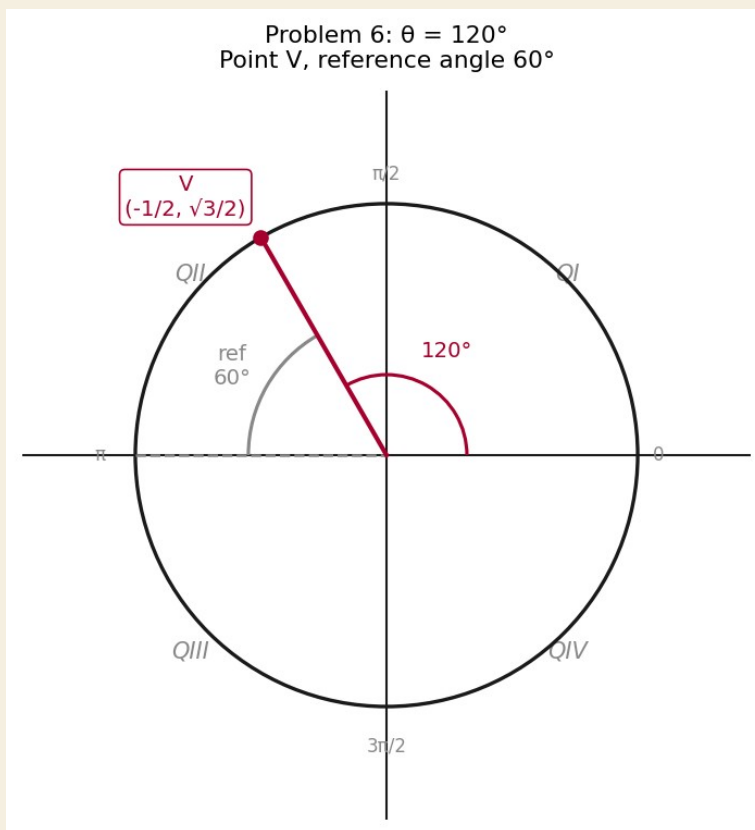
Since $(0, -1)$ is one of the four key axis points, you can name θ straight from memory without setting up $\cos$ and $\sin$ equations at all.

Desmos Tip

Type $x^2 + y^2 = 1$ to draw the unit circle, then type the point $(0, -1)$ to see exactly where it lands, confirming it's the bottom of the circle at $3\pi/2$.

Radians & the Unit Circle — Problem 6 (Medium)

Point V lies on the unit circle $x^2 + y^2 = 1$. The terminal side of a 120-degree angle, measured counterclockwise from the positive x-axis, meets the circle exactly at point V. What are the exact coordinates of point V?



Formula / Rule Used: Reference angle + common angles table + quadrant signs.

Solution

Step 1: 120 degrees is between 90 and 180, so point V is in Quadrant II.

Step 2: Find the reference angle, the acute angle to the nearest x-axis: $180 - 120 = 60$ degrees.

Step 3: Look up the common angle 60 degrees: $\cos(60) = 1/2$ and $\sin(60) = \sqrt{3}/2$.

Step 4: In Quadrant II, the x-coordinate is negative and the y-coordinate is positive.

Step 5: Apply those signs to the reference values: $x = -1/2$ and $y = \sqrt{3}/2$.

Step 6: So the coordinates of point V are $(-1/2, \sqrt{3}/2)$.

Answer: $(-1/2, \sqrt{3}/2)$

Tip

For any angle that isn't a key axis angle, find its reference angle first, look up that reference angle's values, then attach the correct sign for the quadrant.

Trap

A common mistake is using the 120-degree angle itself in the common angles table instead of first finding its 60-degree reference angle.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$(1/2, \sqrt{3}/2)$	Used the correct reference values but forgot to make the x-coordinate negative for Quadrant II.
$(-\sqrt{3}/2, 1/2)$	Swapped which value goes with cosine and which goes with sine for the 60-degree reference angle.
$(-1/2, -\sqrt{3}/2)$	Correctly found the reference angle but applied Quadrant III signs instead of Quadrant II signs.

Shortcut

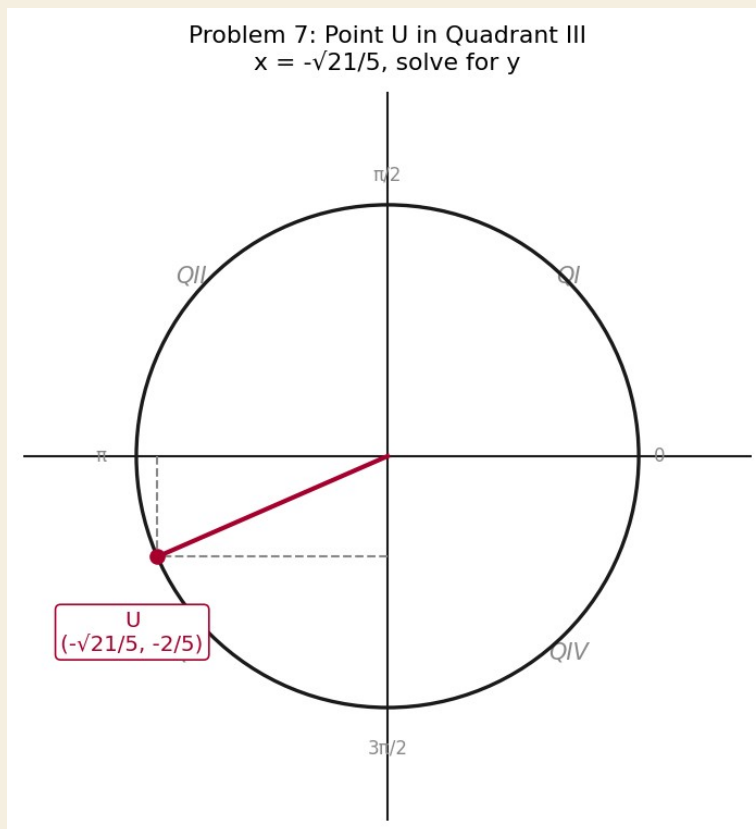
Once you recognize $120 = 180 - 60$, you can skip straight to the memorized 30-60-90 values for 60 degrees and just flip the x-sign for Quadrant II.

Desmos Tip

Type $(\cos(120^\circ), \sin(120^\circ))$ into Desmos to get the decimal coordinates, then compare them to -0.5 and $\sqrt{3}/2$ to confirm the exact values.

Radians & the Unit Circle — Problem 7 (Medium)

Point U lies on the unit circle $x^2 + y^2 = 1$, in Quadrant III (where both coordinates are negative). The x-coordinate of point U is $-\sqrt{21}/5$. What is the y-coordinate of point U?



Formula / Rule Used: Unit circle equation $x^2 + y^2 = 1$, then apply quadrant sign.

Solution

Step 1: Since point U is on the unit circle, its coordinates must satisfy $x^2 + y^2 = 1$.

Step 2: Substitute the known x-coordinate: $(-\sqrt{21}/5)^2 + y^2 = 1$.

Step 3: Square the x-coordinate: $(\sqrt{21})^2/5^2 = 21/25$.

Step 4: So the equation becomes $21/25 + y^2 = 1$.

Step 5: Subtract $21/25$ from both sides: $y^2 = 1 - 21/25 = 4/25$.

Step 6: Take the square root of both sides: $y = \sqrt{4/25} = 2/5$, so $y = 2/5$ or $y = -2/5$.

Step 7: Since point U is in Quadrant III, the y-coordinate must be negative, so choose $y = -2/5$.

Answer: $y = -2/5$

Tip

Whenever you're given one coordinate on the unit circle, plug it into $x^2 + y^2 = 1$ to solve for the other, then use the quadrant to pick the correct sign.

Trap

A common mistake is stopping after $y^2 = 4/25$ and writing $y = 2/5$, forgetting that the quadrant requires the negative root instead.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
$2/5$	Solved $y^2 = 4/25$ correctly but took only the positive square root, ignoring that Quadrant III needs a negative y-coordinate.
$-4/25$	Forgot to take the square root after finding $y^2 = 4/25$ and reported y^2 instead of y .
$-21/5$	Skipped squaring the x-coordinate and used $-\sqrt{21}/5$ directly inside the subtraction instead of $(-\sqrt{21}/5)^2$.

Shortcut

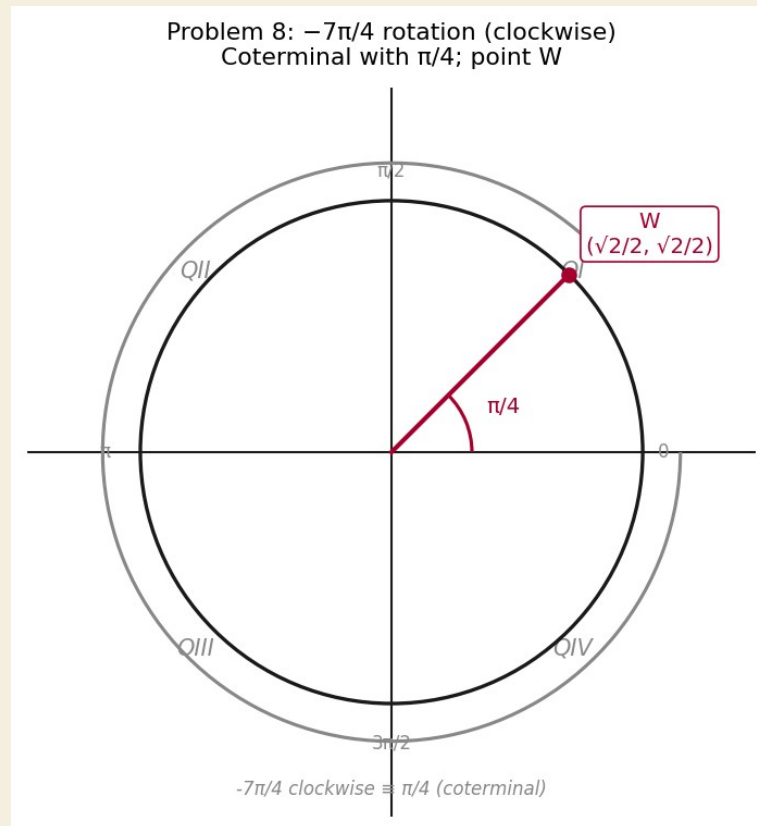
There's no shortcut around solving $x^2 + y^2 = 1$ here — but recognizing $21/25$ and $4/25$ as a clean pair (adding to $25/25$) helps you check your arithmetic quickly.

Desmos Tip

Type $x^2 + y^2 = 1$ to draw the unit circle, then type the point $(-\sqrt{21}/5, -2/5)$ to confirm it lands exactly on the circle in the lower-left quadrant.

Radians & the Unit Circle — Problem 8 (Medium)

At a robotics expo in Eindhoven, a camera gimbal rotates $-7\pi/4$ radians from its starting position (a negative angle means the rotation is clockwise instead of counterclockwise), pointing its lens at target point W on a unit-circle test rig of radius 1 meter. The gimbal motor completes this rotation in exactly 2 seconds. What are the exact coordinates of point W?



Formula / Rule Used: Coterminal angles + common angles table.

Solution

Step 1: The motor's speed (2 seconds) doesn't affect the final direction, so ignore it and focus only on the angle $-7\pi/4$.

Step 2: To bring a negative angle into the standard 0 to 2π range, add 2π : $-7\pi/4 + 2\pi$.

Step 3: Write 2π as $8\pi/4$ so the fractions match: $-7\pi/4 + 8\pi/4$.

Step 4: Add the numerators: $(-7 + 8)\pi/4 = \pi/4$.

Step 5: So the coterminal angle is $\pi/4$ radians, which is one of the common angles (45 degrees).

Step 6: At $\pi/4$, $\cos(\pi/4) = \sqrt{2}/2$ and $\sin(\pi/4) = \sqrt{2}/2$, since $\pi/4$ is in Quadrant I where both are positive.

Step 7: So point W has coordinates $(\sqrt{2}/2, \sqrt{2}/2)$.

Answer: $(\sqrt{2}/2, \sqrt{2}/2)$

Tip

Ignore any extra details that don't involve the angle itself, like time or motor speed, and convert a negative angle to standard range by adding 2π first.

**Trap**

A common mistake is treating the negative sign as meaning the coordinates should be negative, instead of using it to mean clockwise rotation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(-\sqrt{2}/2, -\sqrt{2}/2)$	Assumed the negative angle directly makes both coordinates negative instead of first converting to the positive coterminal angle $\pi/4$.
$(\sqrt{2}/2, -\sqrt{2}/2)$	Treated $-7\pi/4$ as equivalent to $-\pi/4$ (Quadrant IV) instead of precisely adding 2π to find the true coterminal angle $\pi/4$ in Quadrant I.
$(\sqrt{3}/2, 1/2)$	Found the correct coterminal angle $\pi/4$ but then used the 30-60-90 reference values for a different common angle instead.

Shortcut

Once you see the angle is close to 2π , just find how far short of a full rotation it is: $2\pi - 7\pi/4 = \pi/4$, which is faster than adding 2π to a negative fraction.

Desmos Tip

Type $(\cos(-7\pi/4), \sin(-7\pi/4))$ into Desmos to get the decimal coordinates directly, then compare them to $\sqrt{2}/2$ approx 0.707 to confirm point W.

28. Circles — Circumference, Arcs, Sectors, Equations & Theorems

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Circle	Κύκλος
Radius / diameter	Ακτίνα / διάμετρος
Circumference	Περιφέρεια (μήκος κύκλου)
Arc	Τόξο
Sector	Κυκλικός τομέας
Central angle / inscribed angle	Επίκεντρη γωνία / εγγεγραμμένη γωνία
Tangent line	Εφαπτομένη ευθεία

Circles — Circumference, Arcs, Sectors, Equations & Theorems — What It Is

What It Is

- A circle is every point that sits the exact same distance (the radius) from one fixed center point.
- The SAT tests circles because a small set of formulas can answer many different-looking questions.
- Arc length and sector area are always just a fraction of the whole circle's circumference or area.
- Circles also show up as algebra: an equation like $(x-h)^2 + (y-k)^2 = r^2$ describes the same shape.
- A few angle theorems (tangent lines, inscribed angles) connect straight-line angles to the circle itself.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Core Rules

Circumference and Area

$$C = 2\pi r = \pi d. \quad A = \pi r^2$$

- Circumference is the distance all the way around the circle, like a fence around a circular yard.
- Diameter is twice the radius, so $d = 2r$ lets you swap between the two circumference formulas.
- Area measures the space enclosed inside the circle, and only ever needs the radius.
- Example: radius 5 gives $C = 2\pi(5) = 10\pi$ and $A = \pi(5)^2 = 25\pi$.

Arc Length

$$\text{Arc length} = (\theta/360^\circ) \times 2\pi r$$

- An arc is just a piece of the circle's curved edge, cut off by a central angle θ .
- $\theta/360^\circ$ tells you what fraction of the entire circle that arc actually is.

- Multiply that fraction by the full circumference ($2\pi r$) to get just the arc's length.
- Example: a 90° arc on a circle of radius 8 is $(90/360) \times 2\pi(8) = (1/4)(16\pi) = 4\pi$.

Sector Area

$$\text{Sector area} = (\theta/360^\circ) \times \pi r^2$$

- A sector is the pizza-slice-shaped region bounded by two radii and an arc.
- It uses the exact same fraction idea as arc length, applied to area instead.
- Divide the central angle by 360° first, then multiply by the full area πr^2 .
- Example: a 60° sector of a circle of radius 6 is $(60/360) \times \pi(6)^2 = (1/6)(36\pi) = 6\pi$.

Equation of a Circle

$$(x - h)^2 + (y - k)^2 = r^2$$

- This equation lists every point (x, y) that is exactly distance r from the center.
- The center of the circle is the point (h, k) , read straight from the equation.
- Both x and y are always subtracted, even when h or k is itself a negative number.
- Example: center $(-3, 2)$, radius 5 gives $(x + 3)^2 + (y - 2)^2 = 25$, since $x - (-3) = x + 3$.

Completing the Square for a Circle

- Sometimes a circle's equation is given expanded, like $x^2 + y^2 + 6x - 4y - 12 = 0$.
- Group all the x -terms together and all the y -terms together first.
- Complete the square separately for the x -group and for the y -group.
- Whatever number you add to complete each square, add that same number to the other side too.

Tangent Lines

radius $\perp$ tangent line

- A tangent line touches the circle's edge at exactly one single point.
- The radius drawn to that touch point is always perpendicular to the tangent line.
- That perpendicularity creates a right angle, which usually means a right triangle.
- Once you see the right triangle, the Pythagorean theorem can find any missing side.

Inscribed Angle Theorem

$$\text{inscribed angle} = (1/2) \times \text{intercepted arc}$$

- An inscribed angle has its vertex sitting on the circle, formed by two chords.
- Its measure is always exactly half of the arc it 'cuts off,' called the intercepted arc.
- An inscribed angle that intercepts a diameter (spans a semicircle) is always exactly 90° .
- Two different inscribed angles that intercept the same arc are always equal to each other.

How to Recognize This on the Test

- Words like 'circumference,' 'arc,' or 'sector' signal you should reach for the circle formulas directly.
- A central angle in degrees paired with 'arc length' or 'area of the shaded region' means use $\theta/360^\circ$.
- An equation with both x^2 and y^2 terms and matching coefficients is almost always a circle in disguise.
- Phrases like 'tangent to the circle' or 'point of tangency' mean the radius-tangent right angle applies.
- A triangle drawn with all its corners touching the circle usually calls for the inscribed angle theorem.

Common Mistakes — and the Fix

- Using a chord length instead of the actual central angle inside the $\theta/360^\circ$ fraction.
- Forgetting the sign flip in the circle equation, writing $(x + h)$ instead of $(x - h)$ for a positive h .
- Completing the square but forgetting to add the same balancing numbers to the other side of the equation.
- Assuming any angle drawn inside a circle is the central angle, when it's really an inscribed angle.
- Plugging the diameter straight into a formula that actually needs the radius (half the diameter).

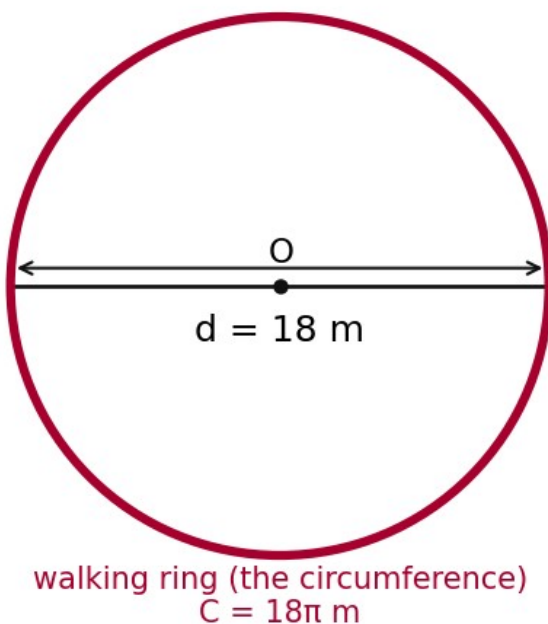
Circles — Circumference, Arcs, Sectors, Equations & Theorems — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Problem 1 (Easy)

A circular fountain in a park in Rotterdam has a diameter of 18 meters. A tiled walking ring traces the fountain's outer edge exactly once. What is the length of the walking ring, in terms of π ?

Fountain ring: diameter 18 m



Formula / Rule Used: Circumference: $C = \pi d$ (equivalently $C = 2\pi r$).

Solution

Step 1: The walking ring traces the circle's edge exactly once, so its length equals the fountain's circumference.

Step 2: Since you're given the diameter, use the circumference formula written in terms of diameter: $C = \pi d$.

Step 3: Substitute the given diameter: $C = \pi(18)$.

Step 4: Multiply: $C = 18\pi$ meters.

Answer: 18π meters

Tip

When a problem gives you the diameter, plug straight into $C = \pi d$ - there's no need to find the radius first.

Trap

It's tempting to cut the diameter in half out of habit and use $C = \pi r$ with $r = 9$, which drops a factor of 2 and gives half the true circumference.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
9π meters	Divided the diameter by 2 to get the radius, then used $C = \pi r$ instead of $C = \pi d$



	$2\pi r$ (or πd), losing a factor of 2.
81π meters	Confused the circumference formula with the area formula and computed $\pi \times 9^2$ using the radius.
324π meters	Used the diameter as if it were the radius in the area formula, computing $\pi \times 18^2$ by mistake.

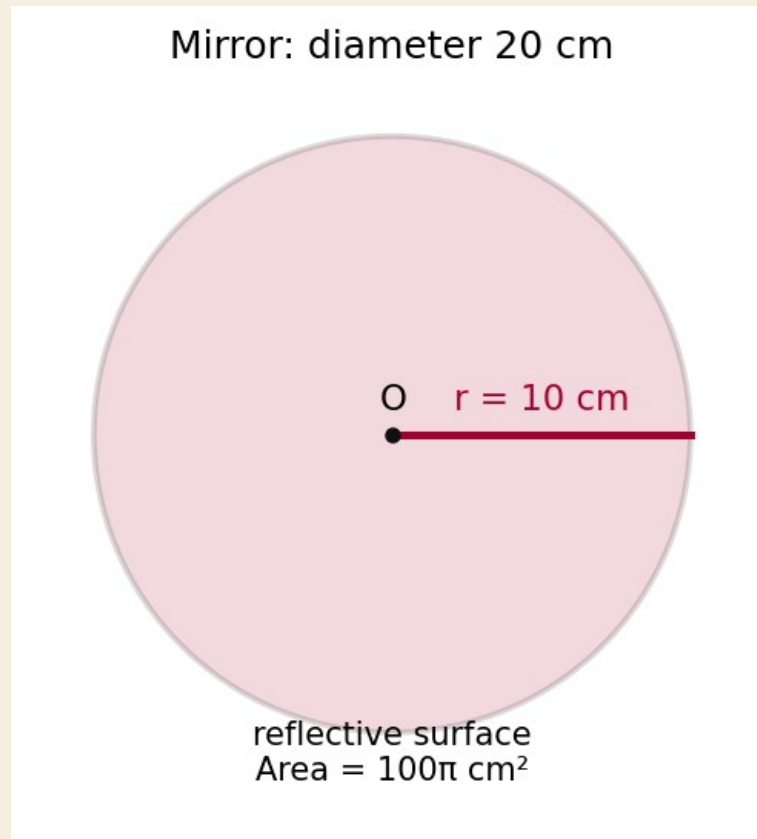
Shortcut

No shortcut needed - $C = \pi d$ applied directly to the given diameter is already the fastest route.

Desmos Tip

Type $18 \times \pi$ into a blank line on Desmos to get the decimal value (about 56.55) and confirm it matches 18π .

A circular mirror in a shop in Milan has a diameter of 20 centimeters. What is the area of the mirror's reflective surface, in terms of π ?



Formula / Rule Used: Area: $A = \pi r^2$ (with $r = d/2$).

Solution

Step 1: Area only needs the radius, so first find the radius from the given diameter.

Step 2: Radius is half the diameter: $r = 20/2 = 10$ centimeters.

Step 3: Plug the radius into the area formula: $A = \pi(10)^2$.

Step 4: Square the radius: $10^2 = 100$.

Step 5: Multiply: $A = 100\pi$ square centimeters.

Answer: 100π square centimeters

Tip

Whenever a circle problem gives the diameter but asks for area, convert to radius first - the area formula only works with r .

Trap

A common slip is plugging the diameter straight into $A = \pi r^2$ as if it were the radius, which makes the area four times too big.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



400π square centimeters	Used the diameter (20) directly as the radius in $A = \pi r^2$ instead of halving it first.
20π square centimeters	Mixed up the area formula with the circumference formula and computed $\pi \times d$ instead of squaring the radius.
10π square centimeters	Forgot to square the radius and multiplied π by r once instead of r^2 .

Shortcut

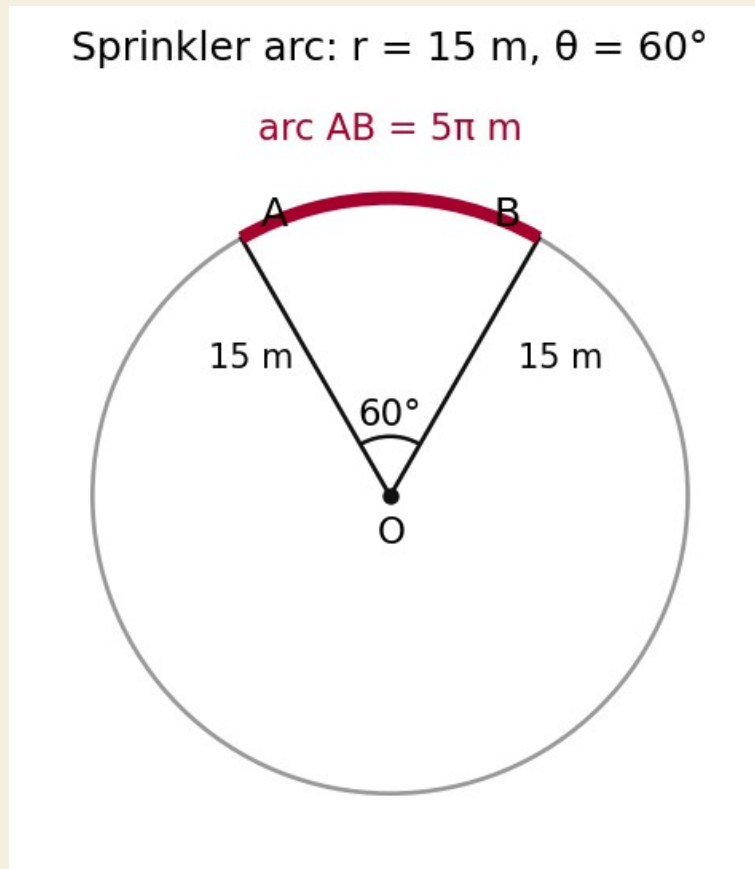
None genuinely faster - halving the diameter, then squaring, is already the quickest path.

Desmos Tip

Type $100 \times \pi$ into Desmos to get the decimal value (about 314.16) as a quick sanity check on your answer.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Problem 3 (Easy)

A sprinkler sits at the center O of a circular garden bed and sprays water out to a distance of 15 meters. The sprinkler rotates through a central angle of 60° before shutting off, wetting an arc-shaped strip along the garden bed's outer edge. Points A and B mark the two ends of the wetted arc, so angle AOB = 60° and OA = OB = 15 meters (the radius). What is the length of arc AB, in terms of π ?



Formula / Rule Used: Arc length = $(\theta/360^\circ) \times 2\pi r$.

Solution

Step 1: The arc length is the fraction $\theta/360$ of the full circumference.

Step 2: Write the fraction using the given central angle: $60/360$.

Step 3: Simplify the fraction: $60/360 = 1/6$.

Step 4: Find the full circumference: $2\pi r = 2\pi(15) = 30\pi$ meters.

Step 5: Multiply the fraction by the circumference: $(1/6) \times 30\pi$.

Step 6: Compute: $30\pi \div 6 = 5\pi$ meters.

Answer: 5π meters

Tip

Start every arc length problem by turning the central angle into a fraction of 360° - that fraction drives everything else.

Trap



It's easy to stop after finding the fraction $1/6$ and forget to multiply it by the full circumference $2\pi r$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2.5π meters	Used the radius alone ($\pi r = 15\pi$) instead of the full circumference ($2\pi r = 30\pi$) before multiplying by $1/6$.
10π meters	Divided the central angle by 180° instead of 360° , treating a straight line as the whole rotation and using $1/3$ instead of $1/6$.
900π meters	Multiplied the angle 60 directly by the circumference 30π instead of first dividing 60 by 360.

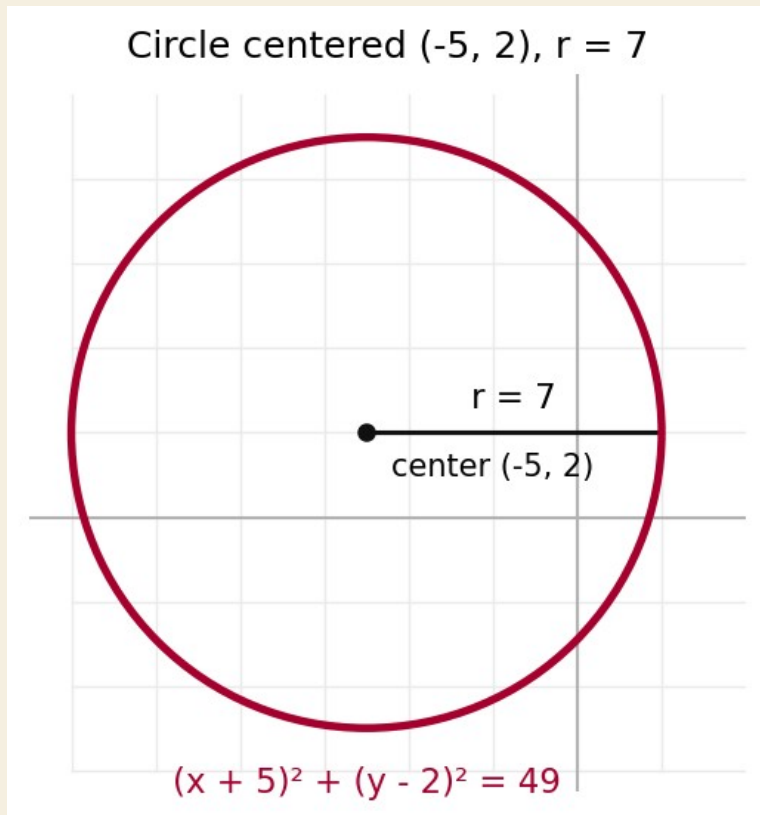
Shortcut

None truly faster - reducing the angle fraction first, then multiplying by $2\pi r$, is already the quickest method.

Desmos Tip

Type $(60/360) * 2 * \pi * 15$ as a single line in Desmos to get the decimal value (about 15.71) and confirm it matches 5π .

A circle is centered at the point $(-5, 2)$ and has a radius of 7. What is the equation of this circle?



Formula / Rule Used: $(x - h)^2 + (y - k)^2 = r^2$.

Solution

Step 1: The standard equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center.

Step 2: Here the center is $(-5, 2)$, so $h = -5$ and $k = 2$.

Step 3: Substitute $h = -5$ into $(x - h)$: $x - (-5)$ becomes $x + 5$, so the first term is $(x + 5)^2$.

Step 4: Substitute $k = 2$ into $(y - k)$: the second term is $(y - 2)^2$.

Step 5: Square the radius: $r^2 = 7^2 = 49$.

Step 6: Combine everything: $(x + 5)^2 + (y - 2)^2 = 49$.

Answer: $(x + 5)^2 + (y - 2)^2 = 49$

Tip

Always substitute the center's coordinates into $(x - h)^2 + (y - k)^2 = r^2$ exactly as written, then simplify the signs afterward.

Trap

Because $h = -5$ is negative, it's tempting to write $(x - 5)^2$ instead of the correct $(x + 5)^2$ - the minus sign in the formula and the negative center combine to a plus.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



$(x - 5)^2 + (y - 2)^2 = 49$	Dropped the sign flip and wrote $(x - h)$ with $h = -5$ as $(x - 5)$ instead of $(x + 5)$.
$(x + 5)^2 + (y - 2)^2 = 7$	Forgot to square the radius and used $r = 7$ directly instead of $r^2 = 49$.
$(x + 5)^2 + (y + 2)^2 = 49$	Flipped the sign on $k = 2$ as well, writing $(y + 2)$ instead of the correct $(y - 2)$.

Shortcut

None needed - direct substitution into the standard form is already the fastest method.

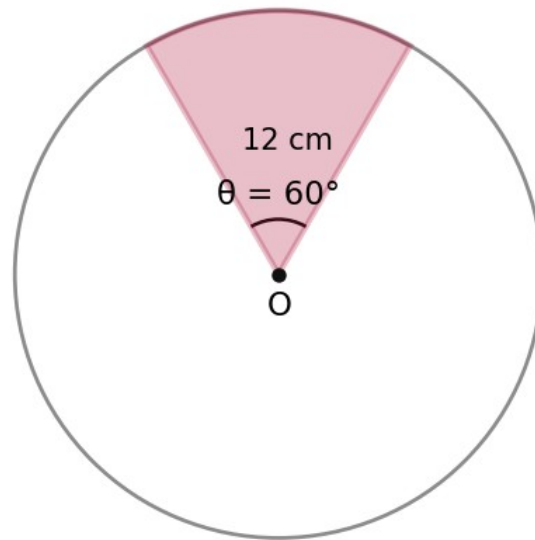
Desmos Tip

Type $(x+5)^2+(y-2)^2=49$ into Desmos to graph the circle and visually confirm the center lands at $(-5, 2)$ with the circle reaching 7 units out in every direction.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Problem 5 (Medium)

A circular clock face in a train station in Brussels has a radius of 12 centimeters. A decorative sector cut from the clock face - like a pie slice bounded by two radii and an arc - has an area of 24π square centimeters. What is the measure of the sector's central angle, in degrees?

Clock face: $r = 12$ cm, sector area = 24π



sector area = 24π cm²

Formula / Rule Used: Sector area = $(\theta/360^\circ) \times \pi r^2$, solved backward for θ .

Solution

Step 1: Start from the sector area formula: sector area = $(\theta/360) \times \pi r^2$.

Step 2: Find the full circle's area first: $\pi r^2 = \pi(12)^2 = 144\pi$ square centimeters.

Step 3: Substitute the known sector area into the formula: $24\pi = (\theta/360) \times 144\pi$.

Step 4: Divide both sides by π to cancel it: $24 = (\theta/360) \times 144$.

Step 5: Divide both sides by 144 to isolate the fraction: $\theta/360 = 24/144$.

Step 6: Simplify the fraction: $24/144 = 1/6$.

Step 7: Multiply both sides by 360 to solve for θ : $\theta = 360 \times (1/6) = 60$.

Answer: 60°

Tip

When you're given the sector's area instead of the angle, set up the same $(\theta/360) \times \pi r^2$ equation and solve backward for θ .

Trap

It's tempting to divide the sector area by the radius (12) instead of by the full circle's area (144π), which skips squaring the radius entirely.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
720°	Divided 24π by 12 (the radius) instead of by $\pi r^2 = 144\pi$, then never checked that a central angle must be under 360°.
180°	Set the fraction up as $24/48$ (using $4r$ instead of $r^2 = 144$) instead of the correct $24/144$.
48°	Divided 24 by 144 correctly but then multiplied by 288 instead of 360, using a miscopied denominator.

Shortcut

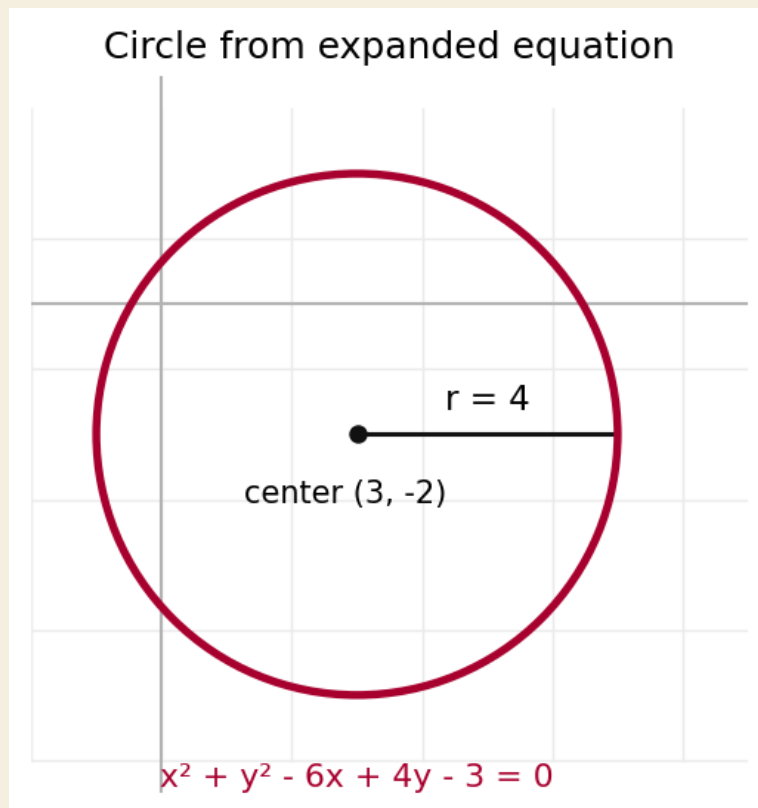
Notice 24π is exactly $1/6$ of the full area 144π at a glance, so $\theta = 360/6 = 60^\circ$ without writing out the full equation.

Desmos Tip

Type $24/144 * 360$ into Desmos as a quick arithmetic check to confirm $\theta = 60$.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Problem 6 (Medium)

A circle is graphed on the xy -coordinate plane. Its equation is given in expanded form: $x^2 + y^2 - 6x + 4y - 3 = 0$. What is the radius of this circle?



Formula / Rule Used: Complete the square to reach $(x - h)^2 + (y - k)^2 = r^2$.

Solution

Step 1: Group the x -terms and y -terms together: $(x^2 - 6x) + (y^2 + 4y) = 3$.

Step 2: Complete the square for the x -group: take half of -6 (which is -3), square it (9), and add 9 inside the x -group.

Step 3: Complete the square for the y -group: take half of 4 (which is 2), square it (4), and add 4 inside the y -group.

Step 4: Add those same numbers (9 and 4) to the right side too, to keep the equation balanced: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 3 + 9 + 4$.

Step 5: Rewrite each group as a squared binomial: $(x - 3)^2 + (y + 2)^2 = 16$.

Step 6: The right side is r^2 , so $r^2 = 16$.

Step 7: Take the square root: $r = 4$.

Answer: 4

Tip

Whenever you see x^2 and y^2 with matching coefficients plus extra x and y terms, it's a circle equation in disguise - complete the square to reveal it.

Trap



It's easy to forget to add the completing-the-square numbers (9 and 4) to the right side too, which throws off the radius.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3	Assumed the constant already on the right side (3) was r^2 without completing the square at all.
16	Correctly completed the square to find $r^2 = 16$ but forgot the final step of taking the square root.
$\sqrt{7}$	Added 4 to complete the square for y but forgot to also add 9 for x , leaving the right side as $3 + 4 = 7$ instead of the correct $3 + 9 + 4 = 16$.

Shortcut

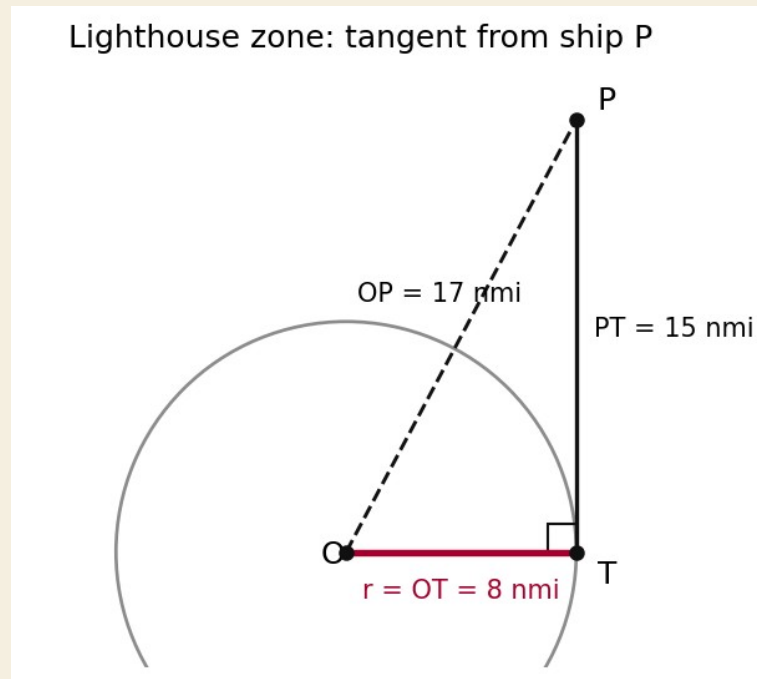
None genuinely faster - completing the square is the only reliable way to reveal the center and radius from expanded form.

Desmos Tip

Type $x^2 + y^2 - 6x + 4y - 3 = 0$ directly into Desmos - it will graph the circle, and you can trace it to confirm the radius spans 4 units from the center.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Problem 7 (Medium)

A lighthouse on a small island in the Aegean Sea sits at the center O of a circular restricted zone with an unknown radius r . A ship is at point P , outside the zone. The sightline from the ship to the nearest edge of the zone touches the circle at exactly one point, T , so segment PT is tangent to the circle at T (meaning OT is perpendicular to PT). The distance from the ship to the lighthouse is $OP = 17$ nautical miles, and the tangent distance $PT = 15$ nautical miles. What is the radius r (that is, the length OT), in nautical miles?



Formula / Rule Used: Radius $\perp$ tangent line, then the Pythagorean theorem.

Solution

Step 1: Since the radius OT is perpendicular to the tangent segment PT , triangle OTP is a right triangle with the right angle at T .

Step 2: OP is the hypotenuse of this right triangle, since it's the side opposite the right angle.

Step 3: Write the Pythagorean theorem: $OT^2 + PT^2 = OP^2$.

Step 4: Substitute the known lengths: $OT^2 + 15^2 = 17^2$.

Step 5: Square the numbers: $OT^2 + 225 = 289$.

Step 6: Subtract 225 from both sides: $OT^2 = 64$.

Step 7: Take the square root: $OT = 8$.

Answer: 8 nautical miles

Tip

Whenever a tangent line and a radius meet at the point of tangency, mark that right angle immediately and set up the Pythagorean theorem.

Trap

It's tempting to treat PT as the hypotenuse since it looks like the longest drawn segment, but OP (from the outside point to the center) is always the hypotenuse of a radius-tangent right triangle.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
32 nautical miles	Added 17 and 15 instead of using the Pythagorean theorem, treating the sides as if they simply combined along one line.
64 nautical miles	Correctly computed $OT^2 = 64$ using the Pythagorean theorem but forgot to take the square root.
about 22.7 nautical miles ($\sqrt{514}$)	Added the squares of 17 and 15 instead of subtracting, mistakenly treating OP as a leg rather than the hypotenuse.

Shortcut

Recognize 8-15-17 as a Pythagorean triple, so once you see legs/hypotenuse 15 and 17 you can jump straight to $OT = 8$ without writing out every arithmetic step.

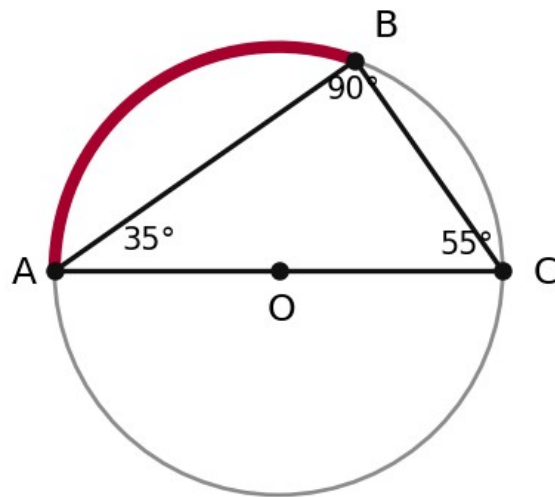
Desmos Tip

Type $\sqrt{17^2 - 15^2}$ into Desmos as a one-line check to instantly confirm $OT = 8$.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Problem 8 (Medium)

In circle O, segment AC is a diameter, and point B lies on the circle so that triangle ABC is formed with all three vertices on the circle. Angle BAC measures 35° . Because AC is a diameter, angle ABC (the angle at vertex B, across from the diameter) is automatically a right angle. What is the measure of arc AB, the arc from A to B that does not contain point C?

Inscribed triangle with diameter AC



$$\text{arc AB} = 110^\circ$$

Formula / Rule Used: Inscribed angle theorem, combined with the triangle angle sum.

Solution

Step 1: Because AC is a diameter, the inscribed angle ABC that intercepts it (spanning the semicircle) is always 90° .

Step 2: The three angles of triangle ABC add up to 180° : angle BAC + angle ABC + angle BCA = 180.

Step 3: Substitute the known angles: $35 + 90 + \text{angle BCA} = 180$.

Step 4: Add 35 and 90: $125 + \text{angle BCA} = 180$.

Step 5: Subtract 125 from both sides: angle BCA = 55.

Step 6: Angle BCA has its vertex at C, and it intercepts arc AB (the arc not containing C).

Step 7: By the inscribed angle theorem, the intercepted arc is double the inscribed angle: arc AB = 2×55 .

Step 8: Multiply: arc AB = 110° .

Answer: 110°

Tip

Whenever a triangle is inscribed in a circle with one side as a diameter, mark that 90° angle first (it's automatic) before doing anything else.

**Trap**

It's tempting to double the given angle BAC (35°) directly since it's the angle you were handed, but BAC intercepts arc BC, not arc AB - you need angle BCA to get arc AB.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
70°	Doubled the given angle BAC = 35° directly, forgetting that BAC intercepts arc BC, not the requested arc AB.
55°	Correctly found angle BCA = 55° using the triangle angle sum but forgot to double it for the inscribed angle theorem.
145°	Computed $180 - 35 = 145$ instead of properly subtracting both 35 and 90 from 180 to find angle BCA first.

Shortcut

Once you know AC is a diameter, angle ABC is 90° for free - that single fact is usually the key that unlocks the rest of the triangle.

Desmos Tip

Graph the circle $x^2 + y^2 = 1$ with A=(-1,0) and C=(1,0) as the diameter endpoints, then use Desmos's angle tool to place B so angle BAC = 35° and read off the resulting arc measure to confirm 110° .

29. Volume, Surface Area & Similarity of Solids

Volume, Surface Area & Similarity of Solids — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Volume	Όγκος
Surface area	Εμβαδόν επιφάνειας
Prism / cylinder / cone / sphere / pyramid	Πρίσμα / κύλινδρος / κώνος / σφαίρα / πυραμίδα
Base area	Εμβαδόν βάσης
Similarity (similar solids)	Ομοιότητα (όμοια στερεά)
Scale factor	Λόγος ομοιότητας

Volume, Surface Area & Similarity of Solids — What It Is

What It Is

- Volume measures how much 3D space is inside a solid, like how much water a container could hold.
- Surface area measures the total area of every outside face of a solid, like wrapping paper needed to cover it.
- The SAT tests these because a handful of formulas can answer almost any 3D-shape question it asks.
- Similar solids are the same shape scaled up or down, so a single scale factor connects their lengths, areas, and volumes.
- Most volume formulas reduce to 'area of the base times height,' sometimes with an extra fraction out front.

Volume, Surface Area & Similarity of Solids — Core Rules

Prisms and Cylinders

$$V = (\text{area of base}) \times \text{height}$$

- A prism or cylinder has two matching, parallel bases connected by straight sides.
- Find the area of one base, then multiply it by how tall the solid is.
- Rectangular prism: $V = l \times w \times h$ (length times width times height).
- Cylinder: $V = \pi r^2 h$, since the base is a circle with area πr^2 .

Pyramids and Cones

$$V = (1/3) \times (\text{area of base}) \times \text{height}$$

- A pyramid or cone comes to a single point instead of having two matching bases.
- It always holds exactly $\frac{1}{3}$ the volume of a prism or cylinder with the same base and height.
- Cone: $V = \frac{1}{3}\pi r^2 h$. Square pyramid: $V = \frac{1}{3}s^2 h$.
- Example: a cone with base area 30 and height 6 has $V = \frac{1}{3}(30)(6) = 60$.

Spheres

$$V = \frac{4}{3}\pi r^3. \quad \text{Surface area} = 4\pi r^2$$

- A sphere is a perfectly round solid, like a ball, and both formulas need only its radius.
- For volume, cube the radius (r^3); for surface area, square the radius (r^2) — don't mix these up.
- A hemisphere is exactly half a sphere, so divide the sphere's volume (or curved surface area) by 2.
- Example: radius 3 gives volume $\frac{4}{3}\pi(27) = 36\pi$ and surface area $4\pi(9) = 36\pi$.

Surface Area of Common Solids

$$\text{Cube: } S = 6s^2. \quad \text{Rect. prism: } S = 2(lw + lh + wh). \quad \text{Cylinder: } S = 2\pi rh + 2\pi r^2$$

- Surface area adds up the area of every single outside face of the solid.
- For a rectangular prism, find the area of each of the 3 unique face-pairs, then double the sum.
- For a cylinder, add the curved side ($2\pi rh$) to the two circular ends ($2\pi r^2$).
- Example: a cube with side 4 has $S = 6(4^2) = 6(16) = 96$.

Composite Solids

- A composite solid is built from two or more basic solids stuck together or cut out of each other.
- Break it into recognizable pieces first, like a cylinder plus a cone, or a cube minus a sphere.
- Find each piece's volume (or surface area) separately using its own formula.
- Add the pieces if they're joined together, or subtract if one piece is hollowed out or removed.

Polygon Interior Angle Sum

$$(n - 2) \times 180^\circ$$

- This finds the sum of all interior angles in any flat polygon with n sides.
- A triangle ($n = 3$) gives $(3-2)(180) = 180^\circ$, and a quadrilateral ($n = 4$) gives $(4-2)(180) = 360^\circ$.
- This can show up when a solid's flat face (like a pentagon base) needs an angle checked.
- Divide the sum by n to find one interior angle, but only if the polygon is regular (all sides equal).

Similarity Ratios

$$\text{length ratio} = \lambda. \quad \text{area ratio} = \lambda^2. \quad \text{volume ratio} = \lambda^3$$

- Two solids are similar when one is an exact scaled-up or scaled-down copy of the other.



- If every length scales by a factor λ , every area scales by λ^2 , and every volume scales by λ^3 .
- This lets you jump straight from a volume or area ratio to a length ratio, without finding real dimensions.
- Example: if $\lambda = 2$ (lengths double), area quadruples ($2^2 = 4$) and volume increases 8 times ($2^3 = 8$).

How to Recognize This on the Test

- Words like 'volume,' 'capacity,' or 'how much can it hold' signal you need a volume formula.
- Words like 'surface area,' 'paint,' 'wrap,' or 'label' signal you need to add up the outside faces.
- Phrases like 'similar solids,' 'scaled version,' or 'same shape, different size' signal a similarity ratio.
- A shape described as two solids joined together (like a cone on top of a cylinder) signals a composite solid.
- A given ratio of volumes or areas, asking for a length or scale factor, signals working backward through λ^3 or λ^2 .

Common Mistakes — and the Fix

- Forgetting the $1/3$ factor for pyramids and cones, and using the prism/cylinder formula instead.
- Plugging the diameter directly into a formula that actually needs the radius (half the diameter).
- Applying a scale factor λ directly to area or volume, instead of squaring or cubing it first.
- For a hemisphere, reusing the full sphere's volume or surface area instead of dividing it by 2.
- In a composite solid, forgetting to add (or subtract) every piece, or double-counting a shared face.

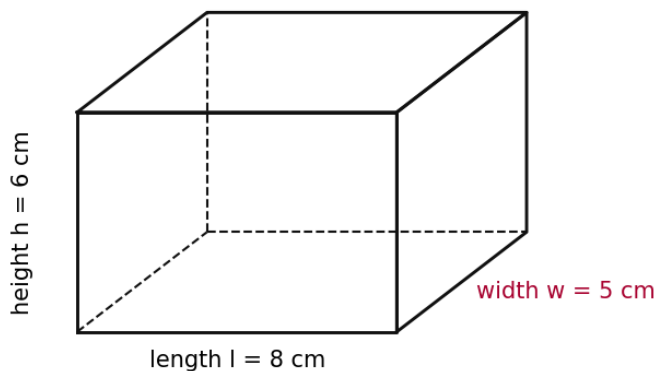
Volume, Surface Area & Similarity of Solids — 8 Practice Problems (4 Easy + 4 Medium)

Work through the Easy problems first to build confidence, then the Medium problems to start training for the twists real SAT/PSAT questions use.

Volume, Surface Area & Similarity of Solids — Problem 1 (Easy)

A shipping company in Rotterdam packs a rectangular cardboard box shaped like a rectangular prism. The box has length 8 centimeters, width 5 centimeters, and height 6 centimeters. What is the volume of the box, in cubic centimeters?

Rectangular Prism — Shipping Box



Formula / Rule Used: Rectangular prism volume: $V = l \times w \times h$.

Solution

Step 1: Write the rectangular prism volume formula: $V = l \times w \times h$.

Step 2: Plug in the given dimensions: $V = 8 \times 5 \times 6$.

Step 3: Multiply the first two numbers: $8 \times 5 = 40$.

Step 4: Multiply that result by the height: $40 \times 6 = 240$.

Step 5: The volume is 240 cubic centimeters.

Answer: 240 cubic centimeters

Tip

Start by writing down all three dimensions from the problem before multiplying, so you don't accidentally leave one out.

Trap

It's tempting to only multiply two of the three dimensions and stop, since that already looks like an 'area' answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
19	Added the three dimensions instead of multiplying them: $8 + 5 + 6 = 19$.
40	Multiplied only length and width ($8 \times 5 = 40$) and forgot to multiply by the height.

236

Accidentally used the surface area formula $2(lw + lh + wh)$ instead of the volume formula.

Shortcut

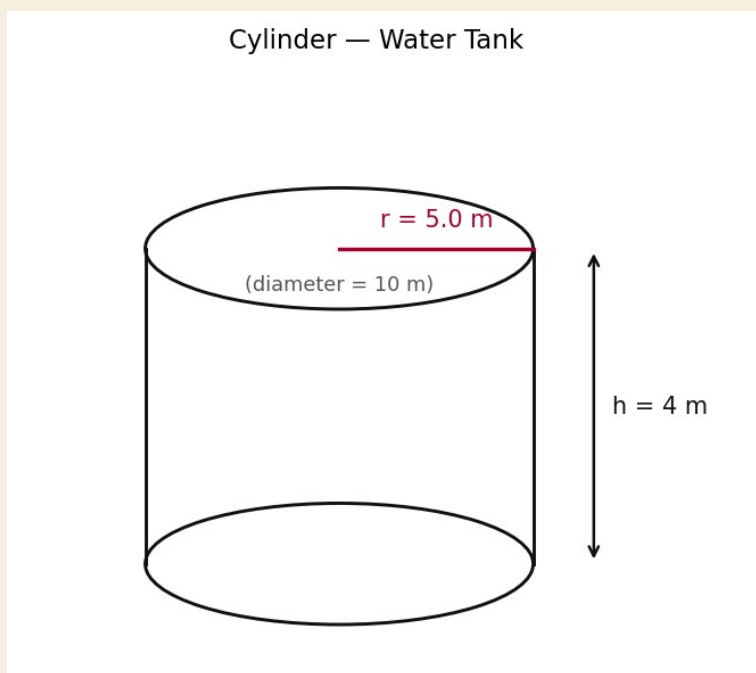
None needed — multiplying all three dimensions straight through is already the fastest method.

Desmos Tip

Type $8 \times 5 \times 6$ into a Desmos calculation line to instantly confirm the product is 240.

Volume, Surface Area & Similarity of Solids — Problem 2 (Easy)

A cylindrical water tank in Lisbon has a diameter of 10 meters and a height of 4 meters. What is the volume of the tank, in cubic meters, in terms of π ?



Formula / Rule Used: Cylinder volume: $V = \pi r^2 h$.

Solution

Step 1: The formula needs the radius, but the problem gives the diameter, so divide by 2 first: $r = 10 / 2 = 5$.

Step 2: Write the cylinder volume formula: $V = \pi r^2 h$.

Step 3: Plug in $r = 5$ and $h = 4$: $V = \pi(5)^2(4)$.

Step 4: Square the radius: $5^2 = 25$.

Step 5: Multiply: $25 \times 4 = 100$.

Step 6: The volume is 100π cubic meters.

Answer: 100π cubic meters

**Tip**

Whenever a cylinder or sphere problem gives you a diameter, divide by 2 to get the radius before touching the formula.

Trap

Plugging the diameter (10) directly into the formula as if it were the radius gives a volume four times too large.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
400π	Used the diameter (10) as the radius directly instead of dividing by 2 first, computing $\pi(10)^2(4)$.
40π	Used the lateral surface area formula $2\pi rh$ by mistake instead of the volume formula $\pi r^2 h$.
20π	Forgot to square the radius, computing $\pi(5)(4)$ instead of $\pi(5)^2(4)$.

Shortcut

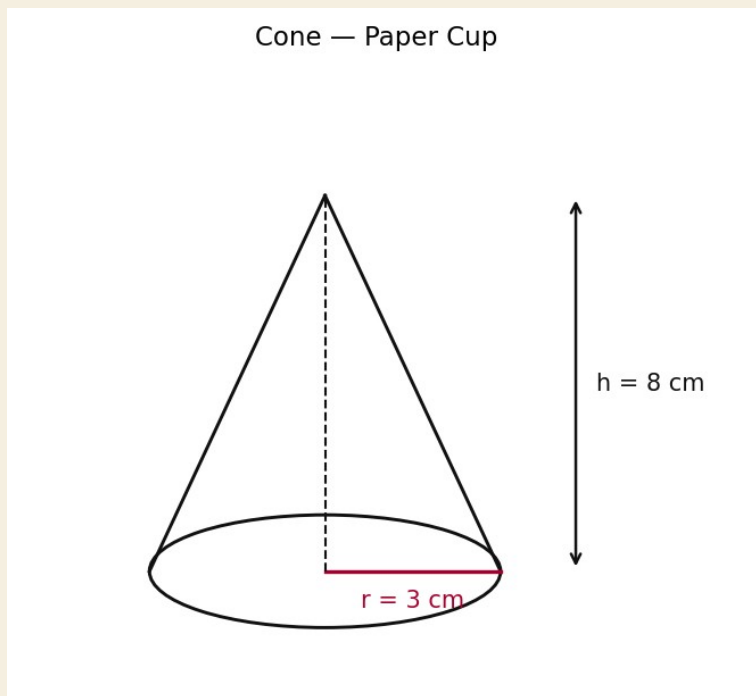
None needed here — finding the radius first and plugging straight into $\pi r^2 h$ is already the fastest path.

Desmos Tip

Type $3.14159 \cdot 5^2 \cdot 4$ in Desmos to get a decimal check of about 314.16, which matches $100\pi \approx 314.16$.

Volume, Surface Area & Similarity of Solids — Problem 3 (Easy)

A cone-shaped paper cup has a base radius of 3 centimeters and a height of 8 centimeters. What is the volume of the cup, in cubic centimeters, in terms of π ?



Formula / Rule Used: Cone volume: $V = (1/3)\pi r^2 h$.

Solution

Step 1: Write the cone volume formula: $V = (1/3)\pi r^2 h$.

Step 2: Plug in $r = 3$ and $h = 8$: $V = (1/3)\pi(3)^2(8)$.

Step 3: Square the radius: $3^2 = 9$.

Step 4: Multiply 9 by the height: $9 \times 8 = 72$.

Step 5: Multiply by $1/3$: $72 \times (1/3) = 24$.

Step 6: The volume is 24π cubic centimeters.

Answer: 24π cubic centimeters

Tip

Whenever a solid comes to a single point, like a cone or pyramid, remember to multiply by $1/3$ at the end.

Trap

Forgetting the $1/3$ factor makes this look like a cylinder problem, giving an answer exactly 3 times too big.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
72π	Forgot the $1/3$ factor entirely and used the cylinder volume formula $\pi r^2 h$ instead of the cone formula.
8π	Used the radius (3) instead of the squared radius (9) in the formula, computing $(1/3)\pi(3)(8)$.

216π

Multiplied by 3 instead of dividing by 3, computing 72×3 instead of $72 \times (1/3)$.

Shortcut

None needed — plugging directly into $(1/3)\pi r^2 h$ is already the fastest way to solve this.

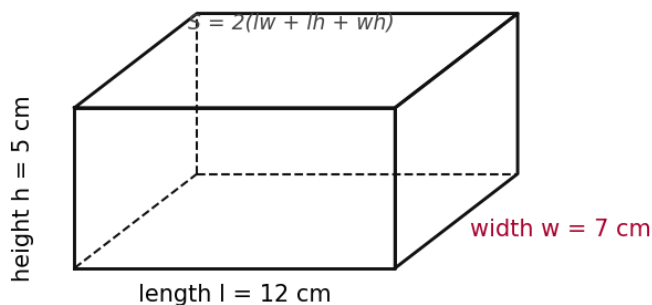
Desmos Tip

Type $(1/3)*3.14159*3^2*8$ in Desmos to get a decimal check of about 75.4, which matches $24\pi \approx 75.4$.

Volume, Surface Area & Similarity of Solids — Problem 4 (Easy)

A gift box in Vienna is shaped like a rectangular prism with length 12 centimeters, width 7 centimeters, and height 5 centimeters. What is the total surface area of the box, in square centimeters, needed to wrap it completely?

Rectangular Prism — Gift Box (Surface Area)



Formula / Rule Used: Rectangular prism surface area: $S = 2(lw + lh + wh)$.

Solution

Step 1: Write the surface area formula: $S = 2(lw + lh + wh)$.

Step 2: Find each face-pair product: $lw = 12 \times 7 = 84$.

Step 3: Next: $lh = 12 \times 5 = 60$.

Step 4: Next: $wh = 7 \times 5 = 35$.

Step 5: Add all three products together: $84 + 60 + 35 = 179$.

Step 6: Multiply the sum by 2: $179 \times 2 = 358$.

Step 7: The surface area is 358 square centimeters.

Answer: 358 square centimeters

**Tip**

Find the three different face-pair products first, add them, and only double the total as your very last step.

Trap

It's easy to compute the three face-pair products correctly, then forget to double the sum before reporting the answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
179	Correctly summed the three face-pair products but forgot to double the sum at the end.
420	Calculated the volume ($12 \times 7 \times 5 = 420$) instead of the surface area.
288	Forgot the width-times-height face pair entirely, computing only $2(lw + lh) = 2(84 + 60)$.

Shortcut

None needed here — adding the three face-pair products and doubling is already the fastest method.

Desmos Tip

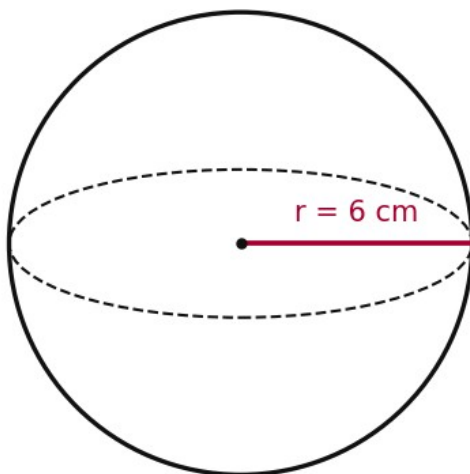
Type $2*(12*7+12*5+7*5)$ into Desmos in one line to confirm the total surface area is 358.

Volume, Surface Area & Similarity of Solids — Problem 5 (Medium)

A spherical stress ball has a surface area of 144π square centimeters. What is the volume of the ball, in cubic centimeters, in terms of π ?

Sphere — Stress Ball

$$\text{Surface area } S = 4\pi r^2 = 144\pi \text{ cm}^2$$



Formula / Rule Used: Sphere surface area ($S = 4\pi r^2$), then sphere volume ($V = (4/3)\pi r^3$).

Solution

Step 1: This problem works backward: set the surface area formula equal to the given value: $4\pi r^2 = 144\pi$.

Step 2: Divide both sides by 4π to isolate r^2 : $r^2 = 144/4 = 36$.

Step 3: Take the square root of both sides to find r : $r = 6$.

Step 4: Now switch to the volume formula and plug in $r = 6$: $V = (4/3)\pi(6)^3$.

Step 5: Cube the radius: $6^3 = 216$.

Step 6: Multiply: $(4/3) \times 216 = 288$.

Step 7: The volume is 288π cubic centimeters.

Answer: 288π cubic centimeters

Tip

When a sphere problem gives surface area but asks for volume, always solve for the radius first, then switch formulas.

Trap

After finding $r^2 = 36$ from the surface area equation, it's tempting to plug 36 straight into the volume formula instead of $r = 6$.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
48π	Used $r^2 = 36$ directly in the volume formula instead of finding $r = 6$ and cubing it, computing $(4/3)\pi(36)$.
864π	Forgot to divide by 3, computing $4\pi(216)$ instead of $(4/3)\pi(216)$.
36π	Treated the radius found (6) as if it were a diameter, halving it to 3 before cubing.

Shortcut

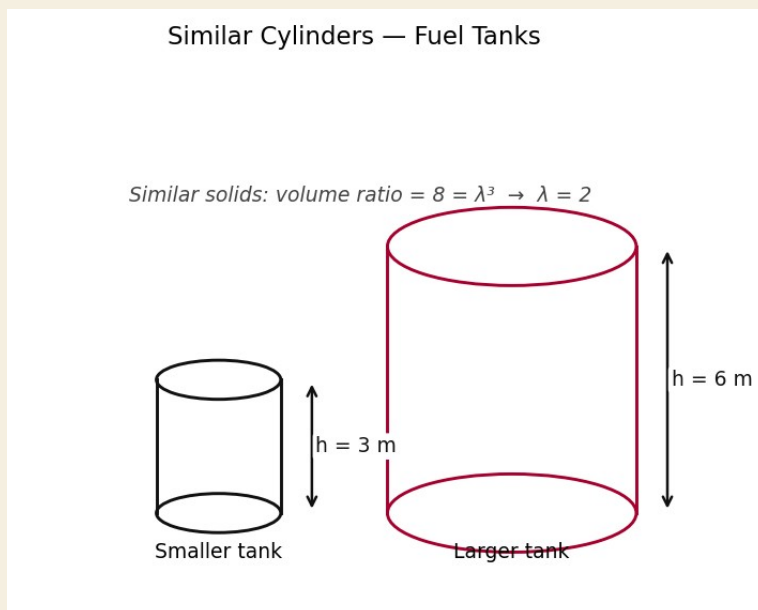
Since $144/4 = 36$ is a perfect square, $r = 6$ comes out cleanly without a calculator.

Desmos Tip

Type $\sqrt{144/4}$ in Desmos to confirm $r = 6$, then type $(4/3) \cdot 3.14159 \cdot 6^3$ to confirm the volume is about 904.8, matching 288π .

Volume, Surface Area & Similarity of Solids — Problem 6 (Medium)

Two similar cylindrical fuel tanks are being compared at a depot in Rotterdam. The larger tank holds exactly 8 times the volume of the smaller tank. If the smaller tank has a height of 3 meters, what is the height of the larger tank, in meters?



Formula / Rule Used: Similarity ratio: volume ratio = λ^3 , length ratio = λ .

Solution

Step 1: The tanks are similar solids, so their volume ratio equals the scale factor cubed: $\lambda^3 = 8$.

Step 2: Take the cube root of both sides to find the scale factor: $\lambda = \text{cube root of } 8 = 2$.

Step 3: Lengths (like height) scale by λ itself, not by λ^3 — this is the key difference from the volume ratio.

Step 4: Multiply the smaller tank's height by the scale factor: $3 \times 2 = 6$.

Step 5: The larger tank's height is 6 meters.

Answer: 6 meters

Tip

Whenever you're given a volume ratio for similar solids, take the cube root first to get the actual length scale factor.

Trap

It's tempting to multiply the height directly by the volume ratio (8) instead of finding the true length scale factor (2) first.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
24	Multiplied the height by the volume ratio 8 directly instead of finding the length scale factor 2 first.
12	Found the correct scale factor 2, but squared it (as if for area) instead of



	using it directly for a length.
1.5	Found the correct scale factor 2, but divided the height by it instead of multiplying.

Shortcut

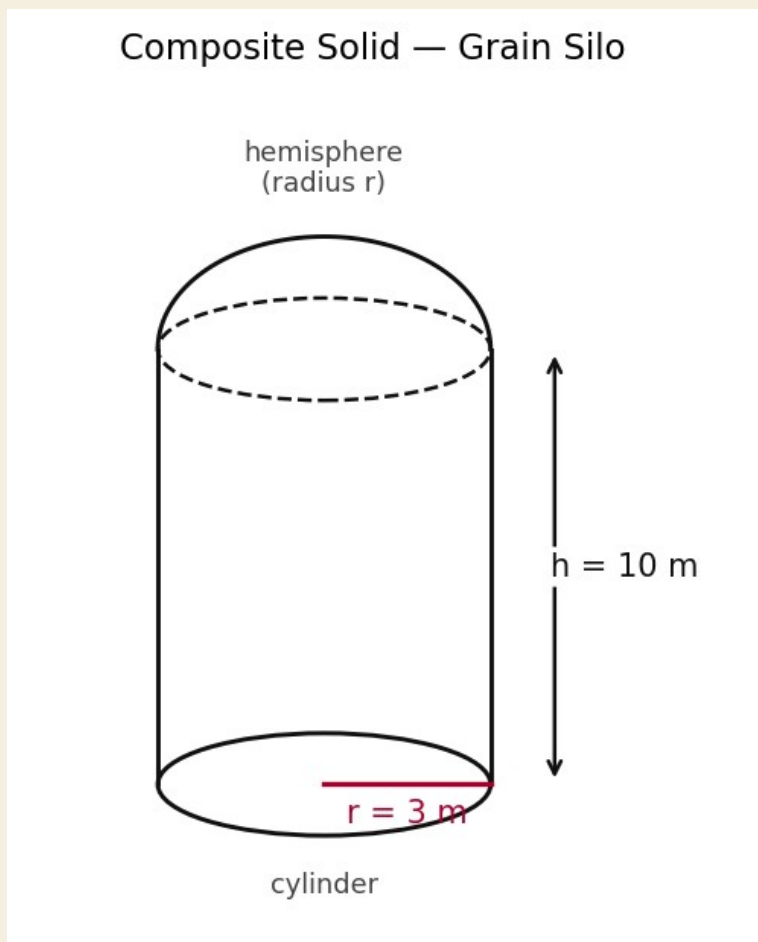
Recognize that 8 is a perfect cube ($2^3 = 8$), so the scale factor 2 can be spotted instantly without long computation.

Desmos Tip

Type 2^3 in Desmos to confirm it equals 8, verifying the scale factor of 2 before multiplying it by the height.

Volume, Surface Area & Similarity of Solids — Problem 7 (Medium)

A grain silo in rural Poland is shaped like a cylinder with a hemisphere sitting on top, sharing the same radius. The cylindrical part has a radius of 3 meters and a height of 10 meters. What is the total volume of the silo, in cubic meters, in terms of π ?



Formula / Rule Used: Composite solid: cylinder volume + hemisphere volume.

Solution

Step 1: Break the silo into two familiar solids: a cylinder on the bottom and a hemisphere on top, both with radius 3.

Step 2: Find the cylinder's volume: $V = \pi r^2 h = \pi(3)^2(10)$.

Step 3: Square the radius and multiply: $3^2 = 9$, and $9 \times 10 = 90$, so $V_{\text{cylinder}} = 90\pi$.

Step 4: Find the hemisphere's volume, which is half a full sphere: $V = (1/2)(4/3)\pi r^3$.

Step 5: Cube the radius: $3^3 = 27$.

Step 6: Multiply: $(1/2)(4/3)(27) = (2/3)(27) = 18$, so $V_{\text{hemisphere}} = 18\pi$.

Step 7: Add the two pieces together: $90\pi + 18\pi = 108\pi$.

Step 8: The total volume is 108π cubic meters.

Answer: 108π cubic meters



For any composite solid, find each recognizable piece's volume separately first, then add them together at the end.

Trap

It's tempting to use the full sphere volume formula for the domed top instead of remembering to cut it in half for a hemisphere.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
126π	Used the full sphere volume formula for the dome instead of halving it for a hemisphere, adding 36π instead of 18π .
90π	Correctly found the cylinder's volume but forgot to add the hemisphere's volume on top entirely.
96π	Used r^2 (9) instead of r^3 (27) when finding the hemisphere's volume, computing $(1/2)(4/3)\pi(9)$.

Shortcut

None faster here — composite solids always require each piece's volume to be found separately before adding.

Desmos Tip

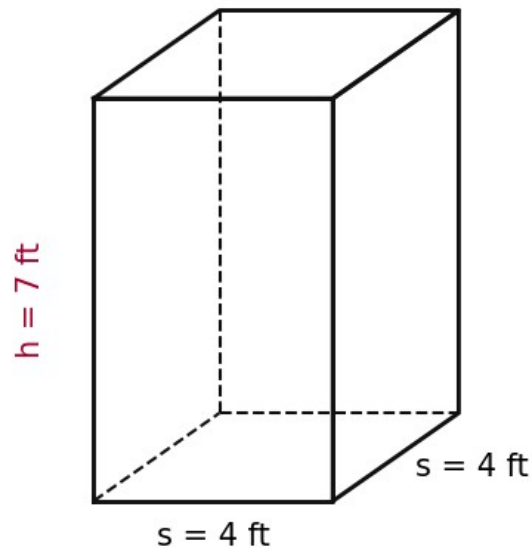
Type $3.14159 \cdot 3^2 \cdot 10 + (1/2) \cdot (4/3) \cdot 3.14159 \cdot 3^3$ in Desmos to confirm the total is about 339.3, matching 108π .

Volume, Surface Area & Similarity of Solids — Problem 8 (Medium)

A rectangular storage locker has a square base with side length 4 feet and an unknown height. The locker's volume is 112 cubic feet. What is the total surface area of the locker, in square feet?

Square-Base Prism — Storage Locker

$$V = s^2h = 112 \text{ ft}^3 \rightarrow h = 7 \text{ ft}$$



Formula / Rule Used: Prism volume ($V = s^2h$) to find height, then surface area ($S = 2(lw + lh + wh)$).

Solution

Step 1: First use the volume formula with the square base: $V = s^2h$, where $s = 4$.

Step 2: Plug in the known values: $112 = (4)^2h$.

Step 3: Simplify the squared term: $112 = 16h$.

Step 4: Divide both sides by 16 to find the height: $h = 112/16 = 7$.

Step 5: Now use the surface area formula with length 4, width 4, and height 7: $S = 2(lw + lh + wh)$.

Step 6: Find each product: $lw = 4 \times 4 = 16$, $lh = 4 \times 7 = 28$, $wh = 4 \times 7 = 28$.

Step 7: Add the three products: $16 + 28 + 28 = 72$.

Step 8: Multiply by 2: $72 \times 2 = 144$, so the surface area is 144 square feet.

Answer: 144 square feet

Tip

When a problem gives volume but asks for surface area, find the missing dimension first, then switch formulas.

Trap



After correctly finding the height (7), it's tempting to stop and report that as the final answer instead of continuing on to surface area.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
7	Correctly found the height but stopped there instead of continuing on to compute the surface area.
72	Correctly found the sum of the three face-pair products (72) but forgot to double it.
480	Divided 112 by the side length (4) instead of the side squared (16) when finding the height, getting $h = 28$ instead of 7.

Shortcut

None shorter — the missing height must be found first before the surface area formula can be applied.

Desmos Tip

Type $112/4^2$ in Desmos to confirm the height is 7, then check each face-area product before adding them together.



You're Ready

Before Test Day

- Revisit the terminology tables for any skill where the vocabulary still feels unfamiliar.
- Redo any Medium problem you got wrong — the twist it tested will very likely reappear.
- Time yourself on a few problems per skill to build speed, not just accuracy.
- Keep practicing on the Digital SAT's actual Desmos calculator, not a different graphing tool.
- This Foundation Course is your on-ramp — CLC's full Digital SAT Theory Companion (Parts 1 & 2) is the next step after the PSAT.